

Atomic Sanskrit

Source & Verification Dossier

Parag Tope

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Preface

This volume is the companion *Source & Verification Dossier* to the printed book *Atomic Sanskrit: The Linguistic Architecture of Sanātan*. The printed book carries one-sentence endnotes — citation anchors, named references, the load-bearing sentence the printed-book reader needs at the point of citation. The dossier carries the full long-form: complete bibliographic citations, primary-source quotes, verification trails, source-history discussions, structural-significance analysis. The two volumes share a single source. The printed book's endnote at “*See expanded endnote for X*” points here.

The dossier is for the reader the printed book respects but does not need to slow down for: the serious reader, the reviewer, the critic, the hostile academic. The case the printed book makes is the case the dossier documents. Every claim, every citation, every primary source. The architecture is on the ground; the dossier is the apparatus that lets a determined reader verify it.

How to navigate

Each entry begins with its **stub-name** as a section heading — a lowercase-hyphenated identifier like `sanskrtam-morphology` or `eleven-pathas`. The same stub-names appear in the printed book's endnotes; both volumes share the identifiers.

Each entry's structure:

- The stub-name as section heading.
- **Short:** *The one-sentence summary that appears in the printed book's endnotes section. Preserved here as a quick preview when scanning.*
- **Deployments:** *Chapter and section references showing where the entry is cited in the printed book.*
- The full long-form content: citation, primary-source quotes, verification trail, source-history discussion, structural-significance analysis.

To find a specific entry: use your PDF reader's search function on the stub-name. Stub-names are stable identifiers; pointers from the printed book land here exactly.

Ordering: entries follow the topical clustering of the underlying source file. The order is not strictly chronological with respect to the printed book's chapter sequence; the *Deployments* line on each entry supplies the chapter-by-chapter reverse index.

Endnotes

samskr̥tam-morphology

Short: *saṃskṛta* (संस्कृत) = *saṃ-* (सम्, totality, completion) + *kr̥ta* (कृत, past participle of $\sqrt{kr}$ (कृ), primary creation rather than combinatorial joining); the dual translation *perfectly synthesized / wholly created* preserves both axes English has no single word for. See Expanded Endnotes for the morphological argument.

Deployments: Preface ¶17; Chapter 1 ¶7 (*both deployments use the same canonical gloss block in the main prose; this endnote provides the underlying grammatical and semantic analysis*).

The compound संस्कृत (*saṃskṛta*) is formed from two morphological elements, each carrying a precise semantic load.

Sam- (सम्) is a Sanskrit *upasarga* — a verbal prefix that attaches to a verbal root and modifies its meaning. The semantic field of *saṃ-* gathers around totality and completeness. Its mathematical sense, preserved most clearly in the English cognate *sum*, is the *whole reached by gathering*. Its completion sense is the *full extent attained*. Its perfection sense, preserved in derivatives like English *summit* (Latin *summus*, the highest, the most complete), is the *finished and proper state*. Greek *syn-* shares the same family of senses. When *saṃ-* attaches to a verbal root in Sanskrit, it pushes the root's action toward this totality / completion / perfection axis — the action carried to its full and proper extent.

The verbal root **kr̥** (कृ), to which *sam-* attaches here, is among the most foundational of all Sanskrit *dhātavaḥ*. It is enumerated by Pāṇini in the *Dhātupāṭha* (the canonical inventory of Sanskrit roots) as a member of the *tanādi* class. Its semantic range is wide but consistent: *make, do, create, produce, perform an action, bring into being*. The English verb closest in semantic range is *make* — covering everything from making physical objects to making a poem to making a sound. The Sanskrit derivatives that radiate from this root show how load-bearing it is for the language’s vocabulary of action: **karma** (the deed, the made-thing, the action), **kāryam** (that which is to be done or made), **kṛti** (a created work, a composition), **kartṛ** (the doer, the agent), **kāraka** (the case-relations to a verb that name the semantic roles of action).

Crucially, **kr̥** does not mean *assemble*. Sanskrit has other verbal roots for that operation: *yuj* (to yoke, to join), *sandhā* (to put together). *Kṛ* names primary creative action — the bringing into being, not the combinatorial joining of pre-existing parts. This distinction is load-bearing for the translation choices argued below.

The past participle **kṛta** (कृत) is the *kṛdanta* formation: *kr̥* + the *-ta* suffix governed by *Aṣṭādhyāyī* 3.2.102 (*niṣṭhā*). The resulting form means *made, done, created, produced, accomplished*. Used adjectivally or substantively, it names the completed state of the *kr̥*-action.

The compound *sam-* + *kṛta* yields **saṃskṛta**. The sandhi transformation — the *m* of *sam-* assimilates to *anusvāra* (ṁ) before the velar *k* — is governed by the rules at *Aṣṭādhyāyī* 8.3.23 and following. The neuter nominative singular form, used when referring to the language, takes the *-am* ending: *saṃskṛtam* (संस्कृतम्).

The compound is *karmadhāraya* — adjective-noun in structure, with the prefix qualifying the past participle’s sense. The result names a state in which the *kr̥*-action has been carried to its *sam-* completion: the made-completely, the created-as-a-whole, the produced-in-its-totality.

saṃskṛta operates across multiple domains of Sanskrit usage. It names the consecrating state produced by the *saṃskāra* rites — the sixteen *ṣoḍaśa-saṃskāra* life-cycle ceremonies that bring a person through stages of ritual preparation, from *garbhādhāna* (conception) through *antyeṣṭi* (the final rite). It names refined materials — the smelted metal, the polished gemstone, the prepared food. It names a cultivated person — one whose conduct has been brought to its proper finished state through training and discipline. And it names the language: the linguistic system that has been made completely, made with nothing missing, made as a unified whole. All these senses operate simultaneously when the language is named *saṃskṛtam*. The polysemy is not coincidence; the architects of Sanskrit chose this name because it carries multiple senses at once.

On the choice of two translations. Two English glosses are offered side-by-side as the closest approximation of the compound's full sense: **perfectly synthesized** and **wholly created**. Each picks up a distinct axis of the original.

Perfectly synthesized picks up the *saṃ-* axis as perfection and the *kr̥* axis as deliberate production. *Perfectly* carries the *saṃ-* sense of *brought to its proper and complete state*. *Synthesized* carries the *kr̥* sense of *produced through deliberate combination* — emphasizing the engineering account. This translation is most useful in passages that argue for Sanskrit's engineered character.

Wholly created picks up the *saṃ-* axis as totality and the *kr̥* axis as primary creation. *Wholly* carries the *saṃ-* sense of *as a complete totality, with nothing missing*. *Created* carries the *kr̥* sense of *brought into being as a primary act, not combined from prior parts*. This translation is most useful in passages that emphasize the comprehensive scope and unified character of the system.

Single-word translations have been considered and rejected:

- **Assembled** is rejected outright. Assembly implies the components existed separately first and were then combined. *Kṛ*

names primary creative action, not combinatorial joining; the components of *saṃskṛtam* are not pre-existing parts brought together but elements brought into being as a system. Translating *saṃskṛtam* as *assembled* imports a semantic field — the workshop, the kit of pre-cut parts — that the original word does not carry.

- **Constructed** is available as a secondary descriptor and the book uses *architecture* and *engineering* in adjacent passages. But *constructed* alone compresses *kṛ* toward a building-trade sense and loses the full primary-creation force the root carries.
- **Refined** captures the polysemy's cultivation axis but loses the engineering axis entirely.
- **Perfected** alone loses the *made* sense and risks introducing a moralistic value-judgment that the Sanskrit compound does not carry.
- **Completed** captures *sam-* but understates *kṛ* — Sanskrit was not just completed; it was *created* and brought to completion.

The dual-translation approach is therefore not aesthetic redundancy. It is necessary because English does not have a single word that carries both the totality / perfection axis (the *sam-* contribution) and the primary-creation axis (the *kṛ* contribution) simultaneously. The two glosses are offered together — *perfectly synthesized or wholly created* — as the closest English approximation of what *saṃskṛtam* says, in its own root structure, about itself.

paspashahnika-apabhramsa-passage

Short: Patañjali's *Paspaśāhnika* (पस्पशाह्निक) — the *Mahābhāṣya* (महाभाष्य) opening; Kielhorn ed. vol. I, p. 2, lines 13–15 — states the *bhūyāṃso 'paśabdāḥ, alpīyāṃsaḥ śabdāḥ* (भूयांसोऽपशब्दाः, अल्पीयांसः शब्दाः) asymmetry maxim and exemplifies it with the *gauḥ* (गौः) → *gāvī* / *goṇī* / *gotā* / *gopotalikā* corruptions in one continuous passage,

split across §5.2 / §5.3 for pedagogical reasons; *apaśabda* (अपशब्द) and *apabhraṃśa* (अपभ्रंश) are deployed as near-synonyms across consecutive clauses.

Deployments: Chapter 5 §5.2 ¶1 (the *bhūyāṃso* asymmetry maxim); Chapter 5 §5.3 ¶1 (the *gauḥ* canonical example with four corruptions) (both citations anchor to the same continuous passage from the *Paspasāhnika* of Patañjali's *Mahābhāṣya*; the chapter's pedagogical split into two sections obscures the textual unity that Patañjali's own connector तद्यथा (*tadyathā*) makes explicit; this endnote restores the full passage).

The two lines cited in §5.2 and §5.3 are one continuous passage from the *Paspasāhnika* — the opening *āhnika* of Patañjali's *Mahābhāṣya*, in which the foundational positions of the *vyākaraṇa* discipline are stated. The chapter quotes the two halves separately to keep each pedagogical move clean; the underlying text is one flowing statement.

The full passage as printed in the Kielhorn standard text (Kielhorn ed. 1880; Kielhorn-Abhyankar BORI revision 1962-1972; *Mahābhāṣya* vol. I, p. 2, lines 13-15):

भूयांसोऽपशब्दाः, अल्पीयांसः शब्दाः । एकैकस्य हि शब्दस्य बहवोऽपशब्दाः ।
तद्यथा — गौरित्यस्य शब्दस्य गावी गोणी गोता गोपोतलिकेत्येवमादयोऽपभ्रंशाः
।

*bhūyāṃso 'paśabdāḥ, alpīyāṃsaḥ śabdāḥ. ekaikasya
hi śabdasya bahavo 'paśabdāḥ. tadyathā — gaur ity
asya śabdasya gāvī goṇī gotā gopotalikety evam ādayo
'pabhraṃśāḥ.*

“Many are the faulty-words (*apaśabdāḥ*); few are the (correct) words. For each one word, indeed, there are many faulty-words (*apaśabdāḥ*). To wit — of the word *gauḥ*, the corruptions (*apabhraṃśāḥ*) are *gāvī, goṇī, gotā, gopotalikā*, and so on.”

Note on Patañjali's term-switch. The maxim's first two clauses use *apaśabda* (अपशब्द, “faulty word, non-word” — the sharper pejorative formed with *apa-* + *śabda*); the *tadyathā* example clause switches to *apabhraṃśa* (अपभ्रंश, “falling-away, corruption” — the more neutral descriptive form from *apa-* + *bhraṃś*). Patañjali deploys both terms within the same continuous passage, treating them as near-synonyms designating the same phenomenon: *apaśabda* foregrounds the “wrong-word” framing; *apabhraṃśa* foregrounds the “fall-from” framing. The chapter's term-of-art is *apabhraṃśa* (which carries the engineering-decay account developed across Chapters 5 and 13); the *apaśabda* / *apabhraṃśa* near-synonymy is itself evidence of the textual unity the passage demonstrates — the same Patañjalian passage labels the *gauḥ* variants with both terms across consecutive clauses.

Three load-bearing observations follow from the unified passage that the split presentation in §5.2 and §5.3 cannot make visible on its own.

First, the *bhūyāṃso* maxim and the *gauḥ* example are not two independent claims that the chapter has aligned for rhetorical purposes. They are one argument made by Patañjali in one passage. The middle clause — *ekaikasya hi śabdasya bahavo 'pabhraṃśāḥ*, “for each one word, indeed, there are many corruptions” — is the structural bridge. It restates the general asymmetry (few correct words, many corruptions) at the per-word level (each correct word has many corruptions of its own). The *gauḥ* example then exemplifies the per-word claim with the four canonical variants. The general claim, the per-word restatement, and the worked example are one demonstrative sequence.

Second, the connector *tadyathā* (तद्यथा) — “to wit,” “as for instance,” “by way of example” — is a standard *Mahābhāṣya* formula used to attach a worked example to a structural claim. Its presence here marks the *gauḥ* variants as Patañjali's chosen exemplification of the per-word asymmetry, not a separate observation. The passage is doing what any rigorous technical exposition does — stating the general principle, restating it at the level of generality the example will bear on, and producing the example.

Third, the prose form is *bhāṣya* — commentarial prose — not metrical *śloka* (verse). The *Mahābhāṣya* is overwhelmingly prose commentary on Pāṇini's *sūtras* and Kātyāyana's *vārttikas*, with embedded *śloka-vārttikas* at certain points. This passage is *bhāṣya* prose. Secondary literature occasionally references the line as a *śloka*; precision matters. The form does not weaken the citation — the *Paspaśāhnikā*'s opening positions carry the full canonical weight of the *vyākaraṇa* discipline regardless of prose-versus-verse form.

The passage is the *locus classicus* for the *apabhraṃśa* phenomenon in the Indic *vyākaraṇa* discipline. Every later reference to *apabhraṃśa* in the *vyākaraṇa* literature — and every later defense of the *siddha* axiom (Ch4 §4.2) against the entropy the *apabhraṃśas* embody — refers back, explicitly or implicitly, to this passage. The chapter splits the passage across §5.2 and §5.3 because the two halves of the argument (quantitative asymmetry; per-word worked example) need to land in sequence, with the three-frames analysis of §5.4 already prepared by the time the *gauḥ* variants are introduced. The textual unity restored here is what the *Mahābhāṣya* itself preserves — a single passage in which the grammarian states the general claim, names the per-word structure, and produces the example, in the order in which the demonstration proceeds.

pre-pie-dictionary-shift

Short: The displacement of Sanskrit from the terminus of Indo-European etymological chains to one daughter alongside others (with PIE inserted as reconstructed terminus) is a two-century arc — Bopp 1816 (Sanskrit as ancestor) → Schleicher 1861 (common source distinct from Sanskrit) → mid-20c (academic references) → 2000s (popular references); *Merriam-Webster's Collegiate 10th* (1993) still showed Sanskrit-at-terminus for *mother*. See Expanded Endnotes for the staged chronology.

Deployments: Chapter 18 §18.1 worked example for *mother* — the dictionary-shift paragraphs showing how the etymological chain's terminus has moved from Sanskrit (early-19th-century philology and popular references into the late 20th century) to reconstructed PIE (the contemporary standard).

The displacement of Sanskrit from the source position of Indo-European etymological chains happened in identifiable stages across roughly two centuries of philological development.

Stage 1 — Early-19th-century philology: Sanskrit as ancestor.

Franz Bopp's *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache* (1816) opened the comparative-philological project by treating Sanskrit verbal morphology as the structural anchor against which Greek, Latin, Persian, and Germanic verbal systems were compared. Bopp's *Vergleichende Grammatik der Sanskrit-, Send-, Armenischen-, Griechischen-, Lateinischen-, Litauischen-, Altslavischen-, Gothischen- und Deutschen* (1833–1852) extended the comparison across the family. August Friedrich Pott's *Etymologische Forschungen auf dem Gebiete der indo-germanischen Sprachen* (1833–1836) carried the etymological work into vocabulary. Throughout this early period, Sanskrit was treated as the ancestor — or as the form closest to a (then-unnamed) common ancestor — across the Indo-European family.

Stage 2 — Mid-to-late-19th-century shift: from Sanskrit-as-ancestor to common-source.

August Schleicher's *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (1861–1862) formalized the family-tree model with a hypothesized common ancestor distinct from Sanskrit. The shift was gradual: Hensleigh Wedgwood's *Dictionary of English Etymology* (1859–1865) and Walter Skeat's *An Etymological Dictionary of the English Language* (1879–1882) gave Sanskrit cognates prominent position but increasingly framed them as siblings rather than ancestors of the European cognates.

Stage 3 — Late-19th to mid-20th-century academic consolidation: PIE reconstruction methodology cements. The neogrammarian project (Karl Brugmann’s *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen*, 1886–1893, revised 1897–1916) systematized the comparative method and built out the apparatus for reconstructing a proto-language distinct from any attested form. *Proto-Indo-European* as a stable term enters the literature by 1905. The *PIE* abbreviation enters routine academic usage in mid-twentieth-century scholarship.

Stage 4 — Mid-20th century: PIE displaces Sanskrit in mainstream academic references. The Oxford English Dictionary (first edition completed 1928), Julius Pokorny’s *Indogermanisches etymologisches Wörterbuch* (1959), and the American Heritage Dictionary’s IE Roots Appendix (1969, prepared by Calvert Watkins) — these standard reference works build the PIE-anchored etymological chain into the academic machinery. In academic-tier references after about 1960, Sanskrit cognates are presented alongside Latin, Greek, and Germanic forms as one daughter language among siblings; the chain’s terminus is the reconstructed PIE form, not Sanskrit.

Stage 5 — Late-20th-century popular-tier residue: 1990s desk dictionaries often retained Sanskrit-at-terminus chains. Popular desk dictionaries (Merriam-Webster’s Collegiate, Webster’s New World Dictionary editions, Random House College Dictionary editions, Collins editions, mass-market reference works) frequently did not carry the academic PIE machinery. Their etymological chains continued to show Sanskrit cognates as the deepest cited form — not because the editors rejected PIE, but because popular references abbreviated the chain at the deepest attested language rather than reconstructing further back.

Verified instance. Merriam-Webster’s Collegiate Dictionary, 10th Edition (Merriam-Webster Inc., Springfield MA, 1993), s.v. “mother”:

moth-er \ 'mə-thər\ n [ME moder, fr. OE modor;

akin to OHG muoter mother, L mater, Gk $m\{\tilde{e}\}t\{\tilde{e}\}r$
 Skt mātr] (bef. 12c)

The chain runs Middle English → Old English with an “akin to” cognate list, and the deepest cited cognate is Sanskrit *mātr*. No starred Proto-Germanic or PIE form appears. The entry for *yoke* in the same edition follows the same pattern with Sanskrit *yuga* as the deepest cited cognate. Both entries are accessible via the Internet Archive scan of the 10th edition. (Notably, the current Merriam-Webster online dictionary still carries the MW10 etymology for *mother* verbatim — Sanskrit-at-terminus, no PIE. The PIE-displaced framing is sharpest against the *American Heritage Dictionary* 3rd edition, 1992, and the online resources that have followed it — etymonline, Wiktionary, dictionary.com.)

Stage 6 — 2000s onward: popular-tier cementing. Online etymological references (etymonline.com, free dictionary aggregators) standardized on PIE-anchored etymologies during the 2000s and 2010s. The Mallory & Adams *Encyclopedia of Indo-European Culture* (1997), Helmut Rix’s *Lexikon der indogermanischen Verben* (first edition 1998; second 2001), Michiel de Vaan’s *Etymological Dictionary of Latin and the Other Italic Languages* (2008), and Robert Beekes’s *Etymological Dictionary of Greek* (2010) — all published in the recent quarter-century — built out the academic reference ecosystem that has cemented PIE as the routine endpoint. The popular tier has followed the academic tier; the 1990s residue of Sanskrit-at-terminus chains has been substantially eliminated in routine reference.

The displacement is therefore not a single event but a two-century arc, with the cementing of the contemporary state visible largely in the past quarter-century.

Verification status. Stages 1–4 are reasonably well-established in the standard history of comparative philology and can be confirmed against any history-of-linguistics reference (e.g., R. H. Robins, *A Short*

History of Linguistics; Anna Morpurgo Davies, *Nineteenth-Century Linguistics*). Stage 5 (the 1990s popular-tier residue) is the subject of an active verification task in `working/as_verification_todo.md`; specific 1990s dictionary entries that show Sanskrit-at-terminus chains need primary-source verification before being cited as canonical evidence. Stage 6 is empirically observable in contemporary references.

schleicher-1868-fable

Short: August Schleicher, “Eine fabel in indogermanischer ursprache,” *Beiträge zur vergleichenden Sprachforschung* 5 (1868): 206–208 — the first complete text composed entirely in reconstructed PIE (*Avis akvāsas ka*), every word marked with the asterisk convention Schleicher had introduced in his 1861 *Compendium*; the fable has been rewritten by Hirt (1939), Lehmann-Zgusta (1979), and Kortlandt (2007) on incompatible reconstructions.

Deployments: Chapter 18 §18.1 ¶3 — the opening establishment of Schleicher’s *Avis akvāsas ka* as the first complete text in reconstructed PIE and the operational debut of the asterisk-before-reconstructed-form convention.

August Schleicher, “Eine fabel in indogermanischer ursprache,” *Beiträge zur vergleichenden Sprachforschung auf dem Gebiete der arischen, celtischen und slawischen Sprachen* 5 (1868): 206–208. The fable carries the title *Avis akvāsas ka* — “The Sheep and the Horses” — and is the first complete piece of connected text composed entirely in a reconstructed proto-language. Every word in the fable is marked with the asterisk convention: **ávis*, **akvāsas*, **kṛnuto*, **w̥lqis*, **ágrom* — each form a reconstruction, none attested in any speaker’s language.

The asterisk-before-reconstructed-form convention itself is Schleicher’s earlier invention, established in his *Compendium der*

vergleichenden Grammatik der indogermanischen Sprachen (Weimar: Hermann Böhlau, 1861; second edition 1866). The *Compendium* is the foundational statement of the *Stammbaumtheorie* (family-tree model) and the systematic application of reconstruction to the Indo-European family. The 1868 fable made the asterisk convention fully operational at literary scale — a working showcase of a reconstructed language run at sentence and paragraph length, demonstrating that the convention could carry the weight of an entire text.

A historical detail worth naming. The fable has been *rewritten* by later philologists multiple times as the reconstruction project's standard reconstructions have shifted: Hermann Hirt produced a revised version in 1939; Winfred Lehmann and Ladislav Zgusta produced another in 1979; Frederik Kortlandt produced a further revision in 2007. The successive versions diverge substantially from Schleicher's 1868 text — different sound laws, different morphology, different lexical choices. The chapter does not dwell on this in the main prose, but the structural fact is worth flagging here: a text that has been rewritten three or four times across a century and a half, by different reconstructors operating on different theoretical commitments, is not a recovered ancient utterance. It is a continuously revised hypothesis dressed up in the genre of literary text. Schleicher's original is cited because it is the original — the first instance of the convention operating at scale — not because it carries any standing as a “correct” reconstruction.

The credit for the asterisk-before-reconstructed-form convention is widely attributed to Schleicher across the standard history of comparative philology (see R. H. Robins, *A Short History of Linguistics*, 4th ed., 1997; Anna Morpurgo Davies, *Nineteenth-Century Linguistics*, in the Routledge *History of Linguistics* series, 1998). Earlier philologists used asterisks for other purposes (marking ungrammatical forms, marking conjectural readings in textual criticism); Schleicher's innovation was the specific repurposing for forms internal to

the reconstruction project — forms whose status is precisely *reconstructed*, not attested.

jakobson-1959-nursery-words

Short: Roman Jakobson, “Why ‘Mama’ and ‘Papa’?” (1959 lecture, published 1960 in *Perspectives in Psychological Theory*) — argues *mama* / *papa* type kinship terms cluster cross-linguistically because of infant phonological universals, not genetic cognation; the deflection touches *mother* but has no purchase on *yoke* (not a kinship term, not phonologically universal), which is why the chapter pairs both worked examples.

Deployments: Chapter 18 §18.5 ¶3 — the *yoke* paragraph, where the nursery-word deflection is named and shown to have no purchase on a non-kinship cognate set.

Roman Jakobson, “Why ‘Mama’ and ‘Papa’?” in Bernard Kaplan and Seymour Wapner (editors), *Perspectives in Psychological Theory: Essays in Honor of Heinz Werner* (New York: International Universities Press, 1960), pages 124–134. The paper was delivered as a 1959 lecture and is conventionally cited by the lecture date in linguistic literature; the published volume carries 1960. Both year-references appear in the secondary literature; the chapter’s “1959” follows the lecture-date convention.

Jakobson’s argument: the cross-linguistic pattern in which the kinship terms for mother and father cluster around CV-syllable forms — *mama* / *māmā* / *mātr̥* / *mater* / *mother* / *mère* for mother; *papa* / *pāpā* / *pit̥r̥* / *pater* / *father* / *père* for father — is not evidence of genetic cognation across the languages involved. It is, on Jakobson’s account, evidence of a phonological universal grounded in infant articulation. Infants across all known languages produce the labial and dental nasal-stop sequences (bilabial /m/ and /p/, dental /t/ and

/d/) at early developmental stages, ahead of more articulatorily demanding sounds. Cultures across the world have taken these early-babble syllables and assigned them as kinship terms — converging on similar surface forms not because the languages share an ancestor but because they share a biology. The implication is that *mama*-type and *papa*-type kinship terms across unrelated language families (Mandarin *māma*, Swahili *mama*, Mongolian *eej* but with the same CV structure in many cases) cannot be used as evidence of common descent or contact.

The deflection has standing in mainstream comparative-philological practice. Where the Sanskrit *mātr* / Greek *mētēr* / Latin *māter* / English *mother* chain is presented as evidence of an Indo-European inheritance, an orthodox linguist can — and routinely does — reply that the *mātr*-type clustering reflects nursery-word universals as much as genetic cognation; the cognate-set evidence from kinship terms is therefore softer than the cognate-set evidence from less universally-attested vocabulary.

The chapter's response — and the load-bearing reason *yoke* is deployed as the secondary worked example after *mother* — is that the deflection has narrow scope. *Yoke* is not a kinship term, not a CV-syllable, not phonologically universal, not an item infants babble. It is an implement-name, an agricultural-technology term, embedded in a specific material practice (the harnessing of draft animals) that is itself a Sanskrit-attested cultural inheritance. The nursery-word framework has nothing to say about *yoke*; the cognate chain Sanskrit *yuga* → Old English *geoc* → Middle English *yok* → Modern English *yoke* is recognized as a regular IE cognate set in every standard etymological reference and cannot be deflected by appeal to phonetic universals. The polemic structural move is to take the strongest deflection the orthodoxy has available and demonstrate that it does not reach. The argument from *mother* alone risks the nursery-word reply; the argument from *yoke* does not. Deploying both — *mother* as the personally-anchored example, *yoke* as the deflection-proof anchor —

closes the rhetorical opening the *mother* case alone leaves.

pie-cementing-recent-decades

Short: PIE has been cemented as the default etymological endpoint in routine reference during the past quarter century (Watkins's *American Heritage* IE Roots Appendix expanded across editions; etymonline launched 2001; Mallory-Adams 1997; Rix 2001; de Vaan 2008; Beekes 2010) — the same window in which dharmic-civilizational discourse first emerged from its long sequence of constraints; an apparatus-level hardening of the *progressive orthodoxy* the chapter argues is not coincidence.

Deployments: Chapter 18 §18.4 close — the cementing-of-PIE-in-recent-decades paragraph that closes the third-pillar diagnosis and prepares the dictionary-shift worked examples that follow in §18.5.

The cementing of PIE as the default endpoint of routine etymological reference has accelerated in roughly the past quarter century. Calvert Watkins's *Indo-European Roots Appendix* in *The American Heritage Dictionary*, present since the first edition in 1969 and substantially expanded across the third (1992), fourth (2000), and fifth (2011) editions, has long been the most prominent IE-anchoring apparatus in any flagship English dictionary. Douglas Harper's *Online Etymology Dictionary* launched in 2001 and grew into the default free online etymological reference for English; it gives PIE as the default endpoint of every etymological chain. The reference ecosystem that anchors this routine usage was substantially built out in the same window: J. P. Mallory and Douglas Q. Adams, *Encyclopedia of Indo-European Culture* (1997); Helmut Rix, *Lexikon der indogermanischen Verben* (2001); Michiel de Vaan, *Etymological Dictionary of Latin and the other Italic Languages* (2008); Robert Beekes, *Etymological Dictionary of Greek* (2010). Each takes PIE as the etymological terminus by de-

fault. Together they form the apparatus that subsequent dictionaries, college texts, and online reference works lean on. The English-speaking reader who looked up a word in *Webster's* in 1995 met etymologies stopping at Latin, Greek, or Sanskrit. The same reader online today meets etymologies that pass through those proximate sources to arrive at PIE. The shift is real, observable, and recent.

The temporal correlation with the broader civilizational moment deserves naming. The window in which PIE was being cemented in routine Western reference is the same window in which India's dharmic-civilizational discourse was first emerging from the long sequence of constraints that had bound it: the centuries of Islamic political dominance that destroyed temples, libraries, and *gurukula* lineages across the subcontinent; the colonial English period that built the philological ecosystem the book's central chapters engage; and the secular Indian establishment that succeeded English rule and continued, in different vocabulary, the work of containment the colonial framework had begun — the framing of dharmic continuity as “communalism,” the structural marginalization of *guru-shishya paramparā* in state education, the confinement of Sanskrit and the *Vedas* to museum status. The chains were beginning to come undone only in the decades the reader has lived through. Indian intellectuals and Sanātan practitioners, no longer required to defer to Mughal or British or Nehruvian-secular orthodoxies, were beginning — for the first time across many generations — to ask the questions the engineered Sanskrit thesis presses. At precisely this moment, and not before, the routine reference ecosystem that anchors English-language etymology to PIE was being substantially hardened.

This temporal coincidence is, on the structural account developed across Chapter 2 §2.5 and the present chapter, not coincidence. The architecture of containment named in Chapter 2 has continued to operate at the level of routine reference, building outward through dictionaries, online resources, and college etymological style during exactly the period when the historical pillars (the Aryan thesis,

the Noachian chronology) were weakening and when alternative accounts of Sanskrit's depth were first becoming articulable. Chapter 3 names the formation that performs this institutional work: the *church of progress* — the academy as institutional carrier — operating its catechetical machinery at the routine-reference level, with its *missionaries of progress* extending the framework into the territories where the dharmic recovery is reaching, hardening the *progressive orthodoxy* at the apparatus level during exactly the window when alternatives are emerging. The third pillar — linear-progress teleology — does its load-bearing defense work not in the contested high theory but in the everyday reference ecosystem the next generation of students, scholars, and engaged readers will inherit. Each PIE-anchored etymology, deployed casually in a search-engine result or a dictionary entry, hardens the ancestor account by exposure. The work is invisible because it is everywhere.

The polemic register here is structural, not personal. The pattern is what an architecture of containment produces when it operates without conscious individual direction — apparatus-level defense at exactly the points where alternatives are emerging. Individual lexicographers, Indo-Europeanists, and etymological-reference editors have made the choices they made for the reasons their disciplines authorize; the cumulative pattern is the perimeter the chapter's analysis predicts. The solidification of PIE in routine reference during the past quarter century is a single observable case of the architecture functioning as Chapter 2 described it.

missionaries-of-progress-precedent

Short: Parag Tope, "Missionaries of 'Progress,'" *Quick Take* (<https://quicktake.wordpress.com/2011/10/29/missionaries-of-progress/>), October 29, 2011 — the author's earlier deployment of the contemporary-NGO-as-missionary structural account that

the *fourth Abrahamic religion* cluster vocabulary in this chapter formalizes.

Deployments: Chapter 3 §3.4 ¶1 — the formal introduction of *missionaries of progress* as one of the six standing cluster terms.

The phrase *missionaries of progress*, used in this chapter as a standing structural term for the function-class that exports the orthodoxy's framework into civilizations that have their own, has antecedents in the author's earlier writing. The structural-religious account of NGO and developmental work as missionary work — Western progress values exported under the cover of universal applicability — was deployed in Parag Tope, "*Missionaries of 'Progress'*", *Quick Take* (<https://quicktake.wordpress.com/2011/10/29/missionaries-of-progress/>), October 29, 2011. The 2011 post argues that contemporary NGOs operate as the modern continuation of missionary work, promoting Western values rather than Jesus, with the structural mechanism preserved across the secularization. The cluster vocabulary formalized across this chapter is the systematization of analytical moves the author has been developing across more than a decade of prior writing.

fourth-abrahamic-eschatology-precedent

Short: Parag Tope, "A Fart Tax and a Pink Revolution Can 'Save the World'," *Quick Take* (<https://quicktake.wordpress.com/2012/12/06/indias-pink-revolution-can-save-the-world/>), December 6, 2012 — earlier deployment of the climate-orthodoxy-as-religious-formation framing (the *GaWD* (*Global Warming Deity*) coinage) that the *fourth Abrahamic religion* cluster vocabulary formalizes.

Deployments: Chapter 3 §3.1 ¶8 — the live-eschatology paragraph naming the fourth Abrahamic religion's contemporary doomsday register.

The account of contemporary climate-progress discourse as a religious formation with its own deity, eschatology, and doomsday cult was deployed in Parag Tope, “A Fart Tax and a Pink Revolution Can ‘Save the World’”, *Quick Take* (<https://quicktake.wordpress.com/2012/12/06/i-pink-revolution-can-save-the-world/>), December 6, 2012. The 2012 post deploys the coinage **GaWD** (*Global Warming Deity*) to name the climate orthodoxy as a religious formation with implicit deity-imputation, and frames climate activism as a “doomsday cult” with the structural template the present chapter formalizes: a named doom (climate catastrophe), prescribed practices for averting it (agricultural emission credits, consumption regulation), and a recalcitrant out-group (Indian agriculture, the developing world more broadly) whose resistance threatens the avoidance. The 2012 post and the 2011 *Missionaries of ‘Progress’* piece together establish the analytical frame the *fourth Abrahamic religion* cluster vocabulary formalizes — a frame the author has been developing across more than a decade.

brahmi-devanagari-structural-identity

Short: Brāhmī (ब्राह्मी) and Devanāgarī (देवनागरी) are structurally identical at the encoding-system level (abugida structure, *varṇamālā* (वर्णमाला) inventory, *sthāna* (स्थान) / *prayatna* (प्रयत्न) organization, conjunct formation) and differ only at the surface (glyph shapes, *śirorekḥā* (शिरोरेखा), stroke ductus, numerals); the architecture both Indic scripts share has no equivalent in Aramaic, so glyph-shape borrowing cannot produce the engineered specification.

Deployments: Appendix Part 3 §3.3 ¶2 — the structural-identity claim that Brāhmī implements the same engineered architecture as Devanāgarī.

The claim that Brāhmī and Devanāgarī are structurally identical op-

erates at the level of the *encoding system* — the phonetic specification the script implements — not at the level of glyph shapes. The two scripts share the following architectural features:

- **Abugida structure.** Each consonant glyph carries an inherent /a/; other vowels attach as positional modifications (diacritics) of the consonant glyph rather than as independent letters. Vowels written initially (without an accompanying consonant) have their own standalone glyphs.
- **The *varṇamālā* inventory.** Both scripts encode the full Sanskrit phonological inventory: the five-by-five *varga* matrix (velar / palatal / retroflex / dental / labial × unvoiced-unaspirated / unvoiced-aspirated / voiced-unaspirated / voiced-aspirated / nasal); the four semivowels (य र ल व); the three sibilants (श ष स) and the aspirate (ह); the vowel inventory (a ā i ī u ū ṛ ṝ ḷ e ai o au); the *ayogavāha* (*anusvāra*, *visarga*).
- **Order by *sthāna* and *prayatna*.** The consonants are organized by place of articulation (from back to front of the mouth) and manner of articulation, exactly as the *Prātiśākhya* / *Śikṣā* literature specifies. The order is the same in both scripts.
- **Conjunct formation.** Consonant clusters form ligatures combining the constituent consonants, with the inherent /a/ suppressed on the non-final members.
- **Writing direction.** Left-to-right, top-to-bottom.

The two scripts differ at the surface level — the level the orthodoxy's glyph-shape comparisons operate on:

- **Glyph shapes.** Brāhmī's letters and Devanāgarī's letters look different. Brāhmī's shapes are simpler and more angular; Devanāgarī's are more elaborate and curved.
- **The headline (*śirorekḥā*).** Devanāgarī's characteristic horizontal line running across the tops of letters is not present in Brāhmī.
- **Stroke ductus and ligature conventions.** The way strokes are formed and the way conjuncts are assembled differ in detail.

- **Numerical signs.** Brāhmī numerals differ from Devanāgarī numerals.
- **Later additions to Devanāgarī.** Devanāgarī acquired a few characters for foreign-language sounds (the dotted forms ञ, ॠ, ॡ, ॢ, ॣ, ।, ॥, ०, १, २, ३, ४) used in Persian and Arabic loanwords. These are not part of the core *varṇamālā* and were added long after Brāhmī's period.

The orthodoxy's Brāhmī-from-Aramaic case relies on glyph-shape resemblances between some Brāhmī letters and some Aramaic letters. Even granting the resemblances at face value, what they could establish is borrowing at the *surface* level — the visible shapes of certain letters — not at the *system* level. The encoding system (the *varṇamālā*, the abugida structure, the *varga* matrix, the vowel-diacritic apparatus) has no equivalent in Aramaic and could not have been borrowed from a source that does not contain it. The shared architecture between Brāhmī and Devanāgarī is the architecture that needed engineering; the shared architecture between Brāhmī and Aramaic does not exist.

Both scripts are visible renderings of the same underlying engineered specification. The specification is the *varṇamālā*. Brāhmī is one rendering of it; Devanāgarī is another. The other Indic scripts (Tamil, Bengali, Gujarati, Oriya, Kannada, Telugu, Malayalam, Sharada, Tibetan adapted, Sinhala, Burmese adapted, the Southeast-Asian descendants) are further renderings — different glyph shapes, the same engineered architecture, with regional variations in the inventory where the local phonology required them (Tamil's reduced inventory; the Tibetan and Southeast-Asian scripts' adaptations for local sounds). The *varṇamālā* is the engineering; the scripts are the visible interfaces. Chapters 14 §13.3 and Appendix Part 3 §§3.2–3.3 are operating on this distinction.

aksara-imperishable-name

Short: *Akṣara* (अक्षर) = *a-* (privative) + $\sqrt{kṣar}$ (क्षर, to flow, to perish) — *the imperishable*; the same word names both the writing-and-utterance primitive and the Upaniṣadic / Vedāntic *Brahman* (*Bhagavad Gītā* 8.3, *akṣaram brahma paramam* / अक्षरं ब्रह्म परमम्); contrasts with Latin *littera* (smear), Greek *γράφω* (scratch), Arabic *ḥarf* (edge), none of which claim non-decay at the level of the unit.

Deployments: Chapter 8 §8.5 (the *akṣara* introduction); Appendix Part 3 §3.4 (the audiograph = *akṣara* coinage).

The Sanskrit term **अक्षर** (*akṣara*) is morphologically *a-* (privative) + $\sqrt{kṣar}$ (to flow, to perish, to wear away) — literally *that which does not flow away* or *that which does not decay*. The same word names both the writing-and-utterance primitive (the syllable rendered as glyph) and the Upaniṣadic / Vedāntic name for *Brahman* itself: अक्षरं ब्रह्म परमम् (*akṣaram brahma paramam*), *the supreme imperishable Brahman* (*Bhagavad Gītā* 8.3, deployed across multiple Upaniṣadic passages). The *Muṇḍaka Upaniṣad* 1.1.5 distinguishes *para vidyā* (higher knowledge) as that by which *akṣaram* (the imperishable) is known; *Maitrāyaṇī Upaniṣad* and *Praśna Upaniṣad* extend the usage. The shared term is not coincidental — the Indic continuum treats the writing-primitive and the metaphysical principle of non-decay as carrying the same name because they share the same property: *that which does not perish*.

The contrast with other major writing traditions' terms is structural. **Latin** *littera* (letter) is from *lino* (to smear, to anoint) — a smeared mark on a surface. **Greek** *γράφω* (*gramma*) is from *γράφω* (*graphō*) (to scratch, to write) — *what is scratched* or *what is written*. **Arabic** *ḥarf* (حرف) means *edge* or *side* — a mark at the edge of a surface. None of these terms carries a non-decay claim; each names the surface mark or the act of marking. The Sanskrit *akṣara*, alone among the major writing traditions' words for the writing-primitive, asserts non-decay at the level of the unit itself. The *sanātan* claim is in the

vocabulary.

The morphological grounding is documentable from Pāṇini's *Aṣṭādhyāyī* (the privative *a-* derivation; *na-Tatpuruṣa* formation), the standard Sanskrit grammatical literature, and the Monier-Williams Sanskrit-English dictionary's principal entries. Document the *Gītā* 8.3 citation in full when the endnote is finalized.

chronology-asymmetry-rationale

Short: The book dates non-Indic figures and texts (Schleicher 1861, Bopp 1816, the Boden Chair 1832, Müller's EIC commission) because those dates are internal to the European-philological enterprise's own documentary record; the refusal to date Indic figures is a precisely targeted strategic position — withholding assent from the one apparatus the book is on trial against — not generalized chronological skepticism. See Expanded Endnotes for the full asymmetry argument.

Deployments: Preface ¶8 — the closing line of the chronology section that names the asymmetry as deliberate rather than as inconsistent mistrust.

The asymmetry between the book's treatment of Indic and non-Indic dating requires a separate framing because the casual reader might otherwise read the position as a generalized skepticism of dating-as-such. It is not. The non-Indic chronologies the book cites — Schleicher's 1861 *Compendium*, Bopp's 1816 *Conjugationssystem*, Müller's *East India Company R̥gveda* commission, the Boden chair endowment of 1832, the *Indogermanisches etymologisches Wörterbuch* of 1959 — are internal to traditions that operate chronologically without distortion. European philology kept its own minutes; the dates are part of the documentary record produced by the enterprise itself. Citing those dates does not import a hostile framework's interpretive judgments; it cites the framework's own self-record. The same applies

to Greek, Roman, Arabic, Tibetan, Chinese, and Egyptian chronologies cited in the book: each is internal to a tradition whose internal documentary practices the book has no quarrel with.

The mistrust is local and structural. It applies to the philological dating of Indic texts and figures because those datings were produced by the same nineteenth-century European-philological enterprise the book prosecutes across its central chapters. The dating is not neutral background; it is the framework's interpretive output — calibrated to produce the chronological sequence (Vedic → Pāṇinian → Classical → Prākṛitic → modern Indic) the framework needs to operate. To accept those dates is to accept the framework that produced them. The refusal is therefore not generalized but precisely targeted: it withholds assent from the one apparatus the book is on trial against. The strategic argument (the position is *refusal*, not counter-construction) is developed in the Epilogue.

bhagavad-gita-1-2-citation

Short: *Bhagavad Gītā* (भगवद्गीता) 1.2 — दृष्ट्वा तु पाण्डवानीकं व्यूढं दुर्योधनस्तदा । आचार्यमुपसङ्गम्य राजा वचनमब्रवीत् ॥ / *dr̥ṣṭvā tu pāṇḍavānikam vyūḍham duryodhanas tadā | ācāryam upasaṅgamyā rājā vacanam abravīt ||* (Sañjaya's narration as Duryodhana surveys the Pāṇḍava battle-formation); the *anuṣṭubh* (अनुष्टुभ्)-metered verse the Preface deploys for the personal-anchor scene.

Deployments: Preface ¶9 — the personal-anchor scene in which the author, as a schoolboy, encountered *Bhagavad Gītā* 1.2 during a lunch-break recitation and was struck by the structural question the verse posed.

The verse is *Bhagavad Gītā* 1.2, the second verse of the first chapter (*Arjuna-Viṣāda Yoga* — “The Yoga of Arjuna's Despondency”):

दृष्ट्वा तु पाण्डवानीकं व्यूढं दुर्योधनस्तदा । आचार्यमुपसङ्गम्य राजा वचनमब्रवीत्
॥

dr̥ṣṭvā tu pāṇḍavānīkaṃ vyūḍhaṃ duryodhanas tadā |
ācāryam upasaṅgamyā rājā vacanam abravīt ||

“Having seen the Pāṇḍava army arrayed in battle-formation, King Duryodhana, approaching his teacher, then spoke these words.”

The verse is spoken by Sañjaya — the narrator-charioteer of the *Mahābhārata* frame — reporting to King Dhṛtarāṣṭra the scene unfolding at Kurukṣetra. The first chapter of the *Gītā* establishes the dramatic situation: the two armies arrayed, Duryodhana surveying the *Pāṇḍava* formation, the dialogue between Kṛṣṇa and Arjuna about to begin. The verse cited here is the second link in the narrative chain that opens the *Gītā* and is among the first verses any student reciting the *Gītā* from the beginning encounters.

The verse’s metrical form is *anuṣṭubh* — the eight-syllable-per-pāda meter that carries most of the *Mahābhārata*’s narrative and most of the *Bhagavad Gītā*’s eighteen chapters (with select passages in *triṣṭubh*). The shape the boy noticed — the way the syllables fall into a rhythm that holds the meaning — is the *anuṣṭubh* doing its work. The structural observation the *Preface* is anchoring (Sanskrit’s poetry is engineered to hold; the engineering is in the meter, the *sandhi*, the word order; the language was built so the poetry could land this way) returns at scale across Chapters 7, 8, 16, and the recitation-lineages material across the back half of the book.

Sources for the verse text: the standard critical edition of the *Mahābhārata* prepared at the Bhandarkar Oriental Research Institute (Poona), volume 7 (*Bhīṣmaparvan*); the *Gītā Press Gorakhpur* edition (the standard popular Indian reference); any of the major commentaries (Śaṅkara, Rāmānuja, Madhusūdana Sarasvatī, Jñāneśvar) that take the verse as it stands. The transliteration here follows the standard IAST conventions used throughout the book.

briggs-1985-ai-magazine

Short: Rick Briggs, “Knowledge Representation in Sanskrit and Artificial Intelligence,” *AI Magazine* 6, no. 1 (Spring 1985): 32–39 (AAAI; written at NASA Ames Research Center) — the first widely-circulated articulation in mainstream English-language scientific literature of the Pāṇinian *vyākaraṇa* (व्याकरण) as a working knowledge-representation system anticipating late-twentieth-century AI semantic-network and frame-based representations.

Deployments: Preface ¶15 — the survey-of-prior-work paragraph that names Rick Briggs’s 1985 paper on Pāṇinian grammar and artificial intelligence as the first widely-circulated articulation of the formal-systems account of Sanskrit in mainstream English-language scientific literature.

Rick Briggs, “Knowledge Representation in Sanskrit and Artificial Intelligence,” *AI Magazine* 6, no. 1 (Spring 1985): 32–39. Published by the Association for the Advancement of Artificial Intelligence (AAAI), the paper was written while Briggs was a researcher at NASA’s Ames Research Center and circulated widely in AI and knowledge-representation circles in the late 1980s.

The argument: Sanskrit grammatical analysis, as developed in the *vyākaraṇa* discipline culminating in Pāṇini’s *Aṣṭādhyāyī*, produces a representational scheme for natural-language sentences that resembles — and in several respects anticipates — the semantic-network and frame-based representations developed in late-twentieth-century artificial intelligence. Briggs walks through the *kāraka* system (Sanskrit’s case-role analysis), the *sūtra*-based generative rules, and the principle that any Sanskrit sentence can be analyzed into an unambiguous case-role specification, and observes that the Sanskrit apparatus is doing — as a working system maintained across the *param-*

parā — what late-twentieth-century AI was attempting to engineer from scratch.

The paper’s reception was, as the *Preface* notes, polite and largely peripheral. *AI Magazine* circulates within the AI research community; the paper was read by computer scientists and computational linguists, occasionally cited in knowledge-representation literature, and never substantially engaged by mainstream Indology. The structural reason — that the implication of Briggs’s account runs counter to the philological orthodoxy’s account of Sanskrit’s character — is the subject the present book develops.

The paper is available through the AAAI’s *AI Magazine* archive (<https://ojs.aaai.org/aimagazine/index.php/aimagazine/article/view/466>), reprinted in several AI anthologies, and widely circulated through Sanātan-aligned scholarly networks. Citation: Briggs, Rick. “Knowledge Representation in Sanskrit and Artificial Intelligence.” *AI Magazine* 6, no. 1 (Spring 1985): 32–39.

kak-paninian-algorithmic

Short: Subhash Kak — across *The Astronomical Code of the Ṛgveda* (Aditya Prakashan, 1994; revised 2000), *Computing Science in Ancient India* (with T. R. N. Rao, USL Press, 1998), *The Architecture of Knowledge* (CSC / Motilal Banarsidass, 2004), and journal papers in *Cryptologia* and *Annals of the History of Computing* — develops the *Aṣṭādhyāyī* (अष्टाध्यायी) as a context-sensitive generative grammar with embedded metarule (*paribhāṣā* परिभाषा) control, anticipating Chomskyan formal-grammar machinery in working form.

Deployments: Preface ¶15 — the survey-of-prior-work paragraph; the second of three named precedents (Briggs, Kak, Staal) for adjacent engineering / formal-systems accounts of the Sanskrit continuum.

Subhash Kak has published from the 1980s through the present on Pāṇinian grammar, Indic mathematics, Vedic astronomy, and the algorithmic character of Indic intellectual systems. The principal book-length anchors: *The Astronomical Code of the Ṛgveda* (Aditya Prakashan, 1994; revised third edition Munshiram Manoharlal, 2000); *Computing Science in Ancient India* (co-edited with T. R. N. Rao; Centre for Advanced Computer Studies, University of Southwestern Louisiana, 1998; later Munshiram Manoharlal reprint); and *The Architecture of Knowledge: Quantum Mechanics, Neuroscience, Computers and Consciousness* (CSC / Motilal Banarsidass, 2004). The Pāṇini-as-algorithmic-system thread runs across these volumes and through a substantial body of journal papers in *Cryptologia*, *Annals of the History of Computing*, and self-archived essays. *The Wishing Tree* (iUniverse, 2008) extends the broader Indic-engineering account into accessible-essay form.

Kak's central claim, restated across the body of work: the *Aṣṭādhyāyī* is structured as a formal computational specification — *sūtras* operating as rewrite rules, with metarules (*paribhāṣās*) determining rule precedence, a base of phonological elements (the *Māheśvara-sūtras*), and a derivational apparatus that produces well-formed Sanskrit utterances from underlying representations. The system is, in formal-systems vocabulary, a context-sensitive generative grammar with embedded metarule control — anticipating in working form the formal-grammar tradition Western linguistics developed from Chomsky onward. The implication Kak draws — and which the *Preface* names — is that Pāṇini was operating within an engineering discipline that had already established the system; the *Aṣṭādhyāyī* documents what the discipline had built, rather than inventing the system itself.

Kak's *Computing Science in Ancient India* (with T. R. N. Rao, USL Press, 1998), *The Architecture of Knowledge* (CSC, 2004), and the various journal papers on Pāṇinian grammar are the principal references. The body of work is large; the *Preface*'s citation gestures at it as a coherent line of argument the present book extends.

staal-formal-systems

Short: J. Frits Staal (1930–2012) — across *Word Order in Sanskrit and Universal Grammar* (Reidel, 1967), *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, 1983; documenting the 1975 Nambūdiri Agnicayana (अग्निचयन) performance), and especially *Rules Without Meaning: Ritual, Mantras and the Human Sciences* (Peter Lang, 1989) — argues that Sanskrit ritual and grammar operate as formal systems with mathematical properties (recursion, embedding, transformations) independent of propositional content, in continuous operation across the *Nambūdiri paramparā* (परम्परा).

Deployments: Preface ¶15 — the survey-of-prior-work paragraph; the third named precedent for engineering / formal-systems accounts of the Sanskrit continuum.

J. Frits Staal (1930–2012), Dutch-American philosopher and scholar of Indic ritual and grammar, worked across more than half a century on the formal-systems character of Sanskrit and Vedic ritual. The relevant works for the *Preface*'s purpose include: *Word Order in Sanskrit and Universal Grammar* (Reidel, 1967); *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, Berkeley, 1983; two volumes — the documentation of the 1975 Nambūdiri Agnicayana performance; Motilal Banarsidass reprint 2001 with CDs); *Universals: Studies in Indian Logic and Linguistics* (University of Chicago Press, 1988); ***Rules Without Meaning: Ritual, Mantras and the Human Sciences*** (Peter Lang, 1989; Motilal Banarsidass reprint as *Ritual and Mantras: Rules Without Meaning*, 1996) — the most direct statement of the formal-systems-without-propositional-content account; *Mantras between Fire and Water: Reflections on a Balinese Rite* (Royal Netherlands Academy of Arts and Sciences, 1992); and his late synthesis *Discovering the Vedas: Origins, Mantras, Rituals, Insights* (Penguin India, 2008).

The structural account Staal developed: Sanskrit ritual and Sanskrit grammar are not merely cultural artifacts to be described ethnographically; they are formal systems with mathematical properties — recursion, embedding, transformations, precise interface rules between sound and meaning — that operate independently of any specific propositional content. Staal’s famous argument (developed across *Mantras between Fire and Water* and the *Agni* documentation) is that Vedic *mantras* are not propositional language at all but rather formal sound-objects in a system that pre-dates and operates orthogonally to propositional speech. The ritual is structured by combinatorial rules that resemble the grammar of language but apply to actions rather than sentences. The system is in continuous operation in the *Nambūdiri* communities of Kerala; Staal documented an *Agnicayana* performance in 1975 (later filmed and published as the 1976 Robert Gardner documentary *Altar of Fire*) precisely to record a working instance of a lineage the orthodoxy was treating as extinct.

The implication of Staal’s account parallels Kak’s: the formal-systems character of Sanskrit is not an external imposition by modern formal linguistics; it is a property of the Sanskrit apparatus as it has operated continuously across the *paramparā*. The *Preface*’s citation names Staal alongside Briggs and Kak as a body of adjacent work; the present book’s contribution is to articulate the full architectural framework — *varṇamālā* + *dhātupāṭha* + *gaṇāḥ* + *upasarga-pratyaya* + recitation redundancy — that the prior work touched in fragments.

patanjali-siddhe-shabdarthasambandhe

Short: Patañjali’s *Mahābhāṣya* (महाभाष्य) opens with *siddhe śabdārthasambandhe* (सिद्धे शब्दार्थसम्बन्धे; Kielhorn ed. vol. I, p. 6) — “the relation between word and meaning being established”; the *siddhe* (सिद्धे, established) asserts at the foundational level of the *vyākaraṇa*

(व्याकरण) discipline's self-description that Sanskrit is a received and operating system whose word-meaning relations are not subjects of analysis but premises of it — the grammarians are *vaiyākaraṇāḥ* (वैयाकरणाः), decoders, not engineers.

Deployments: Preface ¶17 — the methodology paragraph that names *siddhe śabdārthasambandhe* as a foundational axiom the book takes seriously rather than dismissing as theological flourish.

The phrase *सिद्धे शब्दार्थसम्बन्धे* (*siddhe śabdārthasambandhe*) appears at the very opening of Patañjali's *Mahābhāṣya* (the *Paspaśāhnika* — the introductory *āhnika*), positioned as one of the foundational statements of the *vyākaraṇa* discipline's epistemic posture. The phrase forms the locative-absolute opening of a longer sentence; its standard translation: “*The relation between word and meaning being established...*” — that is, *given as already established*, not requiring derivation.

The full clause in context (Kielhorn's standard edition, *Mahābhāṣya* on Pāṇini 1.1.1, *paspaśāhnika*, opening):

सिद्धे शब्दार्थसम्बन्धे लोकतोऽर्थप्रयुक्ते शब्दप्रयोगे शास्त्रेण धर्मनियमः ।

siddhe śabdārthasambandhe lokato 'rthaprayukte śab-
daprayoge śāstreṇa dharmaniyamaḥ.

“*The relation between word and meaning being established*
(by the paramparā, not requiring derivation), and the use
of words for the purpose of meaning being established in the
world, the śāstra (grammatical science) regulates correct
usage.”

The polemic load the phrase carries — and which the *Preface* picks up — is precisely the *siddhe* (*established, accomplished*). The *vyākaraṇa* discipline is not in the business of inventing a relation between words and meanings; it operates on the premise that the relation is already given, already operating, already maintained across the *paramparā*. The grammatical *śāstra* is regulatory, not constitutive. It governs cor-

rect usage of an already-existing system; it does not produce the system.

The implication is structural. Patañjali's opening axiom asserts, at the foundational level of the *vyākaraṇa* discipline's self-description, that Sanskrit is a received and operating system whose word-meaning relations are not subjects of analysis but premises of it. The architects of the system are *not* the grammarians; the grammarians inherited the encoded system and decoded it. This is the same point the book makes across Chapter 8 §8.6 (*heroic erasure* and the orthodoxy's celebration of Pāṇini), Chapter 16, and the foundational chapters on the *varṇamālā* and *dhātupāṭha*: the named figures (Pāṇini, Patañjali, the *Prātiśākhya* compilers) are **vaiyākaraṇāḥ** (वैयाकरणाः) — *decoders, analysts, unfolds-apart* — not engineers. The axiom *siddhe śabdārthasambandhe* is the discipline's own statement of this distinction. *Sanskrit was engineered. Encoded in the Vedas. Decoded by many. Pāṇini's decoding is the finest.*

Source: Kielhorn's edition of the *Mahābhāṣya* (third edition revised by Abhyankar; Bhandarkar Oriental Research Institute, Pune), volume I, p. 6, opening of the *Paspaśāhnika*; standard scholarly references include S. D. Joshi and J. A. F. Roodbergen's translation-commentary series on the *Mahābhāṣya* (Sahitya Akademi / Pune University publications, 1968–2003).

eleven-pathas

Short: The Vedic recitation lineages transmit each *Samhitā* (संहिता) through eleven *pāṭhas* (पाठ) — five *prakṛti* (प्रकृति: *Samhitā*, *Pada* (पद), *Krama* (क्रम), *Jaṭā* (जटा), *Ghana* (घन)) and six *vikṛti* (विकृति: *Mālā*, *Śikhā*, *Rekhā*, *Dhvaja*, *Daṇḍa*, *Ratha*) — operating as a many-to-one error-correcting code: each *pāṭha* recombines the same word-sequence by a different permutation rule, so any phoneme

error propagates inconsistently and is caught by the *guru-shishya paramparā* (गुरुशिष्यपरम्परा). Documented in the *Prātiśākhya* (प्रातिशाख्य) and *Śikṣā* (शिक्षा) literature; UNESCO-recognized 2003.

Deployments: Preface ¶17 — the methodology paragraph that names the eleven *pāṭhas* as a redundancy system rather than cultural ornament; subsequent extended treatment at Chapter 15 §§15.2–15.3.

The Vedic recitation lineages transmit each *Samhitā* text through eleven distinct *pāṭhas* (modes of recitation) operating as a layered error-detection-and-correction scheme. The eleven are conventionally divided into the *prakṛti-pāṭhas* (the *natural* modes — straightforward recitations) and the *vikṛti-pāṭhas* (the *transformed* modes — combinatorial recombinations that act as cross-checks). The full list:

Prakṛti-pāṭhas (5):

1. ***Samhitā-pāṭha*** — the continuous recitation with full *sandhi* applied (the form in which the *Samhitā* texts are conventionally recited).
2. ***Pada-pāṭha*** — the word-by-word recitation with each word in its *pre-sandhi* form, isolated.
3. ***Krama-pāṭha*** — the *step* recitation: each word recited paired with its successor (word₁-word₂, word₂-word₃, word₃-word₄, ...).
4. ***Jaṭā-pāṭha*** — the *braid* recitation: each pair recited forward-and-backward-and-forward (word₁-word₂-word₁, word₂-word₃-word₂, ...).
5. ***Ghana-pāṭha*** — the *dense* recitation: each triad recited through forward, backward, and embedded permutations (word₁-word₂, word₂-word₁, word₁-word₂-word₃, word₃-word₂-word₁, word₁-word₂-word₃, ...).

Vikṛti-pāṭhas (6):

6. *Mālā-pāṭha* — the *garland* recitation: words combined in a long looping pattern.
7. *Śikhā-pāṭha* — the *crest* recitation: triadic groupings rotated.
8. *Rekhā-pāṭha* — the *line* recitation: extended linear permutations.
9. *Dhvaja-pāṭha* — the *flag* recitation: ends-to-beginning pairings.
10. *Daṇḍa-pāṭha* — the *staff* recitation: progressive extension across the text.
11. *Ratha-pāṭha* — the *chariot* recitation: complex multi-axis permutations.

The combinatorial logic is the load-bearing point. Each successive *pāṭha* recombines the same word-sequence according to a different permutation rule. A reciter who has mastered all eleven has effectively held the text in eleven independently-derived forms. Any phoneme-level error in a single word would propagate inconsistently across the eleven, producing detectable mismatches that the *guru-shishya paramparā* uses to localize and correct the error. The system is — in formal-systems vocabulary — a many-to-one error-correcting code operating on the syllable as the imperishable unit. The architects of the recitation lineages built redundancy into the transmission channel; the channel survived across many generations because the redundancy survived.

The *Ghana-pāṭha* mastery confers the title *Ghanapāṭhin* — a reciter recognized as having mastered the densest of the standard *prakṛti-pāṭhas*. The honorific is socially recognized across Vedic recitation lineages (Nambūdiri, Iyengar, Iyer, Gauḍa, Marāṭhī Deśastha, and others maintaining recitation lineages); naming a reciter *Ghanapāṭhin* anchors the speaker's training depth.

The eleven *pāṭhas* are treated extensively in the *Prātiśākhya* literature (the canonical phonetic-recitational *śāstra* attached to each *Veda*: *Ṛk-Prātiśākhya*, *Taittirīya-Prātiśākhya*, *Vājasaneyi-Prātiśākhya*, *Atharvaveda-Prātiśākhya*) and in the *Śikṣā* texts (the auxiliary

phonetics literature including *Pāṇinīya-Śikṣā*, *Nāradya-Śikṣā*, *Yājñavalkya-Śikṣā*. Modern documentation: Wayne Howard, *Sāmavedic Chant* (Yale University Press, 1977); Frits Staal, *The Science of Ritual* (Bhandarkar Oriental Research Institute, 1982); the UNESCO 2003 recognition of *Vedic Chanting* as a *Masterpiece of the Oral and Intangible Heritage of Humanity* (transferred in 2008 to the UNESCO Representative List of the Intangible Cultural Heritage of Humanity). Chapter 15 §§15.2–15.3 develops the redundancy-system analysis in full.

english-sanskrit-loanwords

Short: The cluster *guru, karma, avatar, mantra, yoga, pundit, jungle, nirvana, Aryan* — and the wider open inventory documented in Yule and Burnell’s *Hobson-Jobson* (1886; Crooke ed. 1903), the OED’s etymological entries, and Monier-Williams’s *Sanskrit-English Dictionary* (1899) — establishes the structural fact that English carries an open inventory of Sanskrit words; the reader is closer to Sanskrit than the reader has been told.

Deployments: Chapter 0 ¶3 — the cluster of English words of Sanskrit origin deployed as the chapter’s opening evidence that the reader is already speaking Sanskrit in fragments.

The English words listed in the chapter’s opening — *guru, karma, avatar, mantra, yoga, pundit, jungle, nirvana, Aryan* — are a representative selection of the many hundreds of Sanskrit-origin words that have entered English across colonial and post-colonial contact. The standard reference treatments include:

- The *Hobson-Jobson* dictionary (Henry Yule and A. C. Burnell, *Hobson-Jobson: A Glossary of Colloquial Anglo-Indian Words and Phrases*, John Murray, London, 1886; expanded second edition by William Crooke, 1903) — the foundational colonial-era doc-

umentation of Anglo-Indian and Indic-origin vocabulary in English.

- The *Oxford English Dictionary*'s etymological entries for the relevant words, each of which traces the English form back through transmission paths (typically through Hindustani or directly from Sanskrit) to the Sanskrit source.
- Monier-Williams's *A Sanskrit-English Dictionary* (Oxford University Press, 1899; standard reference), which catalogues the Sanskrit source forms with their precise grammatical formation.

The selection in Chapter 0's opening covers semantic categories the polemic frame uses: *guru* and *pundit* (teaching and learning); *karma*, *yoga*, *mantra* (engineered practice); *avatar* (theological-technical vocabulary); *nirvāṇa* (the goal of the soteriological apparatus); *jungle* (the geographic-environmental); and *ārya* (the contested racial-philological term the book later prosecutes at length in Chapter 16). Each word travels with a specific gloss in the chapter; the gloss is the operational definition the chapter uses, not the full semantic field the Sanskrit term carries.

The wider inventory of English words of Sanskrit origin — including *bandanna*, *cot*, *jodhpurs*, *juggernaut*, *loot*, *pajamas*, *pundit*, *shampoo*, *thug*, *veranda*, *bungalow*, *pundit*, *cheetah*, *catamaran*, *coir*, *curry*, *mongoose*, *chintz*, *toddy*, *cashmere*, *opal*, *patchouli*, and dozens of others — is documented across the *Hobson-Jobson* tradition and traced individually in OED entries. The point the chapter makes does not depend on exhaustive enumeration; the named cluster establishes the structural fact that English carries an open inventory of Sanskrit words, and from that the chapter develops the argument that the reader is closer to Sanskrit than the reader has been told.

place-value-arabic-transmission

Short: The decimal place-value numeral system with zero as a position-holder was developed in the Indic mathematical discipline (Bakhshali manuscript; Āryabhaṭīya (आर्यभटीय); Brahmagupta’s *Brāhma-Sphuṭa-Siddhānta* (ब्राह्मस्फुटसिद्धान्त)) and transmitted to Europe through al-Khwārizmī’s *Kitāb al-Jam‘ wa-l-Tafrīq bi-Ḥisāb al-Hind* (الهند بحساب والتفريق الجمع كتاب; early 9th century — the title names the Indic origin); the *Hindu-Arabic* compound preserves the transmission path in the system’s own name, and *algorithm* / *algorism* derive from al-Khwārizmī.

Deployments: Chapter 0 ¶ (numerals paragraph) — the historical-transmission paragraph that anchors the *Hindu-Arabic numerals* naming convention and traces the system’s path from India through Arabic intermediaries to Europe.

The decimal place-value numeral system with zero as a position-holder was developed in the Indic mathematical discipline. The earliest unambiguous documentation of the place-value system with a symbol for zero is the Bakhshali manuscript (carbon-dated to a range across the early centuries CE; the precise chronology is unsettled and not the point); the system is fully operational in the work of Āryabhaṭa (*Āryabhaṭīya*, with the *bhūta-saṃkhyā* and other numerical conventions), Brahmagupta (*Brāhma-Sphuṭa-Siddhānta*, with the formal arithmetic of zero and negative numbers), and the later Indic mathematical discipline (Mahāvīra, Bhāskara I, Bhāskara II).

The transmission to Europe ran through the Arabic-speaking mathematical tradition. Al-Khwārizmī (the Persian polymath working at the *Bayt al-Ḥikma* in Baghdad, early 9th century by the conventional dating) authored *Kitāb al-Jam‘ wa-l-Tafrīq bi-Ḥisāb al-Hind* (*The Book of Addition and Subtraction According to the Hindu Calculation*) — the standard reference in the Arabic mathematical tradition for the Indic numerical system, with the title itself naming the system’s origin.

The Latin translations of al-Khwārizmī's work that reached medieval Europe (notably *Dixit Algorizmi* and *Algoritmi de numero Indorum*) carried the *al-Hind* / *of the Indians* attribution forward; the word *algorithm* in English derives from al-Khwārizmī's name, and the term *algorism* in medieval European mathematical literature named the Indic place-value calculation method specifically.

The Arabic transmission stage is documented in standard histories of mathematics: Kim Plofker, *Mathematics in India* (Princeton University Press, 2009), Chapter 5; Jens Høyrup, *In Measure, Number, and Weight: Studies in Mathematics and Culture* (SUNY Press, 1994); George Gheverghese Joseph, *The Crest of the Peacock: Non-European Roots of Mathematics* (3rd edition, Princeton University Press, 2010); the *MacTutor History of Mathematics* archive's entries on al-Khwārizmī and the Indic numeral system. The conventional academic-historical framing — *Hindu-Arabic numerals* — preserves the Indic origin in the name, even where the engineering-significance account the chapter develops is muted.

The book's point, anchored at this paragraph, is that the world's numerical practice today operates on an engineered Indic system whose Indic origin is not contested even by the orthodoxy. The *engineering* account the chapter develops is therefore not a hostile claim against established fact; it is a reframing of an already-acknowledged transmission as the operational arrival of an engineered apparatus, rather than as the discovery of a natural inevitability.

ishopanishad-invocation

Short: The *śāntipāṭha* (शान्तिपाठ) opening the *Īśopaniṣad* (ईशोपनिषद्) — ॐ पूर्णमदः पूर्णमिदं पूर्णात्पूर्णमुदच्यते । पूर्णस्य पूर्णमादाय पूर्णमेवावशिष्यते ॥ / *om pūrṇam adaḥ pūrṇam idaṁ pūrṇāt pūrṇam udacyate | pūrṇasya pūrṇam ādāya pūrṇam evāvaśiṣyate ||* (“That is whole. This is

whole. From the whole, the whole emerges. Taking the whole from the whole, the whole alone remains”); placed at the chapter close to land the integrative-whole register — the architecture is not depleted by being articulated.

Deployments: Chapter 0 ¶ (closing) — the *pūrṇam adaḥ pūrṇam idam* invocation that closes the chapter on the integrative-whole register.

The verse is the *śāntipāṭha* (peace invocation) that opens the *Īsopaniṣad* (also known as the *Īśāvāsya Upaniṣad*) — the shortest of the principal Upaniṣads (the eighteen-mantra Upaniṣad attached to the *Vājasaneyī Saṃhitā* of the *Śukla-Yajurveda*).

ॐ पूर्णमदः पूर्णमिदं पूर्णात्पूर्णमुदच्यते । पूर्णस्य पूर्णमादाय पूर्णमेवावशिष्यते ॥

oṃ pūrṇam adaḥ pūrṇam idam pūrṇāt pūrṇam udacyate |
pūrṇasya pūrṇam ādāya pūrṇam evāvaśiṣyate | |

“That is whole. This is whole. From the whole, the whole emerges. Taking the whole from the whole, the whole alone remains.”

The verse is among the most widely-cited *śāntipāṭhas* in the Vedic / Upaniṣadic corpus and is recited at the opening and closing of formal study sessions, particularly those treating *Vedānta*. The verse’s structural logic — wholeness from which wholeness emerges without diminishing the wholeness from which it emerged — is taken by the *Vedānta* discipline as a statement of the ontological non-diminution of *Brahman* under emanation: the manifest world (*idam*) does not deplete the unmanifest source (*adaḥ*); the source remains whole even after the manifest has emerged from it.

Chapter 0’s deployment of the verse picks up the same structural logic at the *evidentiary* register the chapter operates in. The book’s argument — that Sanskrit is an engineered architecture continuously operating across the *paramparā* — does not deplete the architecture by being articulated. The architecture remains whole; the book is one

rendering of it. The closing invocation places the chapter, and by extension the book, inside the same *pūrṇa* register the Sanskrit continuum uses for any operation that draws on the integrative whole without diminishing it.

Source: *Īsopaniṣad*, opening *śāntipāṭha*. The standard editions: *Eighteen Principal Upaniṣads* (V. P. Limaye and R. D. Vadekar, eds., Vaidika Samshodhana Mandala, Pune, 1958); the *Śaṅkara-bhāṣya* on the *Īsopaniṣad* (Gita Press Gorakhpur, standard edition); Swāmī Gambhīrānanda's translation-commentary (Advaita Ashrama, Calcutta). The verse is also embedded in many other devotional and ritual contexts across the *paramparā*.

schleicher-stammbaumtheorie

Short: August Schleicher (1821–1868) formalized the *Stammbaumtheorie* (family-tree theory) across *Die Deutsche Sprache* (1860), the *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (1861), and *Die Darwinsche Theorie und die Sprachwissenschaft* (1863) — languages as *Naturorganismen* (natural organisms growing and decaying like biological species); Schleicher trained as a botanist before turning to linguistics, and the imported biological framing was doctrinal, not casual.

Deployments: Chapter 1 §1.1 ¶1 — the opening establishment of August Schleicher and the *Stammbaumtheorie* (family-tree theory) as the founding metaphor of comparative philology.

August Schleicher (1821–1868), German linguist and Indo-Europeanist, formalized the *Stammbaumtheorie* — the *family-tree theory* of language descent — across his major works of the 1850s and 1860s. The two principal texts:

1. *Die Deutsche Sprache* (Stuttgart: J. G. Cotta'scher Verlag, 1860),

in which Schleicher developed his biological-organic theory of language: languages are *natural organisms* (*Naturorganismen*) that grow, reproduce, decay, and die like biological species. The framing was explicit and load-bearing; Schleicher trained as a botanist before turning to linguistics, and the imported metaphor was not casual but doctrinal.

2. *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (Weimar: Hermann Böhlau, 1861; second edition 1866), the foundational reference work of the comparative method in Indo-European linguistics. The *Compendium* systematized the reconstruction methodology and established the family-tree diagram as the standard visual machinery for representing language descent. Schleicher's diagrams branched a hypothetical *Proto-Indogermanic* into Asiatic and European branches, then into the daughter families and named languages, with each node a hypothetical ancestor.

Schleicher's biological framing was made explicit in his 1863 pamphlet *Die Darwinsche Theorie und die Sprachwissenschaft* (*The Darwinian Theory and the Science of Language*; Weimar: Hermann Böhlau, 1863), in which he argued that Darwin's theory of natural selection applied directly to language change: languages, like species, evolved through descent with modification under selective pressures. The biological-organic frame the *Preface* and Chapter 1 prosecute as *botanical* is Schleicher's own framing, not a polemic ascription.

The family-tree model dominated comparative philology from the 1860s onward and remains the foundational visual machinery of historical linguistics. Subsequent refinements (Johannes Schmidt's *Wellentheorie* — *wave theory* — in his 1872 *Die Verwandtschaftsverhältnisse der indogermanischen Sprachen*, which proposed dialect-continuum diffusion as an alternative to clean family-tree branching; later neogrammarian and structuralist contributions) modified the picture but preserved the underlying biological-descent metaphor. The contemporary apparatus that

displays Indo-European languages as a branching tree descending from reconstructed Proto-Indo-European is Schleicher's apparatus, slightly refined.

Standard references for Schleicher's role: R. H. Robins, *A Short History of Linguistics* (4th edition, Longman, 1997), Chapter 7; Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Volume IV of the Routledge *History of Linguistics*, 1998), Chapters 4–5; Holger Pedersen, *The Discovery of Language: Linguistic Science in the Nineteenth Century* (Indiana University Press, 1962; original Danish 1924); Tomáš Hoskovec et al., eds., *Schleicher's Indogermanische Chrestomathie und Wörterbuch* and related Schleicher reception studies.

hlāfweard-etymology

Short: *Lord* = Old English *hlāfweard* (*hlāf* “loaf” + *weard* “guardian”) → *hlāford* → Middle English *laverd* → *lorde* → Modern English *Lord*; phonetic contraction and compositional opacity across roughly a thousand years — the textbook *apabhraṃśa* (अपभ्रंश) decay arc Chapter 1 contrasts with Sanskrit *dhātu* (धातु) preservation across many more generations.

Deployments: Chapter 1 §1.1 ¶2 — the worked example of *hlāfweard* → *laverd* → *lorde* → *Lord* as the textbook case of botanical decay and semantic shift across a thousand years of English.

The etymological chain of *Lord* runs through documented stages of Old English and Middle English, each visible in attested manuscript and lexicographic evidence:

Stage 1 — Old English (pre-1100): *hlāfweard*. A compound of *hlāf* (loaf, bread) + *weard* (guardian, keeper, protector — cognate with *ward*, *warden*). Literal meaning: *bread-guardian*, *loaf-keeper*. The

compound names a specific social role — the head of a household who provides and guards the food supply for those under his protection. The form is attested in early Old English texts (the *hlāfweard* and the dual *hlāfdige* — *bread-kneader* — for the female counterpart, eventually becoming *lady*).

Stage 2 — Late Old English / Early Middle English (~1000–1200): *hlāford*. The compound contracted as the second element *-weard* lost its independent transparency. The form *hlāford* appears across late Old English and the early Norman period.

Stage 3 — Middle English (~1200–1400): *laverd*, *lavord*, *lord*. The aspirate *h*- fell away, the medial fricative weakened, and the bisyllabic form contracted toward the monosyllabic. The semantic field expanded from *household provisioner* through *feudal landholder* to *nobleman*.

Stage 4 — Late Middle English to Early Modern English (~1400–1600): *lorde*, *Lord*. The form crystallized as *Lord*, used both for the nobility and (with the religious connotation that English Christianity had built up across the same period) for God.

The chain is documented in: the *Oxford English Dictionary*'s entry for *lord* (etymological section); Joseph Bosworth and T. Northcote Toller, *An Anglo-Saxon Dictionary* (Oxford, 1898; supplement 1921); Walter Skeat, *An Etymological Dictionary of the English Language* (Oxford, 1879–1882; revised editions); the *Middle English Dictionary* (University of Michigan Press, 1952–2001).

The structural point the chapter makes from the chain: this is exactly the kind of language history the botanical metaphor describes — a word loses its compositional transparency (a speaker of modern English does not hear *loaf-guardian* in *Lord*); its phonetic form simplifies under articulatory pressure; its meaning extends and shifts under social-historical pressure. *Hlāfweard* → *Lord* is a perfect textbook case of *apabhraṃśa*-style decay operating across a thousand years of a language without engineered preservation. The contrast the chap-

ter immediately develops — Sanskrit *dhātu* preserved across many more generations without comparable decay — is the contrast the book’s central thesis turns on.

dhātu-pre-panini-vedic

Short: *Dhātuḥ* (धातुः) as the technical grammatical term predates Pāṇini — attested in Yāska’s *Nirukta* (निरुक्त) and the *Prātiśākhya* (प्रातिशाख्य) literature, used by Pāṇini as already-given (*Aṣṭādhyāyī* (अष्टाध्यायी) 1.3.1 *bhūvādayo dhātavaḥ*); the same term operates in metallurgy (*sapta-dhātavaḥ* (सप्तधातवः)), Āyurvedic medicine, and *Rasaśāstra* (रसशास्त्र) — the architects chose engineering vocabulary, not the *bīja* (बीज) or *mūla* (मूल) botanical terms available to them.

Deployments: Chapter 1 §1.1 ¶ (closing of the section) — the *dhātu*-not-*bīja* paragraph that names the architects’ deliberate refusal of botanical vocabulary for the foundational grammatical unit.

The term धातुः (*dhātuḥ*) as the name for the foundational grammatical-semantic unit is pre-Pāṇinian and embedded in the *Prātiśākhya* and *Nirukta* disciplines that precede the *Aṣṭādhyāyī*. The relevant attestations:

- ***Nirukta*** of Yāska (the canonical pre-Pāṇinian etymological treatise, attached to the *Veda* as one of the six *Vedāṅgas*). Yāska’s *Nirukta* uses *dhātu* in the technical grammatical sense of *verbal root* — the foundational semantic constituent from which derived forms are produced. Yāska’s framework, anchored at *Nirukta* 1.1–1.14, treats *dhātu* as the operative unit of verbal derivation and presupposes the term’s technical use in the prior *vyākaraṇa* discipline.
- ***Prātiśākhya*** literature (the canonical phonetic-recitational treatises attached to each *Veda*: *Ṛk-Prātiśākhya*, *Taittirīya-Prātiśākhya*, *Vājasaneyi-Prātiśākhya*, *Atharvaveda-Prātiśākhya*).

These texts use *dhātu* in the technical grammatical sense and presuppose the term's establishment.

- **Aṣṭādhyāyī** itself (Pāṇini 1.3.1 — *bhūvādayo dhātavaḥ*, defining the *dhātus* as the elements of the *Dhātupāṭha* beginning with *bhū*). Pāṇini does not invent the term; he uses it in its already-established technical sense and refers to a *Dhātupāṭha* (the canonical enumeration of roots) that he treats as already given.

The term's pre-grammatical use in the *Vedic* register carries the same root-meaning *constituent constant*. In the metallurgical / extractive register, *dhātu* names the foundational metallic constituent extracted from ore (the *sapta-dhātavaḥ* — gold, silver, copper, tin, lead, iron, mercury — across the relevant *Rasaśāstra* literature; the term appears across the *Samhitā*, *Brāhmaṇa*, and post-Vedic ritual and scientific texts). In the *Āyurvedic* biological register, *dhātu* names the seven structural constituents of the body (*rasa*, *rakta*, *māṃsa*, *medas*, *asthi*, *majjā*, *śukra*; see endnote saptadhatu-canonical). The grammatical *dhātu* is the same term operating in the same structural register — the constituent constant of the verbal system — across an engineering vocabulary that uses *dhātu* consistently in this sense across multiple domains (metallurgy, medicine, grammar, chemistry).

The polemic point the chapter establishes: the architects of Sanskrit had *bīja* and *mūla* available — both already operating in the botanical / agricultural register of Sanskrit vocabulary — and could have named the grammatical foundational unit either *vyākaraṇa-bīja* (grammar-seed) or *vyākaraṇa-mūla* (grammar-root) without strain. The choice of *dhātu* — the engineering term — places the grammatical foundational unit in the engineering category. The botanical framing the philological orthodoxy imposes (“root” as in *plant root*, with all the botanical connotations attached) reverses the architects’ own choice. The mistranslation is the imposition the chapter names; the original term registers the engineering placement explicitly.

The point is developed across Chapter 6 (the multi-domain *dhātu*

usage) and Chapter 10 (the *dhātu* as the foundational atomic unit in the periodic-table architecture).

leviticus-slavery-25-44-46

Short: *Leviticus* 25:44–46 (RSV) authorizes the purchase of male and female slaves “from among the nations that are around you,” their treatment as inheritable property bequeathed “to your sons after you to inherit as a possession forever,” with the exclusion at the end that fellow Israelites may not be enslaved with harshness — in-group / out-group framing, not a critique of slavery itself.

Deployments: Chapter 2 §2.2 ¶3 — the Abrahamic-scriptural-sanction cluster naming the Hebrew Bible’s authorization of slavery as inheritable property.

Leviticus 25:44–46 is the passage in which the Hebrew Bible authorizes the purchase of slaves from neighboring peoples and the inheritance of those slaves across generations. The standard text (Revised Standard Version):

“As for your male and female slaves whom you may have: you may buy male and female slaves from among the nations that are around you. You may also buy from among the strangers who sojourn with you and their families that are with you, who have been born in your land, and they may be your property. You may bequeath them to your sons after you to inherit as a possession forever. You may make slaves of them, but over your brothers the people of Israel you shall not rule, one over another with harshness.”
(*Leviticus* 25:44–46, RSV)

The passage’s structural authorization carries the polemic load Chapter 2 deploys it for: slavery is authorized scripturally; the slaves are

property; the property is inheritable; the inheritance is “forever.” The exclusion at the end — that Israelites may not enslave fellow Israelites with harshness — is in-group / out-group, not a critique of slavery itself. The framework authorizes the institution and partitions who may be enslaved by lineage.

Standard references: the *Jewish Publication Society Tanakh* (JPS, 1985); *The New Oxford Annotated Bible* (NRSV, fifth edition, 2018); *The Anchor Bible Commentary on Leviticus* (Jacob Milgrom, three volumes, Yale University Press, 1991–2001) on the slavery passages and their reception. The structural fact — that slavery is scripturally authorized in Hebrew Bible — is uncontested across the relevant biblical and historical scholarship and is not weakened by interpretive efforts to soften the passage’s force.

ephesians-slavery-6-5

Short: *Ephesians* 6:5–8 (with parallel passage at *Colossians* 3:22–25 and the entire *Epistle to Philemon*) instructs slaves to obey their earthly masters “with fear and trembling, in singleness of heart, as to Christ”; the institution is preserved across the New Testament, with moral language operating within it rather than against it — the European intellectual class that constructed AIT did so from inside this scriptural framework, not outside it.

Deployments: Chapter 2 §2.2 ¶3 — the Abrahamic-scriptural-sanction cluster naming the Christian New Testament’s instruction to slaves to obey earthly masters.

Ephesians 6:5 is the New Testament passage in which slaves are instructed to obey their earthly masters with the same diligence with which they obey Christ. The standard text (Revised Standard Version):

“Slaves, be obedient to those who are your earthly masters, with fear and trembling, in singleness of heart, as to Christ; not in the way of eye-service, as men-pleasers, but as servants of Christ, doing the will of God from the heart, rendering service with a good will as to the Lord and not to men, knowing that whatever good any one does, he will receive the same again from the Lord, whether he is a slave or free.” (Ephesians 6:5–8, RSV)

The parallel passage at *Colossians* 3:22–25 carries the same instruction. The *Epistle to Philemon* — the entire book — concerns the return of the slave Onesimus to his Christian master Philemon and is the standard reference for Pauline reasoning about the institution. Across the New Testament, slavery is not denounced as an institution; slaves are instructed to obey their masters, and masters are instructed to treat slaves justly. The institution is preserved; the moral language operates within it rather than against it.

The polemic load Chapter 2 carries from these passages: the Christian colonial enterprise that operated chattel-slavery economies across the Atlantic and into the subcontinent did so with explicit scriptural authorization. The institution had textual sanction at the level of the foundational Christian writings. The European intellectual class that constructed the AIT did so from inside this scriptural framework, not outside it; the master-slave binary the framework projected onto Indic prehistory was internal to the constructors’ own moral universe.

Standard references: *The New Oxford Annotated Bible* (NRSV, fifth edition, 2018); *The Anchor Bible Commentary on Ephesians* (Markus Barth, two volumes, Yale University Press, 1974). Historical treatments: David Brion Davis, *In the Image of God: Religion, Moral Values, and Our Heritage of Slavery* (Yale University Press, 2001); Bernard Lewis, *Race and Slavery in the Middle East* (Oxford University Press, 1990) for the comparative framing.

quran-slavery-citations

Short: The Quranic formula *mā malakat aymānukum* (أَيْمَانُكُمْ مَلَكَتْ مَا) — “what your right hands possess”) authorizes the sexual use of female captives outside marriage across Surah Al-Mu’minūn 23:5–6, Al-Ma’ārij 70:29–30, An-Nisā’ 4:24, and Al-Aḥzāb 33; the framework operated through centuries of subcontinental conquest from the Ghaznavid raids through the Delhi Sultanate and the Mughal imperial harem.

Deployments: Chapter 2 §2.2 ¶3 — the Abrahamic-scriptural-sanction cluster naming the Quran’s sanctioning of the taking of captives as slaves and concubines.

The Quranic passages authorizing the taking of captives — particularly female captives — as slaves and concubines run across multiple chapters. The standard references the chapter cites:

Surah Al-Mu’minūn (23:5–6): > “And those who guard their chastity, except from their wives or those their right hands possess, for indeed, they will not be blamed.” (23:5–6)

Surah Al-Ma’ārij (70:29–30): > “And those who guard their chastity, except from their wives or those their right hands possess, for indeed, they are not to be blamed.” (70:29–30)

Surah An-Nisā’ (4:24): > “And [also prohibited to you are all] married women except those your right hands possess. [This is] the decree of Allah upon you.” (4:24, partial citation)

The phrase “what your right hands possess” (*mā malakat aymānukum* — أَيْمَانُكُمْ مَلَكَتْ مَا) is the standard Quranic locution for slaves, including female captives taken in war. The Quranic framework authorizes the sexual use of female captives by their masters outside the otherwise-required marriage contract. Surahs 33 (Al-Aḥzāb), 70 (Al-Ma’ārij), and 23 (Al-Mu’minūn) preserve the same locution across multiple deployments; the practice is named extensively in the *hadith* literature

(Bukhari and Muslim collections) and in the classical legal compendia (*fiqh*) across the four Sunni schools and the Shi'a tradition.

The polemic load Chapter 2 carries from these passages: the Islamic political tradition that preceded European colonialism in the subcontinent operated on the same master-slave binary at the level of foundational scripture. Centuries of Islamic conquest of the subcontinent — the Ghaznavid raids, the Ghurid invasions, the Delhi Sultanate, the regional Sultanates, and the Mughal Empire — drew on this scriptural authorization. The captive-slave economies, the imperial harem institutions, the systematic taking of women as concubines after military conquest — these were not departures from the framework; they were the framework operating.

Standard references: M. A. S. Abdel Haleem, *The Qur'an: A New Translation* (Oxford World's Classics, 2004); the standard *Sahih International* English translation; the classical commentaries (*tafsir*) of al-Ṭabarī, al-Qurṭubī, and Ibn Kathīr on the relevant verses. Comparative-scholarly treatments: Bernard Lewis, *Race and Slavery in the Middle East* (Oxford University Press, 1990); Murray Gordon, *Slavery in the Arab World* (New Amsterdam Books, 1989); William Gervase Clarence-Smith, *Islam and the Abolition of Slavery* (Oxford University Press, 2006). For the subcontinental application: Andre Wink, *Al-Hind: The Making of the Indo-Islamic World* (three volumes, Brill, 1990–2004); K. S. Lal, *Muslim Slave System in Medieval India* (Aditya Prakashan, 1994).

delhi-sultanate-mamluk

Short: The Delhi Sultanate's Slave Dynasty (*Mamlūk* / *Ghulām* dynasty, 1206–1290) — founded by Quṭb al-Dīn Aybak, a Turkic slave-warrior in the Ghurid military apparatus — was the first of five dynasties ruling Delhi for three centuries; the *Mamlūk* institution was the

dynasty's organizational principle, and the captive-slave economies continued through the Khaljī, Tughluq, Sayyid, Lodī, and Mughal successions.

Deployments: Chapter 2 §2.2 ¶3 — the Islamic-conquest paragraph that names the Delhi Sultanate's Slave Dynasty as the institutional expression of the Quranic master-slave framework operating in the subcontinent.

The *Delhi Sultanate's Slave Dynasty* — known in the standard reference literature as the *Mamlūk Sultanate of Delhi* (1206–1290) or the *Ghulām Dynasty* (Persian *ghulām*, slave) — was the first of the five dynasties that ruled the Delhi Sultanate across approximately three centuries. The founder, Qutb al-Dīn Aybak, was originally a Turkic slave-warrior in the service of Muḥammad of Ghor; he rose through the Ghurid military apparatus to command the Indian campaigns and, after Muḥammad's death, established independent rule from Delhi. His successors — Iltutmish, Razia Sultana, Balban, and others — were either themselves former slaves elevated through the Mamlūk system or were members of the slave-warrior class.

The structural fact the chapter establishes at this passage: the *Mamlūk / Ghulām* dynasty is named for the institution that produced its rulers. Slave-warriors, taken as captives in earlier military campaigns or purchased from slave markets, were trained as elite cavalry and administrators, manumitted, and elevated into ruling positions. The institution was not incidental to the dynasty's character; it was the dynasty's organizational principle. The wider *Mamlūk* institution — operating across the Islamic political world from Egypt to the Caucasus to the subcontinent — was the standing mechanism by which the Quranic captive-slave framework was converted into political and military apparatus.

The Delhi Sultanate that followed (Khaljī, Tughluq, Sayyid, Lodī dynasties; 1290–1526) and the Mughal Empire that succeeded it (1526–1857) continued operating slave-economies of varying intensity. The

Mughal imperial harem, the captive-taking after military conquests, the slave-soldier regiments, the household-slave economies of the imperial and regional courts — these are documented across the Mughal chronicles (*Tarikh-i Firishta*, *Tabaqāt-i Akbarī*, *Ain-i Akbarī*, the *Bābur-nāma*, the *Akbar-nāma*, the *Tūzuk-i Jahāngīrī*) and reconstructed in standard Mughal historiography.

Standard references: K. S. Lal, *Muslim Slave System in Medieval India* (Aditya Prakashan, 1994); Andre Wink, *Al-Hind: The Making of the Indo-Islamic World*, Volume II: *The Slave Kings and the Islamic Conquest, 11th–13th Centuries* (Brill, 1997); Peter Jackson, *The Delhi Sultanate: A Political and Military History* (Cambridge University Press, 1999); Sunil Kumar, *The Emergence of the Delhi Sultanate, 1192–1286* (Permanent Black, 2007); Salim Kidwai, *Sultans, Eunuchs, and Domestic: New Forms of Bondage in Medieval India* (Tulika Books, 1985 — in *Chains of Servitude*, ed. Patnaik and Dingwaney).

assalayana-sutta

Short: *Majjhima Nikāya* 93 — the *Assalāyana Sutta* (अस्सलायन सुत्त), Pali Buddhist Canon — the Buddha tells the young Brahmin Assalāyana that the two-*vaṇṇa* (वर्ण) *ārya* / *dāsa* (आर्य / दास) binary is a feature of foreign bordering nations (*Yona* / *Kamboja* — the Greek-influenced and Central Asian frontier), not of the Indic interior, and that even there the binary is mobile (*ārya* can become *dāsa* and the reverse); the mule-analogy wave (*assatara*) closes the refutation by dismantling heredity-essentialism.

Deployments: Chapter 2 §2.2 ¶3 (closing) — the dharmic primary-source documentation of the two-*varṇa* system as a foreign-bordering-nations feature; Chapter 3 §3.1 (forward deployment, signaled by *assalayana-sutta-fwd*); Chapter 16 §16.2 (extended treatment in the *ārya* / *dāsa* prosecution).

The *Assalāyana Sutta* is sutta 93 of the *Majjhima Nikāya* (the *Middle-Length Discourses* of the Pāli Buddhist Canon). The dialogue is between the Buddha and the young Brahmin Assalāyana, who has been deputized by five hundred Brahmin elders to dispute the Buddha's denial of the four-*varṇa* system's claimed essentialism. Across the course of the dialogue, the Buddha dismantles the racial-essentialist account of *varṇa* through a series of empirical-observational arguments. The relevant passage for Chapter 2's purpose comes in the early movement of the dialogue:

“Assalāyana, have you not heard that in Yona and Kamboja, and in other border-countries, there are only two varṇas — the ārya and the dāsa; and that the ārya can become a dāsa and the dāsa can become an ārya?”

(*Majjhima Nikāya* 93, the Buddha's question to Assalāyana. *Rendering note:* the Pali compound the Buddha uses is *Yonakambojesu aññesu ca paccantimesu janapadesu dveva vaṇṇā — ariyo ceva dāso ca*. The Ñāṇamoli–Bodhi (1995) translation renders the *ariya* / *dāsa* pair as “masters and slaves”; Ṭhānissaro Bhikkhu's translation does the same. The rendering above is the chapter's own, preserving the Pali-Sanskrit cognate forms *ārya* / *dāsa* / *varṇa* — defensible since the Pali *ariya* IS the Sanskrit *ārya* and the Pali *vaṇṇa* IS the Sanskrit *varṇa*, and the chapter's polemic depends on the load-bearing terminological continuity. Where a standard translator's English is wanted, Bodhi's “masters and slaves” is the canonical English.)

The passage carries three load-bearing observations the book's argument turns on.

First, the two-*varṇa* system (*ārya* / *dāsa*) is documented by the Buddha himself — within Sanātan's primary canonical record — as a *foreign* (*paccantimesu janapadesu* — the border-countries, outside the *majjhima-desa*, the *middle country* / Indic interior) feature, not as an Indic one. The named territories are *Yonakambojesu* — the Yonas (the Greek-influenced territories of the northwestern frontier; *Yona*

derives from *Ionia* and names the Indo-Greek-kingdom zone and the broader Hellenistic-cultural communities reaching across Bactria and the Kabul valley) and the Kambojas (the further-northwestern Pamir-Afghan zone, the upper Oxus territories and the Paropamisadae). The implication is that the Indic interior operated a different system — the four-*varṇa* or, more precisely, the functional-occupational categorization the *Bhagavad Gītā* 4.13 articulates (*cāturvarṇyaṃ mayā sṛṣṭam guṇa-karma-vibhāgaśaḥ* — the four *varṇas* distributed by *guṇa* and *karma*, not by birth).

Second, the binary itself is acknowledged as mobile — *the ārya can become a dāsa and the dāsa can become an ārya*. The Buddha names the social mobility within the binary, observing that even in the territories operating the two-*varṇa* system, the categories are not hereditarily fixed. This further undermines the racial-essentialist account the AIT later projects onto Indic prehistory.

Third, the Buddha is *internal* to the dharmic continuum. The observation that the two-*varṇa* binary is a foreign-bordering-nations feature is therefore documented at the dharmic continuum's primary level, by a teacher whose authority the wider Buddhist and dharmic paramparā acknowledges. The AIT's projection of the master-slave binary onto Indic prehistory is contradicted by a primary-source dharmic text that locates the binary precisely where the historical record places it: on the northwestern frontier, in the contact zone with Greek, Persian, and Central Asian polities.

Secondary anchor — the mule-analogy wave. The Yona/Kamboja passage is the *first* of nine successive waves of refutation the Buddha presents to Assalāyana across the dialogue. Each wave demolishes a different aspect of the *brahmin*-essentialist *cāturvarṇya* claim. The structurally strongest secondary anchor for the book's polemic is **Wave 8 — the cross-*vaṇṇa* offspring / mule analogy**. The Buddha asks Assalāyana whether the offspring of a *kṣatriya* father and a *brāhmaṇa* mother (or the reverse) should be called by the father's *vaṇṇa* or the mother's; Assalāyana concedes the child could be called either.

The Buddha then deploys the **mule** (Pali *assatara*, the donkey-mare hybrid): the mule “is neither a horse like its mother nor a donkey like its father; it is called precisely a mule.” Hybrid offspring across a hereditary boundary inherit from both lineages — and the *vaṇṇa*-essentialist binary cannot survive the empirical fact of mixed-*vaṇṇa* offspring. Where the Yona/Kamboja wave demolishes the *geography-essentialism* account, the mule-analogy wave demolishes the *heredity-essentialism* account. The two waves together dismantle the AIT projection from both flanks — the *ārya* / *dāsa* binary as a foreign feature and hereditary essentialism as a structurally unsupportable framing even where the four-*vaṇṇa* system does operate. Chapter 16 §16.2’s extended *ārya* / *dāsa* prosecution deploys both.

The polemic structural move Chapter 2 establishes: when the AIT’s binary is read against the *Assalāyana Sutta*, the binary belongs to the *constructors* of the AIT (the European intellectual tradition operating from inside Abrahamic master-slave scriptural frameworks) and to the *foreign bordering nations* (the Greek-influenced and Central Asian zones the Buddha names), not to the Indic civilizational interior. The AIT did not retroject a universal-human pattern onto Indic prehistory; it retrojected the *Abrahamic* pattern, and the *dharmic primary-source confirmation* the Buddha provides is that the pattern was already documented as foreign to the Indic interior.

Source: *Majjhima Nikāya* 93 (the *Assalāyana Sutta*), *M* ii 147 ff. in the Pali Text Society edition (Robert Chalmers, ed., *Majjhima Nikāya* vol. II, London, 1898). Bhikkhu Ñāṇamoli and Bhikkhu Bodhi, translators, *The Middle Length Discourses of the Buddha* (Wisdom Publications, 1995), with the *Assalāyana Sutta* beginning at page 763. The Vipassana Research Institute *Chaṭṭha Saṅgāyana Tipiṭaka* digital edition (tipitaka.org); the SuttaCentral and Access to Insight digital archives (suttacentral.net/mn93; accesstosight.org/tipitaka/mn/mn.093.than.html) carry parallel translations (Sujato, Horner, Ṭhāṇissaro). The sutta is also extensively discussed in Steven Collins, *Selfless Persons* (Cambridge University Press,

1982); Patrick Olivelle's translations and commentaries on the *Dharmasūtra* literature for the structural context; Vincent Smith, *Aśoka, the Buddhist Emperor of India* (Oxford, 1909) for the historical geography of Yona and Kamboja (and Aśokan Rock Edicts V and XIII for the independent dharmic-imperial confirmation of the Yona-Kamboja frontier-district pairing).

The endnote is deployed once with full prose at the first chapter-deployment (Chapter 2 §2.2 ¶3) and short-cited at later deployments (Chapter 3 §3.1, Chapter 16 §16.2). The assalayana-sutta-fw marker in Chapter 3 §3.1 signals a forward-reference to this canonical endnote.

rosa-law-2013

Short: Rosa's Law (Public Law 111-256, signed by President Obama October 5, 2010) replaced *mental retardation* with *intellectual disability* across U.S. federal health, education, and labor statutes; the diagnostic-register retirement followed in DSM-5 (American Psychiatric Association, 2013) and ICD-11 (WHO, 2018) — the formal endpoint of the term's euphemism-treadmill arc the chapter walks.

Deployments: Chapter 5 §5.6 ¶ (the moron-treadmill paragraph) — the documentary anchor for the *retarded* tier's retirement from the U.S. federal statute book.

Rosa's Law (Public Law 111-256, 124 Stat. 2643), signed into U.S. federal law by President Obama on October 5, 2010, replaced the term *mental retardation* with *intellectual disability* throughout federal health, education, and labor statutes. The law was named for Rosa Marcellino, a child with Down syndrome whose family campaigned for the change. The bill amended specific terminology in the *Individuals with Disabilities Education Act* (IDEA), the *Rehabilitation Act of 1973*, and the *Developmental Disabilities Assistance and Bill of Rights*

Act, replacing *mentally retarded* with *intellectually disabled* and *mental retardation* with *intellectual disability* in each federal statute.

The chapter's date — “Rosa's Law (2010)” — names the year of signing. The diagnostic-register retirement at the *DSM-5* publication carries the parallel: the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition (*DSM-5*), published May 2013, replaced *mental retardation* with *intellectual disability* (intellectual developmental disorder) as the diagnostic term. The International Classification of Diseases (*ICD-11*) followed in 2018, replacing *mental retardation* with *disorders of intellectual development*.

The chapter's deployment of Rosa's Law and *DSM-5* marks the formal retirement of *retarded* from the clinical-statutory register; the term's pejorative trajectory had been complete in popular usage well before. The named example anchors the broader argument the section makes — that English vocabulary for cognitive failure walks the euphemism treadmill across roughly one to two generations per term, and that the cycle is the signature of a calibrant-less language drifting under social-charge pressure.

Standard references: the text of Rosa's Law (Public Law 111-256, available through the U.S. Government Publishing Office); the American Psychiatric Association's *DSM-5* (APA, 2013) at the diagnostic criteria for *Intellectual Disability*; the World Health Organization's *ICD-11* documentation.

pinker-euphemism-treadmill

Short: Steven Pinker, “The Game of the Name,” *The New York Times* op-ed (April 5, 1994), introduces the coinage *euphemism treadmill* — naming the phenomenon by which a euphemism, adopted to escape the stigma of an older term, accumulates the same stigma

within a generation because the stigma attaches to the referent, not the word; extended in *The Stuff of Thought* (Viking, 2007), Chapter 7.

Deployments: Chapter 5 §5.6 ¶ (the same moron-treadmill paragraph) — the citation for Steven Pinker’s naming of the phenomenon as the *euphemism treadmill*.

Steven Pinker, “The Game of the Name,” *The New York Times*, op-ed, April 5, 1994. In this op-ed, Pinker introduces the coinage *euphemism treadmill* to name the recurring phenomenon by which a euphemism, adopted to escape the stigma attached to an older term, accumulates the same stigma across a generation and is itself displaced by a new euphemism. The naming has stuck across cognitive science and linguistic anthropology.

Pinker’s later extended discussions of the phenomenon appear in *The Stuff of Thought: Language as a Window into Human Nature* (Viking, 2007), Chapter 7; *The Better Angels of Our Nature* (Viking, 2011), in the relevant chapters on shifting norms; and across his standing lecture circuit. The structural diagnosis Pinker offers: the stigma attaches to the referent (the underlying social phenomenon being named), not to the word. Whatever new word replaces the old one inherits the stigma within a generation or two because the referent has not changed. The replacement cycle therefore continues indefinitely; only the words change.

Chapter 5 §5.6 deploys Pinker’s coinage to name the phenomenon for the reader and to set up the structural contrast: where English (calibrant-less) walks the treadmill, Sanskrit (calibrant-anchored) does not. The contrast carries the chapter’s argument. The book treats Pinker’s diagnosis as accurate but limited — the diagnosis names the phenomenon without naming the structural condition (calibrant absence) that makes the phenomenon possible.

Standard references: Pinker, *The New York Times* op-ed of April 5, 1994 (accessible through *The New York Times* archive); *The Stuff*

of *Thought* (Viking, 2007), pp. 320–328 (the euphemism-treadmill discussion); secondary discussions across cognitive-linguistic and rhetorical literature.

rasashastra-chemistry-anticipation

Short: *Rasaśāstra* (रसशास्त्र) — the Indic science of mineral and metallic substances, with the *mahā-rasa* (महारस) / *upa-rasa* (उपरस) / *dhātu* (धातु) / *ratna* (रत्न) classifications by reactivity, structural property, and combinatorial bonding behavior that anticipate the modern periodic table’s organizational principles — runs across the *Rasārṇava* (रसार्णव), *Rasaratnasamuccaya* (रसरत्नसमुच्चय) of Vāgbhaṭa, and the broader medieval canonical literature; documented in P. C. Ray, *A History of Hindu Chemistry* (2 vols., 1902–1909).

Deployments: Chapter 6 §6.2 ¶ (the chemical-sciences paragraph) — the citation block for the *Rasaśāstra* and *Rasāyana-shastra* disciplines’ anticipation of modern chemical categories.

The Indic chemical sciences operate across two related but distinct disciplines: *Rasaśāstra* (the science of mineral and metallic substances, anchored in the alchemical, metallurgical, and medicinal-mineral applications) and *Rasāyana-shastra* (the science of *rasāyana* — the rejuvenative and therapeutic preparations of the *Āyurveda* and broader medicinal disciplines). Both disciplines catalogue substances, classify them by reactivity properties, and prescribe their preparation, combination, and therapeutic use.

The *Rasaśāstra* canonical texts include: *Rasārṇava* (the foundational compendium, available in editions across multiple manuscript recensions); *Rasaratnasamuccaya* of Vāgbhaṭa (the standard reference compendium, eleventh through thirteenth century by conventional reckoning); *Rasendracūḍāmaṇi*; *Rasaratnākara* of Nityanātha; *Ānandakanda*. The *Rasāyana* literature is embedded across the

Caraka Saṃhitā and *Suśruta Saṃhitā* (the foundational *Āyurvedic* compendia), with extensive treatments in *Cakradatta*, *Bhāvaprakāśa*, *Śārṅgadhara Saṃhitā*, and the broader medieval *vaidya* literature.

The structural anticipation of modern chemistry the chapter names operates at several levels. The *Rasaśāstra* discipline classifies substances into *mahā-rasa* (major reactive substances — mercury, sulfur, mica, copper-pyrite, magnetite, cinnabar, and others), *upa-rasa* (auxiliary reactives), *sādhāraṇa-rasa* (common reactives), *ratna* (gemstones with prescribed reactive properties), *dhātu* (metals — gold, silver, copper, iron, tin, lead, zinc, mercury), and *upa-dhātu* (auxiliary metals — brass, bronze, *kāṃśya*). The classifications by reactivity, by structural property (atomic-mass-like ordering visible across the metal sequence), and by combinatorial bonding behavior anticipate categories the modern periodic table later formalized.

The chapter's claim — that the work is performed in a register the modern academy has not yet learned to read on the *paramparā*'s own terms — picks up a recurring polemic move across multiple chapters: the orthodoxy treats *Rasaśāstra* and *Rasāyana* as quaint pre-scientific practices or as ethnobotanical curiosities, rather than as engineering operating in its own register. The standard reference literature in *Rasaśāstra* runs to many volumes; the contemporary scholarly reception treats the discipline primarily as an artifact of medical history rather than as a working chemical-engineering system.

Standard references: P. C. Ray, *A History of Hindu Chemistry* (two volumes, Bengal Chemical and Pharmaceutical Works, Calcutta, 1902 and 1909) — the foundational nineteenth-century reconstruction of the *Rasaśāstra* discipline by an Indian chemist who could read the texts on their own terms. Modern editions and translations: *Rasarat-nasamuccaya* with editions across Chaukhamba Sanskrit Series and Krishnadas Academy publications; *Rasārṇava* in the Anandashrama Sanskrit Series; the *History of Science, Philosophy, and Culture in Indian Civilization* (Centre for Studies in Civilizations, multi-volume series, 1997–present) volume on chemistry. The work is large and

continuous; the chapter's citation gestures at the body of work rather than picking out a single canonical passage.

saptadhātu-canonical

Short: The *saptadhātu* (सप्तधातु) — *rasaḥ* (रसः, plasma), *raktam* (रक्तम्, blood), *māṃsam* (मांसम्, flesh), *medas* (मेदस्, fat), *asthi* (अस्थि, bone), *majjā* (मज्जा, marrow), *śukram* (शुक्रम्, generative fluid) — is the *Āyurvedic* (आयुर्वेद) cascade-of-refinement architecture of the body, each stratum produced from the previous through *dhātu-agni* (धात्वग्नि, the *dhātu*-specific digestive fire); canonical across *Caraka Saṃhitā*, *Suśruta Saṃhitā*, and *Aṣṭāṅga Hṛdaya*.

Deployments: Chapter 6 §6.3 ¶ (the *Āyurveda* / *Śarīra-vijñāna* paragraph) — the citation anchor for the seven-fold *dhātu* classification of the body.

The *saptadhātu* — *seven dhātus* — is the canonical classification of the body's structural strata across the *Āyurvedic* discipline. The seven are enumerated, in their standard sequence:

1. रसः (*rasaḥ*) — plasma; the first product of digested food.
2. रक्तम् (*raktam*) — blood; produced from *rasa*.
3. मांसम् (*māṃsam*) — flesh, muscle tissue; produced from *rakta*.
4. मेदस् (*medas*) — adipose tissue, fat; produced from *māṃsa*.
5. अस्थि (*asthi*) — bone; produced from *medas*.
6. मज्जा (*majjā*) — marrow and the related nervous-system substance; produced from *asthi*.
7. शुक्रम् (*śukram*) — generative fluid; the final and most refined product of the cascade.

The classification is anchored across the principal *Āyurvedic* compendia. *Caraka Saṃhitā*, *Sūtrasthāna* chapter 28 and *Cikitsāsthāna* chapter 15 develop the *dhātu* theory in detail. *Suśruta Saṃhitā*,

Sūtrasthāna chapter 14 and *Śārīrasthāna* chapter 4 develop the structural-physiological framework. *Aṣṭāṅga Hṛdaya* of Vāgbhaṭa (the standard reference compendium that consolidates the earlier *Caraka* and *Suśruta* materials) carries the classification across its *Sūtrasthāna* and *Śārīrasthāna*.

The cascade-of-refinement structure the chapter names — each *dhātu* produced from the previous one through a digestive-metabolic process the *Āyurvedic* discipline calls *dhātv-agni* (the *dhātu*-specific digestive fire) — is the standard mechanism the *Āyurvedic* texts describe. The seven-fold cascade takes approximately one month (per the standard textual reckoning) for food to traverse from *rasa* to *śukra* — roughly five days at each level of refinement. The mechanism the discipline describes is functionally analogous to a multi-stage biochemical refinement pipeline, with each stage performing a specific transformation on the output of the previous stage.

The structural point the chapter establishes: the *saptadhātu* is the engineering category-system the *Āyurveda* uses to describe the body. Each *dhātu* is a constituent constant — the structural element that participates in physiological function while persisting as a recognizable category. The *dhātu* terminology in the *Āyurvedic* register is the same *dhātu* terminology operating in metallurgy, chemistry, grammar, and the broader Indic engineering vocabulary: the constituent constant. The body, in the *Āyurvedic* engineering, is built from seven constituent constants in a cascade-of-refinement architecture.

Standard references: *Caraka Saṃhitā* (Chaukhamba Sanskrit Series, with the *Cakrapāṇidatta* commentary; multiple modern translations); *Suśruta Saṃhitā* (Chaukhamba Sanskrit Series, with *Ḍalhaṇa* commentary; multiple modern translations including Bhishagratna and Sharma); *Aṣṭāṅga Hṛdaya* (Krishnadas Academy and Chaukhamba editions). Modern scholarly treatments: Dominik Wujastyk, *The Roots of Ayurveda* (Penguin Classics, 2003); G. Jan Meulenbeld, *A History of Indian Medical Literature* (Egbert Forsten, 1999–2002, five volumes — the standard reference work). The

classification is canonical across the *Āyurvedic* discipline and is not contested within the discipline's literature.

dhatupatha-count-and-ganas

Short: Pāṇini's *Dhātupāṭha* (धातुपाठ) enumerates approximately 2,000 verbal roots organized into ten *gaṇāḥ* (गणाः) — *Bhvādi* (भ्वादि, 1), *Adādi* (अदादि, 2), *Juhotyādi* (जुहोत्यादि, 3), *Divādi* (दिवादि, 4), *Svādi* (स्वादि, 5), *Tudādi* (तुदादि, 6), *Rudhādi* (रुधादि, 7), *Tanādi* (तनादि, 8), *Kryādi* (क्र्यादि, 9), *Curādi* (चुरादि, 10) — each *gaṇa* defined by the specific morphological transformations its roots undergo in conjugation; the operand-set of the *Aṣṭādhyāyī*'s generative apparatus.

Deployments: Chapter 6 §6.5 ¶ (the grammatical-*dhātu* paragraph) — the citation anchor for Pāṇini's *Dhātupāṭha* enumeration of approximately two thousand verbal roots organized into ten *gaṇāḥ*.

The *Dhātupāṭha* is the canonical enumeration of Sanskrit verbal roots organized as an appendix to Pāṇini's *Aṣṭādhyāyī*. The standard count is approximately 1,943 to 2,014 roots depending on the recension (the count varies slightly across the surviving manuscript recensions and across the commentarial recensions; the canonical figure conventionally cited is ~2,000). The roots are organized into ten *gaṇāḥ* (classes), each *gaṇa* defined by the specific morphological transformations the roots undergo in their conjugational forms:

1. *Bhvādi* (the *bhū*-and-following class — class 1) — the largest class, containing roots that take the *-a-* thematic suffix in conjugation. Includes *bhū* (to be), *gam* (to go), *paṭh* (to read), *yaj* (to sacrifice), and the majority of the *Dhātupāṭha*.
2. *Adādi* (the *ad*-and-following class — class 2) — athematic roots, conjugated without the *-a-* suffix; the *ad* (to eat) and *as* (to be) class.

3. **Juhotyādi** (the *hu*-and-following class — class 3) — reduplicating roots in present-tense formation.
4. **Divādi** (the *div*-and-following class — class 4) — roots taking the *-ya-* thematic suffix.
5. **Svādi** (the *su*-and-following class — class 5) — roots taking the *-nu/-no-* suffix.
6. **Tudādi** (the *tud*-and-following class — class 6) — roots taking the *-a-* suffix with thematic accent.
7. **Rudhādi** (the *rudh*-and-following class — class 7) — roots with the *-na-* infix.
8. **Tanādi** (the *tan*-and-following class — class 8) — small class with the *-u-* suffix.
9. **Kryādi** (the *krī*-and-following class — class 9) — roots taking the *-nā-* suffix.
10. **Curādi** (the *cur*-and-following class — class 10) — causative and intensive roots with the *-aya-* suffix.

The classification is morphological — each *gaṇa* is defined by the formal transformations the roots undergo in producing finite verbal forms. The *gaṇa*-classification is not a semantic classification (the semantic field of a root does not determine its *gaṇa*); it is a structural-formal classification governing how the root combines with the tense, mood, voice, person, and number affixes to produce well-formed verbal output.

The structural point the chapter establishes: Pāṇini's enumeration is not a vocabulary list. The *Dhātupāṭha* is a *catalogue of operating elements*, each element specified by its formal-combinatorial properties (the *gaṇa* it belongs to, the meaning it carries, the irregularities it exhibits in specific formations). The catalogue is the input to the *Aṣṭādhyāyī*'s generative apparatus — the rules that operate on these roots, applying the appropriate affixes per *gaṇa* and producing the full verbal-derivative apparatus of the language. The *Dhātupāṭha* is, in formal-systems vocabulary, the operand-set of the generative grammar; the *Aṣṭādhyāyī* is the operator-set. The two together con-

stitute the engineering of Sanskrit’s verbal system. Chapter 10 develops the *varṇa*-to-*dhātu* synthesis in full; Chapter 11 develops the periodic-table account of the full *dhātu* inventory.

Standard references: the *Dhātupāṭha* with the *Kshīratarāṅgiṇī* commentary of Kṣīrasvāmin (the standard early-medieval commentary); the *Dhātupāṭha* in the various standard editions of the *Aṣṭādhyāyī* (Vasu’s *The Aṣṭādhyāyī of Pāṇini*, 1891; Sharma’s *The Aṣṭādhyāyī of Pāṇini*, multi-volume, Munshiram Manoharlal, 1987–2003; Katre’s *Aṣṭādhyāyī of Pāṇini*, University of Texas Press, 1987). Modern scholarly treatments: George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988); Hartmut Scharfe, *Grammatical Literature* (Volume V, Fascicle 2 of Jan Gonda’s *A History of Indian Literature*, Otto Harrassowitz, 1977).

retroflex-substrate-standard-account

Short: The PIE-orthodox account explains the retroflex (*mūrdhanya* (मूर्धन्य)) set as substrate-acquired (Emeneau 1956), contact-induced (Kuiper, *Aryans in the Rigveda*, 1991), or independently developed inside “Indo-Aryan” — all treating the set as peripheral and late; the architectural test reverses the framing — the 5×5 *varga* (वर्ग) matrix places *mūrdhanya* at its geometric center, with the velar / palatal / dental / labial rows organized *around* it — what is structurally central was not acquired peripherally.

Deployments: Chapter 16 §16.2 — the *Substrate-Borrowing Claim* section that names the orthodoxy’s canonical formulation Ch 16 prosecutes; Chapter 17 §17.4 ¶ — the *retroflex core* paragraph that anchors the standard PIE-orthodox account of the *mūrdhanya* development as a substrate/contact/independent-development feature.

The standard PIE-orthodox account of the retroflex (*mūrdhanya*) consonant series in Indic languages runs along three sub-accounts,

each operating within the framework's structural commitment to the migration-from-outside account.

Sub-account 1: Substrate influence. The retroflex set is acquired from a pre-existing non-Indo-European substrate language (or family of languages) spoken across the subcontinent before the framework-required migration. The standard candidates the framework names are “Dravidian” (the framework’s family-tree categorization of the south-Indian languages, all of which possess robust retroflex inventories) or an unnamed pre-Vedic substrate. The seminal statement is Murray Emeneau, “*India as a Linguistic Area*” (*Language* 32, no. 1, 1956: 3–16), which names the retroflex set as one of the areal features (alongside quotative *iti*, dative-of-purpose, and echo-word compounds) Sanskrit allegedly acquired through subcontinental contact; the argument is extended in *Bilingualism and Structural Borrowing* (*Proceedings of the American Philosophical Society* 106, no. 5, 1962) and *Language and Linguistic Area* (Stanford University Press, 1980). Internal dissent within the orthodoxy: Hans Henrich Hock, “*Substratum Influence on (Rig-Vedic) Sanskrit?*” (*Studies in the Linguistic Sciences* 5, no. 2, 1975: 76–125), argues against the substrate hypothesis on its own methodological terms — internal criticism the textbook account has not absorbed.

Sub-account 2: Contact-induced development. The retroflex set arose through contact between the framework’s “incoming” speakers and the subcontinental substrate populations, with bilingual transfer of phonetic categories. F. B. J. Kuiper, *Aryans in the Rigveda* (Rodopi, 1991), develops the contact-induced account in detail; Madhav Deshpande and Peter Hook, eds., *Aryan and Non-Aryan in India* (University of Michigan Press, 1979), collects the relevant comparative work.

Sub-account 3: Independent development. The retroflex set developed inside what the framework calls “*Indo-Aryan*” after the migration, through internal phonological processes (typically described as cluster simplification, retraction of dental stops adjacent to *-r-* or *-i-*, or other context-conditioned shifts). This is the framework’s pre-

ferred account in some recent literature (e.g., the references in the *Encyclopedia of Indo-European Culture*, Mallory and Adams, 1997, s.v. “Indo-Aryan”).

All three sub-accounts share a structural feature: the retroflex set is treated as *peripheral* and *acquired* rather than as *central* and *original* to the language’s phonological architecture. The set is something that happened to Indic phonology, not something built into Indic phonology from the start. The framework needs this account because the alternative (the retroflex set as a designed feature of the *varṇamālā*) would imply an engineered architecture present before the migration the framework requires.

The chapter’s prosecutorial move at this paragraph: the standard account handles the absence in the PIE reconstruction by treating the *mūrdhanya* set as late, peripheral, and from-outside. The architectural test the chapter applies asks the opposite question: *why is the mūrdhanya set structurally central?* The 5×5 *varga* matrix places the *mūrdhanya* row at the geometric center, with the *kaṇṭhya* (velar) and *tālavya* (palatal) above and the *dantya* (dental) and *oṣṭhya* (labial) below. The architecture is built around the *mūrdhanya* set, not around the dental or velar sets. A feature that is structurally central is not a feature acquired peripherally and late.

The two accounts — the standard PIE-orthodox account and the architectural-test account — produce incompatible predictions about where in the *varṇamālā* the *mūrdhanya* set should sit. The orthodox account predicts a peripheral position; the architectural account predicts a central position. The *varṇamālā* itself supplies the test: the *mūrdhanya* set sits at the center. The architectural account is what the data supports; the orthodox account is what the framework needs.

Standard references for the orthodox account: Murray Emeneau, *Language and Linguistic Area* (Stanford University Press, 1980); F. B. J. Kuiper, *Aryans in the Rigveda* (Rodopi, 1991); Madhav Deshpande

and Peter Hook, eds., *Aryan and Non-Aryan in India* (University of Michigan Press, 1979); Bertil Tikkanen, *The Sanskrit Gerund: A Synchronic, Diachronic, and Typological Analysis* (Helsinki, 1987) on the substrate-influence framing; J. P. Mallory and Douglas Q. Adams, *Encyclopedia of Indo-European Culture* (Fitzroy Dearborn, 1997), s.v. “Indo-Aryan” and “Retroflex.”

heavenly-city-becker

Short: Carl L. Becker, *The Heavenly City of the Eighteenth-Century Philosophers* (Yale University Press, 1932; Storrs Lectures, Yale Law School, 1931) — the foundational scholarly statement that the *Enlightenment* philosophes did not abandon Christian eschatology but secularized it, the *heavenly city* reconstituted as the perfected society achieved through reason across a single linear-historical trajectory.

Deployments: Chapter 3 §3.1 ¶ (the introduction of *progressive orthodoxy* as the doctrinal formation) — the canonical-reference anchor for the secularization-of-Christian-eschatology account the chapter develops.

Carl L. Becker, *The Heavenly City of the Eighteenth-Century Philosophers* (Yale University Press, 1932). The book was Becker’s Storrs Lectures, delivered at Yale Law School in 1931 and published the following year. The central argument: the eighteenth-century European philosophes — Voltaire, Rousseau, Diderot, Condorcet, and the broader *Enlightenment* current — did not abandon the Christian eschatology they inherited; they secularized it. The medieval Christian *heavenly city* — the New Jerusalem at the end of history, the kingdom of God, the final reconciliation of human society to divine order — was reconstituted in *Enlightenment* thought as the *heavenly city on earth*: the perfected society to be achieved through reason, science, and political reform across a single linear-historical trajectory. The

structural form of the Christian eschatology was preserved; the eschatology's content was replaced.

Becker's chapters develop the argument with specific reference to the philosophes' rhetoric of *progress*, *perfectibility*, *enlightenment*, and *the future age*. He shows that the philosophes operated, at the structural level of their thinking, in the same eschatological time-frame the Christian tradition had established: a fallen present, a redemptive trajectory, a perfected end-state. The substitution from religious to secular vocabulary obscured the continuity but did not break it.

The chapter's deployment of Becker as canonical reference: Becker is the named scholar who first established the secularization-of-Christian-eschatology account as a serious historiographic position. Subsequent scholarship (Voegelin, Gray, Löwith, and others the chapter also names) extends Becker's diagnosis along specific axes. Becker's contribution is the original framing the chapter operates within.

Standard reference: Carl L. Becker, *The Heavenly City of the Eighteenth-Century Philosophers* (Yale University Press, 1932; 70th-anniversary edition with foreword by Johnson Kent Wright, Yale University Press, 2003). Secondary discussions: Peter Gay, *The Enlightenment: An Interpretation* (Knopf, 1966–1969, two volumes) — the standard mid-twentieth-century reassessment that argues against Becker's continuity-thesis while preserving Becker's framing as the position to be argued with; Karl Löwith, *Meaning in History* (University of Chicago Press, 1949) — parallel-development argument extending Becker's framing to the philosophy of history more broadly.

end-of-history-fukuyama

Short: Francis Fukuyama, *The End of History and the Last Man* (Free Press, 1992), building on “The End of History?” (*The National Interest*, Summer 1989) — argues that Western liberal democracy is the terminal political form after the Cold War; the title and frame are explicit secularizations of Christian eschatology (*end of history* = *end times* without the divine vertical).

Deployments: Chapter 3 §3.1 ¶ (the substitutions paragraph) — the citation anchor for *end of history* as the secular continuation of *end times*.

Francis Fukuyama, *The End of History and the Last Man* (Free Press, 1992). The book’s central argument, building on Fukuyama’s earlier essay “The End of History?” (*The National Interest*, Summer 1989), is that the collapse of the Soviet Union and the triumph of Western liberal democracy as the apparent universal political form mark the *end of history* in the Hegelian sense — the resolution of the dialectical sequence that had carried human political development across competing forms (monarchies, theocracies, fascisms, communisms) and the arrival at the final, stable political form to which all societies will converge.

The structural fact the chapter establishes: the title and conceptual frame are explicit secularizations of Christian eschatological vocabulary. *End of history* is *end times* operating without the divine vertical. The arrival at a final and stable order is *the kingdom* operating without the king. The convergence of all societies onto a single political form is the *universal Christ-confession* operating without Christ. Fukuyama’s framework operates within the same eschatological temporal structure the Christian tradition established; only the eschaton’s content has been swapped.

The chapter’s deployment is precisely on the substitution mechanism, not on the substantive argument about post-Cold-War political con-

vergence. Whether or not Fukuyama's substantive claim about liberal democracy's terminal triumph holds (it has been substantially contested across the subsequent decades), the structural point — that the *end-of-history* frame is a secularization of *end-times* — is preserved across the contestation.

Standard reference: Francis Fukuyama, *The End of History and the Last Man* (Free Press, 1992; reissued Free Press, 2006, with new afterword). The original essay: "The End of History?" *The National Interest* 16 (Summer 1989): 3–18. Secondary discussions: John Gray, *Black Mass* (Penguin, 2007) — which extends the secularized-eschatology diagnosis specifically to Fukuyama; Anderson's *The Ends of History* essay collection; the Fukuyama-Huntington debate around the 1993 *Foreign Affairs* exchange.

voegelin-gnosticism

Short: Eric Voegelin, *The New Science of Politics* (University of Chicago Press, 1952; Walgreen Lectures, Chicago, 1951) — argues that modern political ideologies (Marxism, progressivism, positivism) are *gnostic* religious formations under secular self-description, claiming privileged historical-developmental knowledge and the capacity to engineer immanent perfection; extended across *Order and History* (5 vols., 1956–1987).

Deployments: Chapter 3 §3.1 ¶ (the genealogy-not-metaphor paragraph) — the citation anchor for the gnostic-political-religion analysis of modern secular eschatologies.

Eric Voegelin, *The New Science of Politics* (University of Chicago Press, 1952). The book is Voegelin's Walgreen Lectures at the University of Chicago, delivered in 1951 and published the following year. The central argument relevant to Chapter 3: modern political ideologies (Marxism, progressivism, positivism, Nazism, communism,

democratism) are not properly secular phenomena; they are *gnostic* religious formations operating under secular self-description.

Voegelin's gnostic analysis draws on the early Christian heresy of Gnosticism — the doctrine that the world is fallen, that hidden knowledge (*gnosis*) of the divine order will save the elect, and that the elect can engineer the immanent perfection of the world through application of the secret knowledge. The structural template Voegelin diagnoses: the modern political religions claim privileged access to the historical-developmental knowledge (Marxist dialectical materialism, positivist scientific-progress doctrine, progressive-evolutionary doctrine), claim the capacity to engineer immanent perfection (the classless society, the technological utopia, the perfected democratic order), and structure their politics around the engineering of this immanent end-state. The structural template is gnostic; the secular vocabulary is the cover.

Voegelin's later multi-volume *Order and History* (LSU Press / University of Missouri Press, 1956–1987, five volumes) extends the framework substantially. The relevant volumes for the chapter's purpose are Volume IV (*The Ecumenic Age*, 1974) and Volume V (*In Search of Order*, 1987), which develop the historical sociology of order-and-disorder formations across multiple civilizations.

The chapter's deployment: Voegelin is the canonical reference for the structural diagnosis that modern secular ideologies operate as religious formations under secular self-description. The diagnosis is the structural backbone of the *fourth Abrahamic religion* framing the chapter develops. Voegelin himself did not extend the analysis to the contemporary *progress / modernization / development / climate-eschatology* register the chapter prosecutes; the extension is the chapter's contribution.

Standard reference: Eric Voegelin, *The New Science of Politics: An Introduction* (University of Chicago Press, 1952; reissued with introduction by Dante Germino, 1987). Secondary discussions: Ellis Sandoz,

The Voegelinian Revolution (LSU Press, 1981, revised edition Transaction, 2000); the multi-volume *Collected Works of Eric Voegelin* (University of Missouri Press / Louisiana State University Press, 1989–2009).

black-mass-gray

Short: John Gray, *Black Mass: Apocalyptic Religion and the Death of Utopia* (Allen Lane / Penguin, 2007) — extends the Becker / Voegelin diagnosis into the post-Cold-War political-religious formations; the *black mass* inversion names apocalyptic religious form operating through political vehicles that disavow religion, which makes the framework more dangerous because invisible to its practitioners.

Deployments: Chapter 3 §3.1 ¶ (the *Enlightenment*-as-post-religious paragraph) — the citation anchor for the secularized-utopianism-as-religious-formation argument.

John Gray, *Black Mass: Apocalyptic Religion and the Death of Utopia* (Allen Lane / Penguin, 2007). The book extends the Becker / Voegelin / Löwith line of analysis into the post-Cold-War political-religious formations of the early twenty-first century. The central argument: modern utopian and apocalyptic political movements — from twentieth-century communism and fascism through neo-conservative regime-change interventionism and contemporary liberal-democratic universalism — are *secularized continuations* of the Christian apocalyptic tradition. The *black mass* of the title names the inversion: the apocalyptic religious form operating through political vehicles that explicitly disavow religion.

Gray's specific contribution that the chapter draws on: the apocalyptic-utopian framework, once secularized, becomes *more dangerous*, not less, because its religious character is invisible to its practitioners. The Christian missionary at least named himself as a missionary; the modern liberal-democratic regime-change

advocate or the climate-doom activist operates from the same apocalyptic-utopian template but under the cover of secular policy or scientific consensus. The cover is what makes the framework portable into territories that have their own — the structural move the chapter generalizes from Gray to the broader *missionaries of progress* phenomenon.

Gray's related works extend the diagnosis: *Straw Dogs: Thoughts on Humans and Other Animals* (Granta, 2002) — the critique of secular humanism as a religious formation under self-disavowal; *Heresies: Against Progress and Other Illusions* (Granta, 2004); *The Silence of Animals: On Progress and Other Modern Myths* (Allen Lane, 2013) — extending the *progress-as-myth* analysis directly. The body of work across these volumes is the contemporary continuation of the Becker / Voegelin line.

Standard reference: John Gray, *Black Mass: Apocalyptic Religion and the Death of Utopia* (Allen Lane, 2007; Penguin paperback, 2008). Secondary engagements: the philosophical reviews in *The New Republic* (Mark Lilla), *The New York Review of Books* (Tony Judt), and the *Times Literary Supplement* across 2007–2008.

ambedkar-pakistan-partition-1945

Short: B. R. Ambedkar, *Pakistan, or the Partition of India* (Thacker and Co., Bombay, 1945; expanded from *Thoughts on Pakistan*, 1940) — Chapter X (pp. 330–332) carries the “*Islam is a close corporation...*” passage diagnosing the structural-religious framework that distinguishes Muslims from non-Muslims at the level of fundamental civic identity (*dār al-Islām* / *dār al-ḥarb* / *dhimmī* / *kāfir* / *jizyā* / *jihād*) — the closed-corporation logic the chapter generalizes to all four Abrahamic religions.

Deployments: Chapter 3 §3.1 ¶ (the citation block before the close-

corporation passage) — the citation anchor for the Ambedkar passage on Islam as a closed corporation.

B. R. Ambedkar, *Pakistan, or the Partition of India* (Thacker and Company, Bombay, 1945; the second, expanded edition of *Thoughts on Pakistan*, first edition 1940). The book is Ambedkar's structural analysis of the Pakistan demand, the Muslim League's two-nation theory, and the underlying ideological frameworks at play. The passage Chapter 3 cites verbatim — "*Islam is a close corporation...*" — appears in Chapter X of the 1945 edition (page 330–332 in the standard pagination), in the section analyzing the structural-religious bases of the two-nation theory.

The structural argument Ambedkar develops in the surrounding pages: the Muslim community's political behavior under colonial India operates from inside a religious-doctrinal framework that distinguishes Muslims from non-Muslims at the level of fundamental civic identity, with the brotherhood of Islam reserved for fellow Muslims and the relation to non-Muslims structured around the Islamic legal categories (*dār al-Islam*, *dār al-ḥarb*, *dhimmī*, *kāfir*, *jizyā*, *jihād*). Ambedkar's diagnosis is that this structural-religious framework is incompatible with the universal-citizenship framework a unified post-colonial India would require, and that the two-nation theory's demand for partition is, at the structural level, an honest acknowledgment of the framework's reality.

The chapter's deployment: the passage anchors the chapter's broader structural diagnosis that the Abrahamic political religions — Islam, Christianity in its colonial-evangelical mode, and the secular *fourth Abrahamic religion* the chapter is developing — operate on a closed-corporation logic that distinguishes the in-group (the saved, the developed, the enlightened, the modern) from the out-group (the infidel, the backward, the unenlightened, the resistant) and that this structural distinction is what gives the framework its missionary-and-defensive apparatus. Ambedkar diagnoses the pattern at one of the framework's deployments; the chapter generalizes the pattern across

all four religions.

Standard references: B. R. Ambedkar, *Pakistan, or the Partition of India* (Thacker and Company, Bombay, 1945; reprinted in *Writings and Speeches of Dr. B. R. Ambedkar*, Volume 8, Government of Maharashtra Education Department, Bombay, 1990; available online through the Ambedkar Foundation digital archive). The first edition (*Thoughts on Pakistan*, 1940) carries the structural argument in slightly different form; the 1945 edition expands the analysis substantially in response to the political developments of 1940–1945. Citations and selections: Christophe Jaffrelot, *Dr. Ambedkar and Untouchability* (Permanent Black, 2005); Arundhati Roy, *The Doctor and the Saint* (Verso, 2017, with B. R. Ambedkar's *Annihilation of Caste*, but referencing the *Pakistan* analysis).

rostow-modernization-theory

Short: W. W. Rostow, *The Stages of Economic Growth: A Non-Communist Manifesto* (Cambridge University Press, 1960; subsequent editions 1971, 1990) — the foundational mid-twentieth-century statement of *modernization theory*, prescribing a five-stage developmental sequence (*traditional* → *preconditions for take-off* → *take-off* → *drive to maturity* → *high mass consumption*) terminating in American mass-consumption capitalism; the apparatus by which the developmental-progress framework was deployed against post-colonial territories.

Deployments: Chapter 3 §3.4 ¶ (the missionaries-of-progress paragraph) — the citation anchor for the *modernization theory* doctrine.

W. W. Rostow, *The Stages of Economic Growth: A Non-Communist Manifesto* (Cambridge University Press, 1960). The book is the foundational mid-twentieth-century theoretical statement of *modernization theory* — the doctrine that all societies pass through a sequence

of developmental stages from *traditional* through *preconditions for take-off*, *take-off*, *drive to maturity*, and *the age of high mass consumption*. The stages-of-growth model the title names is the operational core: every society, on Rostow's account, follows the same five-stage developmental sequence, with the technological-industrial Western (and specifically American) form at the terminus.

The book's subtitle — *A Non-Communist Manifesto* — names its strategic position in the Cold War context. Rostow positioned the model as the secular-liberal-democratic answer to the Marxist-communist developmental theory; both frameworks proposed universal developmental sequences with prescribed end-states, but they prescribed different end-states (Soviet communism vs. American mass-consumption capitalism). The structural form of the developmental sequence is identical across the two competing frameworks; the destinations differ.

The chapter's deployment: Rostow's model is the *foundational doctrine of mid-twentieth-century missionary work*. The model is the technical apparatus by which the developmental-progress framework was deployed against the post-colonial territories — including India — as the prescriptive sequence the territories were required to follow. The U.S. State Department, the World Bank in its early decades, and the *Alliance for Progress* in Latin America operated from inside Rostow's framework. The doctrine's descendants — development economics in its early formulations, the *Washington Consensus* of the 1980s and 1990s, contemporary *Sustainable Development Goals* — all operate from inside the same structural template, with the prescriptive sequence updated for the moment but the missionary form preserved.

The book itself is a clear and well-written introductory text; Rostow's later works (*Politics and the Stages of Growth*, Cambridge, 1971; *The World Economy: History and Prospect*, University of Texas Press, 1978) extend the framework. Critical engagements: Frank's *dependency theory* critique (Andre Gunder Frank, *Capitalism and Underdevelopment in Latin America*, Monthly Review Press, 1967); Waller-

stein's *world-systems theory* critique (Immanuel Wallerstein, *The Modern World-System*, Academic Press, 1974); Alexander Gerschenkron's *Economic Backwardness in Historical Perspective* (Harvard University Press, 1962) — which argued, against Rostow, that the developmental sequence varies substantially across societies.

Standard reference: W. W. Rostow, *The Stages of Economic Growth: A Non-Communist Manifesto* (Cambridge University Press, 1960; second edition 1971; third edition 1990). The book's status as the foundational modernization-theory text is uncontested across development economics and the sociology of development.

juvenal-quis-custodiet

Short: Juvenal, *Satires* Book VI, lines 347–348 — ***quis custodiet ipsos custodes?*** (“*who will guard the guards themselves?*”) — originally satirical in context (the futility of placing guards on a suspected wife), but the structural question of accountability infinite-regress has become canonical across political philosophy from Plato's *Republic* (the guardian-class problem) through Madison's *Federalist* separation-of-powers apparatus to the modern administrative state.

Deployments: Chapter 3 §3.5 ¶ (the peer-review-and-the-priesthood paragraph) — the citation anchor for the canonical *quis custodiet ipsos custodes?* question Juvenal raises in the *Satires*.

The Latin phrase ***quis custodiet ipsos custodes?*** — “*who will guard the guards themselves?*” — appears in Juvenal's *Satires*, Book VI, lines 347–348. The full passage:

audio quid veteres olim moneatis amici, “pone seram, cohibe.” sed quis custodiet ipsos custodes? cauta est et ab illis incipit uxor.

“I hear what my old friends have long since warned: ‘Set

a lock, lock her up.’ But who will guard the guards themselves? The clever wife begins her affair with the guards.”

The passage’s immediate context is satirical and domestic — Juvenal is mocking the futility of placing guards on a wife suspected of infidelity, since the guards themselves become the targets of the wife’s manipulation. The structural question the line poses, however, has become canonical across political and institutional philosophy. The recurring question every authority structure must answer: *who watches the watchers? who governs the governors? who guards the guards?* The infinite-regress problem of accountability is the question’s structural content, regardless of Juvenal’s original satirical purpose.

Plato’s *Republic* (Book III, 403e and following; Book IV, 419a and following) raised the parallel question in the discussion of the guardian class — how do you ensure that the philosopher-kings, set above the rest of the city to govern in the city’s interest, do not become the city’s chief predators? Plato’s answer was structural — the guardians are subject to specific educational, dietary, and social arrangements that make corruption difficult. The question runs through the political-philosophical literature: Rousseau, Madison (the *Federalist Papers*’ separation-of-powers apparatus), the *common-law* tradition of judicial review, and into the modern administrative-state literature.

The chapter’s deployment: Juvenal’s two-thousand-year-old question is named as the canonical formulation that the *progressive orthodoxy*’s peer-review apparatus answers procedurally. The orthodoxy’s answer — *the guards guard each other* — is precisely the failure mode the question is designed to surface. Mutual qualification of credentialed insiders is not substantive verification; it is the priesthood verifying itself. Chapter 3 §3.5’s analysis of peer review as a structural mechanism is anchored at this question.

Standard references: Juvenal, *Satires*, Book VI, lines 347–348. The standard Loeb Classical Library edition: Susanna Morton Braund,

translator, *Juvenal and Persius* (Loeb Classical Library 91, Harvard University Press, 2004). Older standard: G. G. Ramsay, translator, *Juvenal and Persius* (Loeb, 1918). For the philosophical reception: Plato, *Republic*, books III–V; Steven Shapin, *A Social History of Truth* (University of Chicago Press, 1994), which extends the *quis custodiet* question to the institutional history of scientific credentialing.

ashtavakra-bandin-mahabharata

Short: The *Aṣṭāvakra* (अष्टावक्र) — *Bandin* (बन्दिन्) episode in the *Mahābhārata*’s *Vana Parva* (*adhyāyas* 132–134, Bhandarkar critical edition) — boy-sage *Aṣṭāvakra*, denied admission to Janaka’s court on grounds of youth and physical deformity, exposes the gatekeeper’s reasoning: *the assembly that judges by external attributes — age, appearance, lineage, credential — is the council of fools*; subsequently defeats *Bandin* in extended philosophical debate. The dharmic continuum’s canonical diagnosis of credential-based gatekeeping as operational failure.

Deployments: Chapter 3 §3.5 ¶ (the Hindu-paramparā’s-verdict paragraph) — the citation anchor for the *Aṣṭāvakra* / *Bandin* episode and the dharmic continuum’s diagnosis of the gatekeeper-failure as operational failure.

The *Aṣṭāvakra* episode is told in the *Vana Parva* (Book of the Forest) of the *Mahābhārata*, specifically in the *Tīrtha-yātrā* sub-section, *adhyāyas* 132–134 in the Bhandarkar critical edition. The narrative is set during the *Pāṇḍavas*’ forest exile, when the sage Lomaśa recounts to Yudhiṣṭhira and the brothers the story of the boy-sage *Aṣṭāvakra* and his confrontation with the court-philosopher *Bandin* (also called *Vandin* in some recensions) at the court of King Janaka.

The narrative summary: *Aṣṭāvakra* — whose name means *eight-bent*, referring to the eight physical deformities he carried — was the son of

the sage Kahoḍa, who had been defeated in a philosophical debate at Janaka's court by Bandin and, per Bandin's standing condition, drowned in the river as the loser. Aṣṭāvakra, learning of his father's fate while still a child, set out for Janaka's court to challenge Bandin.

At the court door, the gatekeeper refused to admit Aṣṭāvakra on grounds of his youth and his physical deformity — the boy could not possibly carry the philosophical authority required to challenge the court's standing philosopher. Aṣṭāvakra's response (the load-bearing moment for Chapter 3's deployment) is to expose the gatekeeper's reasoning as failure-of-judgment: *the assembly that judges by external attributes — age, appearance, lineage, credential — is the council of fools*. The wise are not those who carry the visible markers of authority; the wise are those who carry the substantive capacity.

Aṣṭāvakra eventually gains admission to the court, defeats Bandin in extended philosophical debate, and recovers his father (Kahoḍa, who had not in fact been drowned but had been held by the *nāgas* in the underworld, returns to the surface after Bandin's defeat).

The episode is significant in the dharmic continuum for several reasons: it documents the paramparā's standing critique of credential-based gatekeeping; it establishes the *Aṣṭāvakra-Gītā* (a separate philosophical text in the *advaita* darśana, attributed to the same Aṣṭāvakra and embedded in the Yājñavalkya / Janaka philosophical lineage) in narrative context; and it provides the canonical dharmic diagnosis of the *peer-review* failure mode the chapter generalizes. The verdict the paramparā carries — that judgment-by-external-attributes is operational failure — is unambiguous across the textual reception.

The chapter's deployment: the dharmic continuum has carried this story across many generations precisely because the diagnostic it offers is permanent. Wherever the gatekeeping function in any authority structure is corrupted into credential-checking rather than substantive verification, the paramparā's verdict applies. The *progressive orthodoxy's* peer-review machinery is the contemporary deploy-

ment of the same failure mode; the *Aṣṭāvakra* episode is the dharmic primary-source for the diagnosis.

Standard references: the *Mahābhārata*, *Vana Parva*, *adhyāyas* 132–134 in the Bhandarkar Oriental Research Institute critical edition (Vishnu Sukthankar et al., eds., 1933–1966). The narrative is also extensively reproduced in the *Bhāgavata Purāṇa* and in the *Aṣṭāvakra-Gītā* commentaries. Modern translations: J. A. B. van Buitenen, *The Mahābhārata*, Volume II (University of Chicago Press, 1975) — *Vana Parva*; P. C. Roy, *The Mahabharata of Krishna-Dwaipayana Vyasa* (twelve volumes, Calcutta, 1883–1896, reprinted Munshiram Manoharlal). The *Aṣṭāvakra-Gītā* itself: Swami Nityaswarupananda, *Ashtavakra Samhita* (Advaita Ashrama, 1953); John Richards, *Ashtavakra Gita* (online, 1994).

assalayana-sutta-fwd

Short: Forward-pointer to the canonical assalayana-sutta endnote — *Majjhima Nikāya* 93’s dharmic primary-source documentation of the *ārya* (आर्य) / *dāsa* (दास) binary as a foreign-bordering-nations (*Yona* / *Kamboja*) feature, deployed at Chapter 3 §3.4 (Wilson/Griffith) and §3.6 (contest-of-architectures).

Deployments: Chapter 3 §3.4 (Wilson/Griffith passage); Chapter 3 §3.6 (contest-of-architectures passage). Both deployments are forward-pointers to the canonical *Assalāyana Sutta* endnote drafted at assalayana-sutta above.

The two assalayana-sutta-fwd markers in Chapter 3 reference the same canonical endnote treatment under assalayana-sutta (see above), which carries the full text-and-context analysis of *Majjhima Nikāya* 93 and the dharmic primary-source documentation of the *ārya* / *dāsa* binary as a foreign-bordering-nations feature. The Chapter 3 deployments both point forward to the Epilogue’s full landing

of the citation and to the canonical endnote treatment.

At chapter-lock time, the three deployment locations (Chapter 2 §2.2, Chapter 3 §3.4, Chapter 3 §3.6, and the Epilogue) will be unified to a single endnote citation. The *assalayana-sutta-fwd* markers can either be consolidated to a single *assalayana-sutta* reference or retained as forward-pointer markers if the chapters benefit from the slightly different framing the forward-pointer language provides.

No separate prose is required for *assalayana-sutta-fwd*; the canonical treatment is at *assalayana-sutta*.

shakalya-padapatha

Short: *Śākalya* (शाकल्य) — pre-Pāṇinian grammarian named in the *Aṣṭādhyāyī* and credited across the *paramparā* with producing the *Padapāṭha* (पदपाठ) of the *Ṛgveda* — the word-by-word recitation with *sandhi* (सन्धि) undone and grammatical case and number marked; Pāṇini cites Śākalya at *Aṣṭādhyāyī* 8.4.51 (*visarga* treatment), 1.1.16 (compound boundaries), 6.1.127 (vowel-*sandhi*), and 8.3.18 (*anusvāra*) — recording points where he either follows or overrules his predecessor’s analytical decisions.

Deployments: Chapter 4 §4.1 ¶ (the pre-Pāṇinian grammarians paragraph) — the citation anchor for Śākalya’s *Padapāṭha* on the *Ṛgveda*.

Śākalya (शाकल्य) is one of the pre-Pāṇinian grammarians named in the *Aṣṭādhyāyī* and credited across the *paramparā* with the production of the *Padapāṭha* (पदपाठ) for the *Ṛgveda* — the word-by-word recitation in which each *pada* of the *Samhitā-pāṭha* is decomposed into its pre-*sandhi* constituents and presented in its isolated grammatical form. The *Padapāṭha* is the second of the eleven *pāṭhas* (see endnote eleven-pathas) and operates as the analytical key to the continuous-sandhi *Samhitā* recitation. Producing it requires sub-

stantial grammatical sophistication — the analyst must identify the *sandhi*-internal word boundaries, undo the *sandhi* transformations to recover the underlying word forms, separate compound elements where appropriate, and present each *pada* with its grammatical case and number marked.

The *Padapāṭha* of the *Ṛgveda* attributed to Śākalya is preserved across the surviving *Śākala-śākhā* recension of the *Ṛgveda* (the standard recension transmitted through the *Śākalya* lineage). Pāṇini cites Śākalya by name at multiple points in the *Aṣṭādhyāyī*, including 8.4.51 (regarding *visarga*-treatment in the *Padapāṭha*), 1.1.16 (regarding the *Padapāṭha*'s treatment of certain compound boundaries), 6.1.127 (vowel-sandhi treatment), and 8.3.18 (regarding *anusvāra*-treatment). The citations record points where Pāṇini either follows Śākalya's analytical decisions or, at specific points, overrules them — preserving Śākalya's alternative analysis as a recorded option.

The structural significance the chapter establishes: Śākalya's *Padapāṭha* is a sophisticated grammatical decoding that predates Pāṇini and that Pāṇini's *Aṣṭādhyāyī* engages directly. The Sanskrit *vyākaraṇa* discipline is not founded by Pāṇini; it is *decoded* by him — the activity is **vyākaraṇam** (व्याकरणम्), analysis, the *taking-apart* of a system already in place. The pre-Pāṇinian decoding work is substantial — sufficient to produce a complete *Padapāṭha* of the *Ṛgveda*, which is itself a non-trivial accomplishment — and Pāṇini's contribution sits on top of it rather than initiating it. *Sanskrit was engineered. Encoded in the Vedas. Decoded by many. Pāṇini's decoding is the finest.*

Standard references: the *Ṛgveda Padapāṭha* in the Theodor Aufrecht and F. Max Müller editions of the *Ṛgveda Saṃhitā* (Müller's editio princeps, 1849–1874; Aufrecht's compact edition, 1861–1863); the *Sāyaṇa* commentary preserves the *Padapāṭha* alongside the *Saṃhitā* text. Modern scholarly treatments: Hartmut Scharfe, *Grammatical Literature* (Volume V, Fascicle 2 of Gonda's *A History of Indian Literature*, Otto Harrassowitz, 1977), Chapter 4 on the pre-Pāṇinian disci-

pline; George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988), discussion of Pāṇini's citations of Śākalya and the broader pre-Pāṇinian grammatical apparatus.

panini-cites-pre-paninian-grammarians

Short: Pāṇini's *Aṣṭādhyāyī* (अष्टाध्यायी) cites by name a roster of pre-Pāṇinian grammarians — *Āpiśali* (आपिशलि, 6.1.92), *Kāśyapa* (काश्यप, 1.2.25, 8.4.67), *Gārgya* (गार्ग्य, 7.3.99, 8.3.20), *Gālava* (गालव, 6.3.61, 7.1.74, 8.4.67), *Cākravarmaṇa* (चाक्रवर्मण, 6.1.130), *Bhāradvāja* (भारद्वाज, 7.2.63), *Saunaga* (सौनाग), *Senaka* (सेनक, 5.4.112), and *Sphoṭāyana* (स्फोटायन, 6.1.123) — each citation recording where Pāṇini ratifies, modifies, or preserves alternative analyses; the *Aṣṭādhyāyī* is not the founding moment of Sanskrit grammar but the finest decoding of a discipline with substantial prior depth.

Deployments: Chapter 4 §4.1 ¶ — the cluster citation anchoring the list of nine other pre-Pāṇinian grammarians whose work Pāṇini engages in the *Aṣṭādhyāyī*.

The *Aṣṭādhyāyī* cites by name a roster of grammarians whose work predates Pāṇini's decoding and whose analytical decisions the *Aṣṭādhyāyī* engages — sometimes adopting, sometimes overruling, sometimes preserving as alternatives. The named figures and the relevant Pāṇinian *sūtras*:

- *Āpiśali* (आपिशलि) — cited at *Aṣṭādhyāyī* 6.1.92 regarding a specific *vṛddhi*-formation rule.
- *Kāśyapa* (काश्यप) — cited at 1.2.25 regarding *liṭ*-tense formation; 8.4.67 regarding *anusvāra*.
- *Gārgya* (गार्ग्य) — cited at 7.3.99 regarding accent placement; 8.3.20 regarding *visarga* assimilation.
- *Gālava* (गालव) — cited at 6.3.61 regarding compound formation; 7.1.74 regarding declensional irregularity; 8.4.67 regard-

ing *anusvāra*.

- **Cākravarmaṇa (चाक्रवर्मण)** — cited at 6.1.130 regarding vowel-sandhi.
- **Bhāradvāja (भारद्वाज)** — cited at 7.2.63 regarding *vṛddhi* in specific verbal classes.
- **Saunaga (सौनाग)** — cited in the *Mahābhāṣya* commentarial literature as a pre-Pāṇinian authority.
- **Senaka (सेनक)** — cited at 5.4.112 regarding compound formation.
- **Sphoṭāyana (स्फोटायन)** — cited at 6.1.123 regarding the *sphoṭa*-doctrine of word-meaning, which Pāṇini engages in his discussion of word-form vs. word-meaning relations.

The full count of pre-Pāṇinian grammarians cited in the *Aṣṭādhyāyī* exceeds the nine listed here. The standard reckoning across the Pāṇinian discipline's literature (the *Mahābhāṣya*, the *Kāśikā-vṛtti*, the *Padamañjarī*, and modern scholarly reconstructions) identifies more than a dozen named individuals whose work Pāṇini engages directly. Each citation operates as an acknowledgment that a specific analytical decision has been made by a predecessor; Pāṇini's *sūtra* either ratifies the predecessor's decision, modifies it, or preserves the alternative analysis as a recorded option.

The structural significance the chapter establishes: the *Aṣṭādhyāyī* is not the founding moment of Sanskrit grammar but the finest decoding of a discipline with substantial prior depth. The named figures whose work Pāṇini engages constitute a *guru-shishya paramparā* of grammatical analysis that operated for many generations before Pāṇini's decoding. Pāṇini's contribution is the integration, regularization, and systematic presentation of this prior grammatical apparatus. The architects of Sanskrit — the engineers who designed the *varṇamālā* and the broader system — are upstream of even this pre-Pāṇinian *vyākaraṇa* discipline; the pre-Pāṇinian grammarians are themselves operating on an architecture that was already in place.

Standard references: George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988), Chapter 1 (introduction) and the citations across the analytical chapters; Hartmut Scharfe, *Grammatical Literature* (Otto Harrassowitz, 1977), Chapter 4 on the pre-Pāṇinian discipline; S. M. Katre, *Pāṇinian Studies* (Sarasvati Vihar Series, 1968); Paul Thieme, *Pāṇini and the Veda* (Allahabad, 1935) — the foundational philological-reconstruction work on Pāṇini’s pre-Pāṇinian-grammarian citations.

siddhe-shabdarthasambandhe-mbh

Short: Patañjali’s *Mahābhāṣya* (महाभाष्य) opens with *siddhe śabdārthasambandhe* (सिद्धे शब्दार्थसम्बन्धे) — “the relation between word and meaning being established” — the foundational axiom anchoring Chapter 4’s argument that the *vyākaraṇa* (व्याकरण) discipline operates on the premise the relation is *siddha* (सिद्ध, already established) rather than *sādhya* (साध्य, to be derived). See endnote patanjali-siddhe-shabdarthasambandhe for the full Kielhorn-edition citation and the parallel Preface deployment.

Deployments: Chapter 4 §4.2 ¶ — the citation anchor for the Pāṇinian axiom *siddhe śabdārthasambandhe* as it appears in Patañjali’s *Mahābhāṣya*.

The phrase *सिद्धे शब्दार्थसम्बन्धे* (*siddhe śabdārthasambandhe*) appears at the opening of Patañjali’s *Mahābhāṣya* (the *Paspaśāhnika* — the introductory *āhnika*), as one of the foundational statements of the *vyākaraṇa* discipline’s epistemic posture. See endnote patanjali-siddhe-shabdarthasambandhe for the full text-and-context treatment of the passage; the Chapter 4 deployment is the same canonical citation, focused on the axiom’s load-bearing function in establishing that the *vyākaraṇa* discipline operates on the premise that the word-meaning relation is *siddha* (already

established) rather than *sādhya* (to be derived).

The standard text:

सिद्धे शब्दार्थसम्बन्धे लोकतोऽर्थप्रयुक्ते शब्दप्रयोगे शास्त्रेण धर्मनियमः ।

*siddhe śabdārthasambandhe lokato 'rthaprayukte śab-
daprayoge śāstreṇa dharmaniyamaḥ.*

*“Given that the bond between word and meaning is estab-
lished (in the paramparā, not requiring derivation), and
given that the use of words for the purpose of meaning is
established in the world, the grammatical science regulates
correct usage.”*

The structural axiom Chapter 4 establishes at this passage: the gram-
matical *śāstra* is not in the business of inventing or deriving the re-
lation between words and meanings. It operates on the premise
that the relation is already given, already operating, already main-
tained across the paramparā. The *śāstra* regulates correct usage of
an already-existing system; it does not constitute the system.

Source and standard references: see endnote patanjali-siddhe-shabdarthasambandhe for the full Kielhorn edition reference and the standard scholarly treatments. The two endnotes — patanjali-siddhe-shabdarthasambandhe and siddhe-shabdarthasambandhe-mbh — point at the same pas-
sage with slightly different framing for the two deployment chapters
(Preface vs. Chapter 4); at chapter-lock time they may be consoli-
dated to a single citation.

adi-vadya-voice-as-original-instrument

Short: The Indian classical music continuum treats the human voice
as the *ādi-vādyā* (आदिवाद्य) — *first instrument* — with constructed in-
struments as partial descendants of the voice’s capabilities; anchored
in Bharata’s *Nāṭyaśāstra* (नाट्यशास्त्र) (which classifies *saṅgīta* (सङ्गीत)

into *gīta* (गीत, vocal) / *vādya* (वाद्य, instrumental) / *nṛtya* (नृत्य, dance) with *gīta* foundational) and developed across Śārṅgadeva's *Saṅgīta-ratnākara* (सङ्गीतरत्नाकर) — the conceptual ground for the *varṇamālā* as engineered classification of the voice's output.

Deployments: Chapter 7 §7.1 ¶ (opening) — the citation anchor for the Indian classical continuum's framing of the human voice as the *ādi-vādya* (original instrument) and constructed instruments as partial descendants.

The Indian classical music continuum treats the human voice as the *ādi-vādya* — the *first* or *original* instrument — across multiple textual sources and the standing performance lineages. The framing is anchored in the *Nāṭyaśāstra* of Bharata (the foundational treatise on performance, music, drama, and aesthetics in the Indic classical continuum), which classifies musical sound (*saṅgīta*) into *gīta* (vocal), *vādya* (instrumental), and *nṛtya* (dance), and treats *gīta* as the foundational category from which the others derive. The *Saṅgīta-ratnākara* of Śārṅgadeva (the standard medieval reference work on Indian music) develops the same framing in detail in its *svara-gata-adhyāya* and *vādya-adhyāya* — the chapters on pitch and on instruments.

The structural framing across the paramparā: the voice produces every category of sound that constructed instruments produce — sustained pitches like wind and string instruments, percussive attacks like drums and plucked strings, sympathetic resonances like drone instruments, articulated rhythmic patterns like rhythmic instruments. The voice is therefore the comprehensive instrument; the constructed instruments are partial descendants, each extending one or another of the voice's capabilities into a specialized acoustic register. The classical training lineages reinforce this: tabla players learn the rhythm by *speaking* the *bols* (the rhythmic syllables); singers learn melody before instrumentalists; the *guru-shishya* lineages anchor the music's transmission at the level of the voice.

The standard contemporary articulation of the framing: T. Vishwanathan and Matthew Harp Allen, *Music in South India: The Karṇāṭak Concert Tradition and Beyond* (Oxford University Press, 2004), Chapter 1; Bonnie C. Wade, *Music in India: The Classical Traditions* (Manohar, revised edition 2004); Daniel M. Neuman, *The Life of Music in North India* (Wayne State University Press, 1980, paperback University of Chicago Press, 1990).

The chapter's deployment: the *ādi-vādyā* framing is the conceptual ground for the chapter's argument that the *varṇamālā* operates as a *mouth-map* — a classification of the sounds the voice produces, organized by the anatomical apparatus that produces them. The Indian classical continuum treats the voice as the comprehensive instrument; the *varṇamālā* is the engineered classification of that comprehensive instrument's output.

vocal-tract-cm-modeling

Short: The adult human vocal tract — measured from lips to glottis along the midline — runs ~17 cm in males (range 16–20 cm), ~14–15 cm in females (range 13–16 cm); standard reference figures from Gunnar Fant, *Acoustic Theory of Speech Production* (Mouton, 1960) and Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998). The architects designed the *varṇamālā* (वर्णमाला) for a specific anatomically-measurable instrument; 17 cm is the operational reference.

Deployments: Chapter 7 §7.2 ¶ (opening) — the citation anchor for the empirical vocal-tract length figures.

The human vocal tract — measured from the lips to the glottis along the midline — varies in length across speakers as a function of age, sex, and individual anatomy. The standard reference figures the chapter cites:

- **Adult males:** mean ~17 cm (range ~16–20 cm). Standard reference: Gunnar Fant, *Acoustic Theory of Speech Production* (Mouton, The Hague, 1960); Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998), Chapter 1.
- **Adult females:** mean ~14–15 cm (range ~13–16 cm). The shorter length reflects smaller body size and shorter pharyngeal length; the difference produces the higher fundamental and formant frequencies characteristic of female voices.
- **Children:** shorter still, scaling with body size. Newborns have vocal tracts approximately 6–8 cm; childhood and adolescence carry the length toward the adult range.
- **Cross-adult range:** approximately 13 cm to 20 cm across the human population.

The figures come from standard phonetics-research measurements, primarily X-ray and MRI imaging of the vocal tract during speech production. Reference works: Fant's 1960 *Acoustic Theory* established the modeling framework; Stevens's 1998 *Acoustic Phonetics* consolidates the post-Fant measurements and modeling. The *source-filter theory* of speech production (the model by which the glottis produces a quasi-periodic source signal that the vocal tract filters into the formant pattern producing distinct vowel and consonant qualities) operates on these vocal-tract length and shape parameters.

The chapter's deployment: the vocal-tract length figures anchor the chapter's argument that the *varṇamālā* operates within a measurable, anatomically specific apparatus. The architects of the *varṇamālā* designed for a specific instrument with specific dimensions; the contemporary phonetics literature provides the measurement baseline against which the architects' design can be reconstructed. The 17-centimetre figure is the operational reference the chapter carries forward.

Standard references: Gunnar Fant, *Acoustic Theory of Speech Production* (Mouton, 1960); Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998); Peter Ladefoged and Keith Johnson, *A Course in Pho-*

netics (Cengage, multiple editions, currently 7th edition 2014); Peter Ladefoged and Ian Maddieson, *The Sounds of the World's Languages* (Blackwell, 1996), the standard cross-linguistic survey.

tabla-bols-mouth-to-drum

Short: The tabla is taught from the mouth — the player learns rhythmic compositions (*kāyda* (कायदा), *raulā, gat* (गत), *paran* (परन), *tukṛā*) first as sequences of **bols** (बोल — spoken syllables: *dha, dhin, na, tin, ti, te, ke, ge, ...*) recited in rhythm, before any contact with the drum; the *bol* is the canonical form, the stroke its drum-rendering. Preserved intact across the Delhi, Lucknow, Ajrara, Farrukhabad, Banaras, and Punjab *gharānās* (घराना) — the consonant comes first; the drum is downstream.

Deployments: Chapter 7 §7.5 ¶ — the citation anchor for the tabla *bol* lineages' preservation of the mouth-to-drum analytical-pedagogical sequence.

The tabla — the paired hand-drum of the North Indian classical music continuum — is traditionally taught through the recitation of **bols** (बोल — *spoken syllables*) before any contact with the drum itself. The *bols* are the spoken-syllable representation of the strokes the player will produce. Each stroke has a named *bol*: *dha, dhin, na, tin, ti, te, ta, ke, ge, ga, kat, trkt, dhin-na*, and others. The student learns the rhythmic compositions (*kāyda, raulā, gat, paran, tukṛā*) as sequences of *bols* — speaking them, in rhythm, before learning to produce the corresponding strokes on the drum.

The structural pedagogical principle: the drum is taught from the mouth. The *bol* is the source; the stroke is the rendering of the *bol* on the drum. A stroke that cannot be spoken cannot be played. A rhythmic composition is first a sequence of *bols*; only after the student has mastered the *bol*-sequence in speech does the rendering on the

drum proceed. The *bol* is therefore not a mnemonic device — it is the canonical form of the rhythm; the drum produces a physical analog of what the mouth produces.

The structural significance the chapter establishes: the tabla lineages operationalize, in their own pedagogical practice, the chapter's broader argument — that the voice is the *ādi-vādyā* and constructed instruments are partial descendants of the voice's capabilities. The tabla is the drum that approximates the consonantal striking of tongue against palate (the *vyañjana* of Sanskrit phonology); the player's training proceeds from the consonant (the *bol*) to its drum-rendering. The two domains — phonetic articulation and tabla performance — share a common analytical substrate.

The named tabla lineages (*gharānās*) preserve the *bol*-pedagogy intact across their distinct stylistic conventions: Delhi *gharānā*, Lucknow *gharānā*, Ajrara *gharānā*, Farrukhabad *gharānā*, Banaras *gharānā*, Punjab *gharānā*. Each carries its own *bol*-vocabulary and compositional repertoire, but the mouth-before-drum pedagogical sequence is universal.

Standard references: Robert Gottlieb, *Solo Tabla Drumming of North India* (Motilal Banarsidass, 1993, two volumes); Daniel M. Neuman, *The Life of Music in North India* (Wayne State University Press, 1980); James Kippen, *The Tabla of Lucknow: A Cultural Analysis of a Musical Tradition* (Cambridge University Press, 1988); Sudhir Mainkar, *The Tabla* (Aaroh Publications, 2010). The paramparā is also extensively documented in the broader Hindustani classical music literature and in the standing *guru-shishya* training across the named *gharānās*.

sarangi-closest-to-human-voice

Short: The *sārangī* (सारंगी) — the bowed string instrument of the Hindustani continuum with three to four main bowed strings, thirty-five

to forty sympathetic strings tuned to the *rāga* (राग), cuticle-fingering against fretless apparatus — is treated across the classical music literature as the constructed instrument closest in expressive capacity to the human voice; the bowed string analogizes the vocal cords, the sympathetic drone strings the nasal-cavity resonator. The *gharānās* of *sārangī* performance are anchored to the vocal *gharānās* they accompany.

Deployments: Chapter 7 §7.6 ¶ — the citation anchor for the *sarangi*-as-closest-to-the-voice framing in the Indian classical continuum.

The *sarangi* (सारंगी) — the bowed string instrument of the North Indian classical music continuum — is widely treated across the *Hindustani* classical music literature as the constructed instrument closest in expressive capacity to the human voice. The instrument's distinctive features: three to four main bowed strings (typically gut), thirty-five to forty sympathetic strings (typically metal) tuned to the notes of the *rāga*, fingering with the cuticles of the left hand against the strings (rather than pressing down to the fingerboard), and a continuous-pitch fretless apparatus that allows the player to slide between pitches at any speed.

The traditional accompaniment role: the *sarangi* accompanies vocalists across the *khayāl*, *thumri*, *dādrā*, *ghazal*, and folk-vocal genres. The accompaniment style mirrors the singer's melodic line note-for-note, with the *sarangi* tracking the vocal ornamentation (*gamak*, *meend*, *kan*, *murki*) at the same speed and inflection the singer produces. The accompaniment relationship is so close that the *gharānā* lineages of *sarangi* performance are anchored to the vocal *gharānās* they serve; a Delhi-*gharānā* *sarangi* player learns from the Delhi-*gharānā* vocal lineage.

The structural acoustic similarity the chapter establishes: the bowed string produces a continuous-pitch tone that the player can vary at any speed, with the sympathetic strings producing the resonance characteristic of the nasal-vowel coupling in voice production. The

bowed string is the analog of the vocal cords; the sympathetic drone strings are the analog of the nasal cavity acting as a resonator when the velum is lowered. The instrument's acoustic capacity is, across the relevant features, a direct rendering of the voice's acoustic capacity in a constructed form.

Standard references: Joep Bor, *The Voice of the Sarangi: An Illustrated History of Bowing in India* (National Centre for the Performing Arts, Bombay, 1986–1987); Daniel M. Neuman, *The Life of Music in North India* (Wayne State University Press, 1980), the chapters on sarangi performance practice; Regula Burckhardt Qureshi, “Master Musicians of India: Hereditary Sarangi Players Speak” (in *Lives in Music*, various editions); the documentary film *The Voice of the Sarangi* (Joep Bor, 1990). The paramparā's articulation of the sarangi-voice closeness is consistent across the named sources and across the standing performance lineages.

nadyashastra-four-instrument-taxonomy

Short: Bharata's *Nāṭyaśāstra*'s *Atodya-vidhi* (आतोद्यविधि) establishes the four-fold classification of musical instruments: *tata* (तत, *stretched* — chordophones), *suśira* (सुषिर, *hollow* — aerophones), *avanaddha* (अवनद्ध, *covered with skin* — membranophones), and *ghana* (घन, *solid* — idiophones); exhaustive over the categories of acoustic mechanism, parallel to the Sachs-Hornbostel organological classification (1914) but anticipating it by many generations, with the additional precision that the Sanskrit names refer to the physical mechanism rather than abstract categories.

Deployments: Chapter 7 §7.7 ¶ — the citation anchor for the *Nāṭyaśāstra*'s four-fold instrument taxonomy.

The *Nāṭyaśāstra* (नाट्यशास्त्र) of Bharata — the foundational treatise on performance, drama, music, and aesthetics in the Indian classi-

cal continuum — establishes the four-fold classification of musical instruments at the beginning of its *Atodya-vidhi* (the section on instruments). The four categories:

1. **Tata (तत)** — *stretched*. String instruments (*vīṇā*, *vipañcī*, *citra*, the later *sitar*, *sarod*, *sarangi*, *santūr*, etc.). The defining feature: a stretched string producing tone through vibration.
2. **Suṣira (सुषिर)** — *hollow*. Wind instruments (*veṇu*, *bansuri*, *śaṅkha*, *muralī*, *muraḥa*, the later *shehnai*, *nadasvaram*, etc.). The defining feature: a hollow tube producing tone through air-column resonance.
3. **Avanaddha (अवनद्ध)** — *covered with skin*. Percussion instruments with a membrane (*mṛdaṅga*, *paṇava*, *dardura*, the later *tabla*, *pakhāvaj*, *ḍholak*, etc.). The defining feature: a stretched membrane producing percussive sound through striking.
4. **Ghana (घन)** — *solid*. Idiophonic instruments — bells, cymbals, *ghaṭam* (clay pot), *tāla* (rhythmic cymbals), *jal-taraṅ* (water-bowls), and similar solid-body instruments. The defining feature: the body of the instrument itself produces sound through striking or shaking, without need of a membrane or resonating column.

The classification is structurally significant in two respects. First, it is *exhaustive* over the categories of acoustic mechanism: every constructed instrument in the Indian continuum falls into one of the four categories. Second, it is *parallel to the modern organological classification* developed by Curt Sachs and Erich von Hornbostel in 1914 (chorophones, aerophones, membranophones, idiophones), with the additional precision that the Sanskrit terms refer to the physical mechanism (stretched, hollow, covered-with-skin, solid) rather than to the abstract sound-production category. The Sachs-Hornbostel system later became the standard ethnomusicological classification; the *Nāṭyaśāstra*'s system anticipates the Sachs-Hornbostel by many generations.

The chapter's deployment: the four-fold instrument taxonomy is the broader analytical-classificatory framework within which the Indian continuum catalogued the *original instrument* (the voice) with parallel rigor. The same intellectual apparatus that produced the *Nāṭyaśāstra*'s instrument classification produced the *varṇamālā*'s phonetic classification; both are products of a continuum that catalogued acoustic mechanism with engineering precision.

Standard references: *Nāṭyaśāstra* of Bharata, *Atodya-vidhi* section. The standard editions: M. Ghosh, *The Nāṭyaśāstra* (Manisha Granthalaya, Calcutta, two volumes, 1950 and 1961); Adya Rangacharya, *The Nāṭyaśāstra* (Munshiram Manoharlal, 1996, single-volume English translation); the *Gaekwad Oriental Series* edition (Baroda). Modern scholarly treatments: P. L. Sharma, *Indian Music: A Scientific Study* (Bharatiya Vidya Bhavan); the standard musicology literature anchored in the *Nāṭyaśāstra* lineage.

place-of-articulation-sanskrit-terms

Short: The Sanskrit *vyākaraṇa* discipline names five canonical places of articulation (*sthāna* स्थान — *stations*) from back to front of the mouth: *kaṇṭhya* (कण्ठ्य, velar), *tālavya* (तालव्य, palatal), *mūrdhanya* (मूर्धन्य, retroflex), *dantya* (दन्त्य, dental), and *oṣṭhya* (ओष्ठ्य, labial) — each anchoring one *varga*, geometric one-to-one; plus secondary positions (*nāsikya* (नासिक्य), *dantoṣṭhya* (दन्तौष्ठ्य), *jīhvāmūlīya* (जिह्वामूलीय), *upadhmānīya* (उपध्मानीय)) for the *anusvāra*, *visarga*, semivowels, and sibilants.

Deployments: Chapter 7 §7.7 ¶ — the citation anchor for the Sanskrit terminology for places of articulation; cross-deployed at Chapter 8 §8.2 in the *varga*-matrix discussion.

The Sanskrit *vyākaraṇa* discipline names five canonical places of articulation (*sthāna* — *stations*) for consonantal sound production. The

five, in the standard order from back to front of the mouth:

1. कण्ठ्य (*kaṇṭhya*) — *of the throat*. The velar position — the soft palate (velum) at the back of the mouth. The *ka-varga* consonants (*ka, kha, ga, gha, ṇa*) are produced here.
2. तालव्य (*tālavya*) — *of the palate*. The palatal position — the hard palate. The *ca-varga* consonants (*ca, cha, ja, jha, ña*) are produced here.
3. मूर्धन्य (*mūrdhanya*) — *of the crown*. The retroflex position — the rear of the hard palate, with the tongue tip curled back. The *ṭa-varga* consonants (*ṭa, ṭha, ḍa, ḍha, ṇa*) are produced here.
4. दन्त्य (*dantya*) — *of the teeth*. The dental position — the upper teeth, with the tongue tip in contact. The *ta-varga* consonants (*ta, tha, da, dha, na*) are produced here.
5. ओष्ठ्य (*oṣṭhya*) — *of the lips*. The labial / bilabial position — the two lips. The *pa-varga* consonants (*pa, pha, ba, bha, ma*) are produced here.

These five places are the contact-stations of the *varga* matrix. Each *varga* (consonantal group) is anchored to one place; each place anchors one *varga*. The structural one-to-one mapping is geometric.

In addition to the five *varga*-anchoring places, the Sanskrit discipline names secondary places used for the *anusvāra*, *visarga*, the semivowels, and the sibilants:

- नासिक्य (*nāsikya*) — *of the nose*. The nasal cavity, used for *anusvāra* and the nasal consonants when the velum is lowered.
- दन्तौष्ठ्य (*dantoṣṭhya*) — *of teeth and lip*. The labiodental position, named for the *va* semivowel where the lower lip contacts the upper teeth (though the contemporary articulation is often pure bilabial).
- जिह्वामूल्य (*jihvāmūliya*) — *of the tongue-root*. The pharyngeal / post-velar position, named for certain *visarga* formations.

- उपध्मानीय (*upadhmanīya*) — of the breath. The labial-breath position, named for certain *visarga* formations.

The five *varga*-anchoring places — *kaṇṭhya*, *tālavya*, *mūrdhanya*, *dantya*, *oṣṭhya* — are the architectural backbone of the *varṇamālā*'s consonant organization. The selection is anatomically grounded (each *sthāna* is a specific physical contact-region in the vocal tract) and acoustically distinguishable (each place produces formant-burst patterns and transitional formant contours that listeners reliably discriminate).

The structural significance the chapter establishes: the *sthāna* vocabulary is *engineering vocabulary*, not phenomenological description. Each term names a precise anatomical region that a trained anatomist could point at; each region is associated with a specific *varga* and the four-fold voicing-and-aspiration grid each *varga* runs through. The vocabulary, with this precision, is the survey instrument the *varṇamālā* uses to organize the sound-field.

Standard references: the *Prātiśākhya* literature (the *Ṛk-Prātiśākhya*, *Taittirīya-Prātiśākhya*, *Vājasaneyi-Prātiśākhya*, *Atharvaveda-Prātiśākhya*) for the canonical articulation of the *sthāna* terminology; the *Pāṇinīya Śikṣā* and other *Śikṣā* texts for the broader phonetic-articulatory framework; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953) — the foundational modern philological treatment; Madhav Deshpande, *Samskṛta-Subodhinī: A Sanskrit Primer* (University of Michigan Center for South Asian Studies, multiple editions) for the standard introductory presentation. Cross-deployed at Chapter 8 §8.2 in the *varga*-matrix architectural analysis.

karana-active-articulator

Short: The Sanskrit phonetic discipline pairs the passive contact-station (*sthāna*) with the active articulator (*karāṇa* करण — *the doer*): *jihvāgra* (जिह्वाग्र, tongue-tip — for dental and labiodental), *jihvāmadhya* (जिह्वामध्य, tongue-blade — for palatal), *jihvopāgra* (जिह्वोपाग्र, sub-apex / under-tip — for retroflex, where the underside of the curled-back tongue contacts the rear hard palate), *jihvāmūla* (जिह्वामूल, tongue-root / dorsum — for velar), and *adharoṣṭha* (अधरोष्ठ, lower lip — for labial); the *sthāna* / *karāṇa* pairing yields the complete two-component articulatory specification.

Deployments: Chapter 7 §7.7 ¶ — the citation anchor for the *karāṇa* / active-articulator vocabulary of the Sanskrit phonetic discipline.

The Sanskrit phonetic discipline names the active articulator that moves to make contact with the *sthāna* the *karāṇa* (करण) — literally *the doer*, the *instrument-of-doing*. The *karāṇa* is the moving part; the *sthāna* is the stationary contact-station. The two together specify where contact happens and what makes the contact.

The principal *karāṇas* the discipline distinguishes:

- जिह्वा (*jihvā*) — the tongue, with its parts further specified:
 - जिह्वाग्र (*jihvāgra*) — the tip / apex of the tongue. The *karāṇa* for the dental position (*ta-varga*) and the labiodental position (*va*).
 - जिह्वामध्य (*jihvāmadhya*) — the middle / blade of the tongue. The *karāṇa* for the palatal position (*ca-varga*).
 - जिह्वोपाग्र (*jihvopāgra*) — the under-tip / sub-apex. The *karāṇa* for the retroflex position (*ṭa-varga*), where the underside of the curled-back tongue contacts the rear of the hard palate.
 - जिह्वामूल (*jihvāmūla*) — the root / dorsum of the tongue. The *karāṇa* for the velar position (*ka-varga*).
- अधरोष्ठ (*adharoṣṭha*) — the lower lip. The *karāṇa* for the labial

position (*pa-varga*), contacting the upper lip (*uttaroṣṭha*); and for the labiodental position (*va* in some lineages).

The pairing of *sthāna* and *karaṇa* is the granular description of consonant production. For each consonant, the discipline specifies which *karaṇa* moves to which *sthāna*. The pairing is precise — a velar consonant is *kaṇṭhya-sthāna* with *jihvāmūla-karaṇa*; a retroflex consonant is *mūrdhanya-sthāna* with *jihvopāgra-karaṇa*; a labial consonant is *oṣṭhya-sthāna* with *adharoṣṭha-karaṇa*. The combinatorial specification yields the complete articulatory description.

The structural significance the chapter establishes: the *sthāna-karaṇa* pairing is a *two-component description of articulatory contact* — passive site and active articulator. The Sanskrit framework anticipates the contemporary phonetic distinction between *place of articulation* (passive) and *active articulator* (the moving part), with the additional precision that the Sanskrit names refer to specific anatomical regions rather than to abstract categories.

Standard references: the *Prātiśākhya* literature and the *Śikṣā* texts; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 2; Madhav Deshpande, “Sanskrit Phonetics” in the relevant volumes of the *Encyclopaedia of Indian Wisdom* (Indira Gandhi National Centre for the Arts publications); George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988), discussion of the *Śikṣā* phonetic framework.

allen-1953-phonetics-ancient-india

Short: W. Sidney Allen, *Phonetics in Ancient India* (Oxford University Press, 1953; reprinted 1961, 1965, 1995) — the foundational twentieth-century modern philological reconstruction of the ancient Indian phonetic discipline (the *Prātiśākhya* / *Śikṣā* / *Aṣṭādhyāyī* / *Mahābhāṣya* framework) in contemporary phonetic vocabulary; eight

chapters cover *sthāna* / *karaṇa*, *prayatna* (internal and external), voicing (*ghoṣa* / *aghoṣa*), nasal coupling (*anunāsika*), accent (*svara*), and *sandhi* — the standard gateway reference for the ancient Indian phonetic apparatus.

Deployments: Chapter 7 §7.7 ¶ — the citation anchor for W. S. Allen’s foundational philological treatment of the *Prātiśākhya* / *Śikṣā* phonetic framework.

W. Sidney Allen, *Phonetics in Ancient India* (Oxford University Press, London, 1953). The book is the foundational twentieth-century modern philological reconstruction of the ancient Indian phonetic discipline, drawing primarily on the *Prātiśākhya* literature, the *Śikṣā* texts, the *Aṣṭādhyāyī*, and the *Mahābhāṣya*. Allen, a Cambridge phonetician with substantial Sanskrit training, presented the ancient Indian phonetic framework in the analytical vocabulary of contemporary phonetics, demonstrating that the Indian discipline had developed a multi-axis classification of speech sounds — with terminology for place of articulation, manner of articulation, voicing, aspiration, nasal coupling, and accent — that anticipates, in working form, the categories the International Phonetic Alphabet later codified.

Allen’s chapters develop the analysis in detail:

- Chapter 1: introductory framing and the structure of the ancient Indian phonetic discipline.
- Chapter 2: place of articulation (*sthāna*) and active articulator (*karaṇa*).
- Chapter 3: manner of articulation (*prayatna*) — internal effort (*ābhyaṅtara prayatna*) and external effort (*bāhya prayatna*).
- Chapter 4: voicing (*ghoṣa* / *aghoṣa*) and the phonation distinctions.
- Chapter 5: nasal coupling (*anunāsika*) and the nasal-vowel system.
- Chapter 6: accent (*svara*) and the Vedic *udātta* / *anudātta* / *svarita* system.

- Chapter 7: sandhi and the cross-word phonetic-transformation apparatus.
- Chapter 8: summary and assessment.

The structural significance the chapter establishes: Allen's book is the standard modern philological reference for the ancient Indian phonetic framework. The framework Allen reconstructs — place, manner, voicing, aspiration, nasal coupling, accent, sandhi — is the same framework the chapter is engaging in its analysis of the *varṇamālā* and the broader Sanskrit phonetic apparatus. Where the chapter needs to anchor a specific term in the ancient Indian phonetic discipline's primary-source documentation, Allen's 1953 book is the gateway reference.

Standard reference: W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953; reprinted Oxford 1961, 1965, and 1995). Related works: Allen's *Vox Latina* (Cambridge University Press, 1965; 2nd edition 1978) — the parallel reconstruction of Latin pronunciation; Allen's *Vox Graeca* (Cambridge University Press, 1968; 3rd edition 1987) — the parallel reconstruction of Ancient Greek pronunciation. Subsequent literature: Madhav Deshpande, *The Mahābhāṣya-Dīpikā of Bharṭṛhari* (Pune University, 1985–1991, multi-volume) for the deeper *Mahābhāṣya* phonetic analysis; the multi-volume *Encyclopaedia of Indian Wisdom* contributions on phonetics; the *Sanskrit-Wörterbuch* references for individual terminological entries.

sprista-isatsprista-isatsamvrta-vivrt-constricti

Short: The Sanskrit phonetic discipline classifies *ābhyantara prayatna* (आभ्यन्तर प्रयत्न — internal articulatory effort) into four mutually-exclusive categories that exhaust the phonological inventory: *sprṣṭa* (स्पर्ष्ट, *touched* — stops), *īṣat-sprṣṭa* (ईषत्स्पर्ष्ट, *slightly touched* — semivowels / approximants), *īṣat-samvrta* (ईषत्संवृत,

slightly closed — sibilants / fricatives), and **vivṛta** (विवृत, *open* — vowels); anticipates the contemporary manner-of-articulation classification with anatomically-precise vocabulary.

Deployments: Chapter 7 §7.8 ¶ — the citation anchor for the four-fold *ābhyantara prayatna* classification of constriction types.

The Sanskrit phonetic discipline classifies the *ābhyantara prayatna* (internal effort — the type of constriction at the place of articulation) into four canonical categories. The classification, anchored across the *Prātiśākhya* literature and the *Śikṣā* texts, distinguishes:

1. **स्पृष्ट (sprṣṭa)** — *touched*; full contact. The active articulator makes complete contact with the passive site, blocking the air-flow entirely. Produces *stops*: *ka, kha, ga, gha; ca, cha, ja, jha; ṭa, ṭha, ḍa, ḍha; ta, tha, da, dha; pa, pha, ba, bha*. Also the nasal stops *ṇa, ṇā, ṇa, ma* (when the velum is lowered to allow nasal airflow but the oral contact remains complete).
2. **ईषत्स्पृष्ट (īṣat-sprṣṭa)** — *slightly touched*; light contact. The active articulator approaches the passive site without making full contact, producing an *approximant*. Produces the four *semivowels*: *ya, ra, la, va*. Each semivowel is articulated at one of the *varga*-place positions (palatal *ya*, retroflex *ra*, dental *la*, labial-dental *va*) but with light rather than full closure.
3. **ईषत्संवृत (īṣat-saṃvṛta)** — *slightly closed*. The articulators come close enough to constrict the airflow into a narrow channel, producing turbulence (frication). Produces the three *sibilants* and the aspirate: *śa* (palatal sibilant), *ṣa* (retroflex sibilant), *sa* (dental sibilant), *ha* (glottal fricative).
4. **विवृत (vivṛta)** — *open*; no contact / aspirated. The airflow passes through an open cavity without constriction. Produces the *vowels* — all twelve canonical vowels of the *varṇamālā*: *a, ā, i, ī, u, ū, r, ṛ, ḷ, e, ai, o, au*. The vowel is the limit case of *vivṛta prayatna*.

The four-category classification is structurally significant in two respects. First, it organizes the entire phonological inventory under a single, mutually-exclusive parameter — every sound in the *varṇamālā* falls into exactly one of the four categories. Second, it parallels the contemporary phonetic distinction between *stops*, *approximants*, *fricatives*, and *vowels* — the four canonical categories modern phonetics uses for manner of articulation. The Sanskrit framework anticipates the manner-of-articulation classification, with the additional precision that the Sanskrit terms refer to the physical degree of constriction (touched, slightly touched, slightly closed, open) rather than to abstract phonological categories.

The structural significance the chapter establishes: the *ābhyantara prayatna* classification is one of the two parallel axes of the *prayatna* (effort) dimension, the other being *bāhya prayatna* (external effort — voicing, aspiration, nasal coupling). Together, the two axes specify the manner-of-articulation parameters for every sound in the *varṇamālā*. The combined specification — place (*sthāna*), active articulator (*karaṇa*), internal effort (*ābhyantara prayatna*), external effort (*bāhya prayatna*) — is the full anatomical-functional description of each sound. The *varṇamālā* is organized around this full multi-parameter specification, not around abstract phonological categories.

Standard references: the *Prātiśākhya* literature and the *Śikṣā* texts; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 3 (the *prayatna* analysis); the *Pāṇinīya-Śikṣā* and other *Śikṣā* texts for the standard articulation of the four-category classification.

abhyantara-bahya-prayatna

Short: The Sanskrit phonetic framework runs *prayatna* (प्रयत्न, effort) on two parallel axes: *ābhyantara prayatna* (आभ्यन्तर प्रयत्न — internal effort: constriction type at the *sthāna*) and *bāhya prayatna* (बाह्य

प्रयत्न — external effort: voicing *ghoṣa* (घोष) / *aghoṣa* (अघोष), aspiration *alpa-prāṇa* (अल्पप्राण) / *mahā-prāṇa* (महाप्राण), nasal coupling *anunāsika* (अनुनासिक), glottal state); the 5×5 *varga* matrix is the *ābhyantara* (*sprṣṭa*) axis crossed with the *bāhya* parameters — the master organizational principle of the *varṇamālā*.

Deployments: Chapter 7 §7.9 ¶ — the citation anchor for the *ābhyantara* / *bāhya* prayatna distinction.

The Sanskrit phonetic discipline distinguishes two parallel axes of *prayatna* (articulatory effort): *ābhyantara prayatna* (आभ्यन्तर प्रयत्न) — internal effort, the type of constriction at the place of articulation — and *bāhya prayatna* (बाह्य प्रयत्न) — external effort, the additional parameters layered on top of the constriction (voicing, aspiration, nasal coupling, phonation state).

Ābhyantara prayatna specifies the type and degree of constriction at the *sthāna*. The four canonical categories — *sprṣṭa*, *īṣat-sprṣṭa*, *īṣat-saṃvṛta*, *vivṛta* (see endnote sprista-isatsprista-isatsamvṛta-vivṛta-const. — exhaust this axis.

Bāhya prayatna specifies the additional parameters that distinguish sounds sharing the same *sthāna* and *ābhyantara prayatna*. The principal categories:

- **श्वास** (*śvāsa*) vs. **नाद** (*nāda*) — *breath* vs. *resonance*. Whether the vocal cords are vibrating. The voiced consonants (*ga*, *gha*, *ja*, *jha*, *ḍa*, *ḍha*, *da*, *dha*, *ba*, *bha*; the nasals; the semivowels; the vowels) have *nāda*; the voiceless consonants (*ka*, *kha*, *ca*, *cha*, *ṭa*, *ṭha*, *ta*, *tha*, *pa*, *pha*; the sibilants) have *śvāsa*. This is the *ghoṣa* / *aghoṣa* (voiced / voiceless) dimension.
- **अल्पप्राण** (*alpa-prāṇa*) vs. **महाप्राण** (*mahā-prāṇa*) — *small breath* vs. *great breath*. Whether the consonant is produced with a forceful aspirate release. *Alpa-prāṇa* consonants: *ka*, *ga*, *ca*, *ja*, *ṭa*, *ḍa*, *ta*, *da*, *pa*, *ba*. *Mahā-prāṇa* consonants: *kha*, *gha*, *cha*, *jha*, *ṭha*, *ḍha*, *tha*, *dha*, *pha*, *bha*. This is the aspiration

dimension.

- **अनुनासिक (anunāsika)** — *nasal*. Whether the velum is lowered to couple the nasal cavity. The nasal stops (*ṇa, ña, ṇa, na, ma*) and the nasalized vowels are *anunāsika*; the oral sounds are not.
- **विवृत (vivṛta)** vs. **संवृत (saṃvṛta)** — *open* vs. *closed*, applied at the glottal level rather than the articulator level. The glottis can be open (producing voiceless sounds) or closed (producing voiced sounds); the additional intermediate states (breathy voice, creaky voice) are also documented in the more detailed *Śikṣā* texts.

The structural significance the chapter establishes: the *ābhyantara* / *bāhya prayatna* distinction is the master organizational principle of the *varṇamālā*'s consonant grid. The *varga* matrix — the 5×5 organization of consonants — is structured around the *ābhyantara prayatna* axis (all consonants in the matrix are *sprṣṭa*) crossed with the *bāhya prayatna* parameters (voiced / voiceless × aspirated / unaspirated × oral / nasal). The combinatorial specification — five places × five manner-and-phonation combinations — yields the 25 *varga* consonants. The additional sounds (semivowels, sibilants, *ha*, vowels) are organized along the same parameter axes but extend beyond the 5×5 grid because their *ābhyantara prayatna* parameters differ.

Standard references: see endnote *sprista-isatsprista-isatsamvṛta-vivṛta* and endnote *svasa-nada-vivṛta-samvṛta-phonation*. The two-axis *prayatna* framework is documented across the *Prātisākhya* and *Śikṣā* literature; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 3, provides the standard modern philological reconstruction.

svasa-nada-vivṛta-saṃvṛta-phonation

Short: The Sanskrit phonetic discipline names the glottal-state parameters of *bāhya prayatna*: *śvāsa* (श्वास, *breath* — voiceless, the open glottis with vocal cords abducted) vs. *nāda* (नाद, *resonance* — voiced, the glottis adducted for vibration); *vivṛta* (विवृत, open) and *saṃvṛta* (संवृत, closed) apply the same distinction at the glottal level — engineering vocabulary that names the physical state of the glottis during specific sound production, corresponding exactly to contemporary voicing categories.

Deployments: Chapter 7 §7.9 ¶ — the citation anchor for the *śvāsa* / *nāda* and *vivṛta* / *saṃvṛta* glottal-state distinctions.

The *bāhya prayatna* dimension of voicing and phonation is anchored in the Sanskrit phonetic discipline’s terminology for the state of the glottis during sound production. The principal terms:

- **श्वास (*śvāsa*)** — *breath*. The glottis is open; the airflow passes through unimpeded by vocal-cord vibration. Produces *voiceless* sounds: the unaspirated voiceless stops (*ka, ca, ṭa, ta, pa*), the aspirated voiceless stops (*kha, cha, ṭha, tha, pha*), and the sibilants (*śa, ṣa, sa*) and *ha* (which has its own complex glottal state).
- **नाद (*nāda*)** — *resonance, tone*. The glottis is in vibratory adduction; the vocal cords vibrate to produce a quasi-periodic source signal. Produces *voiced* sounds: the unaspirated voiced stops (*ga, ja, ḍa, da, ba*), the aspirated voiced stops (*gha, jha, ḍha, dha, bha*), the nasal stops (*ṇa, ña, ṇa, na, ma*), the semivowels (*ya, ra, la, va*), and all the vowels.

The additional glottal-state terms:

- **विवृत (*vivṛta*)** — *open*. Applied to the glottis: the *śvāsa* state with the vocal cords abducted.
- **संवृत (*saṃvṛta*)** — *closed*. Applied to the glottis: the *nāda* state

with the vocal cords adducted for vibration.

The four-state classification of glottal aperture is documented across the *Prātiśākhya* and *Śikṣā* texts: *vivṛta* (open / breathy), *saṃvṛta* (closed / glottal-stop-like), *vivṛta-saṃvṛta* (intermediate states), and the additional phonation states the more detailed texts recognize.

The structural significance the chapter establishes: the *śvāsa* / *nāda* distinction corresponds exactly to the contemporary phonetic distinction between *voiceless* and *voiced* sounds. The *vivṛta* / *saṃvṛta* distinction at the glottis corresponds to the contemporary distinction between *open glottis* (associated with voiceless sounds) and *closed glottis* (associated with voiced sounds). The Sanskrit terminology, with this precision, is engineering vocabulary that names the physical state of the glottis during specific sound productions.

The chapter's deployment: the *bāhya prayatna* glottal-state vocabulary is what allows the *varṇamālā* to distinguish the four-fold voicing-and-aspiration array on each *varga* (unvoiced-unaspirated × unvoiced-aspirated × voiced-unaspirated × voiced-aspirated). The four-fold array crossed with the five *sthāna* places yields the 20-cell stop-consonant grid, augmented by the nasals (*nāda* with nasal coupling) to produce the full 25-cell *varga* matrix.

Standard references: the *Prātiśākhya* literature and the *Śikṣā* texts; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 4 (voicing and phonation); for the contemporary phonetic correspondence: Peter Ladefoged, *A Course in Phonetics* (Cengage, multiple editions); Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998), Chapter 2 on voicing and phonation.

ayogavaha-category-pratisakhya

Short: The *ayogavāha* (अयोगवाह — *that which carries without combining*) is the third major class of Sanskrit phonemes beyond *svara* (vowels) and *vyañjana* (consonants); five members enumerated across the *Prātiśākhya* literature — *anusvāra* (अनुस्वार, the nasal resonance ँ), *visarga* (विसर्ग, the voiceless aspiration ः), *jihvāmūlīya* (जिह्वामूलीय, the velar variant of *visarga* before *kavarga*), *upadhmānīya* (उपध्मानीय, the labial variant before *pavarga*), and *yama* (यम, the nasalized release of certain consonant clusters); breath-gesture phonemes the contemporary phonological vocabulary collapses, but the *Prātiśākhya* discipline keeps structurally distinct.

Deployments: Chapter 8 §8.1 ¶ — the citation anchor for the *ayogavāha* category as recognized in the *Prātiśākhya* literature.

The *अयोगवाह* (*ayogavāha*) category is the third major class of phonemic units in the Sanskrit phonological discipline, distinct from *svara* (vowels) and *vyañjana* (consonants). The term itself is a *bahuvrīhi* compound: *a-* (privative) + *yoga-vāha* (combining-carrier). The literal translation: *that which carries without combining* — that is, a phonemic unit that depends on a preceding vowel and modifies how the vowel ends, but does not stand on its own as a syllable. The *Prātiśākhya* literature establishes the category and enumerates its members:

1. *अनुस्वार* (*anusvāra*) — the nasal resonance at the end of a vowel. Written as ँ (a dot above the vowel-bearing letter). The terminating nasal that does not articulate at a specific *sthāna* but resonates through the nasal cavity.
2. *विसर्ग* (*visarga*) — the voiceless aspiration at the end of a vowel. Written as ः. The terminating breath-release that exhales a soft *ha*-color matched to the preceding vowel.
3. *जिह्वामूलीय* (*jihvāmūlīya*) — the velar/pharyngeal voiceless fricative produced as a positional variant of *visarga* before the

kavarga consonants (k, kh).

4. **उपध्मानीय (upadhmānīya)** — the labial voiceless fricative produced as a positional variant of *visarga* before the *pavarga* consonants (p, ph).
5. **यम (yama)** — the nasalized release of certain consonant clusters; the nasal portion of a stop-nasal sequence in close articulation.

The five members are documented across the principal *Prātiśākhya* texts. The *Ṛk-Prātiśākhya* (attached to the *Ṛgveda*, in the *Śākala* recension) treats the category at *Adhyāya* I.5 and following. The *Taittirīya-Prātiśākhya* (attached to the *Taittirīya Saṃhitā* of the *Kṛṣṇa-Yajurveda*) treats it at *Praśna* I.18 and following. The *Vājasaneyī-Prātiśākhya* (attached to the *Vājasaneyī Saṃhitā* of the *Śukla-Yajurveda*) treats it at *Adhyāya* I.5 and following. The *Atharvaveda-Prātiśākhya* and the *Ṛk-Tantra-Prātiśākhya* carry parallel treatments with minor recensional variations.

The structural significance the chapter establishes: the *ayogavāha* category is the architectural recognition that breath-gestures at the close of a vowel are phonologically distinct units, neither vowels nor consonants. The *Prātiśākhya* discipline's classification is more granular than the contemporary phonological vocabulary; modern phonetics groups *anusvāra* under *nasalization* and *visarga* under *glottal fricative*, missing the structural distinction the *Prātiśākhya* discipline makes between the breath-gesture phoneme and the consonant phoneme. The *ayogavāha* category is therefore one of the standing analytical advantages of the Sanskrit phonological framework over its modern Western philological counterpart.

Standard references: the *Prātiśākhya* texts in their standard editions. *Ṛk-Prātiśākhya*: edited by Mangal Deva Shastri (Banaras Hindu University, 1959–1963); Virendra Kumar Verma's edition with English translation (Vishveshvaranand Vishva Bandhu Institute, 1970); Max Müller's earlier German edition (Leipzig, 1869).

Taittirīya-Prātiśākhya: edited by W. D. Whitney (American Oriental Society, 1871; reprinted Motilal Banarsidass, 1973). *Vājasaneyī-Prātiśākhya*: edited by Indu Rastogi (Sanskrit University, Varanasi, 1967). *Atharvaveda-Prātiśākhya*: edited by Surya Kanta (Lahore, 1939). W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 2.

visarga-anusvara-articulation

Short: *Anusvāra* (अनुस्वार) closes the mouth, opens the velum, continues voicing into nasal resonance — the inward-and-upward breath-gesture (*kumbhaka* (कुम्भक) in Mishra’s *prāṇāyāma* framing); *visarga* (विसर्ग) opens the glottis, ceases voicing, exhales a soft aspiration carrying the preceding vowel’s formant color — the outward breath-gesture (*recaka* (रेचक)); the two are inverse and together exhaust the canonical syllable-terminal breath options.

Deployments: Chapter 8 §8.1 ¶ — the citation anchor for the articulatory specification of *anusvāra* and *visarga* as breath gestures at the close of a vowel.

The articulatory specification of *anusvāra* and *visarga* is documented across the *Prātiśākhya* and *Śikṣā* literature. The two breath-gestures specify distinct mechanical operations at the close of a vowel:

***Anusvāra*:** The mouth closes (the lips meet for terminal closure, or the tongue meets the palate at a position determined by the following consonant — see endnote *sandhi-anusvara-assimilation* for the positional-assimilation rule). The velum opens, allowing the air-flow to pass through the nasal cavity rather than the oral cavity. The voicing continues into the nasal resonance. The acoustic result is a humming nasal continuation of the vowel, with the nasal-cavity resonance producing characteristic spectral nasalization (the lowered

second formant and the appearance of nasal anti-resonances). The gesture is *inward* and *upward* — breath continues into the resonant cavities of the head.

Visarga: The glottis opens; the vocal-cord vibration ceases. A soft aspiration is exhaled, carrying the formant color of the preceding vowel. The aspiration is voiceless; the formant pattern of the vowel persists into the aspirated release. The acoustic result is a soft *ha*-coloured breath at the close of the vowel: *aḥ* exhales as a soft *ha*; *iḥ* as a soft *hi* (with a slightly palatalized aspiration); *uḥ* as a soft *hu* (with a slightly labialized aspiration). The gesture is *outward* — breath releases from the body, terminating the syllable in exhalation.

The contemporary articulation by Sampadananda Mishra (see end-note mishra-breath-pedagogy) describes the two breath-gestures as engineering of the speaker's breath pattern: *visarga* as the *recaka* (exhalation) phase of *prāṇāyāma*; *anusvāra* as the *kumbhaka* (retained-breath) phase. The structural framing converts the phonological specification into a breath-pedagogy account that recovers the architects' broader engineering — breath as the substrate the engineered phonology operates on.

The structural significance the chapter establishes: the *anusvāra* and *visarga* specify not merely terminal-phoneme sound qualities but the *manner of breath at the close of a syllable*. The two gestures are inverse: inward-and-upward vs. outward; nasal vs. glottal; voiced vs. voiceless. Together they exhaust the canonical syllable-terminal breath options for a vowel-ending syllable. The architecture of *ayogavāha* is the engineering of breath-termination into the phonological system.

Standard references: the *Prātiśākhya* literature; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 5; the *Pāṇinīya-Śikṣā* and other *Śikṣā* texts. For the contemporary articulation: Sampadananda Mishra's lecture series on Sanskrit phonology (Sri Aurobindo Society, Pondicherry; available across multiple online

lecture recordings and the Sri Aurobindo Society publications).

mishra-breath-pedagogy

Short: Sampadananda Mishra — contemporary Sanskrit scholar and pedagogue at the Sri Aurobindo Society, Pondicherry — articulates the *anusvāra* and *visarga* not as terminal phonemes but as *breath gestures* engineered into the language: *visarga* as the *recaka* (रेचक, exhalation) phase of *prāṇāyāma* (प्राणायाम), *anusvāra* as the *kumbhaka* (कुम्भक, retained breath) phase. The framing recovers the phonological-engineering claim from inside the *paramparā*'s contemporary practice, not from external philological reconstruction.

Deployments: Chapter 8 §8.1 ¶ — the citation anchor for Sampadananda Mishra's contemporary articulation of the *anusvāra* / *visarga* breath-pedagogy framing.

Sampadananda Mishra is a contemporary Sanskrit scholar and pedagogue affiliated with the Sri Aurobindo Society in Pondicherry, with a substantial body of teaching on Sanskrit phonology, grammar, and recitation. His contemporary articulation that Chapter 8 §8.1 draws on: the *anusvāra* and *visarga* are not merely terminal phonemes; they are *breath gestures* engineered into the language. *Visarga* is a literal out-breath at the close of a vowel — a release of breath that mirrors the *recaka* phase of *prāṇāyāma*. *Anusvāra* is a literal internalization — a resonance held inside the body that mirrors the *kumbhaka* phase. The two endings are the engineering of breath into the language's atomic units.

The framing recovers the broader engineering claim: the *ayogavāha* phonemes are not aesthetic flourishes added to the syllable; they specify the breath state at syllable closure. The architects of Sanskrit engineered the breath-termination into the phoneme inventory itself. The framing is anchored in Mishra's contemporary public-lecture

series, his commentary on the *Yoga Sūtras*'s engagement with *prāṇāyāma* in relation to recitation, and his standing teaching at the Sri Aurobindo Ashram's Sanskrit programs.

Mishra's body of work includes: *Vande Mātaram and the Bhāratiya National Anthem* (Sri Aurobindo Society, 2014); *The Wonder That is Sanskrit* (Sri Aurobindo Society, multiple editions); the *Sanskrit Sambhāṣaṇa* lecture series; the *Sanskrit and Sri Aurobindo* paper series. He also delivers extended workshop sessions on Sanskrit phonology, including specific treatment of the *anusvāra* / *visarga* breath-pedagogy at the Sri Aurobindo Ashram's training programs.

Standard references: the Sri Aurobindo Society publications by Sampadananda Mishra; the standing lecture archive at the Sri Aurobindo Ashram (Pondicherry); his contributions to the *Sanskrit Vimarsha* journal of the Rashtriya Sanskrit Sansthan. The book may also cite his contemporary public lectures available through the Sri Aurobindo Ashram's online archive at sriurobindoashram.org and through his standing presence in Sanskrit-pedagogy communities.

The chapter's deployment: Mishra's contemporary articulation provides a living-paramparā anchor for the structural account the chapter develops. The breath-pedagogy framing recovers the phonological-engineering claim from inside the paramparā's own contemporary practice, not from an external philological reconstruction.

visarga-cognate-shadow

Short: The cognate chain *सिन्धुः (Sindhuh)* → Old Persian *Hinduš* (𐎧𐎠𐎧𐎡𐎹𐎡𐎹, with Indo-Iranian *s* → *h*) → Greek *Indós* (Ἰνδός, dropping initial *h*-) → Latin *Indus* — the contact languages preserve the surface shape of the *visarga* (विसर्ग)-bearing ending but lose its breath-specification at each step; the contemporary name *Hindu* descends

through the Iranian rendering, carrying the cognate shadow rather than the Sanskrit *visarga* itself. The *Pratibimba* (प्रतिबिम्ब, *reflection / mirror image*) pattern Chapter 18 §18.6 develops in full.

Deployments: Chapter 8 §8.1 ¶ (worked-example paragraph) — the citation anchor for the *Sindhuḥ* → *Hinduš* / *Indós* / *Indus* cognate-shadow analysis.

The cognate chain from Sanskrit सिन्धुः (*Sindhuḥ*) through Old Persian, Greek, and Latin renderings illustrates the structural pattern Chapter 18 §18.6 develops in full under the *Pratibimba* (प्रतिबिम्ब — *reflection / mirror image*) analysis. The chain:

- **Sanskrit सिन्धुः (*Sindhuḥ*)** — nominative singular of the *Sindhu* river name, with the *visarga* (-ḥ) carrying the engineered breath-release at the close of the name. The full pronunciation includes the *visarga*’s voiceless aspiration; the river-name is *Sindhuḥ*, not *Sindhu*.
- **Old Persian** 𐎧𐎱𐎠𐎿𐎧𐎡𐎹 (*Hinduš*) — the cognate form in Old Persian. The Indo-Iranian *s* → *h* shift (a regular sound change documented across the Iranian languages) converts initial *s* to *h*, yielding *Hindu-* from *Sindhu-*. The terminal -š renders the *visarga* consonantly — phonetically a fricative, no longer a breath-gesture. The breath specification of the *visarga* is lost; the consonantal trace remains.
- **Greek Ἰνδός (*Indós*)** — the Greek rendering, reduced to *Indós* with a plain -os nominative ending. The Greek tradition drops the initial *h-* of Old Persian *Hinduš* (Greek had no productive initial *h-* after the loss of the rough breathing in Hellenistic Greek), giving *Indos*. The terminal -os is the standard Greek masculine nominative singular ending, applied to the loan-name.
- **Latin Indus** — the Latin rendering of the Greek, with the standard Latin -us nominative ending substituted for the Greek -os. The Latin form transmits to the medieval and modern European

languages as the standard name for the river and, by extension, for the land — *Indos / India*.

The structural observation: the contact languages preserve the *surface form* of the visarga-bearing ending. Old Persian renders it consonantly as -š; Greek and Latin substitute their own thematic-vowel nominative endings (-os, -us). What the contact languages cannot carry forward is the *breath specification* the visarga encoded — the voiceless aspirated release at the close of the vowel. The breath is lost in transmission; only the consonantal silhouette of the ending survives. The further from the calibrant, the less of the breath-engineering survives.

This pattern — the *Pratibimba* pattern — is general across Sanskrit-to-Indo-European cognates: the calibrant-anchored Sanskrit form preserves the engineered phonetic specification (breath, accent, syllable structure); the contact-language cognate preserves the *shape* of the form but loses the specification. Chapter 18 §18.6 develops the full *Pratibimba* analysis with multiple worked examples (the *Sindhu* chain; the *yuga / yoke* chain; the *mātr / mater / mother* chain; the *aśva / equus / hippos* chain).

The chapter's deployment at the *Sindhu* example: the *Hindu* name, by which the dharmic civilization is widely known, descends through this very chain — the Old Persian *Hindu-* (with the *s* → *h* shift) entering general non-Indic usage. The contemporary name *Hindu* therefore preserves the Iranian rendering of the visarga rather than the Sanskrit visarga itself. The civilizational name carries the cognate shadow of the engineered original; the engineered original (*Sindhuḥ*, with the visarga's breath-release) remains operational only inside the Sanskrit-anchored register.

Standard references for the cognate chain: Manfred Mayrhofer, *Etymologisches Wörterbuch des Altindoarischen* (Carl Winter, Heidelberg, 1986–2001), s.v. *Sindhu*; Roland G. Kent, *Old Persian: Grammar, Texts, Lexicon* (American Oriental Society, 1953), s.v. *Hinduš*; H.

Frisk, *Griechisches Etymologisches Wörterbuch* (Carl Winter, Heidelberg, 1960–1972), s.v. *Indós*; the *Oxford Latin Dictionary*, s.v. *Indus*; J. P. Mallory and Douglas Q. Adams, *Encyclopedia of Indo-European Culture* (Fitzroy Dearborn, 1997), s.v. *Sindhu*.

varṇamāla-grid-geometry

Short: The five *sparśa* places of the *varṇamālā* (वर्णमाला) sample the vocal tract at well-separated positions: labial (~0 cm from lip-line), dental (~3 cm), retroflex (~7 cm), palatal (~9 cm), velar (~12 cm); the minimum ~2 cm separation between adjacent positions produces acoustic-formant distinguishability — spatial engineering produces acoustic engineering, and the *mūrdhanya* (मूर्धन्य) base at ~7 cm sits structurally central. Reference figures: Ladefoged & Maddieson, *The Sounds of the World's Languages* (Blackwell, 1996); Stevens, *Acoustic Phonetics* (MIT Press, 1998).

Deployments: Chapter 8 §8.2 ¶ — the citation anchor for the spatial-geometry analysis of the *sparśa* grid sampling positions.

The five *sparśa* (stop-consonant) places of articulation in the *varṇamālā* sample five positions along the vocal tract, measured from the lips backward to the velar/uvular region. Standard reference figures from contemporary X-ray and MRI imaging of speech production (Ladefoged and Maddieson, *The Sounds of the World's Languages*, Blackwell, 1996; Stevens, *Acoustic Phonetics*, MIT Press, 1998):

- **Labial (pavarga)** — approximately 0 cm from the lip-line. The lips are the most-forward contact-station.
- **Dental (tavarga)** — approximately 3 cm posterior to the lips. The upper-teeth contact-region.
- **Retroflex (ṭavarga)** — approximately 7 cm posterior to the lips. The rear of the hard palate / front of the soft palate contact-

region, with the tongue tip curled back.

- **Palatal (cavarga)** — approximately 9 cm posterior to the lips. The hard palate contact-region.
- **Velar (kavarga)** — approximately 12 cm posterior to the lips. The soft palate / velum contact-region.

The figures are approximate and vary across individual speakers, but the *relative spacing* between adjacent positions is the structural fact: ~3 cm (labial to dental), ~4 cm (dental to retroflex), ~2 cm (retroflex to palatal), ~3 cm (palatal to velar). The intervals are not strictly equidistant. The grid samples five well-separated positions across the front-to-back range, with adjacent-pair separation of at least ~2 cm.

The structural significance the chapter establishes: the spatial spacing is *not arbitrary*. The minimum ~2 cm separation between adjacent contact-positions is what produces *acoustic distinguishability* in the formant-burst patterns. Sounds produced at positions closer than ~2 cm apart begin to produce confusable formant-burst signatures, which would compromise the *snap-to-grid* principle. The architects of the *varṇamālā* selected positions that are far enough apart in articulatory space to produce distinct acoustic outcomes — the spatial engineering produces the acoustic engineering.

The retroflex base at ~7 cm sits structurally central — the midpoint of the five-position sampling. This is consistent with the architectural centrality of the *mūrdhanya* set the chapter develops at multiple points (see endnotes retroflex-substrate-standard-account and the Chapter 17 §17.4 analysis).

Standard references: Peter Ladefoged and Ian Maddieson, *The Sounds of the World's Languages* (Blackwell, 1996), the chapter on stop-consonant places of articulation; Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998), Chapter 8 on stop consonants; Gunnar Fant, *Acoustic Theory of Speech Production* (Mouton, 1960). The specific cm-figures the chapter cites are within the standard ranges

across these references.

sandhi-anusvara-assimilation

Short: Pāṇini's *Aṣṭādhyāyī* 8.4.58 — *anusvārasya yayi parasavarṇaḥ* (अनुस्वारस्य ययि परसवर्णः) — governs the *anusvāra* (अनुस्वार ँ) assimilation rule: before a *varga* consonant, the terminal nasal takes the homorganic nasal of the following consonant's place ($\text{ṃ} + k\text{-varga} \rightarrow \text{ङ्}$; $+ c\text{-varga} \rightarrow \text{ञ्}$; $+ ṭ\text{-varga} \rightarrow \text{ण्}$; $+ t\text{-varga} \rightarrow \text{न्}$; $+ p\text{-varga} \rightarrow \text{म्}$); the *snap-to-grid* relaxes in governed ways at boundary positions to preserve *varga*-place coherence at every syllable boundary.

Deployments: Chapter 8 §8.2 ¶ — the citation anchor for the *anusvāra* place-of-articulation assimilation rule in *sandhi*.

The Sanskrit *sandhi* rule for *anusvāra* (the terminal nasal ṃ) before a stop consonant: the *anusvāra* assimilates to the place of articulation of the following stop, taking the nasal consonant at that *sthāna*. The transformation, in the standard formal notation of the *Aṣṭādhyāyī*:

- $\text{ṃ} + k\text{-varga}$ (*ka, kha, ga, gha*) $\rightarrow$ ङ्, ङ्ख, ङ्ग, ङ्घ (the velar nasal ṅ)
- $\text{ṃ} + c\text{-varga}$ (*ca, cha, ja, jha*) $\rightarrow$ ञ्, ञ्छ, ञ्ज, ञ्झ (the palatal nasal ṇ)
- $\text{ṃ} + ṭ\text{-varga}$ (*ṭa, ṭha, ḍa, ḍha*) $\rightarrow$ ण्ट, ण्ठ, ण्ड, ण्ढ (the retroflex nasal ṇ)
- $\text{ṃ} + t\text{-varga}$ (*ta, tha, da, dha*) $\rightarrow$ न्त, न्थ, न्द, न्ध (the dental nasal n)
- $\text{ṃ} + p\text{-varga}$ (*pa, pha, ba, bha*) $\rightarrow$ म्प, म्फ, म्ब, म्भ (the labial nasal m)

The rule is governed in the *Aṣṭādhyāyī* by *sūtra* 8.4.58 — *anusvārasya yayi parasavarṇaḥ* — “the *anusvāra*, when followed by a *yay* (any of the *varga* consonants), takes the *parasavarṇa* — the homorganic nasal corresponding to the following consonant's *varga*.”

The structural significance the chapter establishes: the *anusvāra*-assimilation rule is the worked example of the *snap-to-grid* principle relaxing in *governed* ways at boundary positions. The nasal does not stay at one fixed position; it relocates to match the following consonant's row. The grid relaxation is *governed* by the *sandhi* rule; the rule is *general* across all five *vargas* (the nasal always moves to match the following stop's place); the rule is *engineered* (it preserves the *varga*-place coherence at every syllable boundary). The relaxation is not random; it is the controlled loosening the architecture builds in.

The phonetic logic: an unconstrained terminal nasal followed by a stop at a different place would require two articulator movements in close sequence — the nasal articulator moving to one position, releasing, and the stop articulator moving to a different position. The assimilation rule co-locates the two: the nasal pre-positions itself at the stop's place, simplifying the motor-control problem and producing a smoother articulation. The engineering serves both the listener (the assimilated nasal-stop sequence is cleaner acoustically) and the speaker (the co-located articulation is mechanically simpler).

Standard references: Pāṇini's *Aṣṭādhyāyī*, *sūtra* 8.4.58 and the surrounding *sandhi* rules (8.3.5 ff. for *visarga*-assimilation, 8.4.40 ff. for general consonant-cluster *sandhi*); the standard editions of the *Aṣṭādhyāyī*; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 7 (*sandhi*); George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988), discussion of *sandhi* rules.

pre-panini-pratisakhya-classification

Short: The phonetic-classificatory vocabulary (*sthāna*, *karaṇa*, *prayatna*, *prāṇa*, *ghoṣa*, *anunāsika*, *sparśa*, *varga*, *varṇa*) is documented across the *Prātiśākhya* literature (*Rk-Prātiśākhya* (ऋक्प्रातिशाख्य) of Śau-

naka, *Taittirīya-Prātiśākhya* (तैत्तिरीयप्रातिशाख्य), *Vājasaneyī-Prātiśākhya* (वाजसनेयिप्रातिशाख्य) of Kātyāyana, *Atharvaveda-Prātiśākhya*, *Ṛk-Tantra-Prātiśākhya*) and the *Śikṣā* (शिक्षा) texts (*Pāṇinīya-Śikṣā*, *Nārādīya-Śikṣā*, *Yājñavalkya-Śikṣā*, *Āpiśali-Śikṣā*, *Vyāsa-Śikṣā*) — pre-Pāṇinian engineering vocabulary used as already-established technical terminology, presupposed by Pāṇini's *Aṣṭādhyāyī*.

Deployments: Chapter 8 §8.4 ¶ — the citation anchor for the *Prātiśākhya* and *Śikṣā* documentation of the *varṇamālā* vocabulary as pre-Pāṇinian.

The phonetic-classificatory vocabulary the chapter deploys — *sthāna*, *karaṇa*, *prayatna*, *prāṇa*, *ghoṣa*, *anunāsika*, *sparśa*, *swara*, *antaḥstha*, *uṣman*, *sprṣṭa*, *asprṣṭa*, *aghoṣa*, *alpa-prāṇa*, *mahā-prāṇa*, *anupradāna*, *vivṛta*, *saṃvṛta*, *varga*, *varṇa*, *varṇamālā* — is documented across the *Prātiśākhya* and *Śikṣā* literature, which together constitute the canonical phonetic-recitational *śāstra* of the *Vedāṅga* disciplines.

The principal *Prātiśākhya* texts:

1. *Ṛk-Prātiśākhya* (ऋक्प्रातिशाख्य) — attached to the *Ṛgveda* in the *Śākala* recension, attributed traditionally to Śaunaka. Treats the phonetic specification, accent system, and recitational rules of the *Ṛgveda*.
2. *Taittirīya-Prātiśākhya* (तैत्तिरीयप्रातिशाख्य) — attached to the *Taittirīya Saṃhitā* of the *Kṛṣṇa-Yajurveda*. Treats the same phonetic-recitational framework for the *Taittirīya śākhā*.
3. *Vājasaneyī-Prātiśākhya* (वाजसनेयिप्रातिशाख्य) — attached to the *Vājasaneyī Saṃhitā* of the *Śukla-Yajurveda*. Attributed traditionally to Kātyāyana. Treats the phonetic-recitational framework for the *Vājasaneyī śākhā*.
4. *Atharvaveda-Prātiśākhya* (अथर्ववेदप्रातिशाख्य) — attached to the *Atharvaveda*. Two recensions: the *Śaunakīya* and the *Caturādhyāyikā* (also called the *Atharva-Pāriṣiṣṭa*).

5. ***Ṛk-Tantra-Prātiśākhya*** (ऋक्तन्त्रप्रातिशाख्य) — attached to the *Sāmaveda*. Treats the phonetic-recitational framework for the *Sāmavedic śākhā*.

The principal *Śikṣā* texts (the *Vedāṅga* discipline of phonetic-recitational training):

- ***Pāṇinīya-Śikṣā*** (पाणिनीयशिक्षा) — attributed to Pāṇini's discipline (though scholarly opinion varies on the authorship). The standard short introductory text on Sanskrit phonetics.
- ***Nārādīya-Śikṣā*** — the *Sāmaveda* phonetic discipline, with substantial musical-theoretical content.
- ***Yājñavalkya-Śikṣā*** — attached to the *Yājñavalkya śākhā*.
- ***Āpīśali-Śikṣā*** — attached to the pre-Pāṇinian grammarian Āpīśali.
- ***Vyāsa-Śikṣā*** — attributed to Vyāsa.
- and others: *Vasiṣṭha-Śikṣā*, *Kātyāyana-Śikṣā*, *Parāśara-Śikṣā*, *Māṇḍūkī-Śikṣā*.

The texts together document the phonetic-classificatory vocabulary the chapter deploys. The vocabulary is *pre-Pāṇinian* — the *Prātiśākhya* and *Śikṣā* literature presupposes the vocabulary as already-established technical terminology and uses it without justifying its construction. Pāṇini's *Aṣṭādhyāyī* operates on the same vocabulary at points where the *Aṣṭādhyāyī*'s rules engage phonetic specification (the *Māheśvara-sūtras* opening, the *sandhi* rules of *Adhyāyas* 6–8, the various place-of-articulation conditioned rules). The vocabulary is therefore at minimum as old as the *Prātiśākhya* discipline and, in its operational use, presupposed by Pāṇini's decoding.

The structural significance the chapter establishes: the *varṇamālā* vocabulary is not a modern coinage and not a Pāṇinian innovation. It is the engineering vocabulary of the *Vedāṅga* disciplines' phonetic śāstra, in operational use across the *guru-shishya paramparā* before Pāṇini and documented in stable canonical form by the *Prātiśākhya*

and *Śikṣā* literature. The architects of the *varṇamālā* operated within this vocabulary; Pāṇini and the *Prātiśākhya* compilers documented its operation.

Standard references: the *Prātiśākhya* texts in their standard editions (see endnote ayogavaha-category-pratisakhya for the full edition list); the *Śikṣā* texts in their standard editions: the *Pāṇinīya-Śikṣā* edited by Manmohan Ghosh (University of Calcutta, 1938); the *Nāradya-Śikṣā* edited by Sharma (Vishveshvaranand Vishva Bandhu Institute, 1980); Vidyāsāgara's *Śikṣā-Saṃgraha* (Calcutta, 1893) — the compendium of multiple *Śikṣā* texts. Modern scholarly treatments: W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953); Hartmut Scharfe, *Grammatical Literature* (Otto Harrassowitz, 1977), Chapter 3 on the *Vedāṅga* disciplines.

staal-mendeleev-varga-comparison

Short: Frits Staal develops the structural comparison between the *varga* matrix and Mendeleev's periodic table — both place units at unique coordinates in a multi-dimensional parameter space (place × manner × voicing × aspiration for *varga*; atomic mass × valence-electron configuration for chemistry), both combinatorial and predictive — across *The Science of Ritual* (BORI, 1982), *Universals: Studies in Indian Logic and Linguistics* (University of Chicago Press, 1988), and *Discovering the Vedas* (Penguin India, 2008). The chapter accepts the structural parallel; rejects Staal's historical extension that the *varga* system, like the periodic table, was the product of *centuries of analysis*.

Deployments: Chapter 8 §8.5 ¶ — the citation anchor for Frits Staal's *varga-to-Mendeleev periodic-table structural comparison*.

Frits Staal develops the comparison between the *varga* system and Mendeleev's periodic table across several works. The principal refer-

ences:

1. Frits Staal, “The Science of Ritual,” *Bhandarkar Oriental Research Institute, Pune, Post-Graduate and Research Department Series, 15* (Bhandarkar Oriental Research Institute, 1982). The structural comparison is developed in the context of the *Vedic* ritual and grammatical apparatus.
2. Frits Staal, *Universals: Studies in Indian Logic and Linguistics* (University of Chicago Press, 1988), the chapters on the *varga* system and the structural-formal comparison with chemistry.
3. Frits Staal, *Discovering the Vedas: Origins, Mantras, Rituals, Insights* (Penguin India, 2008), the consolidated articulation of the comparison for a general readership.

The structural comparison Staal develops: both the *varga* system and the periodic table place units at unique coordinates in a multi-dimensional parameter space, with the coordinates determined by the unit’s constituent components. The periodic table places elements at (row $\times$ column) coordinates determined by (atomic mass $\times$ valence-electron configuration); the *varga* system places consonants at (row $\times$ column) coordinates determined by (place of articulation $\times$ manner of articulation + voicing + aspiration). Both systems are *combinatorial* — every cell in the matrix is occupied by exactly one unit, and the unit’s properties are predicted by its coordinates. Both systems are *predictive* — gaps in the matrix can be inspected and filled, with the predicted properties matching the eventually-observed properties.

The structural parallel is the *architectural* parallel. The two systems share a deep formal structure: the discretization of continuous parameter space into a finite combinatorial matrix; the assignment of distinct units to cells; the mapping of unit-properties to cell-coordinates.

The chapter’s deployment: the structural comparison Staal de-

velops is correct and load-bearing for the chapter's broader argument that the *varga* matrix is an engineered combinatorial system. The chapter accepts Staal's structural parallel. The chapter, however, rejects Staal's *historical* extension of the comparison (the claim that the *varga* system, *like* the periodic table, was the product of *centuries of analysis*) — see endnote architecture-not-analysis-pratisakhya for the full treatment of the historical-extension issue.

Standard references: Frits Staal, *Universals: Studies in Indian Logic and Linguistics* (University of Chicago Press, 1988), Chapter 2; Staal, "The Sound Pattern of Sanskrit," in the relevant collected volumes of his papers; the *Discovering the Vedas* (Penguin India, 2008) consolidated treatment.

architecture-not-analysis-pratisakhya

Short: Mendeleev's periodic table was assembled by chemists working empirically across decades (Lavoisier → Dalton → mid-19c atomic-mass measurements → 1869 periodic-law formulation → predictive validation), with the historical record documented in journal articles, correspondence, and laboratory notebooks; the *varga* matrix has no such record — the *Prātiśākhya* literature presents it as already-operational vocabulary, without an empirical-analytical reconstruction or documented sequence of trial-and-error. **Architecture, not analysis** — the architects engineered the grid; the *Prātiśākhya* compilers documented it; the historical-analytical projection is an inference, not a documented claim.

Deployments: Chapter 8 §8.5 ¶ (the close of the section, the critique of Staal's historical extension) — the citation anchor for the argument that the *Prātiśākhya* texts present the *varga* system as already-operational vocabulary rather than as the residue of an empirical-

historical analytical project.

The structural claim the chapter establishes at this paragraph: Staal's structural comparison of the *varga* system to the periodic table is correct, but his historical extension (that the *varga* system, like the periodic table, was the product of *centuries of analysis*) is unsupported. The reasoning:

Mendeleev's periodic table was assembled by chemists working empirically over decades of laboratory inquiry. The historical record is documented: Lavoisier's *Traité Élémentaire de Chimie* (1789) and the early establishment of element-chemistry; Dalton's atomic theory (early nineteenth century); the systematic gathering of atomic-mass measurements across the early-to-mid nineteenth century; Mendeleev's *Osnovy khimii* (*Principles of Chemistry*, 1869–1871) with the periodic-law formulation; the predictive validation of the gaps (gallium, scandium, germanium discovered after Mendeleev's predicted properties had been published); the subsequent refinement under Moseley's atomic-number reanalysis (1913). The historical assembly is documented in the chemistry literature of the period — the journal articles, the correspondence, the laboratory notebooks, the textbooks of the era.

The *varga* grid has no such record. The *Prāṭisākhya* texts, the earliest documentation of the *varga* system, present the grid as *already operational vocabulary*. The texts use the terms *sthāna*, *karaṇa*, *prayatna*, *varga*, *sprṣṭa*, *aghoṣa*, etc., without justifying their construction or documenting an empirical-analytical process by which the categories were derived. There is no *Prāṭisākhya* equivalent of Lavoisier or Mendeleev — no documented sequence of empirical observations leading to a tentative classification, followed by refinement under further observation, followed by canonical formulation. The texts present the canonical formulation as the starting point.

Staal's *centuries of analysis* projection is therefore an *inference*, not a documented historical claim. The inference operates from the as-

sumption that any multi-axis combinatorial framework *must* be the residue of empirical inquiry — that there is no other way to arrive at such a framework than by accumulated trial-and-error analysis. The assumption is not necessary. A combinatorial framework can be *engineered* — designed from first principles by architects working out the structure deductively, then refined and documented. The *varga* system’s presentation in the *Prāṭisākhya* literature is consistent with engineering, not consistent with the historical-analytical assembly Staal projects.

The chapter’s broader argument turns on this distinction. *Architecture, not analysis*. The *Prāṭisākhya* discipline documents the engineering — the architecture that the architects produced — without documenting an empirical-analytical history because there was no empirical-analytical history. The system was engineered, presented to the *Vedāṅga* disciplines, documented by the *Prāṭisākhya* compilers, and transmitted across the *paramparā*. The architects’ identity is not documented; the architecture is documented. The two are distinct claims and the chapter holds them distinct.

Standard references: see endnote *staal-mendeleev-varga-comparison* for Staal’s primary references; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953) for the documented presentation of the *Prāṭisākhya* phonetic framework; the *Prāṭisākhya* texts in their standard editions. The chapter’s *architecture-not-analysis* framing is its own analytical position, built on the documentary record the *Prāṭisākhya* discipline provides.

western-linguistic-encounter-sanskrit-1786-1879

Short: Five foundational works carry the European-philological encounter with Sanskrit grammar from 1786 through 1879 — Sir William Jones, “The Third Anniversary Discourse, on the Hindus”

(Calcutta 1786; *Asiatic Researches* 1, 1788) → Friedrich Schlegel, *Über die Sprache und Weisheit der Indier* (Heidelberg, 1808) → Franz Bopp, *Über das Conjugationssystem der Sanskritsprache* (Frankfurt, 1816) → Otto von Böhtlingk, *Pāṇini's acht Bücher grammatischer Regeln* (Bonn, 1839–1840, first European critical edition of the *Aṣṭādhyāyī*) → William Dwight Whitney, *Sanskrit Grammar* (Leipzig / London, 1879) — followed by the IPA founding (1886) and first chart (1888) on a 2D grid structurally identical to the *varṇamālā*.

Deployments: Chapter 8 §8.7 ¶ — the consolidated citation anchor for the standard works of the European Sanskrit-encounter across 1786–1879. The body prose names the chronology without scholar names; this endnote supplies them.

Four foundational works carried the European-philological encounter with Sanskrit grammar from Jones's 1786 Calcutta address through the end of the nineteenth century:

- **Sir William Jones**, “The Third Anniversary Discourse, on the Hindus” (Calcutta, 1786; published *Asiatic Researches* 1, 1788) — the founding recognition of Sanskrit's structural relationship to Greek, Latin, Gothic, Celtic, and Old Persian; the inaugural moment of comparative philology. See endnote [jones-1786-third-anniversary-discourse](#) for the full passage and modern scholarly treatments.
- **Friedrich Schlegel**, *Über die Sprache und Weisheit der Indier* (Heidelberg: Mohr und Zimmer, 1808) — the German introduction of Sanskrit grammatical analysis into European philology; the work that brought Sanskrit's structural categories into the German Romantic-philological imagination and named the *Indogermanisch* category that the subsequent comparative project operated within.
- **Franz Bopp**, *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache* (Frank-

furt am Main, 1816) — the work that established the systematic-comparative method, with Sanskrit verbal morphology positioned as the structural anchor against which the other Indo-European systems were compared. See endnote bopp-1816-conjugationssystem for the full prosecutorial treatment in Appendix Part 1 §1.3.

- **Otto von Böhtlingk, *Pāṇini's acht Bücher grammatischer Regeln*** (Bonn: König, 1839–1840) — the first critical edition of the *Aṣṭādhyāyī* in Europe, putting Pāṇini's grammatical methodology in front of European linguistics in its original *sūtra-by-sūtra* form. Böhtlingk subsequently produced the *Sanskrit-Wörterbuch* (1855–1875, seven volumes, with Rudolph Roth at Göttingen) — the standard Sanskrit-German lexicographical reference for the rest of the nineteenth century.
- **William Dwight Whitney, *Sanskrit Grammar*** (Leipzig: Breitkopf and Härtel; London: Trübner; 1879; second edition 1889) — the standard English-language reference grammar of Sanskrit through the twentieth century. Whitney's *Roots, Verb-Forms, and Primary Derivatives of the Sanskrit Language* (Leipzig, 1885) became the canonical lexicographical reference for the verbal-root inventory.

The chronological sequence the body prose carries: 1786 (Jones / opening) → 1808 (Schlegel / German absorption) → 1816 (Bopp / systematic comparative method) → 1839–40 (Böhtlingk / *Aṣṭādhyāyī* in front of European linguistics) → 1879 (Whitney / standard English-language reference). By 1886 the IPA founding and the 1888 chart land on a 2D grid structurally identical to the *varṇamālā*.

early-19c-comparative-philology-bopp-pott

Short: Three foundational works of the early-19th-century comparative-philological project, positioning Sanskrit at the source position of Indo-European etymological chains: Franz Bopp, *Über das Conjugationssystem der Sanskritsprache* (Frankfurt, 1816); Franz Bopp, *Vergleichende Grammatik der Sanskrit-, Send-, Armenischen-, Griechischen-, Lateinischen-, Litauischen-, Altslavischen-, Gothischen- und Deutschen* (Berlin, 1833–1852, six fascicles); August Friedrich Pott, *Etymologische Forschungen auf dem Gebiete der indo-germanischen Sprachen* (Lemgo: Meyer, 1833–1836; expanded 1859–1876, five volumes). The post-1861 Schleicher inversion moved the anchor from Sanskrit to reconstructed PIE.

Deployments: Chapter 19 §19.X ¶ — the citation anchor for the early-19th-century comparative-philology works that positioned Sanskrit at the source position of Indo-European etymological chains, before the post-1861 Schleicher inversion moved the anchor from Sanskrit to the reconstructed proto-language.

Three foundational works:

- **Franz Bopp**, *Über das Conjugationssystem der Sanskritsprache* (Frankfurt am Main, 1816) — see endnote bopp-1816-conjugationssystem for full treatment.
- **Franz Bopp**, *Vergleichende Grammatik der Sanskrit-, Send-, Armenischen-, Griechischen-, Lateinischen-, Litauischen-, Altslavischen-, Gothischen- und Deutschen* (Berlin, 1833–1852, six fascicles) — the extension of Bopp’s comparative method across the full Indo-European family. Sanskrit remained the structural anchor through this work.
- **August Friedrich Pott**, *Etymologische Forschungen auf dem Gebiete der indo-germanischen Sprachen* (Lemgo: Meyer, 1833–1836, two volumes; second expanded edition 1859–1876, five volumes) — the etymological-research ex-

tension of the comparative method, carrying the structural comparison into vocabulary. Pott's work, alongside Bopp's *Vergleichende Grammatik*, defined the canonical first-generation comparative-philological project. Like Bopp, Pott positioned Sanskrit at or near the source position of Indo-European etymological chains.

Through this entire early period (1816–1860), Sanskrit was treated as the source-language of the family the comparative method was constructing — the analytical anchor against which the Greek, Latin, Persian, and Germanic forms were compared. The post-1861 Schleicher inversion (introducing the reconstructed common ancestor *distinct from* any attested language) moved the anchor from Sanskrit to PIE; the *bake* the Appendix Part 1 polemic prosecutes is the inversion that displaced Sanskrit from this source position.

jones-1786-third-anniversary-discourse

Short: Sir William Jones, “The Third Anniversary Discourse, on the Hindus,” delivered February 2, 1786 at the Asiatic Society in Calcutta, published *Asiatic Researches* 1 (1788): 415–431 — the founding statement of comparative philology: “*The Sanskrit language ... is of a wonderful structure; more perfect than the Greek, more copious than the Latin, and more exquisitely refined than either ... no philologist could examine them all three, without believing them to have sprung from some common source, which, perhaps, no longer exists.*” Opens the European-philological project that across the long 19th century absorbed Sanskrit grammatical analysis into the apparatus that became the IPA framework.

Deployments: Chapter 8 §8.7 ¶ — the citation anchor for William Jones's 1786 Calcutta address recognizing Sanskrit's structural relationship to Greek, Latin, and other languages.

Sir William Jones, “The Third Anniversary Discourse, on the Hindus,” delivered at the Asiatic Society in Calcutta on February 2, 1786. Published in *Asiatic Researches* Volume 1 (Calcutta, 1788), pages 415–431. The passage that became canonical in the history of comparative philology:

“The Sanskrit language, whatever be its antiquity, is of a wonderful structure; more perfect than the Greek, more copious than the Latin, and more exquisitely refined than either, yet bearing to both of them a stronger affinity, both in the roots of verbs and in the forms of grammar, than could possibly have been produced by accident; so strong indeed, that no philologist could examine them all three, without believing them to have sprung from some common source, which, perhaps, no longer exists; there is a similar reason, though not quite so forcible, for supposing that both the Gothic and the Celtick, though blended with a very different idiom, had the same origin with the Sanscrit; and the old Persian might be added to the same family.”

The passage is one of the founding statements of comparative philology — the recognition that Sanskrit, Greek, Latin, Gothic, Celtic, and Old Persian share structural features (in verbal roots and grammatical forms) that cannot plausibly be explained by chance and that imply a common source. Jones’s “common source” hypothesis was later formalized as the *Proto-Indo-European* reconstruction project, anchored in the comparative method.

The structural significance the chapter establishes: Jones’s 1786 address opens the European-philological project that operated continuously across the nineteenth and twentieth centuries to engage Sanskrit grammatical analysis. The chapter’s broader argument — that the IPA framework and the contemporary phonological grid are direct absorptions of the Sanskrit *varṇamālā* architecture by European linguistics across the long nineteenth century — begins at this 1786 moment. Jones recognized Sanskrit’s grammatical structure as more per-

fect than the European languages he knew; the recognition launched the project of bringing Sanskrit grammatical analysis into European scholarship; the project's terminal output (the IPA, the contemporary phonological framework) is structurally identical to the *varṇamālā*'s 2D specification.

Standard references: Sir William Jones, “The Third Anniversary Discourse, on the Hindus,” *Asiatic Researches* 1 (1788): 415–431. Reprinted in *The Works of Sir William Jones* (London, 1799), Volume 1; in *Sir William Jones: A Reader* (edited by Satya S. Pachori, Oxford University Press, 1993); and in the standard history-of-linguistics anthologies. Modern scholarly treatments: Garland Cannon, *The Life and Mind of Oriental Jones* (Cambridge University Press, 1990); Michael J. Franklin, “Orientalist Jones”: *Sir William Jones, Poet, Lawyer, and Linguist, 1746–1794* (Oxford University Press, 2011); Rosane Rocher, *Orientalism, Poetry, and the Millennium: The Checkered Life of Nathaniel Brassey Halhed, 1751–1830* (Motilal Banarsidass, 1983) — the contextual reconstruction of the Calcutta orientalist scene in which Jones operated.

ipa-1886-founding-1888-chart

Short: The International Phonetic Association (*L'Association Phonétique Internationale*) was founded in Paris in 1886 by Paul Passy with Henry Sweet and Daniel Jones; the first International Phonetic Alphabet (IPA) chart was published in 1888 in *Le Maître Phonétique*, organizing consonantal sounds by place of articulation (columns) and manner of articulation (rows) on a 2D grid *structurally identical to the varṇamālā*'s 5×5 *varga matrix* — with European-language consonantal contents substituted in, the Sanskrit anatomical terminology translated into Greek / Latin equivalents, and the *Vedāṅga* engineering provenance silently dropped.

Deployments: Chapter 8 §8.7 ¶ — the citation anchor for the founding of the International Phonetic Association in 1886 and the publication of the first IPA chart in 1888.

The International Phonetic Association (originally *L'Association Phonétique Internationale*, abbreviated AFI) was founded in Paris in 1886 by Paul Passy and a group of French and English phonetic teachers (including the British phoneticians Henry Sweet and Daniel Jones in subsequent years). The founding purpose: to develop a unified phonetic alphabet for the consistent representation of speech sounds across the world's languages, primarily for use in language teaching and phonetic transcription.

The first version of the International Phonetic Alphabet (IPA) chart was published in 1888 in the Association's journal *Le Maître Phonétique*. The chart organized consonantal sounds by place of articulation (columns: labial, dental, alveolar, palatal, velar, glottal) and manner of articulation (rows: plosives, fricatives, nasals, laterals, trills, semivowels) — a 2D grid structurally identical to the *varṇamālā*'s 5×5 *varga* matrix, with additional rows for the consonant-classes the *varṇamālā* organizes as *antaḥstha* (semivowels), *ūṣman* (sibilants and *ha*), and the *ayogavāha* (here grouped under glottal and nasal categories).

The IPA chart has been revised periodically across the subsequent decades — the most significant revisions in 1900, 1932, 1989 (the Kiel Convention), 1993, 1996, and 2005 — to accommodate additional speech sounds, refine the place-and-manner organization, and harmonize with developments in phonetic science. The structural form, however, remains the same: a 2D grid with place of articulation organized along one axis and manner of articulation along the other.

The structural significance the chapter establishes: by 1888 — within a century of Jones's 1786 *Third Anniversary Discourse* and within decades of the European-philological absorption of Sanskrit gram-

mathematical analysis (Schlegel 1808, Bopp 1816, Böhlingk's *Aṣṭādhyāyī* edition 1839–1840, Whitney's *Sanskrit Grammar* 1879) — European phonological science was operating within a framework *structurally identical* to the *varṇamālā*'s 2D specification. The IPA chart is the *varṇamālā*'s architecture with European-language consonantal contents substituted in, the Sanskrit anatomical terminology translated into Greek/Latin anatomical terminology, and the *Vedāṅga*'s engineering provenance silently dropped.

The standard history of the IPA recognizes the European-philological-tradition genealogy back through Bell, Sweet, Passy, and the *Romic* alphabet experiments of the late nineteenth century but does not generally credit the Sanskrit framework as the structural source of the 2D organization. The chapter's analysis recovers the structural-source claim that the orthodox IPA history does not document.

Standard references: International Phonetic Association, *Handbook of the International Phonetic Association: A Guide to the Use of the International Phonetic Alphabet* (Cambridge University Press, 1999) — the standard contemporary reference, with historical-development chapter. Michael K. C. MacMahon, "Phonetic Notation," in Peter T. Daniels and William Bright, eds., *The World's Writing Systems* (Oxford University Press, 1996), pages 821–846. The historical archive of *Le Maître Phonétique* is available through the Association's website and through standard linguistics-history library collections. For the broader history of phonetics: R. H. Robins, *A Short History of Linguistics* (Longman, 4th edition 1997), Chapter 16; Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Routledge, 1998).

history-of-linguistics-sanskrit-influence

Short: The standard history-of-linguistics literature — R. H. Robins, *A Short History of Linguistics* (Longman, 4th ed. 1997); Anna

Morpurgo Davies, *Nineteenth-Century Linguistics* (Routledge, 1998); Hartmut Scharfe, *Grammatical Literature* (Harrassowitz, 1977); George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988); John E. Joseph, *From Whitney to Chomsky* (Benjamins, 2002); Madhav Deshpande, *The Idea of the Independent Word in Pāṇinian Grammatical Tradition* (American Oriental Society, 1992) — documents the substantial European-philological absorption of Sanskrit *vyākaraṇa* (व्याकरण) analytical apparatus across the long 19th century: the phoneme concept, the morpheme concept, the generative-rule format, the 2D phonetic-classification grid.

Deployments: Chapter 8 §8.7 ¶ — the citation anchor for the standard history-of-linguistics treatment of the Sanskrit *vyākaraṇa* discipline's influence on European linguistic analysis.

The standard history-of-linguistics literature recognizes the Sanskrit *vyākaraṇa* discipline as a major source of analytical apparatus and theoretical framework for nineteenth-century European linguistics. The principal references:

1. **R. H. Robins, *A Short History of Linguistics*** (Longman, 4th edition 1997). The standard one-volume history. Chapter 6 (Indian linguistic theory) develops Pāṇini's *Aṣṭādhyāyī* and the Sanskrit *vyākaraṇa* discipline's analytical apparatus. Chapter 7 (Nineteenth-century linguistics) traces the European absorption of the Sanskrit framework across the comparative-philological project.
2. **Anna Morpurgo Davies, *Nineteenth-Century Linguistics*** (Volume IV of the Routledge *History of Linguistics*, 1998). The authoritative academic history of nineteenth-century European linguistics. Chapters 1–3 develop the foundational role of Sanskrit grammatical analysis in establishing the comparative method.
3. **Hartmut Scharfe, *Grammatical Literature*** (Volume V, Fascicle 2 of Jan Gonda's *A History of Indian Literature*, Otto

Harrassowitz, 1977). The standard reference for the Sanskrit vyākaraṇa discipline itself.

4. **George Cardona, *Pāṇini: His Work and Its Traditions*** (Motilal Banarsidass, 1988). The standard modern philological reconstruction of Pāṇini's work and its reception across both the Indian and European traditions.
5. **John E. Joseph, *From Whitney to Chomsky: Essays in the History of American Linguistics*** (John Benjamins, 2002). Treats the specific American-linguistics tradition's engagement with Sanskrit, including Whitney's standard-reference role.
6. **Madhav Deshpande, *The Idea of the Independent Word in Pāṇinian Grammatical Tradition*** (American Oriental Society, 1992). Specialist treatment of the *padam* (word) concept in Pāṇinian analysis and its influence on later linguistic analysis.

The convergent historical claim across these references: the European-philological absorption of Sanskrit grammatical analysis across the long nineteenth century was substantial, structural, and acknowledged. Specific lines of influence:

- **The phoneme concept.** European phonology's distinction between the phoneme as an abstract underlying unit and its phonetic realizations follows the *varṇa* / *ucchāraṇa* distinction the Sanskrit discipline operates with.
- **The morpheme concept.** European morphology's analysis of words into constituent morphemes follows the Sanskrit discipline's analysis of words into root + affix combinations.
- **The generative-rule format.** European generative phonology and generative syntax in the mid-twentieth century (Chomsky, Halle, the early formal-linguistics tradition) explicitly cited Pāṇini's *sūtra* format as the methodological precedent for rule-based linguistic description.

- **The 2D phonetic-classification grid.** The IPA's 2D place × manner organization, as established at the IPA's founding in 1886 and the first chart in 1888, follows the *varga* matrix's 2D structure.

The chapter's deployment: the standard history-of-linguistics literature documents the lines of influence in detail. The chapter's broader argument — that the contemporary phonological framework is structurally absorbed from Sanskrit — is anchored at this standard history. The English terms (labial, dental, retroflex, palatal, velar) have their own Greek and Latin anatomical etymologies, independent of Sanskrit; but the *systematic deployment* of these terms as a unified 2D phonological grid follows the Sanskrit framework absorbed across the nineteenth century.

Standard references as enumerated above. The body of work is large and well-documented; the chapter's citation gestures at the standard history-of-linguistics literature without attempting an exhaustive enumeration.

formants-source-filter-theory

Short: The source-filter theory of speech production (Gunnar Fant, *Acoustic Theory of Speech Production*, Mouton 1960) models speech as a glottal source signal filtered by the vocal-tract cavity geometry, producing the characteristic formant peaks (*F1* primarily tongue height, *F2* primarily tongue front-to-back position) that distinguish vowel qualities; the five *sthāna* positions of the *varṇamālā* sample the formant space at well-separated acoustic points just as they sample the vocal tract at well-separated cm-distances — spatial well-separation produces acoustic well-separation, and *anunāsika* (अनुनासिक) nasal-coupling adds spectral anti-resonances the four oral positions cannot produce.

Deployments: Chapter 8 §8.8 ¶ — the citation anchor for the source-filter theory of speech production and the formant-based acoustic analysis of vowel and consonant production.

The *source-filter theory* of speech production, established in modern phonetics by Gunnar Fant in his 1960 *Acoustic Theory of Speech Production* (Mouton, The Hague), models the speech-production apparatus as two functionally distinct components:

1. **Source** — the glottis, producing a quasi-periodic source signal (the laryngeal buzz) when the vocal cords are vibrating, or a turbulent noise source when the vocal cords are open and air passes through unimpeded.
2. **Filter** — the vocal tract above the glottis (pharynx + oral cavity + nasal cavity when coupled), acting as a frequency-selective filter that emphasizes certain frequency bands (the *formants*) and attenuates others. The filter's frequency response is determined by the vocal tract's shape — its length, cross-sectional area at each point, and any constriction-points the speaker has introduced.

The acoustic signature of any vowel is the pattern of its formants — the first few resonant peaks in the spectrum. Standard formant parameters:

- **F1** (first formant) — primarily determined by tongue height (low vowels like /a/ have higher F1; high vowels like /i/ and /u/ have lower F1).
- **F2** (second formant) — primarily determined by tongue front-to-back position (front vowels like /i/ have higher F2; back vowels like /u/ have lower F2).
- **F3, F4** (higher formants) — primarily determined by lip rounding, vocal tract shape details, and individual anatomical variation.

The F1-F2 plane is the conventional 2D space in which vowel qualities

are plotted. The *cardinal vowels* of the IPA (the canonical reference vowels: *i, e, ε, a, ɔ, o, u*) plot at characteristic positions in the F1-F2 plane that correspond to articulatory tongue positions.

For consonants, the source-filter framework applies with additional parameters: the place of articulation determines the formant-burst pattern at consonant-vowel transitions; the manner of articulation determines the spectral and temporal characteristics of the consonant itself.

The structural significance the chapter establishes: the five *sthāna* positions of the *varṇamālā* sample the formant space at well-separated acoustic points, just as they sample the vocal tract at well-separated cm-distances. Spatial well-separation produces acoustic well-separation. The five places of articulation are five distinguishable cavity geometries; the cavity geometries produce five distinguishable formant signatures; the listener's ear discriminates the formant signatures without effort because the engineering selected positions far enough apart in formant space that confusion is unlikely. The nasal coupling adds anti-resonances — spectral notches — that the four oral positions cannot produce, which is why *anunāsika* is a separate dimension. The *alpa-prāṇa* / *mahā-prāṇa* contrast is a pressure differential at the moment of release — the same physics as overblowing a wind instrument.

The architects of the *varṇamālā* engineered the phonological grid to fall at acoustically distinguishable positions in formant space. The engineering is consistent with what contemporary acoustic phonetics would design from first principles; the architects arrived at the same engineering many generations ago.

Standard references: Gunnar Fant, *Acoustic Theory of Speech Production* (Mouton, 1960); Kenneth N. Stevens, *Acoustic Phonetics* (MIT Press, 1998), the central reference on formant theory and source-filter analysis; James L. Flanagan, *Speech Analysis, Synthesis, and Perception* (Springer, 2nd edition 1972); Ray D. Kent and Charles

Read, *The Acoustic Analysis of Speech* (Singular Publishing, 2nd edition 2002). For the cardinal-vowel framework: Daniel Jones, *An Outline of English Phonetics* (Cambridge University Press, 9th edition 1960).

hrasva-dīrgha-pluta-matra

Short: The Sanskrit phonetic discipline classifies vowel duration in three canonical degrees, measured in *mātrā* (मात्रा) units: *hrasva* (ह्रस्व, short — 1 *mātrā*, ~100 ms; *a*, *i*, *u*, *r*, *l*), *dīrgha* (दीर्घ, long — 2 *mātrās*; *ā*, *ī*, *ū*, *ṛ*, *e*, *ai*, *o*, *au*), and *pluta* (प्लुत, extended — 3+ *mātrās*, used in calling across distance and certain Vedic recitational contexts); governed by *Aṣṭādhyāyī* 1.2.27–28 (*ūkālo 'jjhrasvadīrghaplutaḥ*) — the temporal dimension of the engineered phonological apparatus, preserved by the Vedic recitation lineages with reproducible 1:2:3 timing-ratios.

Deployments: Chapter 8 §8.10 ¶ — the citation anchor for the three-fold vowel-duration classification in the Sanskrit phonetic discipline.

The Sanskrit phonetic discipline distinguishes three canonical durations for vowels:

1. **ह्रस्व (*hrasva*)** — *short*. One *mātrā* (the canonical unit of time-measurement in Sanskrit phonology). The short vowels: *a*, *i*, *u*, *r*, *l*. Approximate duration: ~100 milliseconds in standard recitation, though the absolute value varies with speech rate.
2. **दीर्घ (*dīrgha*)** — *long*. Two *mātrās*. Exactly twice the duration of *hrasva*. The long vowels: *ā*, *ī*, *ū*, *ṛ*, *e*, *ai*, *o*, *au*. The diphthongs *e*, *ai*, *o*, *au* are conventionally classified as *dīrgha* in their standard form.
3. **प्लुत (*pluta*)** — *extended*. Three or more *mātrās*. Used in calling-out across distance, in certain Vedic recitational contexts, and in

moments where the speaker holds the vowel deliberately. The standard Pāṇinian example is *he Devadatta3* — the numeral 3 in some Sanskrit texts marks a *pluta* vowel held for three beats. The use is restricted; *pluta* does not function as a routine vowel-duration in Pāṇinian *bhāṣāyām*.

The three-fold classification is governed in the *Aṣṭādhyāyī* by *sūtras* 1.2.27–28: *ūkālo 'jjhrasvadīrghaplutaḥ; acaśca* — defining the three durations and their applicability. The *Prātiśākhya* and *Śikṣā* texts develop the duration framework in detail, with specific recitational rules for how each duration is to be held in different contextual environments.

The structural significance the chapter establishes: the *mātrā*-based duration framework is the temporal-engineering dimension of the Sanskrit phonological apparatus. Vowels are not only specified by their formant signature (F1, F2 — the place-and-manner dimension) but also by their temporal extent (one, two, or three+ *mātrās* — the duration dimension). The full vowel specification is therefore a three-axis description: place (back-of-mouth vs. front-of-mouth — *a/ā, i/ī, u/ū*, etc.), manner (open vs. close), and duration (one, two, three+ *mātrās*).

The temporal precision the *mātrā* framework imposes is one of the engineering features the Vedic recitation lineages preserve. The reciter holds each vowel for exactly the prescribed duration; the *guru-shishya* training transmits the timing-precision across generations. The contemporary Vedic *śākhā* reciters demonstrate the timing-precision in their recitations, with the duration-ratios (1:2:3+) reproducible across multiple reciters from multiple lineages.

Standard references: Pāṇini's *Aṣṭādhyāyī*, *sūtras* 1.2.27–28; the *Prātiśākhya* literature (*Ṛk-Prātiśākhya* on the *mātrā* framework; *Taittirīya-Prātiśākhya* 1.37 and following); the *Pāṇinīya-Śikṣā* on duration; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 6 (on duration and accent); George Cardona,

Pāṇini: His Work and Its Traditions (Motilal Banarsidass, 1988), discussion of the duration-rules.

vedic-svara-system

Short: The Vedic recitation lineages operate a three-fold accent system specifying syllable pitch: **udātta** (उदात्त, *raised* — high pitch), **anudātta** (अनुदात्त, *not-raised* — low pitch), and **svarita** (स्वरित, *sounded* — the mid-falling-pitch accent that follows an *udātta* syllable); governed across *Aṣṭādhyāyī Adhyāya* 8 and complemented by Śāntanava's *Phīṭ-sūtras* (फिट्सूत्र); the Vedic mode operates the full three-fold system, the *bhāṣyām* carries it in attenuated form. The Indian classical music *swara* system (*sa, ri, ga, ma, pa, dha, ni*) operates the same pitch-categorization framework at fuller temporal range.

Deployments: Chapter 8 §8.10 ¶ — the citation anchor for the Vedic *svara* (accent / pitch) system and its relationship to the broader Sanskrit phonological apparatus.

The Vedic recitation lineages operate on a three-fold accent system that specifies the pitch contour of each syllable. The three canonical accents:

1. **उदात्त (udātta)** — *raised*. The high-pitch accent. In Vedic recitation, the *udātta* syllable is pronounced at a relatively higher pitch than the surrounding syllables. The notation in Vedic *Samhitā* texts varies across the *śākhā* lineages: in the *R̥gveda* (Śākala recension), *udātta* is typically unmarked (the default reading); in the *Taittirīya śākhā*, the *udātta* is marked with a vertical line above the syllable.
2. **अनुदात्त (anudātta)** — *not-raised*. The low-pitch accent. The *anudātta* syllable is pronounced at a relatively lower pitch.

Marked with a horizontal line below the syllable in many *Vedic* manuscript lineages.

3. **स्वरित (svarita)** — *sounded*. The mid-falling-pitch accent that follows an *udātta* syllable. The *svarita* is acoustically distinctive — it begins at the *udātta* high-pitch level and falls during the syllable to the *anudātta* low-pitch level, producing a characteristic downward pitch contour. Marked with a vertical line above the syllable in the standard Vedic manuscript notation, often differentiated from the *udātta* marking by the line's position relative to the syllable's vowel.

The three-fold accent system is governed by Pāṇini's *Aṣṭādhyāyī* across multiple *sūtras* in *Adhyāya* 8 (the accent-specifying rules), with the *Phit-sūtras* of Śāntanava operating as a complementary treatise on Vedic-text accent. The *Prātiśākhya* texts of each *Veda* specify the accent-recitation rules in detail for their respective recensions.

The structural significance the chapter establishes: the *udātta* / *anudātta* / *svarita* system is the *pitch* dimension of the Sanskrit phonological apparatus. The Vedic mode operates on the full three-fold system; the post-Pāṇinian Sanskrit mode (the *bhāṣāyām* — the spoken / classical mode Pāṇini's *Aṣṭādhyāyī* primarily describes) carries the accent system in attenuated form, with the *udātta-anudātta-svarita* distinction collapsed across non-Vedic registers. The Indian classical music continuum's pitch system (the *swara* — *sa, ri, ga, ma, pa, dha, ni*) operates the same pitch-categorization framework at fuller temporal range, with each note held for bars rather than the moment-of-the-syllable duration the speech register uses.

The chapter's deployment: the *swara* concept is structurally continuous across speech (the Vedic three-fold accent), music (the seven-note scale), and the broader sound-engineering of the *varṇamālā*. Two instruments, one apparatus. The same *swara* that Indian classi-

cal music holds for bars and the Vedic mode marks for pitch contour, speech cuts to precise *mātrā* durations.

Standard references: Pāṇini's *Aṣṭādhyāyī*, *Adhyāya* 8 (the accent-rules) and 6.1.158 and following (vowel-accent placement); the *Prātiśākhya* literature on Vedic accent; the *Phit-sūtras* of Śāntanava; W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 6 (on duration and accent); Madhav M. Deshpande, "Vedic Aryans, Non-Vedic Aryans, and Non-Aryans" (in *The Indo-Aryan Controversy*, ed. Edwin Bryant and Laurie Patton, Routledge, 2005); Frits Staal, *Mantras between Fire and Water* (Royal Netherlands Academy of Arts and Sciences, 1992) on the Vedic recitation lineages.

agnimile-rigveda-opening

Short: The opening of the *R̥gveda* (1.1.1, *Śākala* recension) — अग्निमीळे पुरोहितं यज्ञस्य देवमृत्विजम् । होतारं रत्नधातमम् ॥ / *agnim īle puro-hitaṃ yajñasya devam ṛtvijam | hotāraṃ ratnadhātamam* || — places the retroflex lateral ऌ (l) at the structural front of the foundational text, the second consonant of the second word; preserved intact across the Nambūdiri, Mādhyandina, Kāṇva and other surviving *śākhā* lineages — empirical confirmation that the engineered Vedic phonological inventory continues to operate.

Deployments: Chapter 16 §16.2 ¶ — the citation anchor for the opening word of the *R̥gveda* — *agnimīle* — and its placement of the retroflex lateral ऌ at the structural front of the foundational text.

The opening verse of the *R̥gveda* — *R̥gveda* 1.1.1 — is the canonical invocation to *Agni* that opens the *Samhitā* in its standard *Śākala* recension:

अग्निमीळे पुरोहितं यज्ञस्य देवमृत्विजम् । होतारं रत्नधातमम् ॥

agnim īle purohitaṃ yajñasya devam ṛtvijam | hotāraṃ
ratnadhātamaṃ ||

“I invoke Agni, the household-priest of the sacrifice, the divine officiant, the priest who summons, the most-bestower of treasures.”

The verb *īle* — first-person singular present of the root *īḍ* (to praise, to invoke, to extol) — carries the retroflex lateral *ḷ* in the standard Vedic-mode pronunciation. The *Padapāṭha* preserves the form *as'īle* with the retroflex lateral; the *Samhitā-pāṭha* recitation across the *Śākala śākhā* holds the same retroflex articulation.

The retroflex lateral *ḷ* (ḷ) is the *mūrdhanya* lateral — the lateral consonant articulated at the same place of articulation as the retroflex stops *ṭ ṭh ḍ ḍh ṇ*. It is the fifth member of the lateral series alongside the dental *l* (la); the retroflex lateral does not appear in the post-Pāṇinian (*bhāṣāyām*) phonological inventory. It is included in the *chandasi* (Vedic-mode) inventory and bounded out of the *bhāṣāyām* inventory — the two-mode synchronic-parallel distinction Pāṇini's *Aṣṭādhyāyī* operates with (see endnote *chandasi-bhasayam-astadhyayi*).

The structural significance the chapter establishes at this passage: the *Ṛgveda*'s opening word places *ḷ* at the *structural front* of the foundational text — the second consonant of the second word of the first verse of the first hymn of the first *maṇḍala*. The placement is load-bearing. If *ḷ* were a peripheral or late feature of the Vedic phonological inventory, it would not appear at this position. Its presence at this opening position confirms that *ḷ* is a *primary* phonological unit of the Vedic mode, present in the engineered phonological inventory from the start, preserved by the layered recitation apparatus across the depth of the *paramparā*.

The contemporary recitation lineages of the *Ṛgveda* — across the *Nambūdiri*, *Mādhyandina*, *Kāṇva*, and other surviving *śākhā* lineages — preserve the retroflex lateral in the *Samhitā* recitation. The re-

citer trained in any of these lineages produces ऌ at the appropriate positions in the *Samhitā* text, with the retroflex articulation distinguishable from the dental लि. The audible preservation across many generations is the empirical confirmation that the engineered phonological inventory the architects designed continues to operate.

Standard references: the *Ṛgveda* in the standard *Śākala* recension. Editions: Theodor Aufrecht, *Die Hymnen des Rigveda* (Bonn, 1861–1863); F. Max Müller, *Rig-Veda-Samhitā* (Oxford, 1849–1874); the *Vedic Reserve* digital edition; the Vaidika Samshodhana Mandala (Pune) critical-text materials. For the retroflex lateral specifically: W. S. Allen, *Phonetics in Ancient India* (Oxford University Press, 1953), Chapter 2 (place-of-articulation analysis including the retroflex laterals); the *Ṛk-Prātiśākhya* on the inclusion of ऌ in the Vedic phonological inventory.

chandasi-bhasayam-astadhyayi

Short: Pāṇini's *Aṣṭādhyāyī* distinguishes Sanskrit's two modes through the operational pair **chandasi** (छन्दसि, *in the Vedic mode*) and **bhāṣāyām** (भाषायाम्, *in the spoken / generative-analytical mode*) — a synchronic-parallel distinction, not chronological — with specific *sūtras* (1.1.16, 1.2.34, 3.4.6–7, 5.4.31 and across *Adhyāyas* 6–7 for *chandasi*; 1.1.62, 2.3.8, 5.4.150 for *bhāṣāyām*) marking which mode each rule applies to. The Vedic mode preserves features the *bhāṣāyām* mode does not (the retroflex lateral ऌ, the three-fold accent system, the *plutaḥ* extended vowels) — by engineered design, not historical descent.

Deployments: Chapter 16 §16.2 ¶ — the citation anchor for Pāṇini's *chandasi* / *bhāṣāyām* synchronic-parallel-mode distinction in the *Aṣṭādhyāyī*.

Pāṇini's *Aṣṭādhyāyī* distinguishes two modes of Sanskrit through

the operational pair *chandasi* (छन्दसि) — in the Vedic mode — and *bhāṣāyām* (भाषायाम्) — in the spoken / generative-analytical mode. The distinction is not a chronological one (Vedic-then-Classical, with the second descending from the first); it is a synchronic-parallel distinction (two modes operating side-by-side, with different phonological inventories, different morphological options, and different syntactic conventions appropriate to each mode's function).

The *chandasi* rules govern Vedic-mode formations and apply to passages, words, and grammatical forms that operate in the Vedic register. The *bhāṣāyām* rules govern the generative-analytical mode — the spoken / classical / *laukika* mode that Pāṇini's *Aṣṭādhyāyī* primarily describes and that operates as the productive grammatical system for Sanskrit composition.

The pair operates across the *Aṣṭādhyāyī* at multiple points. Specific *sūtras* that mark the distinction:

- *Aṣṭādhyāyī* 1.1.16 (with the *chandasi* context); 1.2.34 (governing Vedic accent placement); 3.4.6–7 (Vedic-mode verb formation); 5.4.31 (Vedic-mode compound formation); and across many other *sūtras* in *Adhyāyas* 6 and 7 that mark Vedic-specific rules with *chandasi* attribution.
- The complementary *bhāṣāyām* attribution appears at *sūtras* like 1.1.62 and 2.3.8 (mode-specific case-rule conditions); 5.4.150 (compound formation in the spoken mode); and across the broader morphological apparatus.

The structural significance the chapter establishes: Pāṇini's *Aṣṭādhyāyī*, in operating with the *chandasi* / *bhāṣāyām* distinction as a structural feature of the grammar's rule-system, decodes a *two-mode parallel* analysis of Sanskrit. The Vedic mode is not historically prior to the generative-analytical mode; the two modes are synchronically parallel, with rules of the grammar marking which mode each rule applies to. The Vedic mode preserves certain phonological features (the retroflex lateral ऌ (*l*), the three-fold accent system, the *plutaḥ*

extended vowels) and certain morphological options (specific Vedic verb forms, specific Vedic compound formations) that the generative-analytical mode does not carry. The two modes share the broader phonological and morphological apparatus; they differ at specific points the grammar explicitly marks.

The framing matters because the conventional account of the *chandasi* / *bhāṣāyām* relationship treats it as a historical relationship — Vedic Sanskrit as the older form, Classical Sanskrit as the descendant — and treats the relevant Pāṇinian rules as documenting the descent. Pāṇini's own framing does not support that account. The *Aṣṭādhyāyī* treats the two modes as synchronically parallel rule-sets within a single grammatical system, with the grammarian operating both modes as features of the decoded language.

Standard references: Pāṇini's *Aṣṭādhyāyī*, with the relevant *chandasi* and *bhāṣāyām* attributions. Standard editions: S. M. Katre, *Aṣṭādhyāyī of Pāṇini* (University of Texas Press, 1987); Rama Nath Sharma, *The Aṣṭādhyāyī of Pāṇini* (Munshiram Manoharlal, 6-volume edition, 1987–2003); Vasu's *The Ashtadhyayi of Panini* (Allahabad, 1891). Scholarly treatments: George Cardona, *Pāṇini: His Work and Its Traditions* (Motilal Banarsidass, 1988); Madhav Deshpande, "The Synchronic and the Diachronic in Pāṇinian Grammar" (in *Papers in Linguistics* 6, 1973); Hartmut Scharfe, *Grammatical Literature* (Otto Harrassowitz, 1977), Chapter 5.

muller-eic-rigveda

Short: Friedrich Max Müller, *Rig-Veda-Saṃhitā: The Sacred Hymns of the Brahmins, together with the Commentary of Sayanacharya* (Oxford University Press, 1849–1874, six volumes) — produced across a quarter-century at Oxford under East India Company patronage; Müller never visited the subcontinent. The broader *Sacred Books of*

the *East* series (Oxford, 1879–1910, fifty volumes) operated within a Christian-Protestant evolution-of-religious-consciousness framework — the *church of progress* in mid-19th-century philological vocabulary, the textual apparatus through which the AIT framework entered Sanskrit scholarship at scale.

Deployments: Chapter 16 §16.5 ¶ — the citation anchor for Max Müller’s East India Company-funded Oxford edition of the *R̥gveda* and the broader *Sacred Books of the East* apparatus.

Friedrich Max Müller (1823–1900), German philologist working at Oxford, produced his critical edition of the *R̥gveda* under the patronage of the English East India Company across a quarter-century at Oxford. The edition:

Max Müller, *Rig-Veda-Saṃhitā: The Sacred Hymns of the Brahmins, together with the Commentary of Sayanacharya* (six volumes, Oxford University Press, 1849–1874). The edition presented the *R̥gveda Saṃhitā* text alongside Sāyaṇa’s medieval commentary, with Müller’s own scholarly apparatus. The work was funded across the quarter-century of production by the East India Company.

Müller never visited the subcontinent. His engagement with Sanskrit was philological, mediated through the textual corpus available in European libraries and the manuscripts the East India Company’s collecting apparatus had brought to Oxford. His broader *Sacred Books of the East* project — fifty volumes published by Oxford University Press from 1879 to 1910 — extended the philological ecosystem to the canonical religious texts of multiple non-Western traditions (Hindu, Buddhist, Jain, Zoroastrian, Confucian, Taoist, Islamic). The series was the foundational scholarly apparatus by which the non-Western religious literatures entered European-academic engagement.

The structural-religious framework Müller’s apparatus operated within: a Christian-Protestant comparative-philological hermeneutic that treated Christianity as the implicit standard against which the other traditions were to be read. Müller’s *Lectures on the*

Science of Religion (1870) and his subsequent works develop the framework explicitly — the religions of the world are stages of the evolution-of-religious-consciousness, with the Vedic religion representing an early stage, the Hindu / Buddhist / Zoroastrian traditions representing middle stages, and the Christian / Lutheran tradition representing the (implicitly highest) culminating stage. The structural assumption — that the religions can be plotted on a developmental sequence with Protestant Christianity at the terminus — is the church-of-progress framework operating in mid-nineteenth-century philological vocabulary.

The polemic load the chapter carries: Müller's edition of the *R̥gveda*, funded by the East India Company, is the textual apparatus through which the AIT framework entered Sanskrit scholarship at scale. The migration thesis, the *ārya*-as-race framing, the *dāsa*-as-aboriginal framing, the dating of the *R̥gveda* to a specific narrow chronological window — these are claims developed in and against Müller's edition's framing. The chapter does not require Müller to have been ill-intentioned; the structural fact is that the apparatus operated as designed, exporting the framework's account into the broader scholarly engagement with the *R̥gveda* across the subsequent century.

Standard references: Max Müller, *Rig-Veda-Saṃhitā* (Oxford, 1849–1874, six volumes); the *Sacred Books of the East* series (Oxford, 1879–1910, fifty volumes). Müller's autobiographical and methodological works: *My Autobiography: A Fragment* (Longmans, Green and Co., 1901, posthumous); *India: What Can It Teach Us?* (Longmans, Green and Co., 1883); *Lectures on the Science of Language* (Scribner, 1861–1864, two volumes); *Lectures on the Science of Religion* (Scribner, 1870). Scholarly treatments: Lourens van den Bosch, *Friedrich Max Müller: A Life Devoted to the Humanities* (Brill, 2002); Sharada Sugirtharajah, *Imagining Hinduism: A Postcolonial Perspective* (Routledge, 2003) on the colonial-philological framing; Edwin Bryant, *The Quest for the Origins of Vedic Culture* (Oxford University Press, 2001)

on the AIT framing.

savarkar-ratnagiri-mleccha

Short: Vinayak Damodar Savarkar (1883–1966), during his internment at Ratnagiri (1924–1937) under British colonial restriction, delivered public speeches at the Patit Pavan Mandir (foundation laid 10 March 1929; *prāṇa-pratiṣṭhā* 22 February 1931) opening with a verse from Samarthā Ramdas’s seventeenth-century *Dasbodh* (दासबोध) — pausing before the loaded *mleccha* (म्लेच्छ) word so the audience, knowing the verse, completed it for him; the speech-and-audience-completion device circumvented colonial surveillance while delivering the dharmic continuum’s standing category for the non-dharmic outsider.

Deployments: Chapter 16 §16.7 ¶ — the citation anchor for the Vinayak Damodar Savarkar speech-and-audience-completion incident in colonial Ratnagiri.

The historical incident the chapter establishes: Vinayak Damodar Savarkar (1883–1966), during his period of internment at Ratnagiri (1924–1937) under British colonial restriction, delivered a public speech in which he opened with a verse from the seventeenth-century saint-poet Samarthā Ramdas’s *Dasbodh* — the famous म्लेच्छ (*mleccha*) verse that the Maratha paramparā associates with the call to resistance against foreign / non-dharmic rule. The verse opens with the word *mleccha*; the audience knew the verse and knew its political force. Savarkar paused before the loaded word — the verse’s structure makes the next word predictable to anyone who knows the verse — and the audience completed it for him.

The colonial monitoring apparatus could not charge Savarkar for a word he had not himself uttered. The speech-and-audience-completion device circumvented the constraint while delivering

the political content the situation called for. The incident is preserved across the Maratha political continuum and the Savarkar biographical literature as a paradigmatic example of constrained-political-speech operating under colonial surveillance.

The dating: the speech-and-incident is anchored across the Ratnagiri-internment period (1924–1937 — Savarkar transferred from the Andaman Cellular Jail to Ratnagiri under colonial-political restrictions in 1924, with the restrictions lifted in 1937). The standard biographical references place it at one of the public meetings Savarkar addressed at the Patit Pavan Mandir (the temple at Ratnagiri that Savarkar established as a site of *asprśya*-inclusive worship and of public political address during his internment; foundation stone laid 10 March 1929 by Shankaracharya Dr. Kurtakoti, with construction funded by Bhagoji Sheth Keer; *prāṇa-pratiṣṭhā* consecration ceremony performed 22 February 1931). The precise date of the *mleccha* speech-and-audience-completion incident varies across the sources; the structural form is consistent.

The structural significance the chapter establishes: the term *mleccha* — derived from the *Sanskrit* root *√mlecch* (to speak indistinctly, to speak barbarously) — is the canonical *Sanskrit* term for the non-dharmic, non-*Sanskrit*-speaking outsider. The term appears across the *Vedic*, *Brāhmaṇa*, *Sūtra*, *Smṛti*, and *Purāṇa* literature as the standing designation for those outside the *Sanskrit*-anchored civilizational frame. Savarkar's deployment of the term in colonial Ratnagiri — and the audience's completion of the verse — placed the British colonial enterprise inside the dharmic continuum's standing category for the non-dharmic outsider. The political force is preserved across the term's continuity across the depth of dharmic textual history.

Standard references: the standard biographies of Savarkar — Dhananjay Keer, *Veer Savarkar* (Popular Prakashan, 1950, with subsequent revised editions); A. S. Bhide, *Savarkar's Journey: From the Andaman Cellular Jail to Hindu Mahasabha President* (in Marathi, multiple editions); Vikram Sampath, *Savarkar: Echoes from a*

Forgotten Past, 1883–1924 (Penguin Viking, 2019) and *Savarkar: A Contested Legacy, 1924–1966* (Penguin Viking, 2021). The incident appears across the Maratha paramparā’s oral and printed accounts of Savarkar’s Ratnagiri period; the precise date and verse-reference vary, with the structural form consistent.

samarth-ramdas-mleccha-verse

Short: Samarth Ramdas (1608–1681) — saint-poet of the Marathi *Bhakti* continuum and *guru* of Chhatrapati Shivaji Maharaj — composed the *Dasbodh* (दासबोध, traditionally dated to his 1654 contemplative retreat; twenty *dasakas* (दशक), ~7,750 *ovīs* (ओवी)), with a *mleccha* (म्लेच्छ)-naming verse in the political-conduct sections that the Ratnagiri audience completed in Savarkar’s pause. The dharmic continuum’s *mleccha* category runs continuously from the *Vedas* through the *Mahābhārata*, *Smṛti*, *Purāṇa* literature, the regional *bhakti* traditions including the *Dasbodh*, and into early-20th-century anticolonial discourse — distinct from and not dependent on the Abrahamic master-slave binary the AIT framework projected.

Deployments: Chapter 16 §16.7 ¶ — the citation anchor for the *Dasbodh* verse from Samarth Ramdas that the Ratnagiri audience completed in Savarkar’s pause.

Samartha Ramdas (1608–1681), saint-poet of the Marathi *Bhakti* continuum and the *guru* of Chhatrapati Shivaji Maharaj, composed the *Dasbodh* — a major work of seventeenth-century Marathi devotional and political-philosophical literature, traditionally dated to the 1654 period of his contemplative retreat. The *Dasbodh* runs to twenty *dasakas* (sections), each containing ten *samāsas* (sub-sections), each containing many *ovīs* (couplets) — approximately 7,750 *ovīs* in total. The work treats *bhakti* (devotion), *jñāna* (knowledge), *karma* (action), *rāja-niti* (political conduct), and the broader dharmic engage-

ment with the moment of seventeenth-century Maratha resistance against the Mughal political-religious framework.

The verse the Ratnagiri audience completed for Savarkar — the *mleccha*-naming verse — is from the *Dasbodh*'s political-conduct sections. The exact *ovi* reference varies across the available sources; the structural form of the verse names the *mleccha* (the non-dharmic outsider) and calls the dharmic community to its standing position of resistance and self-assertion. The verse is preserved across the Marathi devotional and political lineages and is widely known among the audience for any Ratnagiri-period Savarkar speech.

The structural significance the chapter establishes: the *Dasbodh* operates inside the dharmic continuum's standing analytical category for the non-dharmic outsider. Samarth Ramdas, writing in seventeenth-century Marathi from inside the Maratha resistance against Mughal rule, deploys the same Sanskrit-derived term (*mleccha*) the *Vedic* and *Smṛti* literature uses. The continuity of the category across the depth of the dharmic literary continuum — from the *Vedas* through the *Mahābhārata*, the *Smṛti*, the *Purāṇas*, the regional *bhakti* literatures including the *Dasbodh*, and onward into the early-twentieth-century anticolonial discourse Savarkar inherited — is the structural backbone of the chapter's argument that the dharmic continuum operates its own primary-source category-system, distinct from and not dependent on the Abrahamic master-slave binary the AIT framework projected onto Indic prehistory.

Standard references: the *Dasbodh* in its standard editions. The principal editions: the Govt. of Maharashtra Education Department's standard edition; the Samarth Vagdevata Mandir (Sajjangad) edition; the Shantilal Shah edition with annotations. English translations: Bhagat Vivek and others have produced partial translations; the standard scholarly engagement: Eleanor Zelliot, *From Untouchable to Dalit: Essays on the Ambedkar Movement* (Manohar, 1996) for the broader Maharashtra-bhakti continuum; Anne Feldhaus, *Connected Places: Region, Pilgrimage, and Geographical Imagination in*

India (Palgrave Macmillan, 2003) for the regional sacred geography Samarth Ramdas operates within. Biographical: Bhavarthadeepika by Maharashtra State Sahitya Sanskruti Mandal; the standard hagiographical and historical sources for Samarth Ramdas's life and the seventeenth-century Maratha *bhakti*-and-resistance context.

south-indian-mahaprana-loan-only

Short: The major south-Indian literary languages (Tamil, Kannada, Malayalam, Telugu, Tulu) do not natively use the *mahāprāṇa* (महाप्राण, aspirated) consonant series — the aspirated stops appear only in *tat-sama* (तत्सम, Sanskrit-loaned) vocabulary, *tadbhava* (तद्भव, Sanskrit-adapted) forms typically simplify aspiration, and foreign loanwords drop it; the south-Indian architects selected from the broader sub-continental phonetic superset and operated without the *mahāprāṇa* manner-axis distinction — engineering selection difference, not deficiency.

Deployments: Chapter 9 §9.3 ¶ — the citation anchor for the south Indian languages' restriction of *mahāprāṇa* (aspirated) consonants to Sanskrit-loaned vocabulary.

The major literary languages of the south Indian peninsula — Tamil, Kannada, Malayalam, Telugu, Tulu — do not natively use the *mahāprāṇa* (aspirated) consonant series. The aspirated stops (*kha*, *gha*, *cha*, *jha*, *ṭha*, *ḍha*, *tha*, *dha*, *pha*, *bha*) appear in south Indian texts and speech only in:

1. ***Tatsama* vocabulary** — Sanskrit-loaned words that retain their original Sanskrit phonological form when used in south Indian writing and speech. Examples: *dharma* (धर्म), *bhārata* (भारत), *megha* (मेघ), *guru* (गुरु, with the unaspirated *g* but contrast against the *gh*-bearing *jagannātha*); in formal-register south Indian writing, the *tatsama* preserves the aspiration.

2. ***Tadbhava* vocabulary** — words originally from Sanskrit that have been adapted to the local phonological inventory, often with the aspiration simplified to the unaspirated counterpart. Examples: Sanskrit *gandha* (गन्ध, with aspirated *dh*) → Tamil *kandam*, with the aspiration lost.
3. **Foreign-language loanwords** — Persian, Arabic, English, and other loanwords with aspirated phonemes are typically adapted to the south Indian inventory by dropping or simplifying the aspiration.

The standard south Indian scripts handle the *mahāprāṇa* series with dedicated glyphs: Tamil ka-varga, ca-varga, ṭa-varga, ta-varga, pa-varga rows in the script reserve the aspirated-stop column for the rare *tatsama* deployments; Kannada, Telugu, and Malayalam scripts include the full aspirated-stop column with regular use in *tatsama* vocabulary. Tamil's script does not include separate glyphs for the *mahāprāṇa* stops — the inventory simply does not productively distinguish them; the script writes only the unaspirated stops, with *tatsama* loans rendered using approximations or with Grantha-script supplementation in specialized contexts.

The structural significance the chapter establishes: the south Indian languages' architects selected from the broader subcontinental phonetic superset and made a different selection than Sanskrit's architects. The south Indian selection took the five-zone place-of-articulation axis, the retroflex consonant series, the voicing distinction, and the nasal series, but did not take the *mahāprāṇa* manner-axis distinction. The result is a south Indian phonological inventory that operates a robust place-of-articulation system without the four-fold voicing-and-aspiration array on each row that Sanskrit's *varga* matrix produces.

The selection-difference is engineering, not deficiency. The south Indian phonological inventory is fully adequate for the regional speech communities' communicative needs; the absence of the *mahāprāṇa*

dimension reflects the architects' choice to operate on a different parameter axis. The contemporary speech communities preserve the inventory across many generations; the regional literary continua (*Saṅgam* Tamil, classical Kannada, classical Malayalam, classical Telugu) operate within the regional inventory across long textual histories.

Standard references: Bhadriraju Krishnamurti, *The Dravidian Languages* (Cambridge University Press, 2003) — the standard reference on the south Indian linguistic family; Kamil Zvelebil, *The Smile of Murugan* (Brill, 1973); Murray B. Emeneau, *Dravidian Linguistics: Ethnology and Folktales* (Annamalai University, 1967, collected papers). For the loan-phonology treatment specifically: Bh. Krishnamurti, "Loan-Words in Telugu" and related papers in his collected *Dravidian Studies*; the standard *Tamil Lexicon* (Madras University, 1924–1936) on *tatsama* / *tadbhava* loan-word patterns.

bengali-va-ba-merger

Short: Standard Bengali (*kolkātā-bhāṣā*) collapses the labial-region distinction between *va* (ব) and *ba* (ব) into a single bilabial pronunciation — the two phonemes sit *adjacent* in the *varṇamālā*'s 2D place-by-manner grid (same labial column, adjacent *antaḥstha* / *sparsā* rows, the smallest snap-distance in the labial zone); natural-language drift across many generations hits the smallest acoustic-distinguishability gaps first, so the merger confirms rather than contests the structural account of the *varṇamālā* (বর্ণমালা) as engineered.

Deployments: Chapter 9 §9.3 ¶ — the citation anchor for the Bengali phonological collapse of ব (*va*) and ব (*ba*) and the structural-locality account of the merger.

Bengali phonology collapses the distinction between the labial-region semivowel ব (*va*) and the labial-region voiced stop ব (*ba*), with both

written-Bengali grapheme positions producing approximately the same bilabial pronunciation in standard Bengali speech. The merger is one of the distinctive phonological features of standard Bengali (the *kolkātā-bhāṣā* — the Calcutta literary standard — and the broader West Bengal / Bangladesh speech-community phonology).

The phonological-historical mechanism: in the *varṇamālā*'s engineered architecture, ব (va) sits in the *antaḥstha* (semivowel) row at the labial-region position (specifically, the labio-dental contact-station in the strict account, or the bilabial position in regional articulations), while বা (ba) sits in the *sparśa* (stop) row at the labial position. The two are *adjacent* in the *varṇamālā*'s 2D place-by-manner grid — same place column, adjacent manner rows. The acoustic-distinguishability gap between them — the snap-to-grid spacing the architects engineered for — is smaller than the gap between either of them and any non-labial-position phoneme.

The structural account the chapter develops: natural-language drift across many generations of speech-field operation hits the smallest distinguishability gaps in the engineered matrix first. The Bengali ব-বা merger happens at exactly the position the *varṇamālā*'s 2D structure makes the most-drift-vulnerable position in the labial zone — the *closest neighbor* of ব is বা in the labial column, and the engineering snap-distance between them is smaller than the snap-distance between either of them and the next-nearest phonemes. The merger is therefore evidence of natural-language drift operating across the engineering perimeter; it confirms, rather than contests, the structural account the chapter develops of the *varṇamālā* as engineered.

The merger does not happen in all regional Bengali speech-communities equally — the eastern-Bengal (Bangladesh) speech registers preserve a slightly more distinct ব-বা articulation than the Calcutta literary standard. The variation across the Bengali speech-field is consistent with the broader observation that natural-language drift varies across regional speech-communities depending on the local engineering-perimeter strength.

Standard references: Suniti Kumar Chatterji, *The Origin and Development of the Bengali Language* (Allen & Unwin, 1926; reprinted Rupa, 1985); Pabitra Sarkar, *Bangla Dvani Bichar* (Calcutta, 1985, in Bengali) — the standard reference on Bengali phonology; Anvita Abbi, *A Manual of Linguistic Field Work and Structures of Indian Languages* (Lincom Europa, 2001) for the broader subcontinental phonological-survey methodology; James W. Gair and W. S. Karunatilaka, *Literary Sinhala Inflected Forms* (Cornell University, 1968) for parallel observations on regional phonological adaptations across Indic languages.

sindhi-implosives-inventory

Short: Sindhi — the language of Sindh, the Sindhu river region — operates a productive set of *implosive* consonants (ᵐ ɸ/ᵐ bilabial, ᵈ ɟ/ᵈ alveolar, ᵑ ɕ/ᵑ palatal, ᵑ ʁ/ᵑ velar) produced with the glottis closed and lowered to create inward-airflow on release; the *varṇamālā* does not include implosives — a *deliberate* engineering exclusion (the implosive set sits in a narrow articulatory region that does not extend to additional combinatorial dimensions; the architects selected dimensions supporting multi-axis combinatorial expansion).

Deployments: Chapter 9 §9.3 ¶ — the citation anchor for Sindhi’s implosive consonant inventory.

Sindhi — the language of Sindh, the Sindhu river region — operates a phoneme inventory that includes a set of *implosive* consonants not present in most other subcontinental languages. The Sindhi implosive inventory:

- ᵐ — voiced bilabial implosive (the *b*’ — the bilabial implosive). Written in the Sindhi (Perso-Arabic) script as ڀ; in the Sindhi Devanāgarī script as ॡ.
- ᵈ — voiced alveolar implosive (the *d*’). Written as ɟ in Perso-Arabic script; as ॢ in Devanāgarī.

- ɟ — voiced palatal implosive (the *j'*). Written as ڄ in Perso-Arabic script; as ज्ञ in Devanāgarī.
- ɠ — voiced velar implosive (the *g'*). Written as ڳ in Perso-Arabic script; as ण in Devanāgarī.

The articulatory mechanism: an implosive consonant is produced with the glottis closed and lowered during the closure phase of the stop, creating a reduction in air pressure inside the oral cavity (rather than the increase produced by the egressive airflow of an ordinary voiced stop). When the oral closure is released, air rushes inward into the oral cavity, producing the characteristic implosive acoustic signature.

The implosives are phonologically contrastive in Sindhi with the corresponding ordinary voiced stops: *ḍa* contrasts with *ba*; *ḍa* contrasts with *da*; etc. The contrast is fully productive across the Sindhi lexicon and is preserved across the contemporary speech communities (in Sindh, in the Sindhi-speaking diaspora communities of Maharashtra, Gujarat, and elsewhere).

The structural significance the chapter establishes: the implosive set is *observable* in the subcontinental sound-field — Sindhi preserves it as a productive phonological feature today, and the regional speech-field of which the Sindhu river basin is part (the river the *R̥gveda* names among its rivers) supplies the architects' empirical access to the feature. The *varṇamālā* does not include implosives. The omission is *deliberate*. Section 10.6 develops the analysis: the architects observed the implosive set and made an engineering decision to exclude it from the *varṇamālā*'s phoneme inventory, on the grounds that the implosive set sits in a narrow articulatory region (the glottis-lowering mechanism) that does not extend to additional phonological dimensions in the broader system. The implosives are a single feature, not an axis that supports further combinatorial elaboration; the *varṇamālā*'s engineering selected dimensions that support multi-axis combinatorial expansion (the five places $\times$ four manner-and-voicing combinations on each *varga*, the orthogonal nasal and aspirated di-

mensions) rather than narrow single-feature inclusions.

Standard references: A. R. Yegorova, *The Sindhi Language* (Nauka, Moscow, 1971, in Russian; translated as part of the *Languages of Asia and Africa* series); Y. M. Khubchandani, *Sindhi: A Linguistic Description* (Indian Institute of Sindhi Linguistics, 1969); George A. Grierson, *Linguistic Survey of India* (Calcutta, 1903–1928, twelve volumes) — Volume VIII Part 1 on Sindhi; Peter Ladefoged and Ian Maddieson, *The Sounds of the World's Languages* (Blackwell, 1996), the section on implosive consonants and their cross-linguistic distribution.

tamil-alveolar-trill

Short: Tamil includes a distinctive alveolar trill ṛ (ra) and an alveolar nasal ṇ (na) at the alveolar ridge — a third place of articulation between the dental and retroflex stations the *varṇamālā* operates with; Tamil's phonological architecture runs six places (labial, dental, alveolar, retroflex, palatal, velar) where the *varṇamālā* runs five. The architects of each system made different engineering selections from the same subcontinental superset — Sanskrit selected the cleaner 5×5 snap-to-grid spacing, Tamil retained the alveolar distinction at slightly closer front-of-mouth spacing.

Deployments: Chapter 9 §9.4 ¶ — the citation anchor for Tamil's distinctive alveolar trill phoneme ṛ (r).

Tamil's phoneme inventory includes a distinctive alveolar trill — written as ṛ (r in standard IAST-style transliteration, or RR in some Tamil-romanization conventions). The alveolar trill is articulated at a contact-station between the dental and the retroflex positions — the tongue-tip strikes the alveolar ridge (the boundary between the back of the upper teeth and the front of the hard palate), distinct from both the dental contact-station of ṇ (na) / ṭ (ta) and the retroflex contact-station of ṇṇ (ṇa) / ṭṭ (ṭa).

The contrast in Tamil between the three place-of-articulation regions is:

- **Dental** — ந (*na*), த (*ta*) — at the upper teeth.
- **Alveolar** — ன (*na*, the alveolar nasal), ற (*ra*, the alveolar trill) — at the alveolar ridge.
- **Retroflex** — ண (*ṇa*), ள (*ṭa*) — at the rear of the hard palate, with the tongue tip curled back.

The alveolar set is a *third* place of articulation in the dental-region zone, between the canonical dental and retroflex stations of the *varṇamālā*. Tamil's phonological architecture therefore operates six places of articulation (labial, dental, alveolar, retroflex, palatal, velar) where the *varṇamālā* operates five (labial, dental, retroflex, palatal, velar).

The structural significance the chapter establishes: the alveolar set is *observable* in the subcontinental sound-field — Tamil preserves it as a productive phonological feature. The architects of the *varṇamālā* could have included an alveolar row between the dental and the retroflex; they did not. The omission is consistent with the *snap-to-grid* principle the chapter develops. The alveolar contact-station sits structurally between the dental (~3 cm) and the retroflex (~7 cm) — at approximately 4–5 cm from the lip-line. Including an alveolar row would have placed three places of articulation within a ~4 cm front-of-mouth region, compressing the acoustic-distinguishability spacing the *varṇamālā*'s 5×5 architecture protects. The architects' selection holds the spacing at ~2 cm minimum between adjacent places, sacrificing the alveolar distinction in favor of the cleaner *snap-to-grid* spacing.

Tamil's architects, operating on a different engineering balance, chose to retain the alveolar distinction at the cost of slightly closer spacing in the front-of-mouth region. The two engineering decisions reflect two different selections from the same subcontinental superset.

Standard references: Kamil Zvelebil, *The Smile of Murugan* (Brill, 1973); Bhadriraju Krishnamurti, *The Dravidian Languages* (Cambridge University Press, 2003); Sanford B. Steever, ed., *The Dravidian Languages* (Routledge, 1998), the chapters on Tamil phonology; the standard *Tamil Lexicon* (Madras University, 1924–1936); A. Govindankutty, *A Comparative Study of Tulu Dialects* (Dravidian Linguistics Association, 1972). For the alveolar consonant family’s cross-linguistic distribution: Peter Ladefoged and Ian Maddieson, *The Sounds of the World’s Languages* (Blackwell, 1996), section on alveolar and post-alveolar articulations.

ho-mundari-checked-consonants

Short: The languages of the central-eastern forest belt — Ho, Mundari, Santali, Korku, Sora — operate phonemic glottal stops as *checked consonants* (word-final or syllable-final glottal closures distinguishing minimal pairs: Ho *da?* “water” vs. *da* without); the *varṇamālā* does not include a glottal-stop phoneme — *deliberate* engineering exclusion (the glottal stop would have placed two places of articulation in the throat region, compressing acoustic-distinguishability spacing; the architects held the throat-region to a single *kaṇṭhya* place).

Deployments: Chapter 9 §9.4 ¶ — the citation anchor for the glottal-stop / checked-consonant phonological feature of Ho, Mundari, and the related languages of the central-eastern forest belt.

The languages of the central-eastern forest belt — Ho, Mundari, Santali, Korku, Sora, and the related languages spoken across the Chotanagpur plateau, the surrounding regions of Jharkhand, Odisha, Bihar, West Bengal, and the central-eastern forest zones — operate phonemic glottal stops as distinctive phonological features. The glottal-stop phenomenon in these languages is documented in

the linguistic descriptions as *checked consonants* — word-final or syllable-final glottal closures that distinguish minimal pairs.

The mechanism: at the end of certain syllables, the glottis closes abruptly, producing a sharp termination of the vowel rather than a smooth fade or a consonantal closure at a non-glottal place. The closure functions phonologically as a distinct phoneme — minimal pairs are distinguished by the presence or absence of the glottal closure at the syllable's end.

Examples from Ho (the language of the Ho community of Jharkhand and Odisha):

- *da?* (water; with checked-glottal closure) vs. *da* (without).
- *sete?* (hot; with checked-glottal closure) vs. *sete* (without).

The corresponding Mundari, Santali, and Korku speech-communities preserve parallel checked-consonant phonology with regional variations in the glottal-closure articulation and its distribution across the lexicon.

The structural significance the chapter establishes: the glottal stop is *observable* in the subcontinental sound-field — the languages of the central-eastern forest belt preserve it as a productive phonological feature. The *varṇamālā* does not include a glottal-stop phoneme. The omission is *deliberate*. The glottal stop sits at the *kaṇṭhya* (throat / glottis) region, structurally adjacent to but distinct from the velar position the *varṇamālā*'s *ka-varga* occupies. Including a glottal-stop row would have placed two places of articulation in the throat region, again compressing the acoustic-distinguishability spacing. The architects' selection holds the throat-region to a single place (the velar / *kaṇṭhya*), excluding the glottal-stop variant.

The forest-belt architects, operating on a different engineering balance, retained the glottal-stop distinction in their phonological inventory. Their languages are not less sophisticated than Sanskrit; they made a different engineering selection from the same subcontinental

superset.

Standard references: G. A. Grierson, *Linguistic Survey of India*, Volume IV (Munda and Dravidian Languages) (Calcutta, 1906); Norman H. Zide, ed., *Studies in Comparative Austroasiatic Linguistics* (Mouton, 1966); Norman Zide, “Korku” and other entries in the *International Encyclopedia of Linguistics* (Oxford University Press, 1992; second edition 2003); Doris Lessing Anderson, *A Comparative Study of Munda Languages* (in *Languages of Asia*); Arun Ghosh, *Santali Language: A Grammatical Description* (Indian Institute of Advanced Study, 1994). For Ho specifically: Ram Dayal Munda, *Ho Vyākaraṇa* (Ranchi University, 1968, in Ho); David Stampe and others on Munda phonology in the *Oceanic Linguistics* and similar journals. *Munda* here is the people-name and the name of the specific language *Mundari*; the chapter’s broader treatment does not use *Munda* as a taxonomic-family label (per the canonical naming convention).

urdu-persian-arabic-loan-phonemes

Short: Urdu’s Persian-and-Arabic-loaned register adds phonemes the *varṇamālā* did not include — labio-dental fricative *f* (ف / ف), alveolar voiced fricative *z* (ز / ز), uvular voiceless stop *q* (ق / ق), uvular fricatives *kh* (خ / خ) and *gh* (غ / غ) — handled via dotted-Devanāgarī supplementation rather than expansion of the engineered architecture itself; the architects could not have observed contact-introduced phonemes available only after Persian and Arabic contact with the western frontier.

Deployments: Chapter 9 §9.4 ¶ — the citation anchor for the labio-dental fricative (and related loan-phoneme) inventory in Urdu and Persian-and-Arabic-influenced registers.

Urdu — the Persian-and-Arabic-loaned register of Hindustani that operates with the Perso-Arabic script (*nasta’līq*) and that carries sub-

stantial Persian and Arabic vocabulary — includes phonemes that the broader Hindi register does not productively use and that the *varṇamālā* does not include in its base inventory. The principal loan-phonemes:

- **f (labio-dental voiceless fricative)** — written as ف (*fā*) in Urdu. Used in Persian and Arabic loanwords (*faraz*, *firqa*, *fauj*, *falsafa*) and in some words of European origin. The corresponding Hindi register typically renders f as *ph* (the *pavarga* aspirated stop), with the labio-dental fricative articulation lost in standard Hindi speech.
- **z (alveolar voiced fricative)** — written as ز (*ze*) in Urdu. Used in Persian and Arabic loanwords (*zamin*, *zindagi*, *zara*, *zakāt*). Hindi register renders z as *j* (the *cavarga* voiced stop), with the fricative articulation lost.
- **q (uvular voiceless stop)** — written as ق (*qāf*) in Urdu. Used in Arabic loanwords (*qissa*, *qadr*, *qabil*, *qurān*). Hindi register renders q as *k* (the *kavarga* unaspirated voiceless stop), with the uvular articulation lost.
- **kh (uvular voiceless fricative)** — written as خ (*khe*) in Urdu. Used in Persian and Arabic loanwords (*khavar*, *khwāhish*). Hindi register often renders this as the *kavarga* aspirated *kh* (the same grapheme position but with velar rather than uvular articulation).
- **gh (uvular voiced fricative)** — written as غ (*ghain*) in Urdu. Used in Persian and Arabic loanwords (*ghazal*, *ghair*). Hindi register often renders this as the *kavarga* voiced *g* with aspiration loss.

The dotted-Devanāgarī conventions for the loan-phonemes — फ़ (*fa*), ज़ (*za*), क़ (*qa*), ख़ (*kha*), ग़ (*gha*), ङ़ (*ṛa*), ढ़ (*ṛha*) — were introduced to allow Devanāgarī to represent the loan-phonology of Urdu and the Persianate register. The dotted forms are not part

of the core *varṇamālā* and were added long after the architecting period. (See endnote brahmi-devanagari-structural-identity for the broader treatment of post-architecting script additions.)

The structural significance the chapter establishes: the labio-dental fricative *f*, the alveolar fricative *z*, and the uvular stop *q* are *observable* in the broader subcontinental sound-field once Persian and Arabic loan-vocabulary entered the regional speech communities (across many generations of contact between the Persian-and-Arabic-cultural zones and the subcontinent's western frontier). The architects of the *varṇamālā* did not have access to this contact and could not have observed the loan-phonemes. The *varṇamālā* therefore does not include these phonemes in its base inventory. The dotted-Devanāgarī supplementation handled the later need to represent the loan-phonology without expanding the *varṇamālā* itself.

The loan-phoneme set is consistent with the broader structural account: the *varṇamālā* selected from the pre-contact subcontinental sound-field; contact-introduced phonemes (Persian, Arabic, English loans) were handled with script-level additions rather than with extensions of the engineered phonological architecture itself.

Standard references: Christopher Shackle, *Urdu Grammar* (in *The Urdu Concise Dictionary*, Oxford University Press); Bhojeshwar Pandey, *Hindi Phonology* (Kalinga, 1969); Y. Kachru, *Hindi* (John Benjamins, 2006); Anvita Abbi, *A Manual of Linguistic Field Work and Structures of Indian Languages* (Lincom Europa, 2001). For the script-level loan-phoneme handling: the *Unicode Consortium* documentation on Devanāgarī dotted characters; the *Hindi Shabdasaagar* (the standard Hindi dictionary) treatment of *tatsama*, *tadbhava*, and loan-vocabulary categories.

punjabi-tonal-development

Short: Standard Punjabi operates a three-way lexical-tone system (high / mid / low — *kórā* “leper” / *kōṛā* “whip” / *kòṛā* “horse”) developed through tonogenesis: the engineered voiced-aspirated stops (*gh*, *jh*, *ḍh*, *dh*, *bh*) merged with their unaspirated counterparts in regional speech-fields, with the aspiration leaking into vowel pitch contours. Post-engineering regional development — what happens when the Sanskrit calibration perimeter does not extend to a speech-field; the architects’ decision to reserve pitch for *recitation* engineering (*udātta* / *anudātta* / *svarita*) rather than allow it as phoneme-distinction kept the Sanskrit phonological inventory stable.

Deployments: Chapter 9 §9.5 ¶ — the citation anchor for the Punjabi three-way lexical-tone development from the engineered voiced-aspirated row of the *varṇamālā*.

Standard Punjabi — particularly the eastern Punjabi register (the language of the contemporary Indian Punjab and the broader Sikh-cultural sphere) — operates a three-way lexical-tone system that distinguishes minimal pairs by pitch contour. The three tones:

1. **High tone** — pitch rises through the syllable.
2. **Mid tone** — neutral pitch across the syllable.
3. **Low tone** — pitch falls through the syllable.

The minimal-pair distinctions: words that would be otherwise homophonous are distinguished by their tonal contour. Examples (the standard Punjabi phonological literature):

- *kòṛā* (with low tone) — *horse*.
- *kōṛā* (with mid tone) — *whip*.
- *kórā* (with high tone) — *leper*.

The phonological-historical mechanism: the engineered *varṇamālā* operates four voiced-aspirated stops — ञ *jh* (no, *gh*), झ (*jh*), ढ (*ḍh*), ध (*dh*), भ (*bh*) — across the five *vargas*. In the historical regional

speech-fields of Punjab, across many generations, the voiced-aspirated stops merged with their unaspirated voiced counterparts in specific phonological environments. When the consonantal contrast collapsed (the *gh* / *g* merger; the *jh* / *j* merger; etc.), the *aspiration* leaked into the surrounding vowels as a *pitch contour*: the aspirated form's high-frequency aspiration produced a higher pitch on the adjacent vowel; the absence of aspiration produced a lower pitch. The consonantal distinction became a tonal distinction.

The mechanism is documented in the comparative-phonological literature on Punjabi. The standard reference is: Ranjit Singh Bhatia, *Punjabi: A Linguistic Description* (Indian Institute of Punjabi Linguistics, 1973); Ujjal Singh Bahri, *Studies in Punjabi Phonology* (Indian Institute of Languages, 1976); Christopher Shackle, *Punjabi* (Hodder and Stoughton, 1972). The mechanism is also discussed in the broader literature on tonogenesis from segmental contrasts (the cross-linguistic phonological-typological literature on how lexical tone develops from segmental contrasts).

The structural significance the chapter establishes: the Punjabi tone is *post-engineering regional development* — the product of natural-language drift inside speech-fields the calibration perimeter (the Sanskrit calibrant-anchored *bhāṣāyām* mode) did not reach. The Sanskrit calibration preserved the voiced-aspirate row intact across all the generations of its operation; the regional speech-fields outside the calibrant perimeter did not. The Punjabi tone is therefore evidence of exactly what the engineering was designed to prevent in Sanskrit — and exactly what happens, on the timescales of natural-language drift, when the engineering perimeter does not extend to a regional speech-field.

The tonal development confirms the architects' decision to *reserve pitch for the recitation engineering* rather than allowing it to operate as a phoneme-distinction dimension. Where pitch is allowed to operate as phoneme-distinction (as in Punjabi after the voiced-aspirate merger), it accelerates phonological drift across generations. Where

pitch is reserved (as in the Sanskrit calibrant's *bhāṣāyām* mode), the phonological inventory remains stable.

Standard references as enumerated above. Additional: ManjitINDER Singh Gill, *Punjabi Tonemics* (Patiala, 1975); Brij Mohan Khanna, *Studies in Comparative Indo-Aryan and Pakistani Languages* (Indian Linguistic Society publications). For tonogenesis cross-linguistically: Larry Hyman, *Universals of Tone Rules: Evidence from West Africa* (1973); James Matisoff, *The Loloish Tonal Split Revisited* (Center for South and Southeast Asia Studies, Berkeley, 1973); and the broader tonogenesis literature in the *Journal of the International Phonetic Association* and *Phonology* journals.

pahari-tonal-features

Short: The western Pahari languages — Garhwali, Kumaoni, Dogri (across the central Himalayan slopes) — operate *two-way* lexical-tone features on a smaller lexical subset (Garhwali *gōṛā* “horse” with high pitch reflecting historical *gh-*, *goṛā* “foot” with neutral pitch); the same tonogenesis mechanism as Punjabi but at smaller scale, with calibrant influence in the Pahari region slowing but not preventing the development — an intermediate stage on the trajectory the full Punjabi case traverses.

Deployments: Chapter 9 §9.5 ¶ — the citation anchor for the more limited tonal features observable in some western Pahari languages (Garhwali, Kumaoni, Dogri) across the central Himalayan slopes.

The western Pahari languages — Garhwali (spoken across Uttarakhand's Garhwal region), Kumaoni (across the Kumaon region), Dogri (across Jammu and the broader Pahari-speaking zones at the Himalayan slopes' western edge) — show tonal features in their contemporary phonology that are *more limited* in scope than the full Punjabi three-way lexical tone but that arise from the same underlying mech-

anism: voiced-aspirated stops merging with their unaspirated voiced counterparts, with the aspiration leaking into vowel pitch contours.

The Pahari tonal features are typically *two-way* rather than three-way: a high-low tonal distinction operating on specific lexical items where the voiced-aspirated stop has merged with the voiced-unaspirated stop. The mechanism parallels the Punjabi tonogenesis but operates on a smaller lexical subset and with less pervasive phonological grammaticalization.

Examples from Garhwali (the standard phonological literature):

- *ghorā* / *górā* — *horse* (with a high pitch contour reflecting the historical voiced-aspirated *gh-* form).
- *gorā* — *foot* (with neutral pitch).

The tonal contrast is preserved across some Pahari speech-communities and is being lost across others under the influence of standard Hindi (which has not developed lexical tone and which serves as the regional prestige register). The variation across the Pahari speech-field is consistent with the broader observation that contact-induced influence can erode regional phonological developments over generations.

The structural significance the chapter establishes: the Pahari tonal features are the same phenomenon as the Punjabi tonogenesis at a smaller scale. Both are post-engineering regional developments produced by natural-language drift on the voiced-aspirated row of the *varṇamālā*. Both confirm the architects' decision to reserve pitch for the recitation engineering rather than allowing it to operate as a phoneme-distinction dimension. The Pahari case is structurally interesting because it shows the *partial* development of lexical tone — an intermediate stage on the same trajectory the full Punjabi case traverses. The Pahari speech-fields sit closer to the calibrant-anchored *bhāṣāyām* perimeter than the Punjabi speech-fields do (Punjab being further from the Sanskrit-anchored cultural centers of the Gangetic plains than the Garhwal/Kumaon Himalayan slopes); the calibrant in-

fluence in the Pahari region has slowed but not prevented the tonal development.

Standard references: M. R. Sharma, *Garhwali Phonology* (in *Journal of Linguistic Studies*); Anoop Chandola, *A Synopsis of Garhwali Grammar* (Stanford University Press, 1966); Brij Mohan Sharma, *Kumauni Bhasa* (in Hindi, Pant Press, Almora); Veena Kumari, *Dogri* (in the relevant *Pahari Linguistic Studies* volumes); Anvita Abbi, *Languages of India and Linguistic Areas* (Bahri Publications, 1996) for the broader Pahari phonological survey.

retroflex-global-distribution

Short: The retroflex consonant series — tongue tip curled back to contact the rear hard palate — is *near-universal within the subcontinental sound-field* (across every named language from Tamil through Sindhi through Santali) and *rare-to-absent outside it* (Iranian plateau, European zone, Levantine Semitic, East Asian, Central Asian, African, and Native American sound-fields all lack a productive retroflex series; some Australian indigenous languages and Mandarin's post-alveolar approximation are the principal non-subcontinental cases). The subcontinental concentration is itself the evidence the retroflex series belongs to the regional sound-field's deep architecture, not a peripheral substrate-acquired or contact-induced layer.

Deployments: Chapter 9 §9.7 ¶ — the citation anchor for the global-cross-linguistic distribution analysis of the retroflex consonant series.

The retroflex consonant series — produced with the tongue tip curled back to contact the rear of the hard palate — is *near-universal within the subcontinental sound-field* and *rare-to-absent outside it*. The cross-linguistic distribution:

Inside the subcontinent: The retroflex series is present in essen-

tially every named language across south India (Tamil, Kannada, Malayalam, Telugu, Tulu), the central forest belt (Gondi, Kui, Kuvi, Kolami, Kurukh), the western and central Indic languages (Hindi, Marathi, Gujarati, Rajasthani, Bhojpuri, Awadhi, Maithili, Magahi, Bundeli, Haryanvi), the eastern subcontinent (Bengali, Odia, Assamese, Maithili), the northern and Himalayan-frontier languages (Nepali, Kashmiri, Dogri, Garhwali, Kumaoni, Sindhi, Punjabi, Sinhala, Dhivehi), and the central-eastern forest belt (Santali, Mundari, Ho, Korku, Sora — with some variation in the retroflex inventory). The series is operating productively in each named language's lexicon.

Outside the subcontinent: The retroflex series is either absent or present only sparsely as a marginal feature.

- **Iranian plateau** (Old Persian, Avestan, Sogdian, modern Persian, Pashto): no productive native retroflex series. (Pashto carries some retroflex consonants influenced by contact with Indic languages along the western frontier of the subcontinent.)
- **European zone** (Greek, Latin, Germanic, Slavic, Celtic): no retroflex series. (Some modern Swedish, Norwegian, and Sicilian dialects carry occasional retroflex articulations that are regional features rather than systematic phonological dimensions.)
- **Levantine and North African Semitic sound-field** (Arabic, Aramaic, Hebrew, Akkadian, Amharic, Tigrinya): no retroflex series.
- **East Asian sound-field** (Mandarin, Cantonese, other Sinitic varieties, Japanese, Korean, Vietnamese): no retroflex series. (Mandarin's "retroflex" series — the sh, ch, zh, r-initials — is articulated at the post-alveolar position rather than the true retroflex curled-back-tongue position the subcontinental retroflex carries.)
- **Central Asian pastoral region** (Turkic, Mongolic, Tungusic): no retroflex series.

- **Australian indigenous languages:** some Australian languages carry retroflex stops as a productive feature (the Pama-Nyungan family). This is the principal non-subcontinental zone of retroflex prevalence.
- **African continent** (Niger-Congo, Afro-Asiatic, Nilo-Saharan, Khoe-San): no general retroflex series.
- **Native American language families:** no general retroflex series.

The global-cross-linguistic distribution confirms that the retroflex series is *not* a universally-available phonological category that languages happen to use or not use. It is a specific articulatory configuration that is structurally central in the subcontinental sound-field and largely absent elsewhere. The subcontinental concentration is itself the evidence that the retroflex series belongs to the regional sound-field's deep architecture rather than being a feature that diffuses or migrates across regions easily.

The structural significance the chapter establishes: the retroflex series carries the chapter's strongest single-feature claim about the engineering. Where PIE reconstructions famously do not posit a retroflex set in the precursor (see endnote retroflex-substrate-standard-account), and where the standard account attributes the *mūrdhanya* development to subcontinental contact, substrate influence, or independent development — the structural fact is that the retroflex series is *structurally central* to the Sanskrit *varṇamālā*, not peripheral, and that its near-universal presence across the subcontinent's named languages confirms it as a deep architectural feature of the regional sound-field that the *varṇamālā*'s architects engineered into the core of the system.

Standard references: Peter Ladefoged and Ian Maddieson, *The Sounds of the World's Languages* (Blackwell, 1996), the chapter on retroflex consonants and their cross-linguistic distribution; the *World Atlas of Language Structures* (WALS, online at wals.info), the feature page on Uvular Consonants and adjacent retroflex categorizations;

David Crystal, *The Cambridge Encyclopedia of Language* (Cambridge University Press, 3rd edition 2010), the phonological-typological survey; Anvita Abbi, *A Manual of Linguistic Field Work and Structures of Indian Languages* (Lincom Europa, 2001) for the comparative-Indic survey.

kailasa-temple-ellora-engineering

Short: The *Kailāsa Temple* (कैलास मन्दिर) at Ellora (Cave 16, Aurangabad district, Maharashtra) — one of the largest monolithic rock-cut structures in the world, carved top-down from a single basalt cliff-face (~200,000 tons of stone removed to expose a complete temple with *śikhara*, colonnades, surrounding shrines, and continuous figural friezes) — was designed by anonymous architects whose names the historical record does not preserve; the architecture is on the ground, testable, structurally rigorous. The structural analog to the *varṇamālā*: engineering does not require named engineers — the architecture itself is the evidence.

Deployments: Chapter 9 §9.8 ¶ — the citation anchor for the Kailasa temple at Ellora as the architectural analog to the *varṇamālā*'s engineering-with-anonymous-architects situation.

The *Kailāsa Temple* (कैलास मन्दिर) at Ellora (Cave 16 of the Ellora cave-temple complex in the Aurangabad district of Maharashtra) is one of the largest monolithic rock-cut structures in the world. The temple is carved from a single basalt cliff-face, top-down, removing approximately 200,000 tons of stone to produce the temple's exposed structure. The construction is conventionally dated by the establishment historiography to the Rashtrakuta dynasty under King Krishna I (the book operates in *strategic refusal* on Indic chronology and does not affirm the conventional dating).

The temple's structural features:

- **Single-block carving** — the entire temple, including its tall *śikhara* (spire), surrounding cave-shrines, free-standing colonnades, intricately sculpted figural panels, and surrounding courtyard, was carved from a single contiguous block of basalt. There are no joints; there is no assembly of separately-quarried stones. The structure is monolithic.
- **Top-down carving sequence** — the carving proceeded from the top of the cliff face downward, with the temple's roof carved first and progressively excavated downward to expose the structure. The technique required the architects to anticipate, with full structural precision, the temple's internal geometry before any of it had been exposed.
- **Structural-load modeling** — the top-down carving produced a building rather than a pile of rubble because the architects had modeled the structural-load distribution accurately. Each column, each wall, each ceiling-section bears load that the architects had calculated before the carving operation began.
- **Multi-storey shrines** — the temple includes multiple levels of interior shrines, surrounding cave-spaces, and connecting passages that interlink the monolithic structure into a functional ritual complex.
- **Iconographic program** — the temple's sculptural panels depict scenes from the *Rāmāyaṇa*, the *Mahābhārata*, and broader Hindu mythological narratives in continuous figural-narrative friezes that wrap the temple's exterior surfaces.

The architects who designed the Kailāsa temple are anonymous to the historical record. No signed plans, no recorded biographies, no contemporary chronicle attributing the design to specific named individuals. The conventional historiography attributes the construction to royal patronage under Krishna I of the Rashtrakuta dynasty, but the *designers* — the figures who specified the rock-cutting sequence, the load distribution, the proportions, the iconographic program —

are not named.

The structural significance the chapter establishes: the Kailāsa temple is the architectural analog to the *varṇamālā*'s situation. The temple is *the architecture is on the ground*. Any engineer can walk the site, verify the rock-removal sequence, check the structural-load modeling, count the columns, measure the proportions. The architecture is testable. The architecture's designers are anonymous. The architecture does not require the designers to be named for the architecture itself to be evidence of engineering. The *varṇamālā* has the same epistemic shape: the engineered phonetic system is present today, audible in every Sanskrit recitation, with its design constraints recoverable from its structure. The architects are anonymous. The architecture does not require named architects for the architecture itself to be evidence of engineering.

Standard references: M. K. Dhavalikar, *Cultural Imperialism: Indian Influence on Asia* (Books and Books, 1973), on the broader rock-cut temple lineage; Lisa N. Owen, *Carving Devotion in the Jain Caves at Ellora* (Brill, 2012); the Archaeological Survey of India's standard documentation on the Ellora cave-temple complex; Mary B. Cuddy, *Kailasanatha: The Iconology and Architecture of Cave 16 at Ellora* (Routledge, 2017); George Michell, *The Hindu Temple: An Introduction to Its Meaning and Forms* (University of Chicago Press, 1988) for the broader Hindu temple architecture framework; Stella Kramrisch, *The Hindu Temple* (University of Calcutta, 1946) — the classical scholarly engagement. The temple is also extensively documented in the contemporary archaeological-conservation literature and is a UNESCO World Heritage Site (inscribed 1983, as part of the Ellora Caves complex).

smṛti-as-mnemoniture

Short: The *smṛti* (स्मृति, *that which is remembered*) category preserves narrative-and-conceptual content through memory and retelling across generations — the two *Itihāsas* (*Rāmāyaṇa*, *Mahābhārata*), the eighteen *Mahāpurāṇas*, the *Dharmaśāstras* (*Manusmṛti*, *Yājñavalkya-smṛti*, etc.), the classical *kāvya* literature, and the regional *bhakti* retellings (*Tulsīdās Rāmcaritmānas*, *Eknāthi Bhāgavata*, *Kambar Rāmāyaṇam*) — with concept-level rather than phoneme-level fidelity; the plurality of retellings does not compromise the preservation but constitutes it. The Indic counterpart of the book-coined *Mnemoniture*.

Deployments: Chapter 14 §14.2 ¶ — the citation anchor for the *smṛti* category as the Indic counterpart of the book-coined *Mnemoniture* preservation mode.

The *स्मृति* (*smṛti*) category in the Indic continuum names *that which is remembered* — content preserved across generations through memory and retelling, in contrast to *श्रुति* (*śruti*) *that which is heard* (the engineered phonetic preservation of the *Vedic* corpus). The standard *smṛti* corpus includes:

- **The two *Itihāsas*** — the *Rāmāyaṇa* and the *Mahābhārata*.
- **The *Purāṇas*** — eighteen *Mahāpurāṇas* (*Viṣṇu*, *Bhāgavata*, *Mārkaṇḍeya*, *Vāyu*, *Brahmā*, *Padma*, *Liṅga*, *Garuḍa*, *Nārada*, *Kūrma*, *Matsya*, *Skanda*, *Bhaviṣya*, *Brahmavaivarta*, *Vāmana*, *Varāha*, *Agni*, *Brahmāṇḍa*) and many *Upapurāṇas*.
- **The *Dharmaśāstras*** — *Manusmṛti*, *Yājñavalkya-smṛti*, *Nārada-smṛti*, *Parāśara-smṛti*, and the wider *smṛti* literature governing dharmic conduct.
- **The classical *kāvya* literature** — the works of Vālmiki, Vyāsa, Kālidāsa, Bhavabhūti, Bāṇabhaṭṭa, Daṇḍin, and the broader Sanskrit literary continuum.
- **The regional *bhakti* compositions and retellings** — the *Tulsīdās Rāmcaritmānas*, the *Eknāthi Bhāgavata*, the *Kambar*

Rāmāyaṇam, the *Kṛttibāsi Rāmāyaṇa*, and many other regional re-renderings of the *smṛti* content across the named languages.

The defining engineering feature of the *smṛti* category: *exact verbal preservation is not the engineering requirement; concept-and-narrative preservation is*. The Tulsidās *Rāmcaritmānas* (sixteenth-century Awadhi retelling of the *Rāmāyaṇa*) is fully *Rāmāyaṇa* in the *smṛti* sense, even though it is not a verbatim translation of Vālmīki's Sanskrit text. The Eknāthi *Bhāgavata* (sixteenth-century Marathi retelling) is fully *Bhāgavata* in the *smṛti* sense. The plurality of retellings does not compromise the preservation — it constitutes it. Multiple translations and reformulations across languages and art forms preserve the *concept* across the imprecisions of any single retelling.

The chapter's deployment: the *smṛti* category is the Indic counterpart of the book-coined *Mnemonic*. The structural feature — preservation via memory with concept-level rather than phoneme-level fidelity — is what *Mnemonic* names. The Indic continuum operates the *smṛti* preservation across many generations of operation in regional speech-communities, with the storytelling, narrative-painting, and devotional-musical lineages providing the carrier infrastructure.

Standard references: P. V. Kane, *History of Dharmaśāstra* (Bhandarkar Oriental Research Institute, five volumes, 1930–1962); Wendy Doniger, *The Hindus: An Alternative History* (Penguin, 2009), the chapters on *smṛti* literature; J. A. B. van Buitenen, *The Mahābhārata* (University of Chicago Press, three volumes, 1973–1978) for the *Mahābhārata* continuum; Robert Goldman et al., eds., *The Rāmāyaṇa of Vālmīki* (Princeton University Press, multi-volume translation, 1984–present); Ludo Rocher, *The Purāṇas* (Volume II, Fascicle 3 of Gonda's *A History of Indian Literature*, Otto Harrassowitz, 1986).

flexture-natyashastra-dance

Short: The book-coined *Flexture* preservation mode — content preserved through trained gestures, postures, and choreographed sequences — operates in the Indic continuum through *mudrā* (मुद्रा) and *hasta* (हस्त) gesture vocabulary (the *Abhinaya Darpaṇa* of Nandikeśvara enumerates 28 *asaṃyuta* + 23 *saṃyuta hastas*), Bharata’s *Nāṭyaśāstra* (नाट्यशास्त्र) codifying the gesture-and-posture specification, and the classical-dance lineages (*Bharatanāṭyam*, *Kathakālī*, *Kuchipudi*, *Odissi*, *Manipuri*, *Mohiniāṭṭam*, *Kathak*) as practitioner-side carriers with the *rasika* community providing audience-side critical reception and error-correction.

Deployments: Chapter 14 §14.2 ¶ — the citation anchor for the *Nāṭyaśāstra* and Indian classical dance lineages as the Indic counterparts of the book-coined *Flexture* preservation mode.

The *Flexture* preservation mode — content preserved through trained gestures, postures, and choreographed sequences — is operated in the Indic continuum through:

- ***Mudrā* and *Hasta* gesture vocabulary** — the engineered specification of hand-gestures and bodily postures that carry narrative and emotional content. The *mudrā* paramparā specifies dozens of named gestures (the *abhinaya darpaṇa* of Nandikeśvara enumerates twenty-eight *asaṃyuta hastas* — single-hand gestures — and twenty-three *saṃyuta hastas* — both-hand gestures). Each gesture has a specified articulation and a specified narrative-and-emotional vocabulary it carries.
- ***The Nāṭyaśāstra* (नाट्यशास्त्र)** — Bharata’s foundational treatise on performance, drama, music, and aesthetics. The *Nāṭyaśāstra* codifies the gesture vocabulary, the posture vocabulary, the *rasa* (aesthetic emotion) theory, the *bhāva* (psychological state) categorization, and the broader specification of how content is rendered through performance.

- **The Classical Dance Lineages** — *Bharatanāṭyam* (Tamil Nadu), *Kathakalī* (Kerala), *Kuchipudi* (Andhra Pradesh / Telangana), *Odissi* (Odisha), *Manipuri* (Manipur), *Mohiniāṭṭam* (Kerala), *Kathak* (north India), and the regional variations. Each lineage operates as a practitioner-side carrier of the *Flexture* preservation across many generations, with the *guru-shishya* paramparā maintaining the gesture-and-posture vocabulary and the audience-side critical reception (across the *rasika* community) providing the error-correction function.

The structural significance the chapter establishes: the *Flexture* operates with a different sense-and-skill complementarity than *Auditure* or *Mnemoniture*. The *Auditure* pairs vocal articulation (practitioner) with hearing (audience); the *Mnemoniture* pairs retelling (practitioner) with hearing-and-remembering (audience); the *Flexture* pairs choreographed motion (practitioner) with sight (audience). The Indic continuum's *Flexture* — through the *Nāṭyaśāstra*'s specification and the classical-dance lineages' transmission — has operated across many generations with the same engineering rigor the *śruti* and *smṛti* categories operate.

Standard references: *Nāṭyaśāstra* of Bharata. Standard editions: M. Ghosh, *The Nāṭyaśāstra* (Manisha Granthalaya, Calcutta, two volumes, 1950 and 1961); Adya Rangacharya, *The Nāṭyaśāstra* (Munshiram Manoharlal, 1996); the *Gaekwad Oriental Series* edition (Baroda). The *Abhinaya Darpaṇa* of Nandikeśvara: Manomohan Ghosh, *The Mirror of Gesture* (American Oriental Society, 1957). Studies: Kapila Vatsyayan, *Classical Indian Dance in Literature and the Arts* (Sangeet Natak Akademi, 1968); *Indian Classical Dance* (Marg Publications); Avanthi Meduri, *Bharata Natyam: What Are You?* (Asian Theatre Journal, 1988). For specific lineages: Sunil Kothari, *Kathak: Indian Classical Dance Art* (Abhinav, 1989); Phillip Zarrilli, *Kathakali Dance-Drama* (Routledge, 2000).

shruti-as-auditure

Short: The *śruti* (श्रुति, *that which is heard*) category preserves the four *Vedas* (*R̥gveda*, *Yajurveda*, *Sāmaveda*, *Atharvaveda*) plus *Brāhmaṇa*, *Āraṇyaka*, and *Upaniṣad* literature through *exact phoneme-level* preservation — the Indic engineering at its most rigorous; layered apparatus combines the *Prātiśākhya* / *Śikṣā* literature, Pāṇini's *Aṣṭādhyāyī*, the eleven *pāṭhas*, and the *guru-shishya paramparā*. The Indic counterpart of the book-coined **Auditure**.

Deployments: Chapter 14 §14.2 ¶ — the citation anchor for the *śruti* category as the Indic counterpart of the book-coined *Auditure* preservation mode.

The *श्रुति* (*śruti*) category in the Indic continuum names *that which is heard* — content preserved across generations through *exact phonetic preservation*. The standard *śruti* corpus includes:

- **The four *Vedas*** — *R̥gveda*, *Yajurveda* (in its *Kṛṣṇa* and *Śukla* recensions), *Sāmaveda*, *Atharvaveda*.
- **The *Brāhmaṇas*** — the prose theological-ritual literature attached to each *Veda*.
- **The *Āraṇyakas*** — the forest-treatises bridging *Brāhmaṇa* and *Upaniṣad*.
- **The *Upaniṣads*** — the philosophical-contemplative literature attached to each *Veda*.

The four-fold *Vedic* corpus operates the *śruti* preservation through the layered engineering apparatus the book develops across Chapters 7, 8, 15, and 16:

- **The *Prātiśākhya* literature** — the phonetic-specification documentation for each *Veda*'s recension.
- **The *Śikṣā* literature** — the auxiliary phonetic-pedagogical *śāstra* for *Vedic* recitation training.
- **The *Aṣṭādhyāyī*** — Pāṇini's grammar decoding the language the *Vedic* corpus operates in.

- **The eleven *pāṭhas*** — the layered recitation-redundancy system providing error-detection-and-correction across transmission generations.
- **The *guru-shishya paramparā*** — the human-network infrastructure that carries the recitation training across generations.

The structural significance the chapter establishes: the *śruti* category is the Indic engineering at its most rigorous. The preservation operates at the *phoneme* level — the *varṇa-by-varṇa* specification of every syllable across the *Samhitā* texts is preserved exactly across the *paramparā*, with the eleven-fold *pāṭha* system supplying the redundancy that catches and corrects drift. The architects of the *Vedic* corpus engineered the preservation system into the language itself — every feature of the *varṇamālā*, every rule of the *Aṣṭādhyāyī*, every *pāṭha* permutation works together to produce a preservation channel with no analog in any other ancient linguistic lineage.

The *śruti* / *smṛti* distinction is not a hierarchy of importance. It is a distinction of *engineering type*. The *śruti* preserves exact phonetic form; the *smṛti* preserves narrative-and-conceptual content. Both are engineered. The two categories together provide the dual-track preservation infrastructure the Indic continuum operates across the depth of time.

Standard references: see the foundational references at endnotes *agnimile-rigveda-opening*, *chandasi-bhasayam-astadhyayi*, *eleven-pathas*, *pre-panini-pratisakhya-classification*, and *aksara-imperishable-name*. The body of work on the *śruti* preservation is large; the chapter's deployment is anchored across multiple individual references.

masoretic-engineered-preservation

Short: The Hebrew Masoretic apparatus (Tiberian and Babylonian schools, ~6th–10th century CE) operates four canonical layers: *kəṭṭīv* (כתיב, consonantal text) + *niqqud* (ניקוד, vowel pointing) + *ṭeʿamim* (טעמים, cantillation marks) + *Masora* (מסורה, marginal apparatus with letter counts, statistical checks); principal manuscript anchors the *Aleppo Codex* (early 10th c.) and *Leningrad Codex* (1008 CE; basis of *Biblia Hebraica Stuttgartensia*). Sophisticated engineered preservation that the Western philological community already recognizes; the Vedic apparatus operates at phoneme level rather than consonant-and-vocalization level, with deeper redundancy (eleven *pāṭhas*).

Deployments: Chapter 14 §14.4 ¶ — the citation anchor for the Hebrew Masoretic apparatus as the engineered-preservation comparative case.

The Hebrew Masoretic apparatus is the textual-preservation engineering operated by the schools of Jewish scribes at Tiberias and Babylonia from approximately the sixth through the tenth century CE. The apparatus's four canonical layers:

1. **Consonantal text** (*kəṭṭīv* — כתיב) — the consonant-only form of the Hebrew Bible. This text was fixed in approximate form well before the Masoretes; the Masoretic apparatus preserved the consonantal text exactly without modification.
2. **Vowel pointing** (*niqqud* — ניקוד) — the system of vowel-marks added below and above the consonant letters to specify the vocalic articulation. The Tiberian *niqqud* system (developed by the Tiberian Masoretes) became the dominant convention; earlier Palestinian and Babylonian systems are also attested.
3. **Cantillation marks** (*ṭeʿamim* — טעמים) — the layer of musical-recitational marks specifying the melodic contour and syntactic phrasing of each verse. Twenty-seven distinct *ṭeʿamim* are used in the standard Tiberian system, with one set for the twenty-one

“prose” books and another for the three “poetic” books (Psalms, Proverbs, Job).

4. **Marginal machinery (Masora — מסורה)** — the *Masora parva* (small Masorah, in the margins) and *Masora magna* (large Masorah, in the top and bottom margins or in dedicated reference works). The marginal apparatus carries letter counts, word counts, mid-Torah and mid-Bible position markers, references to similar passages elsewhere in the corpus, and statistical checks on the integrity of the consonantal text.

The principal manuscript anchors: the **Aleppo Codex** (Crown of Aleppo, early tenth century CE; held at the Israel Museum, Jerusalem, with substantial portions lost in the 1947 Aleppo riot) and the **Leninrad Codex** (1008 CE; held at the Russian National Library, St. Petersburg; the basis of the standard scholarly editions of the Hebrew Bible — the *Biblia Hebraica Stuttgartensia* and the *Biblia Hebraica Quinta*).

The Western philological community reads the Masoretic apparatus as engineered preservation. The textual-critical literature explicitly treats the Masoretic text as a deliberately fixed specification against which medieval and modern Hebrew Bible manuscripts are evaluated. The terminology of “engineering” applied to textual preservation is the Western field’s own; the book’s chapter operates within that established framing to make the comparative-tradition case.

The structural comparison the chapter establishes: the Masoretic apparatus is sophisticated engineered preservation, operating across roughly four centuries of codification activity. The apparatus is impressive on its own terms. The chapter’s broader claim is that the Vedic preservation apparatus is *more rigorous* — operating at the phoneme level rather than the consonant-and-vocalization-and-cantillation level, across more generations of continuous operation, with deeper redundancy (the eleven *pāṭhas*) and broader independent-lineage cross-verification (see endnotes

nambudiri-vedic-recitation-isolation, cross-shakha-verification-field
unesco-vedic-chanting-2003).

Standard references: Israel Yeivin, *Introduction to the Tiberian Masorah* (Scholars Press, 1980, English translation of his 1968 Hebrew original); Aron Dotan, *The Hebrew Bible (Codex Leningrad B19A): Reproduced in Facsimile* (Eerdmans, 1998); Emanuel Tov, *Textual Criticism of the Hebrew Bible* (Fortress Press, 3rd edition 2012); Page H. Kelley, Daniel S. Mynatt, Timothy G. Crawford, *The Masorah of Biblia Hebraica Stuttgartensia: Introduction and Annotated Glossary* (Eerdmans, 1998). For the Aleppo Codex: Mordechai Glatzer, *Aleppo Codex: Codicological and Paleographic Aspects of the Crown of Aleppo* (Ben-Zvi Institute, 1989).

quranic-engineered-preservation

Short: The Quranic preservation system operates four layers: *tajwīd* (تجويد, recitation rules), *qirāʾāt* (قراءات, seven canonical readings codified by Ibn Mujāhid in the 10th c.; ten by Ibn al-Jazarī in the 15th c.), *isnād* (إسناد, documented chains of transmission), and *ḥifẓ* (حفظ, memorization tradition; the *ḥāfiẓ* / *ḥuffāẓ* preserve the entire Quran). Sophisticated engineered preservation across ~14 centuries — but preserves canonical *variation* across recitations rather than combinatorial re-encoding of a single canonical text the Vedic *pāṭha* system operates.

Deployments: Chapter 14 §14.4 ¶ — the citation anchor for the Quranic preservation architecture as the engineered-preservation comparative case from the Islamic tradition.

The Quranic preservation system operates a multi-layered architecture developed and continuously refined since the seventh century CE codification under the third Caliph Uthmān ibn ‘Affān. The architecture’s principal layers:

1. **Tajwīd** (تجويد) — the body of recitation rules specifying the phonetic and rhythmic features of correct Quranic recitation. The *tajwīd* literature specifies articulation points (*makhārij*), the modifications consonants undergo at junctions (*idgām*, *ikhfā*, *iqḷāb*, *iẓhār*), the elongations (*madd*) and their durations, the pause-and-stop conventions (*waqf*), the nasalization rules (*ghunna*), and the broader recitational specification.
2. **Qirāʾāt** (قراءات) — the canonical recitation traditions. Ibn Mujāhid's tenth-century codification established the canonical *seven readings*; the later canonical settlements (Ibn al-Jazarī's fifteenth-century work) extended the canonical set to *ten readings*. Each *qirāʾat* is named for its founding reciter — Nāfiʿ, Ibn Kathīr, Abū ʿAmr, Ibn ʿĀmir, ʿĀṣim, Ḥamza, al-Kisāʾī (the seven), with Abū Jaʿfar, Yaʿqūb, Khalaf added later (the ten) — and is preserved exactly through documented chains of transmission.
3. **Isnād** (إسناد) — the chain-of-transmission documentation. Every transmitter of a recitation tradition is recorded teacher-by-teacher back to the founding reciter and through them to specific Companions of the Prophet (the *Ṣaḥāba*) and through them to Muhammad. The *isnād* is the documentary apparatus that authenticates each transmission line.
4. **Ḥifẓ** (حفظ) — the memorization tradition. *Ḥāfiẓ* (حافظ; plural *ḥuffāẓ* (حَفَّاء)) is the title for one who has memorized the entire Quran in one or more *qirāʾāt*. The *ḥuffāẓ* have been produced continuously across the Islamic world since the seventh century, providing a distributed verification network against which any written variant of the text can be checked.

The Uthmanic *muṣḥaf* — the standardized written text produced under the third Caliph — operates as the canonical written reference. Multiple manuscript and printed editions descend from the Uthmanic codification; the most widely used contemporary print edition is the

Cairo edition (*muṣḥaf al-Madīna al-Nabawiyya*), first published 1924 and revised editions thereafter.

The structural comparison the chapter establishes: the Quranic preservation system is sophisticated engineered preservation operating across approximately fourteen centuries of continuous operation. Its layered redundancy (memorization + canonical recitations + documented chains-of-transmission + written codification) provides high preservation rigor. The chapter's comparative claim, however, is that the Quranic system does not operate the eleven-fold combinatorial re-encoding that the Vedic *pāṭha* system operates. The *qirā'āt* preserve documented variation; the eleven *pāṭhas* re-encode the same content under combinatorial permutations that produce mutually-checking redundancy. The two preservation strategies are different at the architectural level (see endnote quran-recitation-vs-pathas-comparison for the comparative analysis).

Standard references: Aḥmad 'Alī al-Imām, *Variant Readings of the Qur'an: A Critical Study of Their Historical and Linguistic Origins* (International Institute of Islamic Thought, 1998); Frederik Leemhuis, "Readings of the Qur'an," in *Encyclopaedia of the Qur'ān* (Brill, 2001–2006); Christopher Melchert, "The Relation of the Ten Readings to One Another," *Journal of Qur'anic Studies* 10, no. 2 (2008); Andrew Rippin, ed., *The Blackwell Companion to the Qur'an* (Wiley-Blackwell, 2006). For *tajwīd*: the standard *tajwīd* manuals in the *Riwāyat Ḥafṣ 'an 'Āṣim* and other transmission lines; Khan Sahib Sheikh, *The Recitation of the Qur'an* (Adam Publishers, 2001).

latin-vulgate-engineered-preservation

Short: The Latin Vulgate (Jerome's translation, late 4th c. CE; commission by Pope Damasus I) operates engineered preserva-

tion through monastic-scriptorium copying (Carolingian, Cistercian, Benedictine — with *correctores* verifying against archetype manuscripts), textual-stemma reconstruction, papal authentication (*Sixto-Clementine Vulgate* 1592 under Pope Clement VIII; *Nova Vulgata* 1979), and modern critical-edition scholarship (*Stuttgart Vulgate*, *Vetus Latina*). Sophisticated written-primary preservation with chant supplementation; less robust than oral-primary Vedic preservation because written-text corruption can propagate silently across copying generations until the textual-critical apparatus catches it.

Deployments: Chapter 14 §14.4 ¶ — the citation anchor for the ecclesiastical Latin manuscript canon (the Vulgate tradition) as the engineered-preservation comparative case from the Christian tradition.

The Latin Vulgate is the Latin translation of the Bible substantially produced by Jerome (Eusebius Sophronius Hieronymus, ca. 347–420 CE) under commission from Pope Damasus I in the late fourth century. The translation was completed across the late fourth and early fifth centuries. The preservation apparatus that operated on the Vulgate across the medieval period:

1. *Monastic scriptorium copying* — the manuscript-production activity at organized monastic scriptoria (Carolingian — Aachen, Tours, Corbie, St. Gall; Cistercian — Clairvaux, Cîteaux; Benedictine — Monte Cassino, Cluny; and many others) with established protocols for copying, correction by *correctores* (the scribal-correctors who reviewed each copy against an exemplar), and verification against established manuscript-archetypes.
2. *Textual stemma* — the manuscript-tree reconstruction the textual-critical scholarship has developed to trace the surviving manuscripts back to a smaller number of archetypes. The stemma is the apparatus by which manuscript corruption is

identified and corrected: variants in the extant manuscripts are mapped to the stemma's branching points, and the *archetype* (or close approximation) is reconstructed.

3. ***Papal authentication*** — the Council of Trent (1545–1563), in its fourth session (April 8, 1546), declared the Vulgate the authoritative Latin text of the Roman Catholic Church. The *Sixto-Clementine Vulgate* (the authorized printed edition: 1592 promulgation under Pope Clement VIII, after the 1590 Sixtine edition was withdrawn) operated as the standard for several centuries. The *Nova Vulgata* (1979, promulgated by Pope John Paul II) is the current authorized edition.
4. ***Critical-edition scholarship*** — the modern textual-critical project that has produced the *Biblia Sacra Vulgata* (Stuttgart Vulgate, edited by Robert Weber et al., German Bible Society, multiple editions since 1969); the *Vetus Latina* project at the *Vetus Latina Institut* in Beuron documenting the pre-Vulgate Old Latin tradition; the *Editio Critica Maior* of the Greek and Latin New Testament traditions.

The academic-philological account of the Latin manuscript canon is engineered preservation through institutional means. The field of *textual criticism* operates explicitly on the premise that the engineered specification can be reconstructed from extant manuscripts through stemmatic analysis.

The structural comparison the chapter establishes: the Latin Vulgate tradition is sophisticated engineered preservation operating primarily through *written* transmission, with chant traditions (*Gregorian chant*; the broader plainchant tradition) layered on top as a supplementary oral preservation. The chapter's comparative claim: the written-primary preservation strategy is less robust than the oral-primary preservation the Vedic corpus operates. Written-text corruption can propagate silently across copying generations until the textual-critical apparatus catches it; oral-recitation corruption is caught immediately

by the *guru-shishya paramparā*'s ongoing verification. The two strategies are different at the architectural level.

Standard references: Robert Weber, ed., *Biblia Sacra Vulgata* (Württembergische Bibelanstalt / German Bible Society, 1969, with subsequent revisions; 5th edition 2007 edited by Roger Gryson); H. F. D. Sparks, *On Translations of the Bible* (Athlone Press, 1973); Bruce Metzger, *The Early Versions of the New Testament* (Oxford University Press, 1977); Frans van Liere, *An Introduction to the Medieval Bible* (Cambridge University Press, 2014); Pierre-Maurice Bogaert, "The Latin Bible" in *The New Cambridge History of the Bible*, Volume 1 (Cambridge University Press, 2013).

shiksha-texts-canonical-list

Short: The principal canonical *Śikṣā* (शिक्षा) texts include: *Pāṇinīya-Śikṣā* (पाणिनीयशिक्षा, ~60 verses; standard introductory), *Yājñavalkya-Śikṣā* (याज्ञवल्क्यशिक्षा, ~200 verses), *Vāsiṣṭhī-Śikṣā* (वासिष्ठीशिक्षा), *Āpīśali-Śikṣā* (आपिशलिशिक्षा, pre-Pāṇinian), *Bhāradvāja-Śikṣā* (भारद्वाजशिक्षा), *Nārādīya-Śikṣā* (नारदीयशिक्षा, *Sāmavedic* with musical-theoretical framework), *Vyāsa-Śikṣā*, *Kātyāyana-Śikṣā*, *Parāśara-Śikṣā*, *Māṇḍūkī-Śikṣā* — short training manuals for an oral practice, each lineage-specific, with the multiplicity itself evidence of *Śikṣā* as a serious transmission-engineering discipline.

Deployments: Chapter 15 §15.1 ¶ — the citation anchor for the canonical list of *Śikṣā* texts.

The principal canonical *Śikṣā* texts of the *Vedāṅga* disciplines:

1. *Pāṇinīya-Śikṣā* (पाणिनीयशिक्षा) — attributed to the Pāṇinian discipline (though scholarly opinion varies on whether the text is by Pāṇini himself or by a later author of the school). The standard introductory text on Sanskrit phonetics; ~60 verses.

2. *Yājñavalkya-Śikṣā* (याज्ञवल्क्यशिक्षा) — attached to the *Yājñavalkya śākhā* of the *Śukla-Yajurveda*; ~200 verses.
3. *Vāsiṣṭhī-Śikṣā* (वासिष्ठीशिक्षा) — attached to the *Vasiṣṭha* lineage; develops the *Sāmavedic* recitation framework.
4. *Āpiśali-Śikṣā* (आपिशलिशिक्षा) — attached to the pre-Pāṇinian grammarian *Āpiśali*; one of the earlier canonical *Śikṣā* texts.
5. *Bhāradvāja-Śikṣā* (भारद्वाजशिक्षा) — attached to the *Bhāradvāja* lineage.
6. *Nārādīya-Śikṣā* (नारदीयशिक्षा) — attached to the *Sāmavedic* *śākhā*; develops the musical-theoretical framework for *Sāmavedic* recitation; ~200 verses.
7. *Vyāsa-Śikṣā* (व्यासशिक्षा) — attributed to *Vyāsa*.
8. *Kātyāyana-Śikṣā* (कात्यायनशिक्षा) — attached to the *Kātyāyana* lineage.
9. *Parāśara-Śikṣā* (पाराशरशिक्षा) — attached to the *Parāśara* lineage.
10. *Māṇḍūkī-Śikṣā* (माण्डूकीशिक्षा) — attached to the *Māṇḍūkā śākhā*.

The full set of named *Śikṣā* texts runs into the dozens; the canonical core is typically given as the first five or six above. Each is associated with a specific *śākhā* or pedagogical lineage; each documents the production-rules its lineage trains; each names the points at which the trainee will be checked. The texts are short — verses, not treatises — because they operate as training manuals for an oral practice rather than as analytic descriptions of a linguistic system.

The structural significance the chapter establishes: the multiplicity of canonical *Śikṣā* texts is itself evidence that *Śikṣā* operated as a serious, lineage-specific, transmission-engineering discipline across the *Vedic* corpus. The texts are not redundant copies of the same content; each carries the specific recitational conventions of its lineage,

with detailed phonetic and rhythmic specifications that differ across lineages in calibrated ways.

Standard references: Vidyāsāgara's *Śikṣā-Saṃgraha* (Calcutta, 1893) — the compendium of multiple *Śikṣā* texts; Manmohan Ghosh, *The Pāṇinīya-Śikṣā or the Śikṣā Vedāṅga ascribed to Pāṇini* (University of Calcutta, 1938); Sharma's edition of the *Nāradya-Śikṣā* (Vishveshvaranand Vishva Bandhu Institute, 1980); Hartmut Scharfe, *Grammatical Literature* (Otto Harrassowitz, 1977), Chapter 3 on the *Vedāṅga* disciplines.

shiksha-first-vedanga-priority

Short: The six *Vedāṅgas* (वेदाङ्ग, *Veda-limbs*) in canonical order: *Śikṣā* (शिक्षा, phonetics / recitation), *Chandas* (छन्दस्, prosody / meter), *Vyākaraṇa* (व्याकरण, grammar), *Nirukta* (निरुक्त, etymology / semantics), *Jyotiṣa* (ज्योतिष, astronomy / ritual timing), *Kalpa* (कल्प, ritual procedure); the *Śikṣā*-first priority is not accidental — the recitation is what the *Vedic* text is, and the other five disciplines are built on what *Śikṣā* trains (without trained recitation: meter has no instantiation, grammar has nothing to govern, etymology has no items to analyze, astronomy no ritual to time, procedure no ritual to specify).

Deployments: Chapter 15 §15.1 ¶ — the citation anchor for *Śikṣā* as the first-listed *Vedāṅga*.

The six *Vedāṅgas* (वेदाङ्ग — *Veda-limbs*) are the canonical six auxiliary disciplines attached to the *Vedic* corpus. The standard enumeration, in the conventional order:

1. *Śikṣā* (शिक्षा) — phonetics and recitation.
2. *Chandas* (छन्दस्) — prosody and meter.
3. *Vyākaraṇa* (व्याकरण) — grammar.

4. *Nirukta* (निरुक्त) — etymology and semantics.
5. *Jyotiṣa* (ज्योतिष) — astronomy and astrology (ritual timing).
6. *Kalpa* (कल्प) — ritual procedure.

The conventional ordering places *Śikṣā* first. The traditional canonical ordering is preserved across the *Vedāṅga* literature: the *Pāṇinīya-Śikṣā* opens by listing the six in this order; the *Yājñavalkya-Śikṣā* preserves the same priority; the standard medieval and contemporary scholarly references all preserve the *Śikṣā*-first ordering.

The structural significance the chapter establishes: the priority of *Śikṣā* is not accidental ordering. The *Vedāṅga* canonical sequence reflects the *foundational dependency* of the *Vedic* corpus's operation. The recitation (which *Śikṣā* trains) is what the *Vedic* text is — the corpus exists as recitation primarily and as written-text secondarily. Without trained recitation, the meter (*Chandas*) has no instantiation; without the recitation operating in a linguistic system, the grammar (*Vyākaraṇa*) has nothing to govern; without the recitation carrying the words, the etymology (*Nirukta*) has no items to analyze; without the recitation timing the ritual, the astronomy (*Jyotiṣa*) has no ritual to time; without the recitation enacting the ritual, the procedure (*Kalpa*) has no ritual to procedurally specify. *Śikṣā* is foundational; the other five *Vedāṅgas* are built on the recitation *Śikṣā* trains.

Standard references: the *Pāṇinīya-Śikṣā*'s opening verses; Madhusūdana Sarasvatī's *Prasthānabheda* (a sixteenth-century synthesis of the *darśana* and *Vedāṅga* literature); the standard *Vedāṅga* references — Maurice Winternitz, *A History of Indian Literature* (Calcutta, 1927; English translation), Volume I, the chapters on *Vedāṅga* literature; Jan Gonda, *Vedic Literature* (Volume I, Fascicle 1 of *A History of Indian Literature*, Otto Harrassowitz, 1975).

eleven-pathas-full-list

Short: Forward-pointer to the canonical eleven-pathas endnote — the full enumeration and analytical treatment of the eleven recitation modes (five *prakṛti-pāṭhas*: *Samhitā*, *Pada*, *Krama*, *Jaṭā*, *Ghana*; six *vikṛti-pāṭhas*: *Mālā*, *Śikhā*, *Rekhā*, *Dhvaja*, *Daṇḍa*, *Ratha*) that together constitute the redundancy architecture holding the *Samhitā* texts against drift across the *paramparā*.

Deployments: Chapter 15 §15.2 ¶ — the citation anchor for the full list of the eleven *pāṭhas* with their permutational specifications.

See endnote eleven-pathas (deployed in the Preface) for the full enumeration and analytical treatment of the eleven recitation modes. The Chapter 15 deployment is the same canonical citation, anchored in the extended technical discussion across §§15.2–15.3 that develops each *pāṭha*’s specific permutational pattern. The eleven *pāṭhas* — five *prakṛti-pāṭhas* (*Samhitā*, *Pada*, *Krama*, *Jaṭā*, *Ghana*) and six *vikṛti-pāṭhas* (*Mālā*, *Śikhā*, *Rekhā*, *Dhvaja*, *Daṇḍa*, *Ratha*) — together constitute the redundancy architecture that holds the *Samhitā* texts against drift across the *paramparā*.

Standard references and full discussion: see endnote eleven-pathas.

ghanapathi-title-recognition

Short: The title *Ghanapāṭhī* (घनपाठी — *one who has mastered the Ghana-pāṭha*) is a recognized social-academic honorific in Vedic recitation communities (*Nambūdiri*, *Iyengar*, *Iyer*, *Gauḍa*, *Marāṭhī Deśastha*, *Kashmir Pandit*, *Banaras*, *Allahabad*, *Karnataka*, *Gujarat*, *Rajasthan*) — conferred informally by community recognition rather than formal credentialing, requiring prior mastery of the *Samhitā*, *Pada*, *Krama*, and *Jaṭā* recitations; the title-holder operates as a transmission-anchor in the *guru-shishya* chain and as cross-lineage

verification authority for disputed readings.

Deployments: Chapter 15 §15.2 ¶ — the citation anchor for the *Ghanapāṭhī* honorific as social recognition of mastery of the *Ghana-pāṭha*.

The title **घनपाठी** (*Ghanapāṭhī*) — *one who has mastered the Ghana-pāṭha* — is a recognized social-academic honorific in Vedic recitation communities across the subcontinent. The title carries weight in several respects:

- **Training depth.** To produce the *Ghana-pāṭha* correctly, the reciter must have first mastered the *Samhitā*, *Pada*, *Krama*, and *Jaṭā* recitations of the relevant *Samhitā* text. The *Ghana* is the densest of the *prakṛti-pāṭhas* and the most demanding to produce.
- **Community recognition.** The Vedic communities (*Nambūdiri*, *Iyengar*, *Iyer*, *Gauḍa*, *Marāṭhī Deśastha*, *Kashmir Pandit*, *Bannaras*, *Allahabad*, *Karnataka*, *Gujarat*, *Rajasthan*, and other regional lineages maintaining recitation *paramparā*) acknowledge the *Ghanapāṭhī* title and treat the title-holder as a senior practitioner. The title is conferred informally by community recognition rather than formally by institutional credentialing; the community's collective verification of the reciter's training is what produces the title.
- **Standing in the *paramparā*.** The *Ghanapāṭhī* operates as a transmission-anchor in the *guru-shishya* chain. Junior trainees seek out *Ghanapāṭhīs* for advanced training; the *Ghanapāṭhīs* serve as cross-lineage verification authorities for disputed readings.

The chapter's deployment: the title's standing in the community is itself evidence that the *Vedic* recitation lineages operate as a serious engineering discipline with recognized levels of training and standing. The chapter does not develop the title at length; it deploys the

title as the standing honorific that names the senior practitioners the recitation lineages produce.

Standard references: the various community-paramparā references — Wayne Howard, *Sāmavedic Chant* (Yale University Press, 1977); Frits Staal, *The Science of Ritual* (Bhandarkar Oriental Research Institute, 1982); Mahadev Chakravarti, *The Concept of Rudra-Śiva through the Ages* (Motilal Banarsidass, 1986) for the Nambūdiri śākhā context; the *paramparā* documentation across the Vedic Research Institute publications and the surviving traditional-school records (*Pāṭhaśālā* records across Tamil Nadu, Kerala, Maharashtra, and the broader sub-continent).

six-vikrti-pathas-pattern-list

Short: The six *vikṛti-pāṭhas* (विकृतिपाठ) extend the *prakṛti*-combinatorial logic with further permutational patterns named for the visual shape of the recitation diagram: ***Mālā*** (मालापाठ, *garland* — long looping pattern), ***Śikhā*** (शिखापाठ, *peak* — triadic rotated), ***Rekhā*** (रेखापाठ, *line* — extended linear), ***Dhvaja*** (ध्वजपाठ, *flag* — ends-to-beginning pairings), ***Daṇḍa*** (दण्डपाठ, *staff* — progressive extension), ***Ratha*** (रथपाठ, *chariot* — complex multi-axis); less commonly mastered than the *prakṛti* set, deployed as the deepest verification layer for disputed word-order or *sandhi* questions.

Deployments: Chapter 15 §15.2 ¶ — the citation anchor for the six *vikṛti-pāṭhas* and their permutational patterns.

The six *vikṛti-pāṭhas* extend the *prakṛti-pāṭha* combinatorial logic with further permutational patterns, each named for the visual shape of its recitation diagram when laid out on the page. Brief specifications:

1. ***Mālā-pāṭha*** (मालापाठ — *garland*) — words combined in a long

looping pattern; the *garland* shape emerges from the recitation diagram's repetitive cycling.

2. *Śikhā-pāṭha* (शिखापाठ — *peak*) — triadic groupings rotated through a *peak* pattern; the recitation creates a triangular visual structure when diagrammed.
3. *Rekhā-pāṭha* (रेखापाठ — *line*) — extended linear permutations producing an elongated diagram-shape.
4. *Dhvaja-pāṭha* (ध्वजपाठ — *flag*) — ends-to-beginning pairings producing a *flag*-shaped diagram with characteristic horizontal-and-vertical structure.
5. *Daṇḍa-pāṭha* (दण्डपाठ — *staff*) — progressive extension across the text producing a *staff*-shaped diagram.
6. *Ratha-pāṭha* (रथपाठ — *chariot*) — complex multi-axis permutations producing the most elaborate of the six visual-diagram shapes.

The *vikṛti-pāṭhas* are less commonly mastered than the *prakṛti* set; they exist as the deepest verification layer. A reciter resorts to them when a question of word-order or *sandhi* needs to be settled against a maximally constrained re-encoding.

The exact permutational patterns of each *vikṛti-pāṭha* are documented in the *Prātiśākhya* literature and in the specialized *Pāṭha* literature (the *Pāṭha-vyākhyā* discipline that develops the technical specifications). The standard reference is across the *Ṛk-Prātiśākhya*, the *Taittirīya-Prātiśākhya*, and the auxiliary *Pāṭha* texts in each *śākhā*'s paramparā.

Standard references: see endnote eleven-pathas for the foundational references; Frits Staal, *Mantras between Fire and Water* (Royal Netherlands Academy of Arts and Sciences, 1992) for the structural discussion; Wayne Howard, *Sāmavedic Chant* (Yale University Press, 1977) for the *Sāmavedic* recitation lineages (which preserve variant

pāṭha conventions).

combinatorial-redundancy-comparative

Short: The Vedic eleven *pāṭhas* produce a multi-channel combinatorial-redundancy architecture with no comparable analog in any other ancient preservation tradition: Hebrew Masoretic operates letter-counting and statistical verification (not combinatorial re-encoding); Quranic *qirā'āt* preserve canonical variation across recitations; Latin Vulgate operates manuscript-stemma reconstruction (written-primary); Greek classical Alexandrian textual criticism, Chinese standard-edition tradition, and Buddhist Pāli Canon Council-and-memorization paramparā — sophisticated each on its own architecture, but none operates *permutation-based redundancy* over a single canonical text.

Deployments: Chapter 15 §15.3 ¶ — the citation anchor for the comparative claim that no other ancient preservation tradition operates a comparable combinatorial-redundancy architecture.

The chapter's comparative claim: the eleven *pāṭhas* together produce a multi-channel redundancy architecture of a kind that no other ancient preservation tradition is documented to have built. Brief comparative survey:

- **Hebrew Masoretic apparatus.** Operates on the consonantal text with vowel-pointing, cantillation, and marginal apparatus. The redundancy operates at the level of letter-counting, word-counting, and statistical verification — substantial engineering, but not combinatorial re-encoding of the text under permutational patterns.
- **Quranic preservation system.** Operates the *tajwīd*, *qirā'āt*, *is-nād*, and *ḥifẓ* layers. The *qirā'āt* preserve documented varia-

tion across canonical recitations; the *ḥuffāz* provide distributed memorization verification. The system is sophisticated but does not include eleven-fold combinatorial re-encoding.

- **Latin Vulgate tradition.** Operates manuscript-stemma reconstruction and scriptorium-correction protocols. Primarily written-text preservation with chant supplementation.
- **Greek classical-text transmission.** Operates Alexandrian-school textual criticism (Aristarchus, Zenodotus, the Alexandrian library tradition); medieval Byzantine manuscript copying; Renaissance textual-critical scholarship. Sophisticated written-text preservation; no combinatorial re-encoding system.
- **Chinese classical-text transmission.** Operates the standard-edition tradition (the *Five Classics* in the Han-dynasty codifications; later *Thirteen Classics*); the imperial-examination training that produced memorized standard editions; the Song-dynasty printing standardization. Sophisticated; no combinatorial re-encoding.
- **Buddhist Pāli Canon transmission.** Operates the *Council* convention (the First Buddhist Council, ~5th century BCE; subsequent councils refining the canonical texts) and the monastic-memorization paramparā. Substantial oral-recitation preservation; less elaborated combinatorial re-encoding than the Vedic *pāṭha* system.

The structural claim the chapter establishes: across the major ancient preservation traditions, the Vedic recitation lineages' eleven-fold combinatorial-re-encoding architecture is unique. The other traditions operate sophisticated engineered preservation through different architectural strategies (written-text criticism, distributed memorization, canonical-recitation variants), but none operates the *permutation-based redundancy* the Vedic system operates.

Standard references for the comparative survey: Emanuel Tov, *Textual Criticism of the Hebrew Bible* (Fortress Press, 3rd edition 2012); Frederik Leemhuis, “Readings of the Qur’an” in the *Encyclopaedia of the Qur’ān* (Brill); L. D. Reynolds and N. G. Wilson, *Scribes and Scholars: A Guide to the Transmission of Greek and Latin Literature* (Oxford University Press, 4th edition 2013); Wilt L. Idema and Lloyd Haft, *A Guide to Chinese Literature* (University of Michigan Press, 1997); Steven Collins, *Nirvana and Other Buddhist Felicities* (Cambridge University Press, 1998) for the Pāli Canon transmission.

nambudiri-vedic-recitation-isolation

Short: The Nambūdiri Brahmins of Kerala have preserved the *Ṛgveda* recitation across many generations in geographic, linguistic (Malayalam-speaking), and pedagogical-lineage isolation from the major Vedic-paramparā centers (Maharashtra, Tamil Nadu, Karnataka, Banaras, Allahabad, Kashmir, Gujarat, Rajasthan); despite the independence, the Nambūdiri recitation matches the recitations preserved at those distant centers at phoneme / accent / sandhi / pāṭha-permutation level — empirical confirmation that the engineered preservation has held the *Samhitā* texts against drift across parallel-operating lineages.

Deployments: Chapter 15 §15.4 ¶ — the citation anchor for the geographic-and-cultural isolation of the Nambūdiri Vedic recitation lineage.

The Nambūdiri Brahmins of Kerala have preserved the *Ṛgveda* recitation paramparā across many generations in a *śākhā* lineage geographically and culturally separated from the other major Vedic-recitation lineages of the subcontinent. The relevant separations:

- **Geographic.** Kerala is at the southwestern tip of the subcontinent, separated from the major Vedic-paramparā centers (Ma-

harashtra, Tamil Nadu, Karnataka, Banaras, Allahabad, Kashmir, Gujarat, Rajasthan) by hundreds of miles of intervening territory and, in the case of Maharashtra and the northern centers, by the western and central mountain systems.

- **Linguistic.** The Nambūdiri community's regional speech-community is Malayalam-speaking; the surrounding Tamil, Kannada, Telugu, and the broader Sanskrit-continuum centers operate in different regional speech-communities.
- **Cultural-institutional.** The Nambūdiri community has historically operated as a relatively self-contained Brahmanical community within Kerala, with the *Illam* (Nambūdiri household-and-temple) and *Sabha* (community-council) institutions providing the local infrastructure for *Vedic* training. Cross-community marriage and substantial cross-lineage exchange with the other major Vedic-paramparā centers is documented but limited across the historical record.
- **Pedagogical-lineage independence.** The Nambūdiri *Vedapāṭhī* training operates through *guru-shishya* chains internal to the community, with the *Veda* mantras transmitted from father to son and from established *guru* to chosen *shishya* across generations. Cross-lineage verification with the northern and central paramparā is documented but episodic.

The empirical fact the chapter establishes: despite this isolation, the Nambūdiri Vedic recitation matches the recitations preserved in the Maharashtra, Tamil Nadu, Karnataka, Banaras, Allahabad, Kashmir, and other major Vedic-paramparā centers — at the phoneme level, the accent level, the *sandhi* level, and the *pāṭha* permutation level. The match is not approximate; it is exact at the level of detail the *Prātiśākhya* documentation specifies. The independence of the lineages combined with the precision of the match is the evidence the chapter develops: the layered preservation engineering has held the *Samhitā* texts against drift across the lineages' parallel operation,

across many generations.

Standard references: Frits Staal, *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, 1983, two volumes) — the documentation of the 1975 Nambūdiri *Agnicayana* performance; Wayne Howard, *Sāmavedic Chant* (Yale University Press, 1977) for the Nambūdiri *Sāmavedic śākhā*; Asko Parpola, “The Pre-Aryan and Early-Aryan Sources of the Indus Civilization” (various papers on the Nambūdiri *Yajurvedic śākhā*); the *Veda Rakṣaṇa Nidhi* Trust documentation; the Sri Chinnamayya Mission and Sankara Math archival materials. Modern documentation includes audio-and-video recordings from the Frits Staal expeditions and subsequent fieldwork by Indian and international scholars.

staal-agni-nambudiri-recording

Short: In 1975, Frits Staal led a documentation expedition to Kerala to record a Nambūdiri performance of the *Agnicayana* (अग्निचयन) — the twelve-day *Vedic* fire-altar ritual — producing audio recordings, film footage (edited into the documentary *Altar of Fire*, Robert Gardner and Frits Staal, Film Study Center Harvard, 1976), photographic documentation, and the two-volume scholarly work *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, 1983). Primary scholarly-documentation anchor for cross-lineage Vedic-recitation comparison.

Deployments: Chapter 15 §15.4 ¶ — the citation anchor for the Frits Staal 1975 Nambūdiri *Agnicayana* recording expedition.

In 1975, Frits Staal led a documentation expedition to Kerala to record a performance of the *Agnicayana* — the twelve-day *Vedic* fire-altar ritual conducted according to the Nambūdiri *śrauta* paramparā. The expedition produced:

- **Audio recordings** of the entire twelve-day ritual, including the recitations from all four *Vedas* (*Rgveda*, *Sāmaveda*, *Yajurveda*, *Atharvaveda*) operated by the trained *Vedic* specialists of the Nambūdiri community.
- **Film documentation** of the ritual performance, including the construction of the bird-shaped *Śyenaciti* altar, the consecrations and oblations, the procession sequences, and the closing ceremonies. The film was later edited into the 1976 documentary *Altar of Fire* (Robert Gardner and Frits Staal, 45 minutes, Film Study Center, Harvard University).
- **Photographic documentation** of the ritual implements, the altar architecture, the participants, and the broader ritual context.
- **Scholarly documentation** — the comprehensive two-volume *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, 1983) published the expedition's findings in a 1,500-page work covering the ritual's textual sources, the performance details, the participants' biographies, the ritual implements, and the broader cultural context.

The recordings and documentation are held in the archives of the Asian Humanities Press, the Film Study Center at Harvard, and the various scholarly collections that have inherited Staal's materials. They are available for scholarly comparison with recitations from other Vedic-paramparā lineages.

The chapter's deployment: the Staal recordings provide one of the primary scholarly-documentation anchors for the empirical claim that the Nambūdiri Vedic recitation matches recitations preserved in the geographically and culturally separated lineages. The recordings exist; the recitations are comparable; the matching is testable.

Standard references: Frits Staal, *Agni: The Vedic Ritual of the Fire Altar* (Asian Humanities Press, 1983, two volumes); Robert Gardner

and Frits Staal, *Altar of Fire* (Film Study Center, Harvard University, 1976, 45-minute documentary); the broader Frits Staal archive at the *Bancroft Library* of the University of California, Berkeley (where Staal's papers and materials were deposited after his death in 2012).

cross-shakha-verification-fieldwork

Short: Cross-*śākhā* verification fieldwork extends beyond the Staal 1975 Nambūdiri expedition: the BORI *Mahābhārata* Critical Edition (Sukthankar et al., 1933–1966); Wayne Howard's *Sāmavedic* fieldwork (*Sāmavedic Chant*, Yale University Press, 1977) across Nambūdiri Kerala / Kauthuma Maharashtra / Jaiminiya Tamil Nadu lineages; Madeleine Biarreau's career-long *śrauta* documentation at École Pratique des Hautes Études; archives at the *Veda Rakṣaṇa Nidhi* Trust, *Vaidika Samshodhana Mandala* Pune, Tirupati Sri Venkateswara Vedic University, and Banaras Hindu University Vedic Research Centre. Where lineages diverge, the divergences are catalogued, named, and located at the level the *Prātiśākhya* texts predict — independent witnesses to a common source, with witness testimony agreeing.

Deployments: Chapter 15 §15.4 ¶ — the citation anchor for the cross-*śākhā* verification fieldwork on Vedic recitation.

The subsequent fieldwork on Vedic recitation across multiple *śākhā* lineages has extended the corpus of recordings beyond the Staal 1975 Nambūdiri expedition. The relevant fieldwork:

- **The Mahabharata Project** — Bhandarkar Oriental Research Institute, Pune (Sukthankar et al., 1933–1966) — produced extensive textual-critical scholarship comparing manuscript and recitation lineages across the major *śākhā* paramparā.
- **Wayne Howard's *Sāmavedic* fieldwork** (documented in

Sāmavedic Chant, Yale University Press, 1977) — extensive recordings of *Sāmavedic* recitation across multiple lineages (Nambūdiri Kerala, Kauthuma Maharashtra, Jaiminiya Tamil Nadu) demonstrating the recitation preservation across the geographic spread.

- **Madeleine Biarreau’s fieldwork on Vedic ritual** (across her career at the École Pratique des Hautes Études) extending the documentation of the *Agnicayana* and other major *śrauta* rituals across multiple lineages.
- **The Sangrahalayas of Vedic Recitation** — institutional repositories at the *Veda Rakṣaṇa Nidhi* Trust, the *Krishnamacharya Yoga Mandiram*, the *Sankara Math* archival projects, the *Ma-harshi Vedic University* projects, and the *Indira Gandhi National Centre for the Arts* documentation programs.
- **Contemporary fieldwork:** ongoing recording projects by Indian Vedic-research institutions (Tirupati Sri Venkateswara Vedic University; Banaras Hindu University Vedic Research Centre; Pune Vaidika Samshodhana Mandala) and international collaboration projects.

The empirical observation the chapter establishes: where the lineages diverge in their recitations, the divergences are *catalogued, named, and located* — *this śākhā has this reading at this point; that śākhā has that reading at that point*. The *Prātiśākhya* of each *śākhā* lays out the phonetic constants the *śākhā* preserves; the differences across lineages are at the level the *Prātiśākhya* texts *predict* rather than at the level of free drift. The lineages are *independent witnesses* to a common source; the *witness testimony* agrees.

Standard references: the works enumerated above; the broader literature on Vedic recitation as engineered preservation; the contemporary documentation projects available through the named institutions.

unesco-vedic-chanting-2003

Short: UNESCO recognized *Vedic chanting* on its *Masterpieces of the Oral and Intangible Heritage of Humanity* list in 2003 (Inscription Number 00062; transferred 2008 to the *Representative List of the Intangible Cultural Heritage of Humanity*); citation submitted by the Indian government, referencing the multi-channel preservation engineering (eleven *pāthas*), cross-lineage continuity across geographically separated paramparā, unbroken *guru-shishya* transmission across many generations, and the integration of phonetics, meter, syntax, and recitation in a unified system.

Deployments: Chapter 15 §15.4 ¶ — the citation anchor for the UNESCO 2003 recognition of Vedic chanting.

UNESCO recognized *Vedic chanting* (the *Tradition of Vedic Chanting*) on its list of *Masterpieces of the Oral and Intangible Heritage of Humanity* in 2003, with the citation submitted by the Indian government. The recognition was transferred in 2008 to the *UNESCO Representative List of the Intangible Cultural Heritage of Humanity* when the Masterpieces program was reorganized.

The UNESCO citation explicitly references:

- The multi-channel preservation engineering — the eleven *pāthas* and the broader phonetic-recitational redundancy system.
- The cross-lineage continuity across geographically separated Vedic-paramparā lineages.
- The unbroken transmission across many generations through the *guru-shishya paramparā*.
- The integration of phonetics, meter, syntax, and recitation in a unified system.

The structural significance the chapter establishes: the UNESCO recognition is, in itself, a small thing — UNESCO's list is a Western-establishment's gesture toward an architecture Western establishments have struggled to read for over a century. But the citation matters because UNESCO had to look at what was actually in front of it, and what was in front of it was an engineering accomplishment that has no comparable analog in any other ancient linguistic tradition.

The official UNESCO documentation is available at unesco.org under the Intangible Cultural Heritage list; the inscription nomination file is the *Tradition of Vedic Chanting* (Inscription Number 00062, originally proclaimed 2003, transferred to the Representative List in 2008). The official summary describes the tradition as “the precise and unbroken transmission of the Vedic texts across many generations through *guru-shishya paramparā*.”

Standard references: UNESCO Intangible Cultural Heritage documentation at unesco.org; the nomination file submitted by the Indian government; subsequent scholarly engagement with the UNESCO recognition (the *International Journal of Intangible Heritage*; various conference proceedings).

masoretic-codification-timing

Short: The Hebrew Masoretic apparatus operates on a *consonantal text* (כְּתִיב *kēṭīv*) substantially fixed *before* Masoretic codification began — Dead Sea Scrolls (late Second Temple period) demonstrate the consonantal text operating in approximately its current form well before the Masoretes' work; the Masoretic period proper (~6th–10th c. CE) developed vowel-pointing (*niqqud*), cantillation (*ṭe'amim*), and marginal apparatus (*Masora*) as preservation engineering *retrofitted* onto an already-canonical text. The contrast with the

Vedic case is structural: the eleven-*pāṭha* system is co-architected with the *Samhitā* text, not retrofitted onto a separately-canonical text.

Deployments: Chapter 15 §15.5 ¶ — the citation anchor for the chronology of the Masoretic codification relative to the underlying consonantal text.

The Masoretic apparatus (see endnote *masoretic-engineered-preservation*) operates on a *consonantal text* that was substantially fixed *before* the Masoretic codification activity began. The chronology:

- **Consonantal text** — the Hebrew Bible’s consonant-only form was substantially canonical by the Second Temple period (centuries BCE), with the Dead Sea Scrolls (from the late Second Temple period) demonstrating that the consonantal text was operating in approximately its current form well before the Masoretes’ work.
- **Tannaitic and Amoraic period** (first to fifth centuries CE) — the rabbinic tradition’s textual-critical engagement with the Hebrew Bible, preserved in the Talmudic literature. The consonantal text is treated as fixed at this stage; the rabbinic engagement is interpretive rather than text-establishing.
- **Masoretic period proper** (approximately sixth to tenth centuries CE) — the *Masoretes* at Tiberias and Babylonia developed the vowel-pointing systems, the cantillation marks, and the marginal apparatus. The consonantal text was *not* the object of Masoretic codification; it was the *input* to the Masoretic apparatus.

The structural significance the chapter establishes: the Masoretic apparatus was substantially codified *after* the canonization of the consonantal text. The preservation engineering was retrofitted onto an already-canonical text. The contrast with the Vedic recitation lineages is structural: the Vedic eleven-*pāṭha* system is *not* retrofitted

onto a separately-canonical text; the recitation engineering and the *Samhitā* text are co-architected, with the recitation system being what the *Samhitā* text *is* in its operating form. The chapter's broader comparative claim — that the Vedic preservation engineering is deeper than the Masoretic — rests on this structural distinction.

Standard references: see endnote masoretic-engineered-preservation for the broader Masoretic references; Frank Moore Cross, *From Epic to Canon: History and Literature in Ancient Israel* (Johns Hopkins University Press, 1998) on the canonization of the Hebrew Bible; Lawrence Schiffman, *Reclaiming the Dead Sea Scrolls* (Jewish Publication Society, 1994) on the Second Temple period textual evidence; James Kugel, *How to Read the Bible* (Free Press, 2007) on the rabbinic and Masoretic engagement.

quran-recitation-vs-pathas-comparison

Short: The Quranic *qirā'āt* (قراءات) preserve documented *variation* across canonical recitations — the seven and ten canonical readings are *different* readings of the Quranic text with variations in pronunciation, word-form, and interpretive significance, each internally consistent and exactly preserved across its own transmission line; the Vedic eleven *pāṭhas* do *not* preserve variation but re-encode *the same text* under combinatorial permutational arrangements. The strategies address different preservation problems: variant-recording (Quranic) vs. error-correcting redundancy over a single canonical text (Vedic).

Deployments: Chapter 15 §15.5 ¶ — the citation anchor for the comparative analysis between Quranic recitation traditions and the eleven Vedic *pāṭhas*.

The Quranic preservation system (see endnote quranic-engineered-preservation) operates significant overlap with the Vedic *Auditure* convention — both traditions operate exact-phonetic preservation through trained

recitation, both maintain documented chains of transmission, both produce communities of memorized-text specialists. The structural difference the chapter establishes:

- **The Quranic *qirā'āt* preserve documented variation** across canonical recitations. The seven and ten canonical *qirā'āt* are *different* readings of the Quranic text, with documented variations in pronunciation, occasionally in word-form, and occasionally in interpretive significance. Each *qirā'at* is internally consistent and exactly preserved across its own transmission line; the *qirā'āt* together provide the canonical variant-record.
- **The Vedic eleven *pāṭhas* do not preserve variation; they preserve the same text under combinatorial permutational re-encoding.** The *Samhitā-pāṭha*, *Pada-pāṭha*, *Krama-pāṭha*, *Jaṭā-pāṭha*, *Ghana-pāṭha*, and the six *vikṛti-pāṭhas* all encode the *same* sequence of words and *sandhi* relations under different permutational arrangements. The eleven recitations are not eleven different texts; they are eleven different recordings of one text.

The combinatorial-permutation strategy of the Vedic *pāṭhas* is functionally analogous to error-correcting codes in modern information theory: redundant re-encodings of the same information stream provide error-detection-and-correction at the channel level. A phoneme-level error in one *pāṭha* produces an inconsistency with the other *pāṭhas* that the trained reciter (or the trained verifier) can detect and correct.

The Quranic *qirā'āt*, by contrast, preserve variants. They do not function as error-correcting codes over a single information stream; they function as the documented record of canonical recitation diversity.

The structural significance the chapter establishes: both preservation strategies are sophisticated. The Vedic strategy — combinatorial re-encoding of a single text — produces deeper error-correction redundancy than the Quranic strategy — variant-recording of canonical

readings. The two architectures address different preservation problems: the Vedic addresses *phoneme-level drift in a single canonical text*; the Quranic addresses *documenting canonical variation across recitation traditions*. The chapter does not argue that one strategy is *better*; it argues that the Vedic strategy is *unique* — no other ancient preservation tradition is documented to have built combinatorial re-encoding redundancy.

Standard references: see endnote `quranic-engineered-preservation` for the Quranic references; endnote `eleven-pathas` for the Vedic *pāṭha* references. Comparative literature: Frits Staal, *Mantras between Fire and Water* (Royal Netherlands Academy of Arts and Sciences, 1992); the *Encyclopaedia of the Qurʾān* (Brill, 2001–2006), entries on recitation and variant readings.

conlangs-tolkien-okrand

Short: J. R. R. Tolkien (1892–1973; Merton Professor of English Language and Literature at Oxford) built *Quenya* and *Sindarin* for his Middle-earth legendarium across five decades — Latin-and-Finnish and Welsh-and-Old-English aesthetics respectively, with full phonological inventories, derivational morphology, syntax, and vocabularies running to thousands of words; Marc Okrand (PhD linguistics, UC Berkeley 1977) built *Klingon* (*tlhIngan Hol*) for Paramount Pictures starting with *Star Trek III* (1984), with a deliberate-foreignness phonology and a non-Indo-European typological profile (object-verb-subject word order, suffix-stacking apparatus). Both honest about what they are — invented-from-scratch creative constructions, not recovered ancestor-languages; the honesty Schleicher’s 1868 *Avis akvāsas ka* lacks.

Deployments: Chapter 18 §18.2 ¶ — the citation anchor for the modern constructed-language (*conlang*) tradition’s honest framing as

invented-from-scratch artificial languages.

The constructed-language (*conlang*) tradition in the modern Anglo-phone literary and entertainment sphere operates with explicit acknowledgment that the languages are *invented* — built from scratch for fictional worlds or for specific creative purposes, not descended from any historical language community. The two foundational examples the chapter cites:

J. R. R. Tolkien (1892–1973) built *Quenya* and *Sindarin* — the two principal Elvish languages of his Middle-earth legendarium — across approximately five decades of philological invention. Tolkien was Professor of Anglo-Saxon (Bosworth and Rawlinson chair, 1925–1945) and then Merton Professor of English Language and Literature (1945–1959) at Oxford. His philological training informed the conlang project: Quenya was modeled on a Latin-and-Finnish phonological aesthetic; Sindarin on a Welsh-and-Old-English aesthetic. Both languages have full phonological inventories, derivational morphology, syntax, and vocabularies running to thousands of words. Tolkien's *History of Middle-earth* (twelve volumes, posthumously edited by Christopher Tolkien, HarperCollins, 1983–1996) preserves the extensive philological notes and successive drafts of the language-construction work. The standard references: Christopher Tolkien, ed., *The History of Middle-earth*; the *Vinyar Tengwar* and *Parma Eldalamberon* journals (the official Tolkien-Estate-authorized publications of Tolkien's linguistic papers); David Salo, *A Gateway to Sindarin: A Grammar of an Elvish Language from J.R.R. Tolkien's Lord of the Rings* (University of Utah Press, 2004).

Marc Okrand (born 1948) built *Klingon* (*tlhIngan Hol*) for Paramount Pictures, starting with the language work for *Star Trek III: The Search for Spock* (1984) and continuing through the subsequent *Star Trek* franchise productions. Okrand has a Ph.D. in linguistics from the University of California, Berkeley (1977; dissertation on Mutsun, a now-dormant California native language). His Klingon construction includes a complete phonology with the deliberate-

foreignness aesthetic (heavy aspiration, retroflex articulations, glottal-stop phonemes), agglutinative case-and-aspect morphology with a non-Indo-European typological profile (object-verb-subject word order, suffix-stacking grammatical apparatus), and a vocabulary the Klingon-speaking community has since extended through the *Klingon Language Institute* (founded 1992). The standard references: Marc Okrand, *The Klingon Dictionary* (Pocket Books, 1985; revised editions); *The Klingon Way* (Pocket Books, 1996); *Klingon for the Galactic Traveler* (Pocket Books, 1997); the *Klingon Language Institute* publications at kli.org.

The structural significance the chapter establishes: both conlang projects are *honest* about what they are. Tolkien and Okrand do not claim that Quenya or Klingon descend from any historical language community; they acknowledge the languages as invented-from-scratch creative constructions. The contrast the chapter develops: Schleicher's 1868 Proto-Indo-European fable was *also* an invented-from-scratch construction, but the philological-orthodoxy framing has treated it as the recovered ancestor of a real historical language family. The conlang tradition operates with the honesty the PIE reconstruction project lacks; the chapter's polemic mode at this passage exposes the structural-typological similarity (both projects build artificial languages from scratch) while naming the framing-difference (the conlang tradition acknowledges the invention; the PIE reconstruction project does not).

Standard references as enumerated above.

pie-term-history

Short: The term *Proto-Indo-European* (PIE) as a stable academic reference is *recent* — Schleicher's 1861 *Compendium* and 1868 fable used *indogermanisch* / *Ursprache*; *Proto-Indo-European* enters the

literature around 1905; the *PIE* abbreviation routinizes mid-20th century (Watkins's *American Heritage IE Roots* Appendix 1969; Pokorny's *Indogermanisches etymologisches Wörterbuch* 1959 uses *idg.* but English-secondary literature standardizes on *PIE*); ubiquitous in routine online reference (etymonline 2001, Wiktionary) only in the past quarter century. The apparent solidity is recent consolidation, not ancient certainty.

Deployments: Chapter 18 §18.4 ¶ — the citation anchor for the historical chronology of *Proto-Indo-European* as a stable term.

The term ***Proto-Indo-European*** (PIE) enters the academic literature as a stable terminological-conceptual reference relatively recently in the history of comparative philology:

- **Mid-nineteenth century:** Schleicher's *Compendium* (1861) and his fable (1868) use *indogermanisch* (Indo-Germanic) as the family-name and refer to the reconstructed common ancestor as the *Ursprache* (proto-language) or *indogermanische Ursprache*. The term *Proto-Indo-European* itself does not appear in this form.
- **Late nineteenth century:** The neogrammarian period (Brugmann's *Grundriss*, 1886–1893; the Karl Verner contributions; the broader Junggrammatiker school) consolidates the reconstruction methodology. The terminology stabilizes around *indogermanisch* in German scholarship and *Indo-European* in English-language scholarship; the proto-language is referred to as the *Ursprache* or *the parent language* or *common Indo-European*.
- **Early twentieth century:** The term *Proto-Indo-European* enters the academic literature around 1905 (the conventional dating in standard history-of-linguistics references). The term is used initially in specialized publications and gradually generalizes through the early-to-mid twentieth century.

- **Mid-twentieth century:** The *PIE* abbreviation enters routine academic usage in mid-twentieth-century scholarship. Watkins's contributions to the *American Heritage Dictionary's* Indo-European Roots Appendix (first edition 1969) use *PIE* as the standard abbreviation; Pokorny's *Indogermanisches etymologisches Wörterbuch* (1959) uses *idg.* as the German abbreviation but the English secondary literature standardizes on *PIE*.
- **Late twentieth century onward:** *PIE* becomes ubiquitous in routine academic and reference usage. The Online Etymology Dictionary (etymonline.com, launched 2001), Wiktionary, and the broader online etymological-reference ecosystem standardize on *PIE* as the routine terminological reference.

The structural significance the chapter establishes: the apparent solidity of *Proto-Indo-European* as a stable academic reference is *recent* — substantially a mid-twentieth-century consolidation that became routine fast enough that current students do not realize how recent the routine is. The chapter is not arguing against an ancient certainty; it is arguing against a recent academic-reference-apparatus consolidation that has been cemented in routine reference works during the past quarter century (see endnote *pie-cementing-recent-decades*).

Standard references: R. H. Robins, *A Short History of Linguistics* (Longman, 4th edition 1997); Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Routledge, 1998); Calvert Watkins, *The American Heritage Dictionary of Indo-European Roots* (Houghton Mifflin, multiple editions since 1969); Julius Pokorny, *Indogermanisches etymologisches Wörterbuch* (Francke Verlag, 1959); the history-of-PIE-terminology discussion in: George Cardona, *The Indo-Aryan Languages* (Routledge, 2003); J. P. Mallory and Douglas Q. Adams, *Encyclopedia of Indo-European Culture* (Fitzroy Dearborn, 1997).

thomason-kaufman-1988

Short: Sarah Grey Thomason and Terrence Kaufman, *Language Contact, Creolization, and Genetic Linguistics* (University of California Press, 1988) — foundational reframing of language-contact theory: any feature (structural or lexical) can in principle transfer between languages in contact; the intensity and duration of contact predicts the depth of structural transfer (four-level scale from casual contact to very strong cultural pressure, with predicted contact-transfer depths at each level). The framework supports the chapter's reversal hypothesis: prolonged multi-generational Sanskrit-bearing specialist contact with Central / West Asian languages would produce extreme structural-transfer effects.

Deployments: Chapter 18 §18.5 ¶ — the citation anchor for the Thomason-Kaufman framework on language contact and structural borrowing.

Sarah Grey Thomason and Terrence Kaufman, *Language Contact, Creolization, and Genetic Linguistics* (University of California Press, 1988). The book is the foundational late-twentieth-century reframing of language-contact theory in historical linguistics. The two principal claims the chapter draws on:

First, any feature — structural or lexical — can in principle transfer between languages in contact. The older assumption (going back to the neogrammarian period and earlier) that *grammar is borrowing-proof* — that contact can transfer vocabulary but cannot transfer structural-grammatical features — is definitively rejected. Thomason and Kaufman demonstrate, through extensive cross-linguistic case studies (the Australian situations, the African Pidgin/Creole continuum, the Anatolian-Indo-European contact zone, the Hindi-Persian-Arabic contact in early-modern north India,

the broader contact-typology literature), that grammatical features routinely transfer between contact languages under sufficiently intense and prolonged contact.

Second, the intensity and duration of contact predicts the depth of structural transfer. Light/brief contact transfers vocabulary primarily; intense/prolonged contact transfers vocabulary, phonology, morphology, and syntax. Thomason and Kaufman develop a four-level scale of contact intensity (casual contact; slightly more intense contact; more intense contact; intense contact; very strong cultural pressure) with predicted contact-transfer depths at each level.

The structural significance the chapter establishes: the reversal hypothesis the chapter develops — that the contact-transfer direction in pre-historic Indic-and-Indo-European contact ran from *Sanskrit-bearing specialists* outward to the Central-and-West-Asian natural languages, not inward to Indic — posits *exactly* the contact conditions Thomason and Kaufman identify as producing extreme contact effects. The Mitanni record (see endnote *mitanni-sanskritic-evidence*) is direct documentation of multi-generational specialist contact between Sanskrit-bearing experts and a non-Indic ruling apparatus in northern Mesopotamia. The Thomason-Kaufman framework supports the chapter's analytical move: prolonged, multi-generational engagement of Sanskrit-bearing specialists with the natural languages of Central and West Asia would have produced exactly the kind of structural transfer the reversal hypothesis names.

Subsequent literature extending the Thomason-Kaufman framework: Thomason, *Language Contact: An Introduction* (Edinburgh University Press, 2001); Yaron Matras, *Language Contact* (Cambridge University Press, 2009); Donald Winford, *An Introduction to Contact Linguistics* (Wiley-Blackwell, 2003); the *Journal of Language Contact* (Brill, 2007–present).

Standard reference: Sarah Grey Thomason and Terrence Kaufman,

Language Contact, Creolization, and Genetic Linguistics (University of California Press, 1988).

ross-metatypy-takia

Short: Malcolm Ross, “Contact-Induced Change and the Comparative Method: Cases from Papua New Guinea,” in Durie & Ross, eds., *The Comparative Method Reviewed* (Oxford University Press, 1996): 180–217 — develops the concept of *metatypy* (the most extreme contact-induced outcome: a language’s morphosyntax wholesale restructured to match a *model* language while retaining native vocabulary); canonical case *Takia* (Austronesian / Oceanic, Karkar Island PNG) restructured by contact with *Waskia* (Papuan / Trans-New Guinea), producing a language with Austronesian vocabulary and Papuan grammar. The chapter develops the reversal hypothesis by analogy: Sanskrit as model, contacted Central / West Asian languages as replicas, PIE as backward-projection of the aggregate replica features.

Deployments: Chapter 18 §18.5 ¶ — the citation anchor for Malcolm Ross’s *metatypy* framework with the *Takia*/*Waskia* case study.

Malcolm Ross developed the concept of *metatypy* in his 1996 paper “Contact-Induced Change and the Comparative Method: Cases from Papua New Guinea” (in *The Comparative Method Reviewed*, ed. Mark Durie and Malcolm Ross, Oxford University Press, 1996). The concept names the most extreme outcome of contact-induced structural change: *a language’s morphosyntax is wholesale restructured to match a model language, while the recipient retains its native vocabulary.*

The mechanism Ross developed: under conditions of prolonged bilingualism in a community with strong social/cultural orientation toward the *model* language, the speakers of the *replica* language progressively restructure their language’s morphosyntax to match the

model. The vocabulary remains native to the replica language; the grammar conforms increasingly closely to the model. Ross's central observation: metatypy is typically *asymmetric* — one language is the model, the other is the replica; the replica restructures itself toward the model; the model is structurally untouched.

The canonical case study: **Takia**, an Austronesian (Oceanic) language spoken on Karkar Island in Papua New Guinea, restructured by sustained contact with **Waskia**, a Papuan (Trans-New Guinea) language spoken on the same island. Over generations of bilingualism (most Takia speakers also spoke Waskia; Waskia speakers were largely monolingual or less commonly bilingual), the Takia community restructured Takia's morphosyntax to match Waskia's grammatical organization while preserving Takia's vocabulary. The result: Takia today carries Austronesian vocabulary but Papuan grammar — a language whose typological profile is at the boundary of two major language families.

The structural significance the chapter establishes: the Takia case is the canonical metatypy example because the asymmetry is clean (one model, one replica; structural restructuring; vocabulary preservation). The chapter develops the reversal hypothesis by analogy: *Sanskrit as the model; contacted languages of Central and West Asia as replicas; the aggregate of replica features projected backward by nineteenth-century European philologists as PIE*. The hypothesis is that the apparent grammatical commonalities the European philological project identified across the *daughter Indo-European languages* are not residue of a common ancestor (the PIE reconstruction) but rather residue of *common contact* with a single model language (Sanskrit, operated by the *Saptaṛṣi*-and-successor specialist transmission across the pre-historic period). The chapter develops this thesis at length in §§18.5–18.6 and across Chapter 19's transmission-mechanism analysis.

Standard references: Malcolm Ross, "Contact-Induced Change and the Comparative Method: Cases from Papua New Guinea," in Mark

Durie and Malcolm Ross, eds., *The Comparative Method Reviewed: Regularity and Irregularity in Language Change* (Oxford University Press, 1996), pp. 180–217. Ross’s subsequent papers extend the framework: “Metatypy” entry in the *International Encyclopedia of Linguistics* (Oxford University Press, second edition 2003); “Diagnosing Contact Processes from Their Outcomes,” in *Studies in Language* 37 (2013). For the Takia/Waskia specifically: Malcolm Ross, *Takia, a Language of Karkar Island, Madang Province* (Pacific Linguistics, Australian National University, 2002).

wiktionary-pasyati-suppletion

Short: The Wiktionary entry on Sanskrit पश्यति (*paśyati*, “he sees”) carries the orthodox-philological account that the root is suppletive — *paśyati* derives from PIE **spek-*, the remainder of the paradigm from PIE **derk-*; the chapter’s polemic alternative: the suppletion is *original* and engineered (the architects deliberately used the *paś-* (पश्) root for present-tense forms and the *drś-* (दृश्) root for aorist-tense forms as a productive feature), and daughter-language cognates display partial inheritance via contact-transfer from the operational Sanskrit paradigm, not from a starred two-PIE-root common ancestor.

Deployments: Chapter 18 §18.6 ¶ — the citation anchor for the Wiktionary etymological entry on Sanskrit *paśyati* and the orthodox philological ecosystem’s account of suppletion across the Indo-European verbal paradigm for *to see*.

The Wiktionary entry on the Sanskrit verb पश्यति (*paśyati*) (“he sees”, third-person singular present active of the root *paś-*) carries the following etymological account (as of the version cited):

*“The root is suppletive — although पश्यति (*paśyati*) derives from Proto-Indo-European **spek-*, the remainder of the morphological paradigm is derived from Proto-Indo-European*

**derk-*.”

The structural fact the chapter establishes: the entry posits *two* Proto-Indo-European starred roots (**spek-* and **derk-*) as the underlying source of the Sanskrit verb’s morphological paradigm. The Sanskrit paradigm itself operates a *suppletive* relationship — *paśyati* (present-stem forms) and *adrākṣīt* (aorist-stem forms) derive from different roots in the surface paradigm. The PIE reconstruction attributes the suppletion to *inheritance* from two distinct PIE roots that have been combined into a single Sanskrit paradigm.

The chapter’s polemic account: the suppletion explanation is the PIE reconstruction project’s standard move for any Sanskrit verbal paradigm that does not match a single reconstructable root. *Posit two PIE roots, derive the paradigm’s parts from each, treat the suppletion as inheritance from the proto-language.* The move preserves the framework’s theoretical commitments (every Sanskrit form must derive from some PIE form; the reconstruction must produce the daughter paradigms) at the cost of multiplying entities — two roots, with their reconstructed paradigms, their reconstructed semantic relations, their reconstructed cross-daughter-language correspondences, all to handle the suppletion that the Sanskrit paradigm itself displays.

The alternative account the reversal hypothesis supports: Sanskrit’s suppletive paradigm is the *original* paradigm — engineered by the architects with the suppletion as a productive feature of the verbal-derivation system, with the *paś-* root carrying the present-tense forms and the *drś-* root carrying the aorist-tense forms by design. The daughter Indo-European languages that display partial cognates of one or both Sanskrit roots inherit not from a starred PIE-paradigm but from the operational Sanskrit paradigm via contact-transfer. The reconstruction project’s two-PIE-root postulation is residue of the same operation applied in reverse — the project working backward from the Sanskrit paradigm and projecting two PIE roots that the daughter languages would need to have inherited.

The Wiktionary entry is one of many instances where the chapter's analysis can be tested against routine etymological reference. The book's broader engagement with the routine reference ecosystem (etymonline, Wiktionary, the dictionary etymological notes) develops across Chapter 18 §§18.4–18.6 and across the cementing-of-PIE analysis at endnote pie-cementing-recent-decades.

Standard references: the Wiktionary entry on पश्यति (paśyati) at en.wiktionary.org; the broader PIE-reconstruction literature for the *spek- and *derk- roots; Manfred Mayrhofer, *Etymologisches Wörterbuch des Altindoarischen* (Carl Winter, 1986–2001), s.v. paś-, dṛś-; Helmut Rix, *Lexikon der indogermanischen Verben* (Reichert Verlag, second edition 2001), entries on the relevant PIE verbal roots.

agastya-sources

Short: The *Agastya* (अगस्त्य) paramparā is documented across both northern and southern textual lineages: northern sources (*R̥gveda* hymns 1.165–1.191 attributed to Agastya with Lopāmudrā; *Mahābhārata Vana Parva adhyāyas* 96–108 with the *Vindhya-bowing* episode; *Rāmāyaṇa* references in the *Aranya Kāṇḍa*) and southern Tamil sources (the *Agattiyam* (अगत्तियम्) — first Tamil grammar attributed to Agastya, non-extant today but cited in *Tolkāppiyam* commentaries; the *Velvikkudi* and *Chinnamanoor* copper-plate inscriptions naming Agastya as priest, Tamil teacher, and Pandya-dynasty coronation-performer). Both lineages agree on north-to-south travel, teaching role, and foundational status — the *recalibrant transmission* as absorption, not replacement.

Deployments: Chapter 19 §19.1 ¶ — the citation anchor for the *Agastya* paramparā and the dual northern-and-southern textual evidence for his role.

The *Agastya* paramparā is documented across multiple bodies of textual evidence:

Northern (Vedic and post-Vedic) sources: *Ṛgveda* hymns 1.165–1.191 are attributed to *Agastya* (co-authored with *Lopāmudrā*); the *Mahābhārata* (*Vana Parva*, *adhyāyas* 96–108) develops the *Agastya* narrative substantially, including the famous *Vindhya-bowing* episode (*Agastya* travels south, crosses the *Vindhya*s, and the *Vindhya* mountain bows in respect, agreeing to remain bowed until *Agastya* returns from the south — a narrative the paramparā reads as the metaphorical opening of the southern subcontinent to Vedic transmission); the *Rāmāyaṇa* references *Agastya* at multiple points in the *Aranya Kaṇḍa*.

Southern (Tamil-paramparā) sources: the *Agattiyam* (अगत्तियम्) — traditionally credited as the first Tamil grammar, attributed to *Agastya*. The text is *non-extant today*; it is referenced in fragments and citations by medieval Tamil commentators, including in the *Tolkāppiyam* commentarial literature. The *Velvikkudi* and *Chinamananoor* copper-plate inscriptions of the *Pandya* dynasty record *Agastya* as priest, Tamil teacher, and coronation-performer for the *Pandya* royal lineage; the *Velvikkudi* grant (dated to the eighth century CE by the conventional chronology) names *Agastya* in the genealogical preamble.

The convergence: both northern and southern lineages agree on (a) *Agastya's north-to-south travel*, (b) his *teaching role*, and (c) his foundational status in the southern textual continuum. The differences are in *emphasis*: northern legends emphasize his role in spreading the Vedic corpus; southern lineages emphasize his role in spreading agriculture, irrigation, and Tamil grammar. The two emphases are complementary, not contradictory.

The structural significance the chapter establishes: the *recalibrant transmission* the chapter analyzes — the transmission of analytical apparatus across the linguistic-and-cultural boundary between Sanskrit and Tamil — did not require the destination paramparā to be

erased. It required the destination paramparā to *absorb* a new analytical apparatus carried by an expert. Tamil paramparā records the absorption gracefully — Agastya is *the father of the Tamil language*, not its *replacer*. The Tamil continuum continues to operate, with the absorbed apparatus extending the paramparā's analytical reach rather than displacing the Tamil-internal content.

Standard references: For the northern textual evidence: *R̥gveda* 1.165–1.191; *Mahābhārata*, *Vana Parva*, *adhyāyas* 96–108. For the southern Tamil evidence: Kamil Zvelebil, *The Smile of Murugan* (Brill, 1973); A. Veluppillai, *Epigraphical Evidences for Tamil Studies* (International Institute of Tamil Studies, 1980); the *Velvikkudi* inscription in *Epigraphia Indica* Volume 17; T. P. Meenakshisundaran, *A History of Tamil Literature* (Annamalai University, 1965). For the broader Agastya paramparā: Madeleine Biardeau, *Hinduism: The Anthropology of a Civilization* (Oxford University Press, 1994); Stephanie Jamison and Joel Brereton, translators, *The R̥gveda: The Earliest Religious Poetry of India* (Oxford University Press, 2014), three-volume translation with Agastya-attributed hymns in volume 1.

mitanni-sanskritic-evidence

Short: The Mitanni kingdom (northern Mesopotamia, c. 16th–14th centuries BCE conventional chronology) preserves four distinct lines of Sanskritic-linguistic evidence: (1) the Hittite-Mitanni treaty (CTH 51 / KBo I 1) invokes *Mitra*, *Varuṇa*, *Indra*, and *Nāsatya* as treaty-witness deities; (2) the Kikkuli horse-training treatise (CTH 284–286) uses *aika* (= *eka*, structurally pre-Vedic), *tera* (*tri*), *panza* (*pañca*), *satta* (*sapta*), *na* (*nava*), *vartana* (*vartana*); (3) Mitanni throne names (*Tushratta* = *Tveṣaratha*, *Shattiwaza* = *Sātivāja*, *Indaruda* = *Indrota*, *Artashumara* = *Ṛtasmarā*); (4) the *marya* warrior-class term. Historical-empirical anchor for Wave 1 of the

recalibrant transmission — Sanskritic content as absorbed-from-elsewhere expert apparatus, not inherited indigenous tradition.

Deployments: Chapter 19 §19.2 ¶ — the citation anchor for the Mitanni Sanskritic-linguistic evidence.

The *Mitanni* kingdom of northern Mesopotamia (modern-day eastern Turkey, northern Iraq, and northeastern Syria) — at its peak from approximately the sixteenth to fourteenth centuries BCE by conventional chronology — preserves several distinct lines of evidence for Sanskritic-linguistic content in a non-Indic political-cultural context:

The Hittite-Mitanni treaty. The treaty between Suppiluliuma I (Hittite king) and Shattiwaza (Mitanni king-claimant), preserved on cuneiform tablets *CTH 51* (= *KBo I 1*, the principal tablet with the Hittite-perspective historical prologue) and *CTH 52* (the companion tablet with the Mitanni-perspective prologue) in the Bogazköy archive (the Hittite royal archive at Hattusa), invokes *Mitra*, *Varuṇa*, *Indra*, and *Nāsatya* (the *Aśvins*) as treaty-witness deities — four Vedic gods named explicitly in a non-Indic diplomatic document, in Sanskritic form. The cuneiform transliteration: *mi-it-ra*, *u-ru-wa-na* (also attested *a-ru-na*), *in-da-ra* (also *in-tar*), *na-ša-ti-ya-an-na* (with Hurrian plural-ending *-nna*).

The Kikkuli horse-training treatise. A 184-day, 1080-line manual on horse training composed on four cuneiform tablets (the *Kikkuli* tablets, catalogued as *CTH 284*, *285*, *286* in the Hittite archive; principal Late Hittite copy *CTH 284*, Middle Hittite copies *CTH 285* and *286*; the text dates from approximately the early-to-mid second millennium BCE in the standard Hittite chronology). The text uses Sanskritic numerical terms in specific contexts:

- *aika* = Sanskrit *eka* (one)
- *tera* = Sanskrit *tri* (three)
- *panza* = Sanskrit *pañca* (five)
- *satta* = Sanskrit *sapta* (seven)
- *na* = Sanskrit *nava* (nine)

- *vartana* = Sanskrit *vartana* (turn / rotation)

The form *aika* is structurally pre-Vedic-Sanskritic — Vedic Sanskrit has *eka*, with the contraction *ai > e* — placing the Mitanni Sanskritic layer in a phonologically *pre-Vedic-Sanskritic* position. The Mitanni Sanskritic content therefore represents a Sanskritic-linguistic stratum that precedes the Vedic mode's phonological contractions.

Mitanni rulers' throne names. The Mitanni royal lineage carries Sanskritic-derivable throne names:

- *Tushratta* → *Tveṣaratha* (*tveṣa-ratha* = brilliant-chariot).
- *Shattiwaza* → *Sātivāja* (*sāti-vāja* = victorious-power).
- *Indaruda* → *Indrota* (*indra-uta* = aided-by-Indra).
- *Artashumara* → Mitanni-Aryan *Artasmara* / Vedic cognate *Ṛtas-mara* (*ṛta-smara* = recalling-ṛta / cosmic order).

The *marya* warrior-class. Mitanni warriors are called *marya* — the Sanskrit term for (*young*) *warrior*. The term is documented in the Hittite-Mitanni records and is preserved in the political-administrative vocabulary of the Mitanni state.

The structural significance the chapter establishes: the Mitanni Sanskritic evidence is the historical-empirical anchor for Wave 1 of the *recalibrant transmission* — the carrying of Sanskritic linguistic, cultural, and intellectual apparatus by expert specialists into non-Indic political-cultural zones. The evidence is in a non-Indic diplomatic document, in a non-Indic horse-training manual, in non-Indic royal nomenclature, in non-Indic warrior-class vocabulary. The Sanskritic content is *not* the inherited indigenous tradition of the Mitanni state; it is the absorbed-from-elsewhere expert apparatus the Mitanni ruling elite operated alongside their own Hurrian-language native tradition.

Standard references: Paul Thieme, “The ‘Aryan’ Gods of the Mitanni Treaties,” *Journal of the American Oriental Society* 80.4 (1960): 301–317 — the canonical settlement of the treaty-deity identifications. Manfred Mayrhofer, *Die Indo-Arier im alten Vorderasien: Mit einer an-*

alytischen Bibliographie (Otto Harrassowitz, Wiesbaden, 1966) — the standard bibliographic-analytical reference on the Indo-Aryan content in the Near East; Mayrhofer, *Etymologisches Wörterbuch des Altindoarischen* (Carl Winter, Heidelberg, 1992–2001) for the onomastic corpus. Annelies Kammenhuber, *Die Arier im Vorderen Orient — ein Mythos?* (Carl Winter, Heidelberg, 1968) — the skeptical-polemic companion that the field has continued to engage. Edwin Bryant, *The Quest for the Origins of Vedic Culture: The Indo-Aryan Migration Debate* (Oxford University Press, 2001), the linguistic-evidence chapters on the Mitanni record; Edwin Bryant and Laurie Patton, eds., *The Indo-Aryan Controversy: Evidence and Inference in Indian History* (Routledge, 2005) — the standard edited volume in the debate. Frits Staal, *Discovering the Vedas* (Penguin India, 2008) for the broader contextual analysis. For the Kikkuli text: Annelies Kammenhuber, *Hippologia hethitica* (Otto Harrassowitz, Wiesbaden, 1961); Frank Starke, *Ausbildung und Training von Streitwagenpferden: Eine hippologisch orientierte Interpretation des Kikkuli-Textes* (Studien zu den Boğazköy-Texten 41, Otto Harrassowitz, 1995); Peter Raulwing, *The Kikkuli Text: Hittite Training Instructions for Chariot Horses* (open-access via LRGAF, 2009) — modern interdisciplinary survey. For the *mariyannu* institutional account: Eva von Dassow, *State and Society in the Late Bronze Age: Alalah under the Mittani Empire* (CDL Press, 2008). For the broader linguistic positioning: Alexander Lubotsky, “The Indo-Iranian Substratum,” in C. Carpelan et al., eds., *Early Contacts between Uralic and Indo-European* (Mémoires de la Société Finno-Ougrienne, 2001).

behistun-inscription

Short: The Behistun Inscription (*Bīsotūn*; multilingual rock relief at Mount Behistun, western Iran; commissioned by Darius I, c. 522–486 BCE; in Old Persian, Elamite, Akkadian; deciphered by Henry Rawl-

inson 1835–1847) — when transliterated into Devanāgarī, the Old Persian text reads with a fluent student of Sanskrit, while a fluent modern Persian speaker does *not* have the same experience (modern Persian has drifted substantially while Sanskrit preserves the engineered form). The smaller worked-example anchor for the calibrant’s anti-entropy function: Sanskrit preserves, contact-zone languages of Iran-and-beyond drift.

Deployments: Chapter 19 §19.2 ¶ — the citation anchor for the Behistun inscription as the secondary empirical anchor for the Sanskritic-Persian cognate proximity.

The *Behistun Inscription* — *Bīsotūn* in modern Persian — is a multilingual rock relief carved on a cliff face at Mount Behistun in western Iran, commissioned by the Achaemenid king Darius I (reigned 522–486 BCE). The inscription is carved in three languages: *Old Persian* (the principal language), *Elamite*, and *Akkadian* (Babylonian). The text narrates Darius’s accession to the throne and his suppression of rebellions across the Achaemenid Empire.

The decipherment of the Old Persian text by Henry Rawlinson across 1835–1847, building on earlier work by Grotefend, was foundational to nineteenth-century Iranian and Mesopotamian philology. The inscription’s structural feature relevant to the chapter:

When transliterated into Devanāgarī, the Behistun inscription can be read today by a fluent student of Sanskrit. The grammar, the root structures, much of the vocabulary — all of it sits within recognizable distance of the language a modern Indian student is trained in. The Old Persian forms display systematic phonological correspondences to Sanskrit forms (the Indo-Iranian *s* → *h* shift; the Old Persian preservation of certain consonant clusters Sanskrit simplifies; the Old Persian-Sanskrit cognate sets for common vocabulary).

The structural significance the chapter establishes: a fluent *modern* speaker of Persian, presented with the same text, does *not* have the same experience. Modern Persian has drifted substantially from the

Old Persian form — vocabulary cycled, phonology shifted, grammar simplified. Modern Persian carries the *cognate-shadow* of the Old Persian; Sanskrit preserves the engineered specification that Old Persian operated on as a contact-language. The two languages — Old Persian and Vedic Sanskrit — were close kin in their attested forms; they have ended up in radically different relationships to their own past. Sanskrit operates today as the calibrant-anchored language that preserves the engineered form; Persian operates today as the natural-language descendant of its Old Persian form, with the descent visible in the drift.

The Behistun observation is the chapter's smaller worked-example anchor for the broader claim about the recalibrant-transmission asymmetry: Sanskrit preserves; the contact-zone languages of Iran-and-beyond drift; the asymmetry is the calibrant's anti-entropy function operating across the depth of time.

Standard references: Roland G. Kent, *Old Persian: Grammar, Texts, Lexicon* (American Oriental Society, second edition 1953); Rüdiger Schmitt, *The Bisitun Inscriptions of Darius the Great: Old Persian Text* (Corpus Inscriptionum Iranicarum, Part I, Volume I, Texts I; School of Oriental and African Studies, 1991); the broader Old Persian philological literature in *Acta Iranica* and the *Corpus Inscriptionum Iranicarum* series; J. Wiesehöfer, *Ancient Persia* (I.B. Tauris, 2001) for the historical-political context.

dionysius-thrax-techne

Short: *Dionysius Thrax* (Διονύσιος ὁ Θραῖξ, c. 170–90 BCE), Alexandrian grammarian, composed the *Téchnē Grammatikē* (Τέχνη Γραμματική — *Art of Grammar*) — the first complete systematic Greek grammar with phonology, the eight parts of speech, morphological paradigms, and analytical apparatus; active during

the period of sustained Greco-Indic contact (Megasthenes c. 302 BCE; Seleucid-Mauryan exchanges; Greco-Bactrian and Indo-Greek kingdoms; Ashokan bilingual edicts; the Buddhist mission to the Hellenistic world). Earlier Greek engagements with language (Plato's *Cratylus*, Aristotle, the Stoics) produced no complete formal description; the complete formal description appears in Alexandria, in the post-contact period — what the recalibrant-transmission hypothesis predicts.

Deployments: Chapter 19 §19.3 ¶ (Greek-grammar paragraph) — the citation anchor for Dionysius Thrax's *Téchnē Grammatikē* as the first systematic Greek grammar.

Dionysius Thrax (Διονύσιος ὁ Θρᾷξ) (c. 170 BCE – c. 90 BCE), Alexandrian grammarian, composed the *Téchnē Grammatikē* (Τέχνη Γραμματική — *Art of Grammar*) in Alexandria during the late Hellenistic period. The text is the first complete systematic Greek grammar — covering phonology, morphology (the eight parts of speech, with their case-and-tense paradigms), syntax in introductory form, and the broader analytical apparatus of grammatical description.

The dating context: Dionysius Thrax was active in Alexandria during the period of sustained Greco-Indic contact following Alexander's campaigns (334–323 BCE), the Mauryan-Seleucid exchanges (Megasthenes's residence at the Mauryan court c. 302 BCE; the Seleucid dynastic intermarriage with the Mauryans; the broader Hellenistic-Mauryan diplomatic apparatus), the Greco-Bactrian and Indo-Greek kingdoms (200 BCE onward), Ashoka's Greek-language edicts (the bilingual Aramaic-Greek edicts at Kandahar, third century BCE), the Buddhist mission to the Hellenistic world (third century BCE forward), and the well-documented intellectual circulation between Alexandria and the subcontinent.

The Greek philosophical tradition before Dionysius Thrax's codification had been speculating about language for centuries: Plato (in the

Cratylus) asked whether names are conventional or natural; Aristotle (in the *De Interpretatione* and the *Categories*) made grammatical observations in passing; the Stoics developed a parts-of-speech analysis. None of these earlier engagements produced a complete formal description. The complete formal description appears in Alexandria, in the post-contact period.

The structural significance the chapter establishes: the timing is not proof of Indic transmission — but it is what the recalibrant-transmission hypothesis *predicts*. The Pāṇinian analytical apparatus had been operating in the subcontinent for many generations before Dionysius Thrax. The hypothesis posits that the methodology — the systematic-formal grammar with parts of speech, morphological paradigms, and rule-based descriptions — was transmitted from the Pāṇinian discipline through the contact apparatus into the Alexandrian intellectual milieu, where Dionysius Thrax's text consolidated it for the Greek-language tradition.

Standard references: the *Téchnē Grammatikē* in its standard editions. *Grammatici Graeci* Volume I (G. Uhlig, ed., Teubner, 1883) is the standard scholarly edition. Translation: Alan Kemp, "The Tekhne Grammatike of Dionysius Thrax" (in *Historiographia Linguistica* 13, 1986). Modern scholarly treatments: Vivien Law and Ineke Sluiter, eds., *Dionysius Thrax and the Technē Grammatikē* (Nodus Publikationen, 1995); Andrew Robins, *A Short History of Linguistics* (Longman, 4th edition 1997), Chapter 2; Daniel J. Taylor, *Declinatio: A Study of the Linguistic Theory of Marcus Terentius Varro* (John Benjamins, 1974) for the broader Greco-Roman grammatical tradition.

donatus-priscian-grammars

Short: *Aelius Donatus* (c. 320–380 CE) composed the *Ars Maior* (and *Ars Minor*) — the standard Latin grammars codifying Latin anal-

ysis on the Greek model with the eight parts of speech and Latinized Greek terminology; **Priscian** (Priscianus Caesariensis, early 6th c. CE) composed the *Institutiones Grammaticae* in Constantinople in eighteen books covering Latin phonology, morphology, syntax, and rhetorical-stylistic analysis — the standard reference text for Latin grammatical study across the medieval European period. Both transparently downstream of Greek methodology — three diffusion-steps from Pāṇinian methodology (Pāṇini → Greek → Latin → medieval European).

Deployments: Chapter 19 §19.3 ¶ (Latin-grammar paragraph) — the citation anchor for Donatus’s *Ars Maior* and Priscian’s *Institutiones Grammaticae* as the foundational Latin grammars transparently downstream of Greek grammatical methodology.

Aelius Donatus (c. 320 – c. 380 CE), Roman grammarian, composed the *Ars Maior* (and the shorter *Ars Minor*) — the standard Latin grammars of the late Roman period. Donatus’s works codified Latin grammatical analysis on the Greek grammatical model, using Greek terminology adapted to Latin material. The eight parts of speech of the Greek tradition (the *Téchnē Grammatikē*’s framework) become the eight parts of speech of Donatus’s Latin grammatical apparatus.

Priscian (Priscianus Caesariensis) (early sixth century CE), Latin grammarian active in Constantinople, composed the *Institutiones Grammaticae* in approximately the early-to-mid sixth century. The work is the most extensive Latin grammar produced in antiquity — eighteen books covering Latin phonology, morphology, syntax, and rhetorical-stylistic analysis. Priscian’s work was the standard reference text for Latin grammatical study across the medieval European period.

Both texts are transparently downstream of Greek grammatical methodology. They use Greek terminology (often Latinized but recognizable), Greek analytical categories (the eight parts of speech, the case-system framework, the morphological-paradigm

presentation), and Greek analytical structure (the discrete-rules-applied-to-categories format) applied to Latin material. The Greek grammatical tradition's influence is acknowledged in the texts themselves — Donatus and Priscian both cite Greek grammarians as their methodological antecedents.

The structural significance the chapter establishes: if Greek grammar is downstream of Pāṇinian methodology (the chapter's hypothesis at endnote dionysius-thrax-techné), Latin grammar is *doubly so*. The European grammatical tradition that becomes the foundation of medieval and Renaissance linguistic thought — the tradition that produced Schleicher's family-tree model and the entire nineteenth-century apparatus of comparative philology — is the *great-grandchild* of Pāṇini, three diffusion-steps removed (Pāṇinian → Greek → Latin → medieval European) and still carrying the methodological DNA of the original engineered framework.

Standard references: *Donati Ars Grammatica* in *Grammatici Latini* Volume IV (H. Keil, ed., Teubner, 1864). Translation: Wayland Johnson Chase, *The Ars Minor of Donatus* (University of Wisconsin Press, 1926); Louis Holtz, *Donat et la tradition de l'enseignement grammatical* (Editions du CNRS, 1981). For Priscian: *Prisciani Institutiones Grammaticae* in *Grammatici Latini* Volumes II–III (H. Keil, ed., Teubner, 1855–1859); Marc Baratin, *La naissance de la syntaxe à Rome* (Editions de Minuit, 1989); R. H. Robins, *Ancient and Mediaeval Grammatical Theory in Europe* (Bell, 1951). For the broader influence chain: Vivien Law, *The History of Linguistics in Europe: From Plato to 1600* (Cambridge University Press, 2003).

thonmi-sambhota-tibetan-grammars

Short: *Thonmi Sambhoṭa* — traditionally credited founder of Tibetan grammatical analysis and originator of the Tibetan script;

dispatched to India by Tibetan king **Songtsen Gampo** (Srong-btsan sgam-po, reigned c. 618–649 CE) to study Sanskrit grammatical methodology. Returned with: (a) the Tibetan script (Brāhmī-derived, mirroring the *varṇamālā*’s *sthāna* organization), (b) the foundational Tibetan grammars **Sum cu pa** (*Sum-cu-pa* — *The Thirty Verses*) and **Rtags kyi ’jug pa** (*rTags kyi ’jug pa* — *The Application of Signs*), both explicitly Pāṇinian in methodology (*sūtra* format, systematic phonological-morphological categorization, abbreviation conventions, metarule apparatus). Documentary anchor for direct Pāṇinian transmission across a major linguistic-cultural boundary.

Deployments: Chapter 19 §19.3 ¶ (Tibetan-grammar paragraph) — the citation anchor for Thonmi Sambhoṭa’s foundational works of Tibetan grammatical analysis as explicitly Pāṇinian.

Thonmi Sambhoṭa (Thon-mi Sam-bho-ṭa, transliterated variously) is the traditionally-credited founder of Tibetan grammatical analysis and the originator of the Tibetan script. Traditional Tibetan-historiographical accounts (preserved in the standard Tibetan religious-historical chronicles — the *Bka’ chems ka khol ma*, the *Mani bka’ ’bum*, the broader Tibetan-language historical literature) record:

- In the seventh century CE, the Tibetan king **Songtsen Gampo** (Srong-btsan sgam-po, reigned c. 618–649 CE) dispatched a mission to India specifically to study Sanskrit grammatical methodology and to develop a writing system for the Tibetan language.
- Thonmi Sambhoṭa led the mission. He spent several years in India studying with Sanskrit grammarians (the specific teachers named vary across the sources; the *Lalitavistara* tradition and various Tibetan-historiographical accounts cite Indian *paṇḍitas* whose Sanskrit names are preserved in Tibetan transliteration).
- On his return, Thonmi Sambhoṭa is credited with: (a) the design of the Tibetan script (modeled on Brāhmī, with adaptations

for the Tibetan phonological inventory); and (b) the authorship of the two foundational Tibetan grammatical works, *Sum cu pa* (*Sum-cu-pa — The Thirty Verses*) and *Rtags kyi 'jug pa* (*rTags kyi 'jug pa — The Application of Signs*).

Both Tibetan grammatical works are explicitly Pāṇinian in methodology — the sūtra-based rule format, the systematic phonological-and-morphological categorization, the abbreviation-conventions for compact rule-statement, the metarule-and-precedence apparatus for handling rule interactions. Both are foundational to every subsequent work in the Tibetan grammatical tradition; the medieval Tibetan grammatical commentators (across the *Sa-skyā*, *Bka'-brgyud*, *dGe-lugs*, and *rNying-ma* schools' grammatical commentaries) work within the framework Thonmi Sambhoṭa's foundational texts established.

The Tibetan script is *Brāhmī-derived*. The script's structural organization mirrors the *varṇamālā*'s organization: consonants by *sthāna* (place of articulation), with the velar / palatal / retroflex / dental / labial sequence preserved, with the four-fold voicing-and-aspiration array on each *varga* preserved in the Tibetan adaptation, with the vowel system adapted for the Tibetan phonological inventory.

The structural significance the chapter establishes: the Tibetan case is the documentary anchor for direct Pāṇinian transmission across a major linguistic-cultural boundary. The Tibetan tradition is *not* coy about the transmission — Tibetan-historiographical accounts explicitly name India as the source of the grammatical methodology and the writing system. Thonmi Sambhoṭa's mission is documented; the Tibetan grammatical works' Pāṇinian methodology is documented; the Brāhmī-derivation of the Tibetan script is documented. The chapter's broader argument about Pāṇinian methodology's transmission outward into non-Indic intellectual contexts is *directly* anchored at the Tibetan case.

Standard references: Roy Andrew Miller, *Studies in the Grammati-*

cal Tradition in Tibet (John Benjamins, 1976); the standard Tibetan grammatical-historical references: Karma sNang-rdor, *Bod kyi brda sprod rig pa'i lo rgyus* (in Tibetan, multiple editions); David Snellgrove, *The Hevajra Tantra* (Oxford University Press, 1959) for the broader Tibetan philological context; Rolf Stein, *Tibetan Civilization* (Stanford University Press, 1972) for the historical-cultural context of Songtsen Gampo's reign. For Thonmi Sambhoṭa specifically: Joseph Edward Walters, "An Introduction to the Tibetan Grammar" (in *The Indian Journal of Buddhist Studies*); the standard *Bka' chems ka khol ma* historical chronicle.

sibawayh-al-kitab

Short: *Sibawayh* (Sībūyih, c. 760–796 CE), Persian linguist in the Basran intellectual milieu of the early Abbasid period, composed *Al-Kitāb* (الكتاب — *The Book*) — the foundational and definitive Arabic grammar, basis of Arabic grammatical science to this day. Methodological signature: formal abbreviation conventions (analogous to Pāṇini's *pratyāhāra*), substitution-based analysis (analogous to *adeśa*), integrated multi-level phonological-and-morphological treatment. The Barmakid family (Buddhist-Bactrian-origin Persian viziers) oversaw translation programs bringing Indic mathematical, medical, and philosophical works into Arabic at exactly the moment Sibawayh composed *Al-Kitāb* — the *parallel development* claim must explain why grammar would be the one Indic export Basra somehow developed independently.

Deployments: Chapter 19 §19.3 ¶ (Arabic-grammar paragraph) — the citation anchor for Sibawayh's *Al-Kitāb* as the foundational Arabic grammar and its methodological-and-contextual proximity to the Pāṇinian-Basran intellectual milieu.

Sibawayh (Persian: Sībūyih; Arabic: Sībawayh) (c. 760 – c. 796 CE),

Persian linguist working in the Basran intellectual milieu in the early Abbasid period, composed *Al-Kitāb* (الكتاب — *The Book*) — the foundational and definitive Arabic grammar. The work remains the basis of Arabic grammatical science to this day, with the subsequent Arabic grammatical tradition (the *Kūfan* school's responses, the *Baṣran* school's continuations, the medieval and modern grammatical commentaries) operating within the framework *Al-Kitāb* established.

The methodological signature of *Al-Kitāb*:

- **Formal abbreviation conventions** — Sibawayh uses a system of formal abbreviations and notation conventions to compact the grammatical rules into terse, *sūtra*-like formulations, analogous to Pāṇini's *pratyāhāra* and metarule conventions in the *Aṣṭādhyāyī*.
- **Substitution-based analysis** — the grammar operates through systematic substitution rules (one form replaces another under specified conditions), analogous to Pāṇini's *adeśa* (substitution) rule-format.
- **Multi-level treatment of phonological and morphological structure** — the work is organized in a layered manner, with phonological analysis (the *taṣrīf* tradition) operating on top of morphological analysis (the *naḥw* tradition), in parallel to the *Aṣṭādhyāyī*'s integrated phonological-and-morphological apparatus.

Multiple scholars (including Carter, Versteegh, Owens) have noted that *Al-Kitāb*'s methodological signature shares specific features with Pāṇinian methodology. The Western philological orthodoxy's response to the observation has been *parallel development* — the claim that Sibawayh independently arrived at a methodology structurally similar to Pāṇini's because both languages have certain typological similarities that make the methodology natural.

The chapter's polemic position on the parallel-development claim:

Sibawayh was working in a milieu where Indic intellectual material was being *systematically translated*. The Barmakid family — the powerful Persian vizierial family of Buddhist Bactrian origin who served as Abbasid viziers — oversaw the translation programs that brought Indic mathematical, medical, and philosophical works into Arabic. The same period and milieu gave the Arabic-Islamic world: the Arabic numerals (Indic in origin); the decimal positional system (Indic); the numeral zero (Indic); the Indic astronomical-mathematical works (translated into Arabic from Sanskrit sources). Given this context — that Indic intellectual material was being translated into Arabic at exactly the moment Sibawayh was composing *Al-Kitāb* — the burden of proof should sit with the parallel-development claim. *Why would grammar be the one Indic export Basra somehow developed independently?*

Standard references: *Al-Kitāb* in its standard editions. The standard scholarly edition: Sibawayh, *Al-Kitāb*, edited by ‘Abd al-Salām Muḥammad Hārūn (Maktabat al-Khānjī, Cairo, multiple editions). Translation: Michael G. Carter, “Sibawayhi” entry in *Encyclopaedia of Islam* (second edition); Kees Versteegh, *The Arabic Linguistic Tradition* (Routledge, 1997); Carter, *Sibawayhi* (Oxford University Press, 2004). For the Pāṇinian-methodological proximity: J. Owens, *The Foundations of Grammar: An Introduction to Medieval Arabic Grammatical Theory* (John Benjamins, 1988); R. H. Robins, *A Short History of Linguistics* (Longman, 4th edition 1997), Chapter 5. For the broader Indic-Arabic intellectual transmission in the Abbasid period: Dimitri Gutas, *Greek Thought, Arabic Culture* (Routledge, 1998).

medieval-hebrew-grammarians

Short: Hebrew grammar was codified mediievally under explicit Arabic grammatical influence: *Saadia Gaon* (Sa’adyāh ben Yōsēf al-

Fayyūmī, 882–942 CE, Egyptian-born Gaon of Sura; the *Ha-Egron* dictionary and *Kutub al-Lugha* — earliest systematic Hebrew grammars, in Judeo-Arabic); **Judah ben David Hayyuj** (c. 945–1000 CE, Cordoba; three-letter-root analysis); **Jonah ibn Janah** (c. 990–1055 CE, Cordoba / Saragossa; *Kitāb al-Tanqīḥ*); **Abraham ibn Ezra** (1089–1164 CE, carried the tradition to Christian Europe in Hebrew); **David Kimhi** (Radak, c. 1160–1235 CE, Provence; *Sefer Mikhlol*). If Arabic grammar is downstream of Pāṇinian methodology, Hebrew grammar is *triply* downstream — Indic → Arabic → Hebrew.

Deployments: Chapter 19 §19.3 ¶ (Hebrew-grammar paragraph) — the citation anchor for the medieval Hebrew grammarians who codified Hebrew grammar under explicit Arabic grammatical influence.

Hebrew grammar was codified medievally, in the period from the tenth century CE onward, under explicit Arabic grammatical influence. The principal medieval Hebrew grammarians:

- **Saadia Gaon** (Saʿadyāh ben Yōsēf al-Fayyūmī) (882–942 CE), Egyptian-born Jewish scholar serving as Gaon of the Sura academy in Babylonia. Saadia’s grammatical works — the *Ha-Egron* (the first Hebrew dictionary, in two parts) and *Kutub al-Lugha* (Books of the Language, in Judeo-Arabic) — are the earliest systematic Hebrew grammars, composed in Judeo-Arabic with explicit references to Arabic grammatical methodology.
- **Judah ben David Hayyuj** (Yehuda ben David Hayyūj) (c. 945 – c. 1000 CE), Cordoban scholar of the Andalusian Jewish intellectual milieu. Hayyuj produced the foundational works on Hebrew verbal morphology that established the systematic three-letter-root analysis of Hebrew verbs. His works *Kitāb al-Afāl Dhawāt Ḥurūf al-Līn* (Book of the Verbs with Weak Letters) and *Kitāb al-Afāl Dhawāt al-Mithlayn* (Book of the Verbs with Doubled Letters) are written in Judeo-Arabic and explicitly model their analytical approach on Arabic grammatical methodology.

- **Jonah ibn Janah** (Yonāh ibn Janāḥ; also Abū al-Walīd Marwān ibn Janāḥ) (c. 990 – c. 1055 CE), Cordoban-and-Saragossan scholar of the Andalusian Jewish intellectual milieu. Ibn Janah's *Kitāb al-Tanqīḥ* (Book of Refinement) is the comprehensive medieval Hebrew grammar-and-dictionary, comprising the *Kitāb al-Luma'* (Book of Splendid Things, the grammar) and the *Kitāb al-Uṣūl* (Book of Roots, the dictionary). The work is composed in Judeo-Arabic with extensive citations of Arabic grammatical works.
- **Abraham ibn Ezra** (1089 – 1164 CE), Andalusian-born Jewish polymath who carried the Hebrew grammatical tradition northward into Christian Europe through his travels and his Hebrew-language grammatical works. *Sefat Yeter* (Excess of Language), *Moznayim* (Scales), *Tsahot* (Eloquence), *Sefer ha-Shem* (Book of the Name) — all written in Hebrew (rather than Judeo-Arabic) — translated the Andalusian Hebrew grammatical tradition for the broader medieval European Jewish readership.
- **David Kimhi** (Radak) (c. 1160 – c. 1235 CE), Provençal Jewish scholar. *Sefer Mikhlol* (the comprehensive Hebrew grammar) and *Sefer ha-Shorashim* (the dictionary) became the standard reference works for Hebrew grammar across the late-medieval and early-modern Jewish world.

The chapter's structural claim: the medieval Hebrew grammarians cite Arabic grammarians, use Arabic methodology adapted to Hebrew material, and produce the first systematic grammars of Hebrew. If Arabic grammar is downstream of Pāṇinian methodology (the chapter's hypothesis at endnote *sibawayh-al-kitab*), Hebrew grammar is *triply* downstream — Indic → Arabic → Hebrew. The methodology reaches the Abrahamic intellectual core through the chain *three diffusion steps* later than Greek to Latin but no less continuously.

Standard references: Geoffrey Khan, ed., *Encyclopedia of Hebrew Lan-*

guage and Linguistics (Brill, 2013, four volumes) — the standard reference; David Téné, “Hebrew Linguistic Literature” entry in the *Encyclopaedia Judaica* (multiple editions); Aaron Maman, *Comparative Semitic Philology in the Middle Ages: From Sa‘adiah Gaon to Ibn Barūn (10th–12th c.)* (Brill, 2004); Kees Versteegh, *Greek Elements in Arabic Linguistic Thinking* (Brill, 1977) for the broader medieval Arabic-Hebrew grammatical context.

jones-1786-anniversary-address

Short: Forward-pointer to the canonical jones-1786-third-anniversary-discourse endnote — Sir William Jones, “The Third Anniversary Discourse, on the Hindus,” delivered February 2, 1786 at the Asiatic Society of Bengal, Calcutta (*Asiatic Researches* 1, 1788: 415–431); deployed in Appendix Part 1 as the structural moment *before the inversion* — European philology still acknowledging Sanskrit’s depth as source-language of the family being constructed, before the mid-19th-century Schleicher inversion displaced Sanskrit from source position into daughter-language position in the reconstructed PIE-anchored framework. The bake had not yet been baked.

Deployments: Appendix Part 1 §1.1 ¶ — the citation anchor for Sir William Jones’s 1786 anniversary address at the Asiatic Society of Bengal, framing the founding direction of the European philological project before the inversion that produced PIE.

Sir William Jones, “The Third Anniversary Discourse, on the Hindus,” delivered February 2, 1786, at the Asiatic Society of Bengal, Calcutta. Published in *Asiatic Researches* Volume 1 (Calcutta, 1788), pages 415–431.

The Appendix deployment is the same canonical citation drafted at endnote jones-1786-third-anniversary-discourse. The Appendix’s framing of the citation differs from Chapter 8’s: where

Chapter 8 deploys the address as the founding moment of the European-philological absorption of Sanskrit grammatical analysis (the IPA-and-phonological-grid trajectory the chapter develops), the Appendix deploys it as the structural moment *before the inversion* — when European philology was still acknowledging Sanskrit’s depth as the source-language of the family being constructed, before the inversion in the mid-nineteenth century displaced Sanskrit from the source position into the daughter-language position in the reconstructed PIE-anchored framework.

The Appendix’s structural account: Jones’s 1786 address was the founding direction of the European philological project. Within five decades of that direction, the project’s professional descendants would *invert* the direction — replacing Sanskrit-as-ancestor with PIE-as-reconstructed-ancestor and Sanskrit-as-one-daughter-language-among-siblings. The inversion was not made visible to the contemporary Indian Sanskritists who supplied the textual, lexicographical, and grammatical material in the post-Jones generation; they were operating Sanskrit-internal work for *paramparā*-internal purposes, not anticipating that their work was being reverse-engineered into a starred-ancestor framework that would displace Sanskrit from its source position.

The chronological-and-structural sequence the Appendix develops: 1786 (Jones acknowledges Sanskrit’s depth) → 1816 (Bopp’s *Conjugationssystem* still positions Sanskrit as the ancestor or close to it) → 1861 (Schleicher’s *Compendium* introduces the family-tree model with a reconstructed common ancestor *distinct from* Sanskrit) → 1868 (Schleicher’s PIE fable; the operational completeness of the manufactured-ancestor framework) → late nineteenth century and onward (cementing of the PIE reconstruction as the standard reference). The Appendix’s polemic structure prosecutes this sequence as the *bake* — the manufacturing operation that produced PIE — with Jones’s 1786 address as the founding moment *before* the bake began.

Standard references: see endnote jones-1786-third-anniversary-discours

for the full reference ecosystem.

boden-chair-1832-evangelical-purpose

Short: The *Boden Chair of Sanskrit* at the University of Oxford (endowed 1832 from the will of Lieutenant Colonel Joseph Boden, 1751–1811, Bombay Native Infantry) carries a charter directing the chair to enable “*the conversion of Indians to Christianity*” through the agency of the Sanskrit-literate; Boden’s will preserved the explicit framing. Chairholders: *Horace Hayman Wilson* (1832–1860, *Sanskrit-English Dictionary* 1819); *Sir Monier Monier-Williams* (1860–1899, elected in the famous 1860 contested election against Max Müller — the chair’s evangelical charter decisive in Williams’s favor; the *Monier-Williams Sanskrit-English Dictionary* 1872 / 1899 remains standard today). The lexicographical apparatus of Sanskrit-English scholarship was produced in a chair endowed for the conversion of Indians to Christianity.

Deployments: Appendix Part 1 §1.1 ¶ — the citation anchor for the Boden Chair of Sanskrit at the University of Oxford and its evangelical-conversion charter.

The *Boden Chair of Sanskrit* at the University of Oxford was endowed in 1832 with funds left in the will of *Lieutenant Colonel Joseph Boden* (1751–1811), an officer of the Bombay Native Infantry. Boden’s will specifically directed that the chair be used to enable *the conversion of Indians to Christianity* through the agency of the Sanskrit-literate. The full intent, as set out in Boden’s will (preserved in the documentary record):

“... my particular wish is to promote the translation of the Scriptures into Sanskrit; so as to enable his countrymen to proceed in the conversion of the natives of India to the Christian religion.”

The chair was established at Oxford after Boden's bequest had accumulated sufficient endowment income (across the two decades following his death). The chair has been held continuously since 1832, with the chairholders:

- **Horace Hayman Wilson** (1832–1860). Wilson had served in India (Calcutta, with the Asiatic Society of Bengal and the East India Company's medical service) and brought substantial Sanskrit-textual expertise to the chair. His *Sanskrit-English Dictionary* (1819, with subsequent revised editions) was the standard reference.
- **Monier Williams (later Sir Monier Monier-Williams)** (1860–1899). Williams was elected to the chair in the famous 1860 contested election against Max Müller — an election in which the explicitly-evangelical character of the chair's charter was decisive in Williams's favor over the more philologically-rigorous Müller, whose Christian-conversion commitments were less politically reliable. Williams's *Sanskrit-English Dictionary* (1872, with the comprehensive 1899 revised edition co-authored with Ernst Leumann and Carl Cappeller) became the standard reference and remains so today as the *Monier-Williams Sanskrit-English Dictionary*.
- Subsequent chairholders across the twentieth century (Arthur Anthony Macdonell; Frederick William Thomas; Arthur Berriedale Keith; the post-Independence-era chairholders) continued the Sanskrit-lexicographical and philological work at Oxford under the chair's institutional umbrella.

The structural significance the Appendix establishes: Sanskrit lexicography in the English-reading world was substantially produced by men whose institutional charter was the *evangelical conversion of India*. The polemic register has been preserved in the documents the institutional successors continue to acknowledge — the *Boden in Boden Chair of Sanskrit* is on the public record at Oxford; the chair's

evangelical-conversion charter is part of the documentary apparatus the chair operates under, even as the contemporary chairholders may distance themselves from that original purpose.

The Appendix's polemic move: the institutional charter of the chair is *not* an embarrassing historical artifact to be quietly dropped; it is the structural origin of the lexicographical apparatus that the contemporary academic Indology continues to operate. The *Monier-Williams Dictionary* — the standard reference for Sanskrit-English lexicography today — was produced by the second Boden Professor, in a chair endowed for the conversion of Indians to Christianity. The lexicographical apparatus carries the institutional charter's framing forward in subtle ways the chapter develops.

Standard references: the documentary record of Joseph Boden's will (preserved in the Oxford University archives); Linda Colley, *Captives: Britain, Empire and the World, 1600–1850* (Pimlico, 2003) for the broader colonial-military-and-evangelical context; Christopher A. Bayly, *Empire and Information: Intelligence Gathering and Social Communication in India, 1780–1870* (Cambridge University Press, 1996) for the East India Company's intelligence-and-conversion apparatus; the institutional history of the Boden Chair in *Oxford University Calendar* and the *Encyclopaedia Britannica's* entries on Wilson and Williams.

deccan-college-founding-arc

Short: Deccan College Pune's institutional arc: founded 1821 as Sanskrit *Pāṭhaśālā* / *Hindoo College* under **Mountstuart Elphinstone** (Governor of Bombay Presidency 1819–1827), redirecting the ***Dakṣiṇā*** (दक्षिणा) charitable endowment that Peshwa Bajirao II had used to subsidize Sanskrit pundits in Pune; renamed *Poona College* (1851), *Deccan College* (1864), reconstituted post-Independence

(1948) as the *Deccan College Post-Graduate and Research Institute* — institutional home of the *Encyclopaedic Dictionary of Sanskrit on Historical Principles* (1948–present). The moment of institutional capture: Maratha-state patronage apparatus for *paramparā*-internal Sanskrit teaching converted into a hybrid colonial-educational institution.

Deployments: Appendix Part 1 §1.1 ¶ — the citation anchor for the founding arc of Deccan College, Pune.

Deccan College at Pune is the institution that names the Appendix (*The Encyclopaedic Confirmation* — Chapter Zero of the Appendix engages the *Encyclopaedic Dictionary of Sanskrit on Historical Principles* produced at Deccan College since 1948). The founding arc:

- **1821:** Founded as a Sanskrit *Pāṭhaśālā* (also referred to as the *Hindoo College*) under **Mountstuart Elphinstone**, Governor of the Bombay Presidency (1819–1827). The institution was established with funds from the ***Dakṣiṇā charitable endowment*** that the Peshwa Bajirao II (the last Peshwa of the Maratha confederacy, 1796–1818) had used to subsidize Sanskrit pundits in Pune. Elphinstone took the endowment that had supported the *paramparā* of Sanskrit teaching and redirected it into a college that would teach Sanskrit *and* English to the same student body. The structural redirection is the moment of institutional capture — the colonial administrator taking the Maratha-state’s traditional patronage apparatus for Sanskrit-paramparā transmission and converting it into a hybrid institution that operated for the colonial educational apparatus.
- **1851:** Renamed the ***Poona College***. The institution by this date had been reorganized as a more general college with Sanskrit-and-English-and-other-subject offerings.
- **1864:** Renamed the ***Deccan College***. By this date the institution had been substantially reorganized as a graduate research institution at the Bombay Presidency’s direction. The colonial-

administrative apparatus had reshaped the institution toward research-and-graduate-instruction functions.

- **1939:** The Bombay Presidency closed the Deccan College's undergraduate operations; the institution became a graduate-and-research institute.
- **1948** (post-Independence): Reconstituted as the *Deccan College Post-Graduate and Research Institute*, with the *Encyclopaedic Dictionary of Sanskrit on Historical Principles* project established as its flagship scholarly enterprise — see the Appendix for the full institutional polemic.

Across the entire nineteenth century, Deccan College was the western Indian Sanskrit-knowledge enterprise's institutional home. The pundits trained there read Sanskrit at a depth no European philological machinery could match. Their work was Sanskrit-internal: textual editing, manuscript collation, grammatical commentary, the architecture of *vyākaraṇa* and *darśana* studies the *paramparā* had carried across the ages. The institution's structural problem the Appendix prosecutes: the pundits' depth was operated in service of an institutional apparatus that was producing the upstream-philological materials the European-philological project would reverse-engineer into PIE. The local Sanskrit depth and the upstream colonial-academic operation were two faces of the same institutional machinery.

The post-Independence continuation: the institution carried the colonial-philological framework forward across the post-1948 period without substantial structural reorientation. The *Encyclopaedic Dictionary of Sanskrit on Historical Principles* project, beginning in 1948 and continuing to the present, operates on the same comparative-philological methodology the colonial framework established — with the predictable consequence that the *historical principles* operating in the dictionary's framing are the *Western philological* historical principles, projecting their developmental-sequence assumptions onto the Sanskrit corpus the dictionary catalogs.

Standard references: K. C. Varadachari, *History of Sanskrit Education in Bombay Presidency* (Bombay, 1959); the *Deccan College Post-Graduate and Research Institute Centenary Volume* (Pune, 1964); the institutional documentation at the Deccan College website (deccan-collegepune.ac.in); A. M. Ghatage, ed., *An Encyclopaedic Dictionary of Sanskrit on Historical Principles*, Volume I (Deccan College, 1976; subsequent volumes continuing). For the broader colonial-educational context: Gauri Viswanathan, *Masks of Conquest: Literary Study and British Rule in India* (Columbia University Press, 1989); Krishna Kumar, *Political Agenda of Education: A Study of Colonialist and Nationalist Ideas* (Sage, 1991).

shabdakalpadruma-deb-1858

Short: *Sir Rādhākānta Deb* (1784–1867), Calcutta-Bengali scholar of the Asiatic Society of Bengal milieu, compiled the *Śabdakalpadruma* (शब्दकल्पद्रुम — *The Wishing-Tree of Words*) — comprehensive Sanskrit lexicographical work in eight volumes, completed 1858, the deepest Sanskrit-internal lexicographical work of the 19th century, produced entirely from within the *paramparā* with entries in Sanskrit organized according to the *vyākaraṇa* discipline’s analytical categories. Deb was knighted by the British colonial state; his generation continued to engage the European philological project before the inversion that produced PIE was visible — the fraud had not yet been baked.

Deployments: Appendix Part 1 §1.2 ¶ — the citation anchor for Rādhākānta Deb’s *Śabdakalpadruma*.

Sir Rādhākānta Deb (1784–1867), Calcutta-Bengali scholar and intellectual of the Asiatic Society of Bengal milieu, compiled the *Śabdakalpadruma* (शब्दकल्पद्रुम — *The Wishing-Tree of Words*) — a comprehensive Sanskrit lexicographical work in eight volumes,

completed in 1858. The *Śabdakalpadruma* is the deepest Sanskrit-internal lexicographical work of the nineteenth century, produced entirely from within the *paramparā*. The work:

- Catalogs Sanskrit vocabulary with detailed grammatical, etymological, and textual-citation information.
- Operates in Sanskrit itself, with the entries written in Sanskrit and the explanatory apparatus organized according to the *vyākaraṇa* discipline's analytical categories.
- Draws on the full breadth of Sanskrit textual sources — the *Vedas*, the *Mahābhārata*, the *Rāmāyaṇa*, the *Purāṇas*, the *Kāvya* literature, the *Śāstra* literature, the *Vedānta* and *Darśana* corpus, the *Smṛti* literature — with citations to specific passages.
- Reflects the *paramparā*-internal analytical apparatus across its eight-volume scope.

Deb completed the work across decades of personal scholarly effort and patronage of pundit collaborators. He was knighted by the British colonial state for his scholarly accomplishments — a recognition the Appendix's polemic register reads as the *priests-of-progress* elevation mechanism: a deep Sanskrit-internal scholar elevated by the colonial honors system to function as the Indian-scholarly imprimatur for the broader colonial-philological enterprise.

The structural significance the Appendix establishes: Deb was operating Sanskrit-internal work for *paramparā*-internal purposes. The *Śabdakalpadruma* was not a strategic contribution to the European-philological project; it was a deep lexicographical resource produced from within the *paramparā* for the *paramparā*'s own continuity. The work's later instrumentalization by the European-philological ecosystem — being mined for cognate-evidence to feed the PIE reconstruction — was not visible to Deb at the time of composition. The fraud had not yet been baked. Deb's generation of senior Indian Sanskritists who continued to engage the European philological project did so before the inversion that produced PIE was visible.

Standard references: Rādhākānta Deb, *Śabdakalpadruma* (Calcutta, eight volumes, 1822–1858; reprinted Nag Publishers, Delhi, 1967, five volumes); the standard scholarly references on nineteenth-century Bengali intellectual history: David Kopf, *British Orientalism and the Bengal Renaissance: The Dynamics of Indian Modernization, 1773–1835* (University of California Press, 1969); Sumit Sarkar, *Modern India, 1885–1947* (Macmillan, 1983) for the broader Bengali intellectual context.

vacaspatyam-taranatha-1873

Short: *Tārānātha Tarkavācaspati* (1812–1885), Calcutta-Bengali scholar of the second generation engaging the European-philological project, compiled the *Vācaspatyam* (वाचस्पत्यम् — *That Which Belongs to the Lord of Speech*) — six-volume Sanskrit lexicographical work completed 1873, organized according to *vyākaraṇa* and *nyāya* (न्याय) discipline categories, with extensive citations from the *Vedic*, *Mahābhārata*, *Purāṇa*, *Kāvya*, *Śāstra*, *Darśana*, and *Smṛti* literature. Tarkavācaspati was associated with the Calcutta Sanskrit College; alongside Deb’s *Śabdakalpadruma*, the work constitutes the deep Sanskrit-internal scholarly resource the European-philological project drew on for the PIE-reconstruction project.

Deployments: Appendix Part 1 §1.2 ¶ — the citation anchor for Tārānātha Tarkavācaspati’s *Vācaspatyam*.

Tārānātha Tarkavācaspati (1812–1885), Calcutta-Bengali scholar of the second generation of Indian Sanskritists engaging the European-philological project, compiled the *Vācaspatyam* (वाचस्पत्यम् — *That Which Belongs to the Lord of Speech*) — a comprehensive Sanskrit lexicographical work in six volumes, completed in 1873. The work is the second major nineteenth-century Sanskrit-internal lexicographical compilation, alongside Deb’s *Śabdakalpadruma*. The

Vācaspatyam:

- Catalogs Sanskrit vocabulary with substantial grammatical, etymological, and textual-citation information.
- Operates in Sanskrit, with the analytical apparatus organized according to the *vyākaraṇa* and *nyāya* (logical-philosophical) disciplines' categories.
- Carries extensive citations from the *Vedic*, *Mahābhārata*, *Purāṇa*, *Kāvya*, *Śāstra*, *Darśana*, and *Smṛti* literature.
- Reflects Tarkavācaspati's training as a scholar of *nyāya* (Indian logic) and *vyākaraṇa* (Sanskrit grammar), with the dictionary's organization reflecting the *paramparā*-internal analytical lineages of these disciplines.

Tarkavācaspati was associated with the Calcutta Sanskrit College and the Asiatic Society of Bengal during the period of the dictionary's composition. The Appendix's structural account parallels the Deb case: a deep Sanskrit-internal scholar producing a *paramparā*-internal resource that was later instrumentalized by the European-philological ecosystem for the PIE-reconstruction project, without the scholar himself having anticipated this instrumentalization at the time of composition.

The combination of Deb's *Śabdakalpadruma* and Tarkavācaspati's *Vācaspatyam* — together with the broader nineteenth-century Bengali, Banaras, and Pune lexicographical-and-grammatical work — constitutes the deep Sanskrit-internal scholarly resource pool that the European-philological project drew on across the second half of the nineteenth century. The Appendix's polemic structure: the *paramparā*-internal scholarship was honest; its upstream instrumentalization by the philological-orthodoxy apparatus was not.

Standard references: Tārānātha Tarkavācaspati, *Vācaspatyam* (Calcutta, six volumes, 1873; reprinted in multiple modern editions including Chowkhamba Sanskrit Series, Varanasi, 1962 and Rashtriya

Sanskrit Sansthan editions). For Tarkavācaspati's broader biographical context: the standard nineteenth-century Bengali intellectual-history references; the *Asiatic Society of Bengal* journals from the relevant period.

rg-bhandarkar-honors

Short: *Sir Rāmakṛṣṇa Gopāla Bhāṇḍārkar* (1837–1925), western-Indian Sanskritist; M.A. from Elphinstone College Bombay (1866), Professor of Sanskrit at Elphinstone (1869–1881) and Deccan College Pune (1882–1893); imperial honors *CIE* (1889) and *KCIE* (1911); Supreme and Bombay Legislative Council appointments (1903–1905); honorary doctorates from Göttingen (1885 — the Indo-European philology center), Edinburgh, Bombay, Calcutta; sustained correspondence with Max Müller, Albrecht Weber, Theodor Aufrecht. The historical-record exemplar of the post-1860s generation's *priests-of-progress* elevation pattern — *paramparā*-internal authority converted to orthodoxy-sanctifying imprimatur through the colonial honors system (Ch3 §3.4 outward-absorption mechanism).

Deployments: Appendix Part 1 §1.3 ¶ — the citation anchor for Sir Rāmakṛṣṇa Gopāla Bhāṇḍārkar's institutional honors and elevation in the post-1860s generation.

Sir Rāmakṛṣṇa Gopāla Bhāṇḍārkar (1837–1925), western-Indian Sanskritist and philologist. The detailed institutional-honors record the Appendix cites:

- **Training:** M.A. from Elphinstone College, Bombay (1866) — one of the earliest Indians to take a Western-style university degree in Sanskrit.
- **Appointments:** Professor of Sanskrit and Oriental Languages at Elphinstone College, Bombay (1869–1881); Professor of San-

skrit at Deccan College, Pune (1882–1893). Bhāṇḍārkar's institutional career operated across the two principal western-Indian colonial-academic institutions for Sanskrit study.

- **Imperial honors:** *CIE (Companion of the Order of the Indian Empire)* in 1889; *KCIE (Knight Commander of the Order of the Indian Empire)* in 1911. The KCIE was the imperial state's highest scholarly-distinction honor at the time, marking Bhāṇḍārkar as one of the most distinguished Indians within the colonial honors system.
- **Council appointments:** Member of the Supreme Legislative Council (1903–1904); Member of the Bombay Legislative Council (1904–1905). Bhāṇḍārkar served on the political-administrative councils of the colonial state, contributing to its legislative-and-administrative apparatus.
- **International honorary degrees:** Honorary Ph.D. from the University of Göttingen (1885) — a particularly significant honor as Göttingen was the German Indo-European philology center at the time. Honorary LL.D. from the University of Edinburgh. Honorary LL.D. from the University of Bombay (1904). Honorary Ph.D. from the University of Calcutta.
- **Scholarly exchange:** Bhāṇḍārkar was in continuous correspondence with the leading German and British Indologists of his generation — *Max Müller* (Oxford), *Albrecht Weber* (Berlin), *Theodor Aufrecht* (Edinburgh), and the broader Indo-European philology community.
- **Institutional naming:** The *Bhandarkar Oriental Research Institute (BORI)* was established at Pune on July 6, 1917 — Bhāṇḍārkar's eighty-first birthday — in his honor. BORI continues to operate as one of the most significant Indian scholarly institutions for Sanskrit-textual research, including the multi-decade *Mahābhārata Critical Edition* project (1933–1966 in its first phase).

The structural significance the Appendix establishes: Bhāṇḍārkar's generation operated as the *priests of progress* of the local Indian machinery. Their Sanskrit-internal credentials were unimpeachable on their own terms — Bhāṇḍārkar's textual-critical work on the *Mahābhārata* and his philological studies of Sanskrit grammar stand as serious scholarship. But the *structural use* to which those credentials were put — by the upstream ecosystem that conferred their honors, and by the wider Anglophone reading public the ecosystem addressed — was to *sanctify* the European philological account of Sanskrit as authoritative.

The mechanism: the KCIE was the imperial state saying *this man is now a peer of our scholarly enterprise*. The Royal Asiatic Society fellowships and the Göttingen-Edinburgh-Bombay-Calcutta honorary doctorates were the trans-imperial enterprise saying *this man is one of us*. What flowed back, structurally, was the Indian-scholarly imprimatur the orthodoxy needed: *the leading Indian Sanskritist of his generation is in continuous scholarly correspondence with our Indologists and concurs with our account*. The Western philological orthodoxy's account of Sanskrit had been sanctified by the paramparā's own elevated scholars.

Standard references: V. S. Sukthankar, ed., *The Mahābhārata for the First Time Critically Edited* (Bhandarkar Oriental Research Institute, Pune, 1933–1966) — the project Bhāṇḍārkar's intellectual descendants conducted; the *Bhandarkar Oriental Research Institute Annual Reports* and Centenary publications; T. G. Mainkar, ed., *Writings and Speeches of Sir R. G. Bhandarkar* (BORI Pune, 1933); Krishna Kumar, *Political Agenda of Education* (Sage, 1991) for the broader colonial-academic-institutional analysis; Vinay Lal, *The History of History: Politics and Scholarship in Modern India* (Oxford University Press, 2003) for the structural critique of the colonial-academic machinery.

bopp-1816-conjugationssystem

Short: *Franz Bopp* (1791–1867; trained at Paris under Antoine-Léonard de Chézy and at London under Henry Thomas Colebrooke), *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache* (Frankfurt am Main: Andreaeische Buchhandlung, 1816) — *where the operation begins*. The work treats Sanskrit verbal morphology as the structural anchor against which Greek, Latin, Persian, and Germanic verbal systems are compared, positioning Sanskrit *at the top* of the comparative machinery as the form closest to (sometimes treated as) the common ancestor. The methodology — comparing daughter-language forms against a privileged anchor — was developed with Sanskrit as the anchor; the post-1861 Schleicher inversion preserved the methodology but moved the anchor from Sanskrit to PIE.

Deployments: Appendix Part 1 §1.5 ¶ — the citation anchor for Franz Bopp’s 1816 *Conjugationssystem*.

Franz Bopp (1791–1867), German philologist trained at Paris under Antoine-Léonard de Chézy (one of the earliest European Sanskrit scholars) and at London under Henry Thomas Colebrooke (the most distinguished British-Sanskritist of the early nineteenth century, then at the East India Company College in Calcutta). Bopp published *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache* in 1816 (Frankfurt am Main: Andreaeische Buchhandlung).

The work’s structural-methodological contribution: it treats Sanskrit verbal morphology as the *structural anchor* against which Greek, Latin, Persian, and Germanic verbal systems are compared. The title itself names the comparison: *On the Conjugation System of the Sanskrit Language, in Comparison with That of the Greek, Latin, Persian, and Germanic Languages*. Bopp positions Sanskrit at the top of the

comparative machinery — as the form *closest to* what would later be called the common ancestor, sometimes treated as the ancestor itself. The comparison is between Sanskrit's verbal-paradigm structure and the verbal-paradigm structures of the European-and-Iranian daughter languages, with the Sanskrit forms positioned as the analytical template.

Bopp's subsequent *Vergleichende Grammatik der Sanskrit-, Send-, Armenischen-, Griechischen-, Lateinischen-, Litauischen-, Altslavischen-, Gothischen- und Deutschen* (1833–1852, in six fascicles) extended the comparison across the family. Sanskrit was still the anchor; the comparative method was being built around it. Through the first half of the nineteenth century the comparative-philological project read Sanskrit as the *source-language* of the family it was constructing.

The structural significance the Appendix establishes: Bopp's 1816 work is *where the operation begins*. The methodology — comparing daughter-language forms against a privileged anchor form — was developed with Sanskrit as the anchor. The subsequent inversion (Schleicher's 1861 *Compendium's* family-tree model with the reconstructed common ancestor *distinct from* any attested language) preserved Bopp's methodology but moved the anchor from *Sanskrit* to *the reconstructed proto-language*. The mechanism of the comparison stayed the same; the anchor's identity changed. Sanskrit was demoted from source to daughter-among-siblings.

The Appendix's polemic structure prosecutes the chronology: Bopp 1816 (Sanskrit as anchor) → Schleicher 1861 (PIE as anchor, Sanskrit demoted) → Brugmann 1886 (PIE consolidated, Sanskrit fully relegated to daughter-language status) → twentieth-century cementing (the routine reference ecosystem solidifies PIE as the default endpoint). The Appendix names this sequence as the *bake* — the manufacturing operation that produced PIE as a finished consumer good.

Standard references: Franz Bopp, *Über das Conjugationssystem der*

Sanskritsprache (Frankfurt am Main, 1816); *Vergleichende Grammatik der Sanskrit-, Send-, Armenischen-, Griechischen-, Lateinischen-, Litauischen-, Altslavischen-, Gothischen- und Deutschen* (Berlin, 1833–1852, six fascicles). Modern scholarly treatments: Salvatore Settis, ed., *The Classical Tradition* (Harvard University Press, 2010) for the broader nineteenth-century philological context; R. H. Robins, *A Short History of Linguistics* (Longman, 4th edition 1997), Chapter 7; Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Routledge, 1998), Chapters 2–3; Lourens van den Bosch, *Friedrich Max Müller: A Life Devoted to the Humanities* (Brill, 2002), for the broader Bopp-Müller-Schleicher generation context.

schleicher-1861-compendium

Short: *August Schleicher* (1821–1868), *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (Hermann Böhlau, Weimar, 1861; 2nd ed. 1866) — *the inversion moment*. The work introduces the *Stammbaumtheorie* (family-tree theory) explicitly: a single common ancestor *distinct from any attested language* (the *Ursprache*) branching into daughter Indo-European languages; Sanskrit, previously the anchor, is demoted to *one daughter language among siblings*. The asterisk-before-reconstructed-form convention — Schleicher’s notational invention — gives the machinery a typographic mark for *forms posited but not attested*. Schleicher’s 1863 pamphlet *Die Darwinsche Theorie und die Sprachwissenschaft* made the biological-organic framing explicit. *The bake* operating at its central moment.

Deployments: Appendix Part 1 §1.5 ¶ — the citation anchor for August Schleicher’s 1861 *Compendium*.

August Schleicher (1821–1868), German philologist at Jena, published the *Compendium der vergleichenden Grammatik der*

indogermanischen Sprachen (Weimar: Hermann Böhlau, 1861, with a second edition 1866). The work introduces the *Stamm-baumtheorie* (family-tree theory) explicitly: a single common ancestor, *distinct from any attested language*, branching into the daughter Indo-European languages.

The structural shift Schleicher's *Compendium* effects: Sanskrit, which had been positioned as the ancestor or close to it in Bopp's 1816 work, is now *one daughter language among siblings*. The reconstructed common ancestor (the *Ursprache*) sits above all the attested languages — including Sanskrit — and the comparative method's task is to reconstruct the ancestor from the daughter-language evidence. The asterisk-before-reconstructed-form convention — Schleicher's notational invention — gives the machinery a typographic mark for the central object the *bake* had produced: *forms that were posited but not attested*.

The biological-organic framing the *Compendium* operates within: Schleicher trained as a botanist before turning to linguistics, and the imported metaphor was not casual but doctrinal. Languages, in Schleicher's framing, are *Naturorganismen* (natural organisms) that grow, branch, reproduce, decay, and die like biological species. The family-tree diagram is the biological-organism's genealogy applied to linguistic descent. Schleicher's 1863 pamphlet *Die Darwinsche Theorie und die Sprachwissenschaft* (*The Darwinian Theory and the Science of Language*; Weimar: Hermann Böhlau) made the Darwinian framing explicit — Darwin's theory of natural selection, on Schleicher's account, applied directly to language change: languages evolved through descent with modification under selective pressures.

The structural significance the Appendix establishes: Schleicher's 1861 *Compendium* is the *inversion moment*. Before 1861, the comparative-philological project still treated Sanskrit as the source. From 1861 onward, the project's structural commitment is to a reconstructed proto-language distinct from any attested language — including Sanskrit. The inversion is the *bake* operating at

its central moment: the manufacture of an ancestor-form from comparative-method work on the daughter languages, with the comparative-method's internal consistency standing in for empirical evidence the operation could not provide and did not require.

The 1868 fable (*Avis akvāsas ka* — see endnote schleicher-1868-fable) was the operational follow-up to the *Compendium*: Schleicher produced a complete text composed entirely in the reconstructed proto-language, every word starred, demonstrating that the apparatus could carry the weight of an extended text. The *bake had produced its first finished good*.

Standard references: August Schleicher, *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (Hermann Böhlau, Weimar, 1861; second edition 1866); *Die Darwinsche Theorie und die Sprachwissenschaft* (Hermann Böhlau, Weimar, 1863); *Die Deutsche Sprache* (J. G. Cotta'scher Verlag, Stuttgart, 1860). See endnote schleicher-stammbaumtheorie for the broader Schleicher reference ecosystem.

brugmann-grundriss-1886

Short: *Karl Brugmann* (1849–1919), of the Leipzig *Junggrammatiker* (Neogrammarian) school, *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen* (Karl J. Trübner, Strasbourg, 1886–1893; revised 1897–1916, with Berthold Delbrück's syntactic volumes) — *the cementing moment*. The work systematized the comparative method around the *Ausnahmslosigkeit der Lautgesetze* doctrine (sound laws operate without exception, licensing reverse-engineering) and consolidated a comprehensive reconstructed PIE — phonology, morphology, lexicon — on the basis of comparative-method work. The reconstructed apparatus operated as a self-validating system in which methodology

and output were mutually reinforcing. *The bake's industrial-scale production line.*

Deployments: Appendix Part 1 §1.5 ¶ — the citation anchor for Karl Brugmann's *Grundriss* and the Neogrammarian school's consolidation.

Karl Brugmann (1849–1919), German philologist of the *Leipzig school* — the *Junggrammatiker (Neogrammarians)* — carried the comparative-method operation through to its mature form. The Leipzig group through the 1870s and 1880s systematized the comparative method around the *Ausnahmslosigkeit der Lautgesetze* doctrine — *sound laws operate without exception*, the central methodological claim that licensed reverse-engineering. The doctrine, in operational terms: when a sound-change-rule is posited (e.g., PIE **p* → Germanic *f*), the rule is presumed to operate without exception across the relevant phonological environment. Any apparent exception is explained as a *secondary development* — a subsequent sound change, a borrowing, a morphological-analogical pressure, an environmental conditioning — rather than as evidence against the rule.

The mature statement of the Neogrammarian apparatus is **Karl Brugmann**, *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen* (1886–1893, with revised editions 1897–1916; published by Karl J. Trübner, Strasbourg, in multiple volumes — *Lautlehre* 1886–1892; *Wortbildungslehre* 1889–1892; *Stammbildungs- und Flexionslehre* 1888–1892; *Syntax* with Berthold Delbrück 1893; with various revised editions thereafter). The work is the regime's *mature statement*: the reconstructed proto-language now had a phonology, a morphology, a vocabulary — all of them assembled from comparative-method work on the daughter languages, with the ecosystem's own internal consistency standing in for empirical evidence the operation could not provide and did not require.

Brugmann's collaborators in the Neogrammarian project: Hermann Osthoff, Hermann Paul (whose 1880 *Prinzipien der Sprachgeschichte* — *Principles of Language History* — is the standard theoretical-methodological statement of the Neogrammarian school), August Leskien, Eduard Sievers, and the broader Leipzig-and-associated group. The school dominated Indo-European philology from the late 1870s through the early twentieth century.

The structural significance the Appendix establishes: Brugmann's *Grundriss* is the *cementing moment*. The Neogrammarian apparatus, by the time the *Grundriss* completed its initial run (1893), had produced a comprehensive reconstructed PIE — phonology, morphology, lexicon, all on the basis of comparative-method work. The reconstructed apparatus had its own internal consistency; the apparatus operated as a self-validating system in which the methodology and the output were mutually reinforcing. The *bake* had produced not just a finished good (Schleicher's 1868 fable) but an entire *industrial-scale production line* (Brugmann's *Grundriss*) — the reconstructed PIE as a comprehensive scholarly-reference ecosystem.

The Appendix's polemic close: the *bake* the cooking-vocabulary cluster names is the operation Brugmann's *Grundriss* completes. The reconstructed PIE — fully cooked, packaged, distributed — became the standard reference for the subsequent century-plus of Indo-European philology. The contemporary cementing of PIE in the routine reference ecosystem (see endnote *pie-cementing-recent-decades*) is the late-stage continuation of the operation Brugmann's *Grundriss* established.

Standard references: Karl Brugmann (and Berthold Delbrück for the syntactic volumes), *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen* (Karl J. Trübner, Strasbourg, multiple volumes, 1886–1916). Translation of the early phonology: Joseph Wright, *Elements of the Comparative Grammar of the Indo-Germanic Languages* (Karl J. Trübner, Strasbourg, 1888–1895, four volumes —

English translation of the first edition). Modern scholarly treatments: R. H. Robins, *A Short History of Linguistics* (Longman, 4th edition 1997), Chapter 7; Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Routledge, 1998), Chapters 5–6; Konrad Koerner, ed., *Historiographia Linguistica*, the journal that has documented the Neogrammarian-school historiography across multiple issues. For the broader methodological context: Hermann Paul, *Prinzipien der Sprachgeschichte* (Halle, 1880; English translation by H. A. Strong, *Principles of the History of Language*, Longmans, Green, 1890); August Leskien, *Die Declination im Slavisch-Litauischen und Germanischen* (Hirzel, Leipzig, 1876).

dhātupatha-empirical-distribution

Short: Empirical statistics in Ch 10 §§10.4–10.6 and Appendix Part 5 are computed against a machine-readable Pāṇinian *Dhātupāṭha* (2,168 entries across the ten *gaṇāḥ*) with standard *anubandha* stripping per *Aṣṭādhyāyī* 1.3.2, 1.3.3, 1.3.5; source data from the open-source *sanskrit/vyakarana* project (github.com/sanskrit/vyakarana). Reproducibility bundle at [analysis/dhātupatha/](#) — self-contained source CSV, Devanāgarī decompositions, Python analysis scripts, README, LICENSE. Empirical claim: 1,795 of 2,168 *dhātavaḥ* (82.8%) are 1 akṣara; CVC dominant at 919 (42.4%); three-and-four-particle *dhātavaḥ* together 79.7%; five-particle 7.2%; six-or-more 1.9% — the compression-principle distribution as empirical signature of an engineered atomic inventory.

Deployments: Chapter 10 §10.4 ¶ (the structural-pattern distribution); Chapter 10 §10.5 (the thermodynamic-threshold distribution); Chapter 10 §10.6 (the Architecture-Deeper-Down findings — cost × distinguishability, OCP, /r/-prominence, cell-level engineering); Appendix Part 5 — *The Architecture by the Numbers* (the full empirical

work with all predictions, data tables, and verdicts including falsifications).

The empirical statistics cited in §§10.4–10.6 are computed against a machine-readable Pāṇinian *Dhātupāṭha* (2,168 entries across the ten *gaṇāḥ*) with the standard *anubandha* stripping applied per *Aṣṭādhyāyī* 1.3.2, 1.3.3, and 1.3.5. A derived companion file — `data/derived/dhatupatha_decomposed.md` — renders every dhātu in Devanāgarī with its varṇa-level decomposition (e.g., कृ = क् + ऋ for *kr̥*; गम् = ग् + अ + म् for *gam*; स्कन्द् = स् + क् + अ + न् + द् for *skand*).

A full reproducibility bundle accompanies the book at the repository subdirectory `analysis/dhatupatha/`. The bundle is self-contained — the source CSV, the derived Devanāgarī decomposition, all the Python analysis scripts, a README with full attribution and methodology notes, and a LICENSE file — and is structured for public sharing (e.g., as a GitHub repository). Any reader can reproduce every empirical claim in Chapter 10 and Appendix Part 5 by running the scripts against the source data: `python3 scripts/analyze_dhatupatha.py`, `python3 scripts/analyze_varga_distribution.py` [*gaṇa*], etc. Requirements: Python 3.10+ with no external dependencies.

Source data. The digital *Dhātupāṭha* used here is `data/dhatupatha.csv` in the book’s repository, sourced from the open-source *sanskrit/vyakarana* project (<https://github.com/sanskrit/vyakarana> — file `data/dhatupatha.csv`). The CSV has three columns: *gaṇa*-number, position-within-*gaṇa*, dhātu in SLP1 transliteration with Pāṇinian accent markers (~, \, ^). The count of 2,168 sits within the conventional Pāṇinian range (~1,940 to ~2,200 depending on recension); other published *Dhātupāṭha* recensions — Bhattoji Dīkṣita’s *Siddhāntakaumudī*, the *Mādhaviya Dhātuvṛtti*, the *Kṣīrasvāmin* commentary — yield comparable totals with minor recensional variation in marginal entries.

Anubandha-stripping methodology. Three *it-saṃjñā* rules are applied algorithmically before structural classification:

1. **Aṣṭādhyāyī 1.3.2 — *upadeśe 'janunāsika it.*** In the citation form (*upadeśa*), a final *anunāsika*-marked short vowel is an *anubandha*. In the Pāṇinian citation convention, the trailing short *-a* / *-i* / *-u* after a consonant carries this status implicitly. The stripping rule applied: a trailing short *-a*, *-i*, or *-u* immediately following a consonant is stripped, *provided that at least one other vowel remains in the form*. The “vowel must remain” condition prevents over-stripping of genuine CV-pattern roots like *ji* जि (to conquer), *hu* हु (to sacrifice), *sru* स्रु (to flow), *ki* कि (to know), *ru* रु (to roar), where the short vowel is root-final, not *anubandha*. Long-vowel finals (*-ā*, *-ī*, *-ū*, *-ṛ*, *-e*, *-ai*, *-o*, *-au*) are root-final and not stripped — these are the *kṛ*, *bhū*, *dā*, *jñā*, *pā* class.
2. **Aṣṭādhyāyī 1.3.5 — *ādir niṭuḍavaḥ.*** The initial two-character sequences *ñi* (SLP1: Ji), *ṭu* (wu), *ḍu* (qu) in dhātu citation forms are *anubandhas* and are stripped from the front.
3. **Aṣṭādhyāyī 1.3.3 — *halantyam.*** A trailing single-consonant *anubandha* — typically the *ñit* (Y), *ñit* (N), *lit* (l), *ṣit* (S/z), *ṭit* (w), or *ḍit* (q) marker signaling grammatical properties such as ātmanepadī conjugation or vowel-shift behavior — is stripped when it sits immediately after a vowel. The classic case is the citation form *ḍukṛñ* (SLP1: qukf\Y) for the *kṛ* dhātu: the initial *ḍu* is stripped by 1.3.5, the trailing *ñ* by 1.3.3, leaving the bare root *kṛ*. 36 dhātus across the corpus (1.7%) carry trailing post-vowel consonant *anubandhas* of this kind; without this rule they would mis-classify as having an extra consonant.

Accent markers (*~*, **, *^*) in the SLP1 encoding indicate *udātta*, *anudātta*, and *svarita* respectively and are stripped before structural classification (they are recitational, not structural).

Computational details. The classification is done by the analysis

script `scripts/analyze_dhatupatha.py` in the book’s repository. The script:

- Reads `data/dhatupatha.csv`
- Strips accent markers (`~`, `\`, `^`) and applies the *it-saṃjñā* stripping rules above
- Maps each remaining SLP1 character to V (vowel) or C (consonant) using the standard SLP1 inventory (vowels: a A i I u U f F x X e E o O; consonants: the 33 stops + semivowels + sibilants + h + visarga + anusvāra)
- Classifies each *dhātu* by structural pattern (CV, CVC, CCVC, CVCC, CCVCC, etc.), particle count (number of V+C constituents), and akṣara count (number of vowel-nuclei)
- Produces summary statistics by gaṇa, by structural pattern, by particle count, and by akṣara count

The classification is reproducible: re-running `python3 scripts/analyze_dhatupatha.py` from the repository root regenerates the figures cited in the chapter.

Edge cases and limitations. Seven entries (~0.3%) classify as bare-vowel V-pattern structures after stripping — these are special-case Pāṇinian-citation forms (e.g., the bare-vowel roots *i* = ३ “to go”, *ṛ* = २ “to go”, *f* in the SLP1 encoding) and are correctly retained as 1-akṣara dhātus. A more granular Pāṇinian analysis would also handle the *cuṭū* (*Aṣṭādhyāyī* 1.3.7) and *laśakvataddhite* (1.3.8) rules for initial-consonant anubandhas in *pratyayas*, but these rules do not affect dhātu citation specifically and are out of scope for the structural analysis here.

Cross-validation. The Sanskrit Heritage Platform’s `parts.csv` (at <https://github.com/sanskrit/data/blob/master/sanskrit-heritage-site/parts.csv>) provides ~11,570 verb stems with their underlying root forms (the *root* column gives the anubandha-stripped form per the Sanskrit Heritage convention). Spot-checking the structural-analysis output against this independent lexicon confirms that the *Aṣṭādhyāyī* 1.3.2 + 1.3.3 + 1.3.5 rules implemented here recover

the standard underlying roots for the vast majority of *Dhātupāṭha* entries — including canonical cases like *kṛ* (ḍukṛñ → kṛ), *brū* (brūñ → brū), *śri* (śriñ → śri).

The empirical claim therefore holds at: 1,795 of 2,168 *dhātavaḥ* (82.8%) are 1 akṣara. CVC is the dominant single pattern at 919 entries (42.4%); CV at 152 (7.0%); together with CCV, VC, CVCC, CCVC, the single-akṣara forms cover the substantial majority of the inventory. Three-particle and four-particle *dhātavaḥ* together account for 1,727 entries (79.7%). Five-particle *dhātavaḥ* are 7.2%; six-or-more is the cliff at 1.9%. The compression-principle distribution is the empirical signature of an engineered atomic inventory.

vyakarana-etymology

Short: *Vyākaraṇam* (व्याकरणम्) = *vi-* (वि-, *apart*) + *ā-* (आ-, *toward*) + *kṛ* (कृ, *to do, to make*) — formed via the verbal stem *vi-ā-kṛ* through the abstract-noun derivation, with literal sense *the act of taking-apart, unfolding-apart, separating into constituents* — i.e., **analysis, de-composition**, not composition. Foundational for the book’s polemic against the orthodoxy’s *codification* vocabulary: Sanskrit’s own name for what Pāṇini did is the opposite operation — the *taking-apart* of a system already present. Sources: Monier-Williams 1899, Apte 1890, Yaska’s *Nirukta*, the *Mahābhāṣya*.

Deployments: Ch1 §1.1 (the Bakers’ Story / standing polemic phrase); Ch4 §4.1 (the *vyākaraṇam* / *vaiyākaraṇa* paragraph); Ch6 §6.3 (the *Dhātupāṭha* as documentation of an inherited inventory); Ch8 §8.6; Ch13 §13.3; Ch14 §14.5; Claim #2.

The Sanskrit word *व्याकरणम्* (**vyākaraṇam**) decomposes morphologically as *वि* (*vi-*, “**apart, asunder**”) + *आ* (*ā-*, “**toward, fully**”) + *कृ* (*kṛ*, “**to do, to make**”), formed via the verbal stem *vi-ā-kṛ* through the abstract-noun derivation that produces *vyākaraṇa* from the corre-

sponding finite-verb forms. The literal sense is *the act of taking-apart, unfolding-apart, separating into constituents* — i.e., **analysis**. The operation the term names is *de-composition*, not composition: the analytical procedure by which a given system (the language) is decomposed into its parts, not the constructive procedure by which a system is built.

This etymology is foundational for the book’s polemic against the orthodoxy’s *codification* vocabulary. *Codification* implies *construction* — the production of structure where none was. Sanskrit’s own name for what Pāṇini did is the opposite operation: the *taking-apart* of a system already present. The morphological decomposition is consistent across the Sanskrit nirukta discipline (Yaska’s *Nirukta* and the *Mahābhāṣya*’s paratextual remarks), in classical lexicons (Monier-Williams 1899: *vyākaraṇa*, “separation, distinction, analysis”; Apte 1890: *vyākaraṇa*, “explanation, analysis, grammar”), and in modern Sanskrit grammatical scholarship (Cardona 1976, *Pāṇini: A Survey of Research*; Kiparsky 1979).

Source: Standard Sanskrit lexicographical references (Monier-Williams 1899; Apte 1890); confirmed by Yaska’s *Nirukta* and the *Mahābhāṣya*’s own paratextual usage.

vaiyakarana-role-title

Short: *Vaiyākaraṇaḥ* (वैयाकरणः, *one who performs vyākaraṇam*; plural *vaiyākaraṇāḥ* वैयाकरणाः) — the Sanskrit role-title for the *grammarian-as-analyst*, derived by the *aṇ-taddhita* suffix with *vṛddhi* of the first syllable. Pāṇini, Kātyāyana, Patañjali, and the pre-Pāṇinian grammarians Pāṇini cites by name (Śākalya, Āpiśali, Kāśyapa, Gārgya, Gālava, Cākravarmaṇa, Bhāradvāja, Saunaga, Senaka, Sphoṭāyana) are all designated *vaiyākaraṇāḥ*. The discipline does *not* call them *sthapati* (architect), *nirmātr* (constructor),

or any cognate of *engineer* — the role-attribution is uniformly *decoder*, *one who unfolds-apart*, contesting the orthodoxy's *codification* claim from inside the *paramparā*'s own vocabulary.

Deployments: Ch4 §4.1 (the role-title paragraph); Claim #2.

The Sanskrit role-title for the practitioner of *vyākaraṇam* is **वैयाकरणः (vaiyākaraṇaḥ)** — *one who performs vyākaraṇam*, **the grammarian-as-analyst**. The plural is **वैयाकरणाः (vaiyākaraṇāḥ)**** — *the grammarians*. The derivation is by the *aṇ-taddhita* suffix (per Pāṇini's own *Aṣṭādhyāyī* — the suffix indicating “*the one who is associated with X*”), with *vṛddhi* of the first syllable yielding *vyā-* → *vai-* and so *vaiyākaraṇa* from *vyākaraṇa*. The Pāṇinian derivational pattern itself produces the role-title; the discipline is internally self-consistent in naming the activity and the agent.

Pāṇini, Kātyāyana, Patañjali, and the pre-Pāṇinian grammarians Pāṇini cites by name — Śākalya, Āpiśali, Kāśyapa, Gārgya, Gālava, Cākravarmaṇa, Bhāradvāja, Saunaga, Senaka, Sphoṭāyana — are all designated *vaiyākaraṇaḥ* by the discipline. So are the later commentators: Bhartṛhari (author of the *Vākyapadīya*), Helārāja, Kaiyaṭa, Nāgeśa Bhaṭṭa. Yaska's primary designation is *nairukta* (etymologist) — but the *vaiyākaraṇa* and *nairukta* disciplines are closely allied, with the *Nirukta* engaging the grammatical analysis and the *Mahābhāṣya* engaging the etymological one.

Critically, the discipline does *not* call any of these figures **स्थपति (sthapati)** (*architect*, the *Vāstu-śāstra* term for a builder of physical structures), **निर्माता (nirmātr)** (*constructor*), or anything cognate with *engineer*. The role-title is uniformly *vaiyākaraṇa* — *the decoder*, *the one who unfolds-apart*. This is the *paramparā*'s own role-attribution: not as engineers of a new system, but as decoders of an encoded one. The standing polemic phrase — *Sanskrit was engineered; encoded in the Vedas; decoded by many; Pāṇini's decoding is the finest* — restates the *paramparā*'s own self-description in the polemic register that contests the orthodoxy's *codification* claim.

Source: Standard Sanskrit lexicographical references (Monier-Williams 1899); Cardona 1976 (*Pāṇini: A Survey of Research*) on the *vaiyākaraṇa* commentarial lineage; Kunjunni Raja 1963 (*Indian Theories of Meaning*) on the broader *śābdika* / *vaiyākaraṇa* discipline.

panini-no-preface

Short: The *Aṣṭādhyāyī* contains no preface, no introductory discourse, no first-person statement of authorial intent — it opens directly with *sūtra* 1.1.1 *vr̥ddhir ādaic* (वृद्धिर् आदैच्) and runs roughly four thousand *sūtras* through to the final *sūtra* 8.4.68 (*a a*) without first-person address; the silence on purpose is consistent with the documenter role (a documenter has nothing to motivate; the document is its own purpose) and inconsistent with the engineer role (an engineer would state design intent). The traditional *why* answer comes one commentarial generation later, in Patañjali's *Mahābhāṣya* — see endnote *prayojanani-paspashahnika*.

Deployments: Ch4 §4.1 (the no-preface observation paragraph).

The *Aṣṭādhyāyī* contains no preface, no introductory discourse, no first-person statement of authorial intent. The text opens directly with *sūtra* 1.1.1 — *vr̥ddhir ādaic* (वृद्धिर् आदैच्) — and runs roughly four thousand *sūtras* through to the final *sūtra* 8.4.68 (*a a*) without any first-person address. There are no statements of design intent, no enumeration of purposes, no naming of the audience, no explanation of the methodology — just the *sūtras* themselves, in their compact technical register.

This silence is significant in the context of the book's polemic. An engineer presenting a new construction would presumably state design intent: what the construction is for, why this design and not another, what problem it solves. A documenter describing an existing system has nothing to motivate, because the document is its own purpose:

the activity is *enumerate-and-describe*, not *design-and-justify*. Pāṇini's silence on purpose is consistent with the documenter role and inconsistent with the engineer role.

The traditional answer to “why was the *Aṣṭādhyāyī* written?” comes one commentarial generation later, in Patañjali's *Mahābhāṣya* (see endnote prayojanani-paspashahnika). Patañjali's five-fold answer (*rakṣā*, *ūha*, *āgama*, *laghu*, *asaṃdeha*) describes grammar's purposes as *meta-operations on an existing language* — preservation, modification, scriptural enjoining, mastery, doubt-removal — not as the act of constructing the language itself. The paramparā's own purpose-statement is therefore second-hand, post-hoc, and the activity-noun is *vyākaraṇam* — *analysis*.

Source: Pāṇini, *Aṣṭādhyāyī* (Bhattoji Dīkṣita's *Siddhāntakaumudī* ordering; Bohtlingk 1839–40 critical edition; Vasu 1891 English translation; Cardona 1976 *Pāṇini: A Survey of Research* on the structural silence of the text).

prayojanani-paspashahnika

Short: Patañjali's *Mahābhāṣya* opens (the *Paspaśāhnika*) with ***kim prayojanam vyākaraṇasya*** (किं प्रयोजनं व्याकरणस्य — “*what is the purpose of grammar?*”) and answers with the ***pañca prayojanāni*** (पञ्च प्रयोजनानि — *five purposes*): ***rakṣā*** (रक्षा, preservation of the *Vedas*), ***ūha*** (ऊह, ritual-context modification), ***āgama*** (आगम, scriptural injunction), ***laghu*** (लघु, brevity / efficient mastery), ***asaṃdeha*** (असंदेह, removal of doubt). All five presuppose an *already-existing* engineered language to preserve, modify, study, condense, clarify; none says “to engineer a new language” or “to codify a drifting one” — the *paramparā*'s own answer to *why grammar?* runs the documenter framing in its very first move.

Deployments: Ch4 §4.1 (the Patañjali five-prayojanāni paragraph);

Claim #2.

The opening of Patañjali's *Mahābhāṣya* — the *Paspaśāhnika*, the introductory *āhnika* — raises the question किं प्रयोजनं व्याकरणस्य (**kim prayo-janam vyākaraṇasya**) — “*what is the purpose of grammar?*” — and answers with five canonical purposes, the पञ्च प्रयोजनानि (**pañca pray-ojanāni**):

1. रक्षा (**rakṣā**) — preservation: of the *Vedas*, by knowing the correct forms. Knowledge of grammar enables the practitioner to preserve Vedic recitation against drift.
2. ऊह (**ūha**) — modification / substitution: ritual contexts require substituting variant forms (when a different deity is invoked in a *mantra*, the case-form must adjust accordingly), and grammar supplies the rules for these substitutions.
3. आगम (**āgama**) — scriptural injunction: the *Vedas* themselves enjoin the study of grammar; learning grammar is a Vedic obligation, not an optional refinement.
4. लघु (**laghu**) — brevity / efficiency: mastering grammar efficiently is itself a virtue; grammar provides the compact rules from which the full language can be regenerated.
5. असंदेह (**asaṁdeha**) — removal of doubt: grammar resolves ambiguity in usage and interpretation.

Every one of the five presupposes an *already-existing* engineered language. *Rakṣā* presupposes a *Veda* to preserve; *ūha* presupposes ritual forms to modify; *āgama* presupposes scripture that enjoins study; *laghu* presupposes correct usage to master efficiently; *asaṁdeha* presupposes ambiguities in existing usage to resolve. None of the five says “to engineer a new language” or “to codify a drifting one.” The paramparā’s own answer to *why grammar?* runs the documenter framing in its very first move.

Source: Patañjali, *Mahābhāṣya*, *Paspaśāhnika* opening — Kielhorn’s edition (third edition revised by Abhyankar, Bhandarkar Oriental Research Institute, Pune), volume I, p. 1; Joshi & Roodbergen’s

translation-commentary series on the *Mahābhāṣya* (Sahitya Akademi / Pune University publications, 1968–2003).

varnavada-presupposes-engineering

Short: The Sanskrit *vyākaraṇa* discipline’s **varṇa-vāda** (वर्णवाद) school — running across Yaska’s *Nirukta* (निरुक्त), Bhartṛhari’s *Vākyapadīya* (वाक्यपदीय), and the *Mīmāṃsā* (मीमांसा) etymological discipline — holds that *varṇas* carry **varṇa-śakti** (वर्णशक्ति, semantic potency); *dhātus* are compositions whose component *varṇas* align with their meaning. The position *only makes sense* if Sanskrit is engineered: for *varṇas* to carry stable, distinguishable, composable semantic charges, the phonetic inventory must be **discrete** (snap-to-grid), **distinguishable** (cost × distinguishability), **stable** (anti-entropy), and **composable** (combinatorial-assembly engineering). The paramparā’s own internal debates *presuppose* the engineering thesis.

Deployments: Ch10 §10.8 (the *varṇa-vāda* synthesis section).

The Sanskrit *vyākaraṇa* discipline’s **varṇa-vāda** (वर्णवाद) school — running across Yaska’s *Nirukta*, Bhartṛhari’s *Vākyapadīya*, and the *Mīmāṃsā* etymological discipline — holds that *varṇas* carry **varṇa-śakti** (वर्णशक्ति), the semantic potency of the *varṇa*. The position is that *dhātus* are not arbitrary assemblies of phonetic atoms but compositions whose component *varṇas* align with their meaning. Yaska’s *Nirukta* methodology — recovering a word’s meaning by decomposing it into *dhātu* and analyzing the constituent *varṇas* — operates on this premise.

The book’s deeper observation in Ch10 §10.8: the *varṇa-vāda* position *only makes sense* if Sanskrit is engineered. For *varṇas* to carry stable, distinguishable, composable semantic charges across the language, the phonetic inventory must be **discrete** (*varṇas* are units, not a continuous spectrum) — requires snap-to-grid engineering; **distin-**

guishable (each varṇa's charge has to be reliably differentiated) — requires cost × distinguishability engineering; **stable** (charges have to hold across the corpus without drift) — requires anti-entropy engineering; **composable** (charges have to sum predictably in *dhātu* assemblies) — requires combinatorial-assembly engineering. A natural-organic language drifting in the way the orthodoxy claims Sanskrit drifts could not support a *varṇa-vāda* position; the position would be incoherent on a drifting substrate.

The implication: Sanskrit's own internal debates — *varṇa-vāda*, *sphoṭa-vāda*, Yaska's etymological method, the *Mīmāṃsā* discipline's *śabda-pramāṇa* — all *presuppose* the engineering thesis. The Western philological orthodoxy's account of Sanskrit as just another natural language is inconsistent with the paramparā's own self-understanding of what Sanskrit is. The book sides with the paramparā. The remaining debate (between intrinsic-charge and architect's-freedom mechanisms — Ch10 §10.8 develops this) is an *internal debate within the engineering thesis*, not between engineering and non-engineering.

Source: Yaska, *Nirukta* (Lakshman Sarup's edition, Motilal Banarsidass 1920–27); Bhartṛhari, *Vākyapadīya* (K. A. Subramania Iyer's edition, Deccan College Pune, 1965); Kunjunni Raja 1963, *Indian Theories of Meaning* (Adyar Library Series), for the *varṇa-vāda* / *sphoṭa-vāda* structural history; Houben 1995 (*The Saṃbandha-samuddeśa*) on Bhartṛhari's signal-word ontology.

productivity-inversion-natural-language

Short: Natural languages display the *frequency-irregularity correlation* — high-frequency forms tend toward suppletive idiosyncrasy (English *be* / *have* / *do*; Latin *esse* / *ire* / *ferre*; Greek *eimi* / *oida* / *phēmi*); the correlation is one of the most-replicated findings in

natural-language morphology, explained by high-frequency forms being mastered as wholes and resisting analogical regularization. Sanskrit shows the *opposite* pattern: the most-productive *dhātus* (*kr*, *bhū*, *gam*, *sthā*, *dā*, *dhā*, *jñā*, *hr*, *nī*) are the most structurally *minimal* (CV / CVC) *and* the most paradigmatically *regular*. Empirical signature in the *Dhātupāṭha* curated sample: Spearman $\rho = -0.485$ between productivity and particle count; CV pattern mean productivity $2.9\times$ the CCVCC pattern's; top-20 productivity ranks dominated by minimal-particle patterns. The architects engineered both axes at once.

Deployments: Ch10 §10.7 (the productivity section); Appendix Part 5 §5.3.11; Claim #21.

The book's productivity-inversion claim — that Sanskrit shows the opposite of the natural-language frequency-irregularity correlation — rests on a cross-linguistic typological observation widely documented in modern morphological theory: in natural languages, high-frequency forms tend toward *idiosyncratic irregularity*. The canonical examples:

- **English** *be* / *have* / *do*: paradigmatically broken — *be* has the suppletive paradigm *am* / *are* / *is* / *was* / *were* / *been*; *have* has irregular past *had* and third-person *has*; *do* has irregular past *did* and third-person *does*.
- **Latin** *esse* / *ire* / *ferre*: similarly suppletive — *sum* / *es* / *est* / *fui* / *esse*; *eo* / *is* / *it* / *ivi* / *ire*; *fero* / *fers* / *fert* / *tuli* / *latum* / *ferre*.
- **Greek** *eimi* / *oida* / *phēmi*: same pattern.
- **Russian** *byt'* (to be): near-absence of present-tense forms.
- **Spanish, French, German**: same correlation.

The frequency-irregularity correlation is one of the most-replicated findings in natural-language morphology. The explanation in the natural-language framework: high-frequency forms are mastered as wholes (and so resist analogical regularization pressure) while low-frequency forms are subject to analogical regularization across

generations. Frequency drives idiosyncrasy because frequency-of-use shapes the form — the language is *natural*.

Sanskrit shows the opposite pattern. The most-productive *dhātus* — *kr̥*, *bhū*, *gam*, *sthā*, *dā*, *dhā*, *jñā*, *hr̥*, *nī* — are the most structurally minimal (CV / CVC patterns) AND the most paradigmatically regular. There is no idiosyncrasy at the top. *Kr̥* generates *karoti*, *karma*, *karṭr̥*, *kāraṇa*, *kāryam*, *saṃskāra*, *prakṛti*, *vikṛti* through the standard *vibhakti* and *kṛdanta* paradigms; no suppletive forms, no paradigm-breaking irregularities.

The empirical signature in the *Dhātupāṭha* curated sample (138 *dhātus*; full data in `analysis/dhatupatha/data/dhatu_productivity.csv`): Spearman $\rho = -0.485$ between productivity and particle count; CV pattern mean productivity $2.9\times$ the CCVCC pattern's; top-20 dominated by minimal-particle patterns (11 of 20 are CV).

The architectural inference: in natural languages, high-frequency drives idiosyncrasy because the language is *natural* — frequency shapes the form. In Sanskrit, high-frequency is associated with structural minimum AND paradigmatic regularity because the language is *engineered* — the architects engineered both axes at once. The pattern is a signature natural-language drift does not produce, has never been observed to produce, and the engineering thesis predicts directly.

Source: Bybee 1985, *Morphology: A Study of the Relation between Meaning and Form* (Benjamins); Haspelmath 2002, *Understanding Morphology* (Hodder); Greenberg 1966, *Language Universals* (Mouton); Bybee & Hopper 2001 (eds.), *Frequency and the Emergence of Linguistic Structure* (Benjamins) — these document the natural-language frequency-irregularity correlation. Monier-Williams 1899 and V. S. Apte 1890 for the Sanskrit productivity counts; reproducibility bundle at `analysis/dhatupatha/` for the curated sample and the analysis script.

bakers-story-seven-moves

Short: The seven-move standard story of textbook Indo-European linguistics was built by four named European comparativists across the 19th century: **Franz Bopp** (1791–1867), *Über das Conjugationssystem der Sanskritsprache* (Frankfurt, 1816) — the systematic comparative method’s founding work; **August Schleicher** (1821–1868), *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (Weimar, 1861) — the family-tree theory formalized; **Max Müller** (1823–1900) — Oxford’s pedagogical machinery, the *R̥gveda* edition (1849–1874), the *Sacred Books of the East* (1879–1910); **Karl Brugmann** (1849–1919), *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen* (Strasbourg, 1886–1893) — the neogrammarian synthesis. The contemporary softened orthodoxy preserves the seven moves while softening the vocabulary; the *codified* concession at the Pāṇini level operates in current register.

Deployments: Ch1 §1.1 (the *Bakers’ Story of Sanskrit* — the seven-move standard orthodox narrative the book contests).

The seven-move standard story Ch1 §1.1 names is the load-bearing structure of textbook Indo-European linguistics. The named European comparativists who built it across the nineteenth century:

- **Franz Bopp** (1791–1867), *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache* (Frankfurt am Main: Andräische Buchhandlung, 1816) — the founding work of the systematic comparative method.
- **August Schleicher** (1821–1868), *Compendium der vergleichenden Grammatik der indogermanischen Sprachen* (Weimar: Hermann Böhlau, 1861; 2nd ed. 1866) — the family-tree theory (*Stammbaumtheorie*) formalized.
- **Max Müller** (1823–1900) — the Boden Chair at Oxford (1860–1899) and the pedagogical machinery: the editions

of the *R̥gveda* (1849–1874), the *Sacred Books of the East* series (1879–1910), the *Lectures on the Science of Language* (1861–1864).

- **Karl Brugmann** (1849–1919), *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen* (Strasbourg: Trübner, 1886–1893; revised 1897–1916) — the neogrammarian synthesis that consolidated PIE reconstruction methodology.

These four are the named figures. The seven-move story is the cumulative product of the framework they built across the nineteenth century. Subsequent scholarship (twentieth-century neogrammarians, structuralist linguistics, modern Indo-Europeanists from Calvert Watkins through Michiel de Vaan) refined the details without altering the underlying seven-move structure.

The contemporary softened orthodoxy — Cardona, Houben, Pollock, Kiparsky and adjacent scholarship — preserves the moves while softening the vocabulary. The *codified* concession at the Pāṇini level is the contemporary register; the rest of the seven moves operate as background assumptions in the working framework rather than as explicitly defended claims. The book's polemic, accordingly, contests both the explicit nineteenth-century version and the softer contemporary version with the same seven-move enumeration.

Source: Standard histories of Indo-European linguistics — Holger Pedersen, *The Discovery of Language: Linguistic Science in the Nineteenth Century* (1931, English trans. 1962); Anna Morpurgo Davies, *Nineteenth-Century Linguistics* (Longman, 1998); Pedro Ramat & Anna Giacalone Ramat (eds.), *The Indo-European Languages* (Routledge, 1998); Trautmann, *Aryans and British India* (University of California Press, 1997) for the colonial-philological context.

vedic-variation-eight-claims

Short: Eight specific orthodox claims about variation inside the Vedic corpus, with canonical sources: (1) *Ṛgveda* vs. *Atharvaveda* differences (Macdonell, *A Vedic Grammar*, 1916; Witzel); (2) Mandala-by-Mandala variation in the *Ṛgveda* (Oldenberg 1888; Witzel 1997); (3) *Samhitā* / *Brāhmaṇa* / *Āraṇyaka* / *Upaniṣad* stratification (Olivelle); (4) *sandhi* variation across Vedic schools (Whitney’s *Atharva-Veda Prātiśākhya*; Macdonell); (5) accent system “erosion” (Wackernagel, *Altindische Grammatik*); (6) word-form variants (Whitney’s *Sanskrit Grammar*); (7) *Śākala* vs. *Bāṣkala* recensional differences; (8) Vedic-Avestan parallels (Mayrhofer’s *Etymologisches Wörterbuch des Altindoarischen* 1986–2001). Ch 5 §5.6’s response: *not drift, but engineered design choices within the same architecture* — Pāṇini’s *chandasi* / *bhāṣāyām* framework.

Deployments: Ch5 §5.6 (the “*What the Orthodoxy Calls ‘Variation in the Vedas’*” section — eight orthodox claims for internal Vedic drift, each with engineering response).

The eight specific orthodox claims about variation inside the Vedic corpus, with their canonical sources:

1. **Rgveda vs. Atharvaveda differences** — Macdonell, *A Vedic Grammar for Students* (1916); Witzel’s extensive Harvard work on Vedic-dialect variation across the four corpora.
2. **Mandala-by-Mandala variation in the Rigveda** — Hopkins, *The Composition of the Mahābhārata* and the family-books vs. Mandalas 1/8/9/10 chronology; Witzel, “The Development of the Vedic Canon” (1997); Oldenberg, *Die Hymnen des Rigveda* (1888).
3. **Samhitā / Brāhmaṇa / Āraṇyaka / Upaniṣad stratification as evolutionary** — Olivelle’s translation series; the orthodoxy’s default chronology read backward into linguistic stratification.
4. **Sandhi variation across Vedic schools** — Whitney, *Atharva-Veda Prātiśākhya*; Macdonell on cross-recension sandhi differ-

ences.

5. **Accent system “erosion”** — Wackernagel, *Altindische Grammatik* (1896–); standard accounts of “accent loss” from Vedic to Classical Sanskrit.
6. **Word-form variants** — Whitney, *Sanskrit Grammar* on alternant forms; Cardona on grammatical optionality.
7. **Śākala vs. Bāṣkala recensional differences** — standard text-critical work on the *Ṛgveda*’s recensional history.
8. **Vedic vs. Avestan parallels** — Mayrhofer, *Etymologisches Wörterbuch des Altindoirischen* (EWAia, 1986–2001); the entire “Proto-Indo-Iranian” reconstruction framework.

Ch5 §5.6’s engineering response to each: not drift, but engineered design choice. The unifying observation: *the orthodoxy reads all variation as drift; the engineering thesis reads all variation as engineered design choices within the same architecture*. The optional Appendix Part 6 (P2 deferred per working/as_todo.md) is where the per-claim technical detail would land if reader review calls for it.

Source: Standard Vedic-studies scholarly references for the orthodox claims (as enumerated above); the book’s engineering response is the chapter’s own argumentative work, anchored in Pāṇini’s *chandasi* / *bhāṣāyām* categorical framework (see endnote *chandasi-bhashayam-mode-markers*).

chandasi-bhashayam-mode-markers

Short: Pāṇini’s *Aṣṭādhyāyī* deploys two register-markers throughout its ~4,000 *sūtras*: **chandasi** (छन्दसि, locative of *chandas* — “in the metrical corpus,” “in the Vedic register”) for rules applying to the Vedic mode (preserving *udātta* / *anudātta* / *svarita*, ऌ, specific verb-form distinctions), and **bhāṣāyām** (भाषायाम्, locative of *bhāṣā* — “in speech,” “in the everyday register”) for rules applying to the spoken / Classi-

cal mode (the *śiṣṭa-bhāṣā* शिष्ट-भाषा the *Aṣṭādhyāyī* operates on by default). Anchor *sūtras* include 3.2.108 *bhāṣāyām sadavaśruvaḥ* and 6.1.34 *viprativiṣayāṇām kalyavakalyādīnām chandasi*, with hundreds of further deployments. **Register markers, not temporal markers** — Pāṇini does not say the language *used to be* Vedic and is *now* Classical; he marks the distinction categorically. The empirical disproof of the two-versions claim sits inside the very text the orthodoxy treats as the codification event.

Deployments: Ch1 §1.1 Move 7 (the *two-versions* claim’s refutation); Ch5 §5.6 claim 5 (accent system “erosion”); Ch14 §14.5 (the calibration matrix’s two-mode operation).

Pāṇini’s *Aṣṭādhyāyī* deploys two register-markers throughout its ~4,000 *sūtras* to indicate which mode a given rule applies to:

- **छन्दसि (chandasi)** — locative of *chandas* (meter / metrical corpus). Glosses as “*in the metrical corpus*,” “*in the Vedic register*,” “*in the recitational mode*.” Pāṇini uses this marker for rules that apply to the Vedic mode — the corpus that requires preservation of accent (*udātta/anudātta/svarita*), the retroflex lateral ऌ (ḷ), specific verb-form distinctions, and other features that the productive Classical mode does not deploy.
- **भाषायाम् (bhāṣāyām)** — locative of *bhāṣā* (speech / the everyday spoken language). Glosses as “*in speech*,” “*in the spoken language*,” “*in the everyday register*.” Pāṇini uses this marker for rules that apply to the Classical mode — the spoken Sanskrit of the educated population (the *śiṣṭa-bhāṣā* शिष्ट-भाषा) the *Aṣṭādhyāyī* operates on as its default.

The two markers appear across the *Aṣṭādhyāyī* hundreds of times. Specific anchor *sūtras* include 3.2.108 (*bhāṣāyām sadavaśruvaḥ*), 6.1.34 (*viprativiṣayāṇām kalyavakalyādīnām chandasi*), and many others. Cardona 1976 (*Pāṇini: A Survey of Research*, ch. 3) catalogues the systematic register-distinction in detail.

The load-bearing structural fact: these are **register markers, not**

temporal markers. Pāṇini does not say “the language used to be like Vedic and is now like Classical”; he says “rule *X* applies in the metrical corpus, rule *Y* applies in speech.” Both modes are present, both are operative, both are part of the same synchronic framework Pāṇini documents. If Pāṇini had thought Vedic was an evolutionary predecessor of Classical, he would have used temporal markers (*pūrvam*, *prāk*); he marks the distinction categorically.

The orthodoxy’s “two versions of Sanskrit” claim — *Vedic Sanskrit and Classical Sanskrit are two different languages* — is therefore inconsistent with Pāṇini’s own framework as documented in the *Aṣṭādhyāyī*. The empirical disproof of the two-versions claim sits inside the very text the orthodoxy treats as the codification event.

Source: Pāṇini, *Aṣṭādhyāyī* (Bohtlingk 1839–40 critical edition; Vasu 1891 English translation); Cardona 1976, *Pāṇini: A Survey of Research* (Mouton), ch. 3 (the *bhāṣāyām* / *chandasi* register distinction); S. D. Joshi & J. A. F. Roodbergen, *The Aṣṭādhyāyī of Pāṇini* (Sahitya Akademi / Sahitya Akademi translation-commentary series, 1991–).

three-deployments-framework

Short: The book’s three-deployments framework names three successive forms in which the same engineered architecture has been transmitted: (1) **Corpus form** — Wave 1 (pre-Pāṇinian); Sanskrit as the *Vedas* perform it, engineered into every recitation rule and preservation form, operative in the corpus the calibration matrix preserves; (2) **Documented form** — Wave 2 (post-Pāṇinian); Sanskrit as Pāṇini’s *Aṣṭādhyāyī* makes it explicit, the *Trimuni Vyākaraṇam* as the methodological framework civilizations across the world imitated; (3) **Recovered form** — Wave 3 (contemporary); the engineered Sanskrit thesis stated in a register the modern academy can read, the book itself a Wave 3 instrument. *Not three codifications — successive*

deployments of the same architecture across different audiences and registers. Architecture constant; form (corpus / document / recovery) varies.

Deployments: Ch19 §19.4 (the *Wave 3 — Forward-Pointer* section; the *three-deployments* framework that replaced the earlier *three-codifications* framing in Session 10).

The book's three-deployments framework names three successive forms in which the same engineered architecture has been transmitted across the depth of time:

1. ***Corpus form*** — Wave 1 (pre-Pāṇinian). Sanskrit as the *Vedas* perform it: implicit but operative, engineered into every recitation rule and every preservation form. The architecture is present in the corpus the calibration matrix preserves; the engineering is not yet stated as explicit *sūtras*, but it is operative in the form of the corpus itself.
2. ***Documented form*** — Wave 2 (post-Pāṇinian). Sanskrit as Pāṇini's *Aṣṭādhyāyī* makes it explicit: the engineered architecture restated as explicit *sūtras*, the *Trimuni Vyākaraṇam* as the methodological framework civilizations across the world imitated. The architecture moves from corpus-implicit to *sūtra*-explicit; the documenter discipline (*vaiyākaraṇāḥ*) writes it down for the first time in formal register.
3. ***Recovered form*** — Wave 3 (contemporary). The engineered Sanskrit thesis stated in a register the modern academy can read. The architecture, having been obscured by the Western philological orthodoxy's *codification* vocabulary across the past century and a half, is recovered in the language the contemporary global discourse can engage. The book itself is a Wave 3 instrument.

The three deployments are not three successive *codifications* (the earlier framing the chapter used) — the word *codification* implies a transition from drifting-before to structured-after, which the engineering

thesis specifically denies. The three forms are *successive deployments of the same engineered architecture* across different audiences and registers. The architecture itself is constant; the form it takes (corpus, document, recovery) varies.

The framework is internal to the book's own polemic and does not have an external scholarly source. Its closest precedents are Sanskrit's own categorical distinctions: *śruti* (the corpus form, *that which is heard*) and *smṛti* (the remembered form), with the contemporary recovery as a third register the paramparā has not previously had occasion to name.

Source: Internal to the book — Ch19 §19.4 names the framework; Ch1 §1.1 names the *codification* contest and lands the four-term polemic stack; Claim #2 deploys the standing polemic phrase *Sanskrit was engineered; encoded in the Vedas; decoded by many; Pāṇini's decoding is the finest* across the three-deployments arc.
